MYP

Mathematics

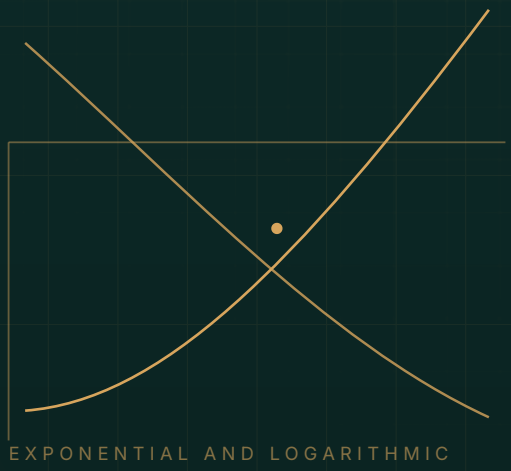
5 STUDENT COURSEBOOK

MYP 5 · YEAR 5 · TWELVE CHAPTERS AND A MODELLING CAPSTONE IN FIVE PARTS

A concept-driven inquiry course in algebra, functions, sequences, inequalities, geometry, vectors, statistics and probability. You investigate, justify and model, then reason, communicate and evaluate at the level the transition to DP Mathematics asks of you.



POLYNOMIALS, THE FACTOR THEOREM AND ALGEBRAIC FRACTIONS · FUNCTIONS: TRANSFORMATIONS, COMPOSITES AND INVERSES · EXPONENTIAL AND LOGARITHMIC FUNCTIONS · SEQUENCES AND SERIES · QUADRATIC INEQUALITIES AND LINEAR PROGRAMMING · CIRCLE GEOMETRY AND GEOMETRIC PROOF · SINE AND COSINE RULES; TRIGONOMETRY IN THREE DIMENSIONS · VECTORS AND MATRICES · STATISTICS: DISPERSION, CORRELATION AND REGRESSION · PROBABILITY: CONDITIONAL PROBABILITY AND COUNTING · CAPSTONE: A MATHEMATICAL MODELLING PROJECT · CUMULATIVE REVISION AND THE TRANSITION TO DP



HOW THIS BOOK WORKS

Every chapter opens with a task. You investigate, notice a pattern, state a mathematical conclusion and justify it before you meet the formal explanation.

MYP ASSESSMENT CRITERIA

CRITERION A	Knowing and understanding
CRITERION B	Investigating patterns
CRITERION C	Communicating
CRITERION D	Applying mathematics in real-life contexts

SADIKOU MOUKODDIM

MYP 5 · STUDENT COURSEBOOK

AUTHOR SADIKOU MOUKODDIM

About this edition

This is the second edition of the MYP 5 mathematics coursebook. It was rebuilt from the pre-rewrite MYP 5 audit, which recorded the chapter scope, the key and related concepts, the global contexts and the inquiry statements of the earlier edition, and identified where the teaching had to be strengthened.

The edition keeps that chapter scope and the visual identity of the collection, and rebuilds the teaching sequence around inquiry: engagement, inquiry questions, investigation, observation, mathematical conclusion, formal development, worked examples, extensive visual practice, and application, assessment and reflection. The register is followed section by section, so the sine and cosine rules, bearings, matrices and determinants, dispersion and standard deviation, and counting with permutations and combinations are all taught here rather than deferred.

The earlier edition carried four confirmed defects, and all four are repaired in the student text rather than only in the answer key. The sine rule's ambiguous case is corrected: two triangles occur when $b \sin A < a < b$, exactly one when $b \sin A = a$, and none when $a < b \sin A$. The matrix product that the capstone asked students to compute had its order mismatched with the question, and is now consistent. The open-top cylinder problem stated its constraint ambiguously, and now states which dimension is fixed. The raw LaTeX that appeared in the Chapter 11 modelling text is gone; every expression in this book is typeset.

All classroom data sets are simulated and are labelled as such. Real data would require provenance, so no claim in this book rests on unverified statistics.

No publisher, ISBN, print specification or commercial print-readiness claim is made in this file. The book is produced as an A4 review copy: fonts are embedded, pagination is continuous and every page has been checked for overflow.

MYP 5 · STUDENT COURSEBOOK

How to use this book

Each chapter follows the same sequence, so you always know what is coming next. The sequence is deliberate: you investigate before you are told the rule, because a rule you have discovered is a rule you can defend — and defending it is what Year 5 asks of you.

The chapter sequence

1. **Engagement.** A short situation that raises the question the chapter will answer.
2. **Inquiry questions.** Factual, conceptual and debatable questions that frame the chapter.
3. **Investigation.** A task you complete before the formal teaching: record evidence in a table and look for a pattern.
4. **Observation and discovery.** What you noticed, stated in your own words and compared with a partner.
5. **Mathematical conclusion.** The rule that the evidence supports, written as a general statement you can justify.
6. **Formal development.** The explanation, definitions and notation, with the reasoning made explicit.
7. **Worked examples.** Problem, visual representation, mathematical reasoning, step-by-step solution, final answer and interpretation.
8. **Practice.** Separated into Criterion A, B, C and D practice, with visual representations throughout.
9. **Application, assessment and reflection.** Real-context tasks, criterion-based assessment practice, and a reflection that returns to the statement of inquiry.

MYP 5 · STUDENT COURSEBOOK

How to use this book

The four MYP mathematics criteria

A · Knowing and understanding

Apply knowledge to familiar and unfamiliar problems, and check results.

B · Investigating patterns

Investigate relationships, describe patterns, formulate a conjecture, test it and justify the generalisation — at Year 5 level a justification is a general argument, not a list of cases.

C · Communicating

Use correct mathematical language, notation and complete representations, and organise reasoning into a coherent structure that a reader can follow.

D · Applying mathematics in real-life contexts

Apply mathematics to authentic situations, identify relevant information, and comment on the accuracy and sense of the result.

Working with this book

- Practice is separated by criterion so that you can see which skill a task is training. Criterion B practice is additional genuine pattern investigation, not a repeat of the teaching.
- Visuals are part of the mathematics. Where a task supplies a diagram, grid, table, number line or model, use it; where it asks you to sketch, your drawing is part of the answer.
- Answers and student-facing reasoning are collected at the back of the book, with a formula bank, a glossary and references.
- Every chapter ends with project connections: a Service as Action opportunity, a Community Project connection and a Personal Project connection for MYP 5.
- Chapter 11 is the capstone: a single sustained modelling task that uses mathematics from every part of the year. Chapter 12 adds no new content; it revises what you know and states what the transition to DP Mathematics expects.

MYP 5 · STUDENT COURSEBOOK

MYP mathematics and this coursebook

The IB Middle Years Programme asks students to inquire into concepts through global contexts, to develop approaches to learning (ATL) skills, and to act on what they learn through service as action. In mathematics this means more than practising procedures: students investigate patterns, justify generalisations, communicate with precise language and apply mathematics to situations that matter to them.

MYP mathematics is organised into four branches — number, algebra, geometry and trigonometry, and statistics and probability — and the programme recommends at least 50 hours of teaching per subject each year. This coursebook covers the Year 5 content of those branches across twelve chapters, grouped into five parts.

Year 5 is the extended-leaning level and the last year of the programme. The register for this year records the same four branches as every other year, and the chapters here follow the branch recorded for each of them: number and abstract reasoning, thinking with models, spatial reasoning and reasoning with data, with the modelling capstone and the transition to DP Mathematics at the end.

Assessment in MYP mathematics uses four equally weighted criteria, each with a maximum of 8: A knowing and understanding, B investigating patterns, C communicating, and D applying mathematics in real-life contexts. This book trains each criterion explicitly, and says so on the page where the training happens.

MYP 5 · STUDENT COURSEBOOK

MYP mathematics and this coursebook

Projects, service and action

Service as Action

Each chapter includes a service opportunity with a community need, a mathematical connection, a proposed action, an expected impact and a student reflection. Completing an exercise is not service: service requires action that benefits others and reflection on that action.

Community Project

MYP students completing the programme in Year 3 or Year 4 complete a community project. Each chapter offers a student-led project idea with a research question, the mathematics involved, an action or outcome, beneficiaries and evidence. Follow your school's requirements and supervision.

Personal Project (MYP 5)

Students completing MYP in Year 5 complete a personal project. Each chapter offers an adaptable MYP 5-oriented extension with a personal interest, a learning goal, a product, the mathematics involved, a process and evidence. It is deliberately different from the community project and from service as action.

What this book does not claim

- Programme requirements quoted here follow the official IB MYP curriculum, mathematics and projects pages and the MYP general regulations. Everything else — the chapter sequence, the investigations, the practice sets and the project ideas — is author-designed material built to serve those requirements, and is not an IB publication.
- No ISBN, publisher, imprint or print specification is claimed. The file is an A4 review copy with embedded fonts and continuous pagination, produced for proofing and classroom review rather than for commercial print production.

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Application and assessment · Review · Projects and service

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Application and assessment · Review · Projects and service

Every chapter follows the same sequence: engagement, inquiry questions, investigation, observation, mathematical conclusion, formal development, worked examples, Criterion A–D practice, application, assessment and reflection, and project connections.

CHAPTER 1 · PART I · NUMBER AND ABSTRACT REASONING

01

Polynomials, Factor Theorem and Algebraic Fractions

A polynomial can be written as a sum of powers or as a product of factors. Each form is the same polynomial, and each makes a different question easy.

ENGAGE

Two students are asked to describe $x^3 - 7x - 6$. One writes it as a sum of three terms; the other writes $(x - 3)(x + 1)(x + 2)$. Neither has changed the polynomial. Which description tells you where the graph crosses the x-axis, and which tells you how steep it is?

Framing the inquiry

Key concept: Form
Related concepts: Equivalence · Generalisation · Simplification
Global context: Personal and cultural expression
Global context exploration: the language of algebra as a way of making structure visible

Statement of inquiry: The structure of a polynomial — its factors, roots and remainders — determines both what it can be simplified to and what its graph will do.

Questions to guide us

- **Factual:** What is the degree of a polynomial, and what do the remainder and factor theorems say?
- **Conceptual:** Why does the factored form of a polynomial reveal its roots, and what does an excluded value protect a simplification from?
- **Debatable:** When a calculation produces a solution that the original equation rejects, has the method failed, or has it done its job?

ATL · CRITICAL THINKING. In order for a student to choose the most useful form of an algebraic expression, students must compare an expanded form with a factorised form of the same polynomial, identify what each form reveals, and justify the choice of form for a stated purpose.

CHAPTER 1 · INVESTIGATION 1.1

Polynomials and their behaviour

INQUIRY QUESTION

What can be read off a polynomial from the form it is written in?

YOUR TASK

Take four polynomials of your own: one of degree 1, one of degree 2, one of degree 3 and one you are unsure about. Add and multiply each pair, record the degree of each result, and then try to state a rule that predicts the degree of the answer before you calculate it.

first	degree	second	degree	operation	result degree
$2x + 1$	1	$x^2 - 3$	2	add	2
$2x + 1$	1	$x^2 - 3$	2	multiply	3
$x^3 + x$	3	$x - 1$	1	add	3
$x^3 + x$	3	$x - 1$	1	multiply	4
$x^2 + 1$	2	$x^2 - 1$	2	multiply	4
5	0	$x^4 + x$	4	multiply	4

CHAPTER 1 · INVESTIGATION 1.1

Polynomials and their behaviour

What do you notice?

- Adding two polynomials gave the larger of the two degrees, unless the leading terms cancelled.
- Multiplying two polynomials always gave the sum of the two degrees.
- A constant has degree 0 and left every degree unchanged when it was multiplied in.
- The only surprise came from addition, where cancellation can lower the degree.

> **Observation.** The degree of a product is the sum of the degrees, and the degree of a sum is at most the larger of the two degrees.

MATHEMATICAL CONCLUSION

The polynomials are closed under addition, subtraction and multiplication, and multiplication of degrees is exact: the degree of a product is the sum of the degrees of the factors. This is why a factorised form shows the roots while an expanded form shows the degree.

Your own words

CHAPTER 1 · 1.1

Polynomials and their behaviour

A polynomial in x is a sum of terms of the form ax^N where a is a real coefficient and n is a non-negative integer. The degree is the highest power present, and the leading coefficient is the coefficient of that power. The two numbers together describe how the polynomial behaves for very large and very small x.

Polynomials are closed under addition, subtraction and multiplication: performing any of these on two polynomials gives another polynomial. Division is the exception, which is why division is the operation that needs its own technique.

Two polynomials are equal when their corresponding coefficients are equal, not when they look alike. Expanding $(x + 1)(x + 2)$ gives $x^2 + 3x + 2$, and the two forms name the same function. Equivalence of form is the idea that makes factorising useful, and the generalisation behind it — that any polynomial can be rewritten in any equivalent form — is what makes algebra a language rather than a set of tricks.

The standard form of a polynomial lists its terms in descending powers with a single coefficient for each power. Rewriting into standard form is the first step in almost every polynomial calculation, because it makes the degree and the leading coefficient visible.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 1 · 1.1 · WORKED EXAMPLE 1

Polynomials and their behaviour

WORKED EXAMPLE 1.1.1

Problem. Write $3 - 2x^4 + x^2$ in standard form, state its degree and leading coefficient, and find its value at $x = -1$.

form	polynomial	what it shows
as given	$3 - 2x^4 + x^2$	no obvious order
standard	$-2x^4 + x^2 + 3$	degree 4, leading -2
at $x = -1$	$-2(1) + 1 + 3$	value 2

Standard form puts the degree and the leading coefficient on view.

Mathematical reasoning. Standard form orders the terms by descending power, and the degree and leading coefficient are read directly from it.

1. Reorder: $-2x^4 + x^2 + 3$.
2. The highest power is 4, so the degree is 4.
3. The leading coefficient is -2.
4. Substitute $x = -1$: $-2(-1)^4 + (-1)^2 + 3 = -2 + 1 + 3$.
5. The value is 2.

Final answer. $-2x^4 + x^2 + 3$, degree 4, leading coefficient -2, value 2 at $x = -1$

Interpretation. Standard form is not cosmetic: it is the form in which the two numbers that govern the polynomial's behaviour can be read off.

CHAPTER 1 · 1.1 · WORKED EXAMPLE 2

Polynomials and their behaviour

WORKED EXAMPLE 1.1.2

Problem. Show that the product of two polynomials is always a polynomial, and explain why the degree of the product is the sum of the degrees.

factor	degree	contribution
$3x^2 - 1$	2	two powers of x
$x^3 + 5x$	3	three powers of x
product	5	2 + 3

Multiplying adds the powers, so the degrees add.

Mathematical reasoning. Each term of the first polynomial multiplies each term of the second, and every product is again a constant times a power of x.

1. Let the polynomials be p and q with degrees m and n.
2. Every term of p is a constant times x raised to a power i with i at most m, and every term of q is a constant times x raised to a power j with j at most n.
3. A product term is the product of the two constants times x raised to i + j, and i + j is at most m + n.
4. The two leading terms multiply to give a non-zero constant times x raised to m + n, so the degree of the product is exactly m + n.
5. Hence the product is a polynomial of degree m + n.

Final answer. the product is a polynomial of degree m + n

Interpretation. One general argument settles every pair of polynomials, which is why an algebraic proof is worth more than a list of examples.

CHAPTER 1 · INVESTIGATION 1.2

Dividing polynomials

INQUIRY QUESTION

How is one polynomial divided by another, and how is the answer checked?

YOUR TASK

Divide $x^4 - 1$ by $x - 1$, then $x^4 - 1$ by $x + 1$, then $x^4 - 1$ by $x^2 + 1$. Record each quotient and remainder, reconstruct the dividend from your answers, and then explain what the three divisions have in common.

divisor	quotient	remainder	reconstruction
$x - 1$	$x^3 + x^2 + x + 1$	0	$(x - 1)(x^3 + x^2 + x + 1) = x^4 - 1$
$x + 1$	$x^3 - x^2 + x - 1$	0	$(x + 1)(x^3 - x^2 + x - 1) = x^4 - 1$
$x^2 + 1$	$x^2 - 1$	0	$(x^2 + 1)(x^2 - 1) = x^4 - 1$
$x^2 - 1$	$x^2 + 1$	0	$(x^2 - 1)(x^2 + 1) = x^4 - 1$
$x - 2$	$x^3 + 2x^2 + 4x + 8$	15	$(x - 2)(\ldots) + 15 = x^4 - 1$
$x + 2$	$x^3 - 2x^2 + 4x - 8$	17	$(x + 2)(\ldots) + 17 = x^4 - 1$

CHAPTER 1 · INVESTIGATION 1.2

Dividing polynomials

What do you notice?

- Three of the six divisors divided $x^4 - 1$ exactly, leaving no remainder.
- The two divisors that left a remainder were not factors of $x^4 - 1$.
- Every reconstruction reproduced the dividend, which is the check that each division was done correctly.
- The divisors that worked were all built from the factors of $x^4 - 1$.

> **Observation.** A divisor that leaves no remainder is a factor, and the quotient is the other factor.

MATHEMATICAL CONCLUSION

Dividing $p(x)$ by $x - a$ leaves remainder zero exactly when $x - a$ is a factor of $p(x)$. The reconstruction dividend = divisor × quotient + remainder is the check on every division.

Your own words

CHAPTER 1 · 1.2

Dividing polynomials

Polynomial long division follows the same logic as long division of numbers: match the leading term of the divisor to the leading term of the dividend, write the quotient term, subtract, and repeat on what is left.

The division identity is dividend = divisor × quotient + remainder, with the remainder either zero or of lower degree than the divisor. This identity is not a convention; it is the statement that the division has been done correctly.

A missing power in the dividend must be kept as a zero placeholder. Dividing $x^3 + 1$ by $x^2 + 1$ requires a $0x$ term, and omitting it is the commonest source of error.

Synthetic division is a compressed arrangement of the same process for a divisor of the form $x - a$. It is faster but it hides the reasoning, so the long division is used first and the short form afterwards.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 1 · 1.2 · WORKED EXAMPLE 1

Dividing polynomials

WORKED EXAMPLE 1.2.1

Problem. Divide $2x^3 + 3x^2 - x + 5$ by $x + 2$, and check the answer.

step	working	remainder
lead	$2x^3 \div x = 2x^2$	$2x^3 + 4x^2$
subtract	$(2x^3 + 3x^2) - (2x^3 + 4x^2)$	$-x^2$
next	$-x^2 \div x = -x$	$-x^2 - 2x$
subtract	$(-x^2 - x) - (-x^2 - 2x)$	x
last	$x \div x = 1$	$x + 2$
subtract	$(x + 5) - (x + 2)$	3

Each step matches the leading term and subtracts.

Mathematical reasoning. Match leading terms, subtract, and repeat until what is left has lower degree than the divisor; then verify with the division identity.

1. $2x^3 \div x = 2x^2$. Multiply $x + 2$ by $2x^2$ to get $2x^3 + 4x^2$ and subtract: $(2x^3 + 3x^2) - (2x^3 + 4x^2) = -x^2$.
2. Bring down $-x$. Now $-x^2 \div x = -x$. Multiply $x + 2$ by $-x$ to get $-x^2 - 2x$ and subtract: $(-x^2 - x) - (-x^2 - 2x) = x$.
3. Bring down $+5$. Now $x \div x = 1$. Multiply $x + 2$ by 1 and subtract: $(x + 5) - (x + 2) = 3$.
4. The quotient is $2x^2 - x + 1$ and the remainder is 3.
5. Check: $(x + 2)(2x^2 - x + 1) + 3 = 2x^3 - x^2 + x + 4x^2 - 2x + 2 + 3 = 2x^3 + 3x^2 - x + 5$.

Final answer. quotient $2x^2 - x + 1$, remainder 3

Interpretation. The division identity is the check: if the reconstruction gives back the dividend, the division was correct.

CHAPTER 1 · 1.2 · WORKED EXAMPLE 2

Dividing polynomials

WORKED EXAMPLE 1.2.2

Problem. Divide $x^3 + 8$ by $x^2 + 2x + 1$, keeping every placeholder.

dividend terms	x^3	$0x^2$	$0x$	8
divisor	x^2	$+2x$	$+1$	—
quotient	x	-2	—	—
remainder	0	$3x$	$+10$	—

The zero placeholder keeps the powers aligned.

Mathematical reasoning. A missing power is a zero term, and writing it keeps the subtraction aligned.

1. Write the dividend as $x^3 + 0x^2 + 0x + 8$.
2. $x^3 \div x^2 = x$. Subtract $x(x^2 + 2x + 1) = x^3 + 2x^2 + x$ to get $-2x^2 - x + 8$.
3. $-2x^2 \div x^2 = -2$. Subtract $-2(x^2 + 2x + 1) = -2x^2 - 4x - 2$ to get $3x + 10$.
4. $3x$ has degree 1, below the divisor's degree 2, so the division stops.
5. Quotient $x - 2$, remainder $3x + 10$. Check: $(x^2 + 2x + 1)(x - 2) + 3x + 10 = x^3 + 2x^2 + x - 2x^2 - 4x - 2 + 3x + 10 = x^3 + 8$.

Final answer. quotient $x - 2$, remainder $3x + 10$

Interpretation. Zero placeholders are not decoration: without them the powers drift and the subtraction goes wrong.

CHAPTER 1 · INVESTIGATION 1.3

The remainder and factor theorems

INQUIRY QUESTION

What does the remainder of a division tell you about the polynomial?

YOUR TASK

Take the polynomial $p(x) = x^3 + 2x^2 - 5x - 6$. Evaluate p at $x = -3, -2, -1, 0, 1, 2$ and 3, record which values are zero, divide out one factor you have found, and then factorise p completely. State how many roots a cubic can have and how many you found.

x	p(x)	zero?	factor found
−3	0	yes	x + 3
−2	4	no	—
−1	2	no	—
0	−6	no	—
1	−8	no	—
2	0	yes	x − 2
3	24	no	—

CHAPTER 1 · INVESTIGATION 1.3

The remainder and factor theorems

What do you notice?

- Two of the seven values of x made p(x) exactly zero.
- Those two values gave the two factors x + 3 and x − 2.
- A cubic has at most three roots, so one root was still missing.
- Dividing by one factor lowered the degree, making the rest easier.

Observation. A root of the polynomial is a value that makes it zero, and each root gives a linear factor.

MATHEMATICAL CONCLUSION

The factor theorem is a generalisation: p(a) = 0 exactly when x − a is a factor of p(x), for every polynomial p and every number a. Finding roots and factorising are therefore the same activity. A cubic has at most three roots, and here p(x) = (x + 3)(x − 2)(x + 1) with roots −3, −1 and 2.

Your own words

CHAPTER 1 · 1.3

The remainder and factor theorems

The remainder theorem states that when p(x) is divided by x − a, the remainder is p(a). Evaluating the polynomial at a single number replaces the whole division.

The factor theorem is the special case: x − a is a factor of p(x) exactly when p(a) = 0. So a is a root of the polynomial exactly when x − a divides it exactly. This is the generalisation that turns a search for roots into a search for factors.

A polynomial of degree n has at most n real roots, and if it has n distinct roots then it has exactly n linear factors. Counting roots therefore counts factors, which is how a cubic is factorised completely.

The theorems work together: find one root by inspection, divide out the corresponding factor, and the quotient is a polynomial of one degree lower, which is easier to solve. This is the standard route to solving a cubic or quartic.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 1 · 1.3 · WORKED EXAMPLE 1

The remainder and factor theorems

WORKED EXAMPLE 1.3.1

Problem. Find the remainder when $p(x) = x^3 - 4x^2 + 5x - 7$ is divided by $x - 2$, and hence decide whether $x - 2$ is a factor.

a	p(a)	remainder on dividing by x − a	factor?
1	−5	−5	no
2	−5	−5	no
3	−1	−1	no
−1	−17	−17	no

The remainder is the value of the polynomial at a.

Mathematical reasoning. By the remainder theorem the remainder is $p(a)$, so evaluate rather than divide.

1. The divisor is $x - 2$, so $a = 2$.
2. $p(2) = 2^3 - 4(2^2) + 5(2) - 7 = 8 - 16 + 10 - 7$.
3. $8 - 16 = -8$, $-8 + 10 = 2$, $2 - 7 = -5$.
4. The remainder is -5.
5. Since $p(2) \neq 0$, $x - 2$ is not a factor of $p(x)$.

Final answer. remainder −5, so x − 2 is not a factor

Interpretation. One substitution answers a question that would otherwise need a whole division, and the same substitution also tests whether the divisor is a factor.

CHAPTER 1 · 1.3 · WORKED EXAMPLE 2

The remainder and factor theorems

WORKED EXAMPLE 1.3.2

Problem. Show that $x - 1$ is a factor of $p(x) = x^3 - 6x^2 + 11x - 6$, find the other factors, and hence solve $p(x) = 0$.

x	p(x)	comment
0	−6	negative
1	0	root, so x − 1 is a factor
2	0	root, so x − 2 is a factor
3	0	root, so x − 3 is a factor

Three roots for a cubic means three linear factors.

Mathematical reasoning. Find one root, divide out its factor, then solve the quadratic that is left.

1. $p(1) = 1 - 6 + 11 - 6 = 0$, so $x - 1$ is a factor.
2. Divide $p(x)$ by $x - 1$: the quotient is $x^2 - 5x + 6$.
3. Check: $(x - 1)(x^2 - 5x + 6) = x^3 - 6x^2 + 11x - 6$.
4. Factorise the quadratic: $x^2 - 5x + 6 = (x - 2)(x - 3)$.
5. So $p(x) = (x - 1)(x - 2)(x - 3)$, and the roots are $x = 1$, 2 and 3.

Final answer. $p(x) = (x - 1)(x - 2)(x - 3)$; roots $x = 1, 2, 3$

Interpretation. The factor theorem turns a cubic into a quadratic, and a quadratic is already solved.

CHAPTER 1 · INVESTIGATION 1.4

Algebraic fractions

INQUIRY QUESTION

When may an algebraic fraction be cancelled, and what must be stated first?

YOUR TASK

Take the four expressions $(x^2 - 9)/(x - 3)$, $(x - 3)/(x^2 - 9)$, $1/(x + 3) + 1/(x - 3)$ and $(x + 3 + x - 3)/(x + 3)$. Simplify each, state the excluded values, and then explain why two of them give the same answer and one of them cannot be simplified at all.

expression	simplified	excluded values	same as
$(x^2 - 9)/(x - 3)$	$x + 3$	$x \neq 3$	—
$(x - 3)/(x^2 - 9)$	$1/(x + 3)$	$x \neq 3, -3$	third row, left
$1/(x + 3) + 1/(x - 3)$	$2x/(x^2 - 9)$	$x \neq 3, -3$	—
$(x + 3 + x - 3)/(x + 3)$	$2x/(x + 3)$	$x \neq -3$	—

CHAPTER 1 · INVESTIGATION 1.4

Algebraic fractions

What do you notice?

- Two of the four simplified to expressions with the same denominator.
- The reciprocal of a simplified fraction is the fraction with numerator and denominator swapped.
- The last expression could not be cancelled at all, because $x + 3$ is not a factor of $2x$.
- Every excluded value came from a factor of a denominator.

> **Observation.** Cancellation needs a common factor; without one, the fraction is already in its simplest form.

MATHEMATICAL CONCLUSION

An algebraic fraction is simplified by cancelling common factors and nothing else, and the excluded values are exactly the values that made the cancelled factors zero. Stating them is part of the answer.

Your own words

CHAPTER 1 · 1.4

Algebraic fractions

An algebraic fraction is cancelled by dividing the numerator and the denominator by a common factor. Cancelling a common term is not cancellation: $(x + 3)/x$ does not become 3, because x is not a factor of $x + 3$.

Before any cancellation, the values that make the denominator zero must be excluded. The fraction $(x^2 - 1)/(x - 1)$ simplifies to $x + 1$ only for $x \neq 1$, and the excluded value is part of the answer, not a footnote.

Multiplying fractions multiplies numerators and denominators; adding them requires a common denominator. The two operations are not interchangeable, and the commonest error is to add numerators and denominators separately.

Simplifying a complex fraction means simplifying the numerator and the denominator first, then dividing. The division is multiplication by the reciprocal, and the reciprocal of a sum is not the sum of the reciprocals.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 1 · 1.4 · WORKED EXAMPLE 1

Algebraic fractions

WORKED EXAMPLE 1.4.1

Problem. Simplify $(x^2 - 5x + 6)/(x^2 - 4)$, stating the excluded values.

expression	factorised	cancelled	left
numerator	$(x - 2)(x - 3)$	$\div (x - 2)$	$x - 3$
denominator	$(x - 2)(x + 2)$	$\div (x - 2)$	$x + 2$
excluded	$x = 2$ and $x = -2$	—	$x \neq 2, -2$

Cancel common factors, and state what was cancelled away.

Mathematical reasoning. Factorise both parts, cancel the common factor, and record the values that made the cancelled factor zero.

1. Factorise: $x^2 - 5x + 6 = (x - 2)(x - 3)$ and $x^2 - 4 = (x - 2)(x + 2)$.
2. The denominator is zero when $x = 2$ or $x = -2$, so both are excluded.
3. Cancel the common factor $x - 2$: the fraction becomes $(x - 3)/(x + 2)$.
4. State the result as $(x - 3)/(x + 2)$ for $x \neq 2$ and $x \neq -2$.

Final answer. $(x - 3)/(x + 2)$, for $x \neq 2$ and $x \neq -2$

Interpretation. The excluded values are the values the cancellation removed, and stating them is what makes the simplification honest.

CHAPTER 1 · 1.4 · WORKED EXAMPLE 2

Algebraic fractions

WORKED EXAMPLE 1.4.2

Problem. Simplify $2/(x + 1) + 3/(x - 1)$ as a single fraction.

fraction	× (x − 1)	× (x + 1)	combined
$2/(x + 1)$	$2(x - 1)$	$2(x - 1)$	$2x - 2$
$3/(x - 1)$	$3(x + 1)$	$3(x + 1)$	$3x + 3$
sum	—	—	$(5x + 1)/(x^2 - 1)$

A common denominator, then add the numerators.

Mathematical reasoning. Build a common denominator by multiplying each fraction by the missing factor, then add.

1. The common denominator is $(x + 1)(x - 1) = x^2 - 1$.
2. $2/(x + 1) = 2(x - 1)/(x^2 - 1) = (2x - 2)/(x^2 - 1)$.
3. $3/(x - 1) = 3(x + 1)/(x^2 - 1) = (3x + 3)/(x^2 - 1)$.
4. Add: $(2x - 2 + 3x + 3)/(x^2 - 1) = (5x + 1)/(x^2 - 1)$.
5. The excluded values are $x = 1$ and $x = -1$.

Final answer. $(5x + 1)/(x^2 - 1)$, for $x \neq 1$ and $x \neq -1$

Interpretation. Adding fractions is never adding numerators and denominators: it is rewriting both over a common denominator first.

CHAPTER 1 · INVESTIGATION 1.5

Rational equations and applications

INQUIRY QUESTION

Why can solving a rational equation produce an answer the equation rejects?

YOUR TASK

Solve $(2)/(x) + (3)/(x + 1) = 1$. Before you clear the fractions, write down the values x may not take. Solve, then substitute every candidate back into the original equation and say which are genuine and which are extraneous.

step	working	result
excluded	$x \neq 0$ and $x \neq -1$	—
clear	$2(x + 1) + 3x = x(x + 1)$	—
expand	$5x + 2 = x^2 + x$	—
rearrange	$x^2 - 4x - 2 = 0$	—
solve	$x = 2 \pm \sqrt{6}$	$2 + \sqrt{6} \approx 4.45$, $2 - \sqrt{6} \approx -0.45$
check first	$2/4.45 + 3/5.45 \approx 0.449 + 0.550$	≈ 0.999
check second	$2/(-0.45) + 3/0.55 \approx -4.44 + 5.45$	≈ 1.01

CHAPTER 1 · INVESTIGATION 1.5

Rational equations and applications

What do you notice?

- Both candidates were allowed, because neither is 0 nor −1.
- Both candidates satisfied the original equation when substituted.
- The equation became a quadratic after clearing, so two answers were possible.
- Had a candidate been 0 or −1, it would have been extraneous.

> **Observation.** Clearing the fractions can introduce candidates that the original equation rejects, and only substitution finds them.

MATHEMATICAL CONCLUSION

A rational equation is solved by clearing the fractions, but every candidate must be tested in the original equation: a value that makes a denominator zero is extraneous and is discarded.

Your own words

CHAPTER 1 · 1.5

Rational equations and applications

A rational equation is solved by multiplying both sides by a common denominator to clear the fractions. The multiplication is only valid for values that do not make that denominator zero.

A solution produced by this method must be substituted back into the original equation. If it makes a denominator zero, it is extraneous: it belongs to the cleared equation but not to the original one.

Applications fall into two families: rate problems, where the rates add, and work problems, where the times add reciprocally. In both, the equation is built from the meaning of the quantities, not from a pattern.

Every application ends with a check for sense. A negative time, a fraction of a person or a speed that exceeds the stated limit are all signals that the model or the algebra needs revisiting.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 1 · 1.5 · WORKED EXAMPLE 1

Rational equations and applications

WORKED EXAMPLE 1.5.1

Problem. Solve $(x + 1)/(x - 2) = 3$, and check for extraneous solutions.

step	working	note
clear	$(x + 1) = 3(x - 2)$	multiply by $x - 2$
expand	$x + 1 = 3x - 6$	—
collect	$7 = 2x$	—
solve	$x = 3.5$	candidate
check	denominator $x - 2 = 1.5 \neq 0$	accepted

Clear the fractions, solve, then check the candidate.

Mathematical reasoning. Multiply by the denominator, solve the resulting linear equation, then test the candidate in the original equation.

1. The denominator is zero when $x = 2$, so $x \neq 2$.
2. Multiply both sides by $x - 2$: $x + 1 = 3(x - 2)$.
3. Expand: $x + 1 = 3x - 6$, so $7 = 2x$ and $x = 3.5$.
4. Check: $x - 2 = 1.5$, which is not zero, so the solution is valid.
5. Verify in the original: $(3.5 + 1)/(3.5 - 2) = 4.5/1.5 = 3$.

Final answer. x = 3.5, valid

Interpretation. The check is not a formality: it is the step that separates a solution of the original equation from a solution of the cleared one.

CHAPTER 1 · 1.5 · WORKED EXAMPLE 2

Rational equations and applications

WORKED EXAMPLE 1.5.2

Problem. Two pipes fill a tank in 4 hours and 6 hours respectively. Working together, how long do they take?

pipe	time (h)	rate (tanks per hour)
first	4	1/4
second	6	1/6
together	t	1/4 + 1/6 = 5/12
answer	12/5 = 2.4	—

Rates add; times do not.

Mathematical reasoning. Rates add, so the combined rate is the sum of the individual rates, and the time is its reciprocal.

1. The first pipe fills 1/4 of the tank per hour and the second 1/6.
2. Together they fill 1/4 + 1/6 = 3/12 + 2/12 = 5/12 per hour.
3. If the time is t hours, then (5/12)t = 1.
4. So t = 12/5 = 2.4 hours.
5. Check: 2.4/4 + 2.4/6 = 0.6 + 0.4 = 1 tank.

Final answer. 2.4 hours, that is 2 hours 24 minutes

Interpretation. Adding the times would give 10 hours, which is longer than either pipe alone and is obviously wrong: rates add, times do not.

CHAPTER 1 · INVESTIGATION 1.6

Investigation: factors, roots and sign

INQUIRY QUESTION

How does the factored form of a polynomial decide where it is positive?

YOUR TASK

Take $p(x) = (x + 3)(x - 1)^2(x - 5)$. Draw a sign line, test one value in each interval, and solve $p(x) > 0$ and $p(x) \le 0$. Then sketch the graph from the factored form and mark which roots are crossings and which are touchings.

interval	$x + 3$	$(x - 1)^2$	$x - 5$	product
$x < -3$	−	+	−	+
$-3 < x < 1$	+	+	−	−
$1 < x < 5$	+	+	−	−
$x > 5$	+	+	+	+

CHAPTER 1 · INVESTIGATION 1.6

Investigation: factors, roots and sign

What do you notice?

- The squared factor was positive in every interval, so it never changed the sign.
- The sign changed at $x = -3$ and at $x = 5$, both simple roots.
- The sign did not change at $x = 1$, the double root.
- The product was negative on two separate intervals that met at $x = 1$.

> **Observation.** Only simple factors change sign; a factor with an even power leaves the sign unchanged.

MATHEMATICAL CONCLUSION

The factored form of a polynomial decides its sign completely: mark the roots, count the negative factors in each interval, and the inequality is solved. A double root is a touching point and is included when the inequality is not strict. Here $p(x) > 0$ gives $x < -3$ or $x > 5$, and $p(x) \le 0$ gives $-3 \le x \le 5$ with $x = 1$ included.

Your own words

CHAPTER 1 · 1.6

Investigation: factors, roots and sign

The sign of a polynomial between its roots is constant, because a polynomial can change sign only by passing through zero. This is why a sign line is a complete method for solving a polynomial inequality.

For a product of factors, the sign is decided by counting the negative factors. Each simple factor changes sign as x passes its root; a factor with an even power does not.

Solving p(x) > 0 means finding the intervals on which the count of negative factors is even. The roots themselves are excluded when the inequality is strict.

A sketch of the graph confirms the algebraic answer: the curve crosses the axis at a simple root and touches it at a double root, and the sign pattern follows the crossing behaviour.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 1 · 1.6 · WORKED EXAMPLE 1

Investigation: factors, roots and sign

WORKED EXAMPLE 1.6.1

Problem. Solve $(x - 1)(x + 2)(x - 4) > 0$.

interval	$x - 1$	$x + 2$	$x - 4$	product
$x < -2$	−	−	−	−
$-2 < x < 1$	−	+	−	+
$1 < x < 4$	+	+	−	−
$x > 4$	+	+	+	+

Three simple roots give four sign intervals.

Mathematical reasoning. Mark the roots on a sign line, test one value in each interval, and read off where the product is positive.

1. The roots are $x = -2$, 1 and 4, and they divide the line into four intervals.
2. For $x < -2$: all three factors are negative, so the product is negative.
3. For $-2 < x < 1$: only $x + 2$ is positive, so the product is positive.
4. For $1 < x < 4$: only $x - 4$ is negative, so the product is negative.
5. For $x > 4$: all three are positive, so the product is positive.
6. The solution is $-2 < x < 1$ or $x > 4$, with the roots excluded.

Final answer. $-2 < x < 1$ or $x > 4$

Interpretation. One sign line settles every polynomial inequality, however many factors there are.

CHAPTER 1 · 1.6 · WORKED EXAMPLE 2

Investigation: factors, roots and sign

WORKED EXAMPLE 1.6.2

Problem. Solve $(x + 1)^2(x - 3) \geq 0$, and explain the difference from the previous example.

x	$x + 1$	$(x + 1)^2$	$x - 3$	product
-2	-1	1	-5	-5
-1	0	0	-4	0
0	1	1	-3	-3
3	4	16	0	0
4	5	25	1	25

A squared factor never changes sign.

Mathematical reasoning. A squared factor is never negative, so only the simple factor decides the sign, and the double root is included when the inequality is not strict.

1. $(x + 1)^2 \geq 0$ for every x, and it is zero only at $x = -1$.
2. So the sign of the product is the sign of $x - 3$, except at $x = -1$ where the product is zero.
3. For $x > 3$ the product is positive; for $x < 3$ it is negative.
4. At $x = 3$ the product is zero, and at $x = -1$ it is also zero.
5. The solution of ≥ 0 is $x = -1$ or $x \geq 3$.

Final answer. $x = -1$ or $x \geq 3$

Interpretation. An even power does not change sign at its root, so the graph touches the axis instead of crossing it, and that root is included.

CHAPTER 1 · INVESTIGATION 1.7

Mixed practice

INQUIRY QUESTION

Which form of each expression makes the next step easiest?

YOUR TASK

Choose a method for each of these four tasks and carry it out: divide $x^4 - 16$ by $x^2 - 4$; find the remainder when $x^4 - 16$ is divided by $x - 3$; simplify $(x^4 - 16)/(x^2 - 4)$; and solve $(x^4 - 16)/(x^2 - 4) = 5$. For each, write down which tool you used and why.

task	tool used	answer	why that tool
divide $x^4 - 16$ by $x^2 - 4$	long division	$x^2 + 4$, remainder 0	a complete quotient is needed
remainder on dividing by $x - 3$	remainder theorem	65	only one value is needed
simplify $(x^4 - 16)/(x^2 - 4)$	cancel common factors	$x^2 + 4$	the numerator and denominator are already factored
solve $(x^4 - 16)/(x^2 - 4) = 5$	simplify then solve	$x = \pm 1$	clearing fractions after simplifying

CHAPTER 1 · INVESTIGATION 1.7

Mixed practice

What do you notice?

- Two of the four tasks used the same underlying division but asked different questions about it.
- The remainder theorem replaced a whole division with one substitution.
- Simplifying before clearing the fractions made the last equation linear rather than quartic.
- The excluded values $x = 2$ and $x = -2$ mattered only in the last two tasks.

Observation. The tool is chosen by the question, not by habit: a full quotient needs division, a single remainder needs substitution, and a fraction needs cancellation.

MATHEMATICAL CONCLUSION

The four tools of this chapter are interchangeable views of one idea — dividing $p(x)$ by $x - a$ — and choosing the right view is what makes the work short. $x^4 - 16 = (x^2 - 4)(x^2 + 4)$, so the quotient is $x^2 + 4$, $p(3) = 81 - 16 = 65$, and $(x^4 - 16)/(x^2 - 4) = 5$ gives $x^2 + 4 = 5$, so $x = \pm 1$.

Your own words

CHAPTER 1 · 1.7

Mixed practice

Choosing a form is a skill in itself. Before calculating, decide whether the expanded form, the factored form or the single-fraction form will answer the question with the least work.

The four tools of this chapter — long division, the remainder theorem, the factor theorem and cancellation of common factors — are selected by the shape of the question. A single evaluation calls for the remainder theorem; a complete factorisation calls for division.

Every rational equation ends with a substitution check, and every simplification ends with a statement of excluded values. These two habits are what separate a correct method from a correct answer.

Mixed practice is where the tools are chosen rather than given, so the reasoning behind the choice is part of what is being practised.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 1 · 1.7 · WORKED EXAMPLE 1

Mixed practice

WORKED EXAMPLE 1.7.1

Problem. Solve $(x^2 - 9)/(x^2 + 4x + 3) = 1$.

expression	factorised	cancelled?
numerator	$(x - 3)(x + 3)$	$x + 3$
denominator	$(x + 1)(x + 3)$	$x + 3$
excluded	$x \neq -1$ and $x \neq -3$	—

Factorise both parts before clearing the fractions.

Mathematical reasoning. Simplify the fraction first, then clear the denominator and check the candidates.

1. Factorise: $(x - 3)(x + 3)/((x + 1)(x + 3))$.
2. Exclude $x = -1$ and $x = -3$, then cancel $x + 3$ to get $(x - 3)/(x + 1)$.
3. The equation becomes $(x - 3)/(x + 1) = 1$, so $x - 3 = x + 1$.
4. This gives $-3 = 1$, which is false, so there is no solution.
5. Note that $x = -3$ would have made the original denominator zero, so it is excluded anyway.

Final answer. no solution

Interpretation. An equation can be consistent and still have no solution, and recognising that is part of solving it.

CHAPTER 1 · 1.7 · WORKED EXAMPLE 2

Mixed practice

WORKED EXAMPLE 1.7.2

Problem. The polynomial $p(x) = x^3 + ax^2 + bx - 12$ has $x - 2$ as a factor and leaves remainder 8 when divided by $x - 1$. Find a and b.

condition	equation
$x - 2$ is a factor, so $p(2) = 0$	$8 + 4a + 2b - 12 = 0$
remainder 8 on dividing by $x - 1$, so $p(1) = 8$	$1 + a + b - 12 = 8$
simplified	$2a + b = 2$ and $a + b = 19$

Two conditions give two equations in a and b.

Mathematical reasoning. Each condition is a value of the polynomial, and two values give two linear equations.

1. Since $x - 2$ is a factor, $p(2) = 0$: $8 + 4a + 2b - 12 = 0$, so $4a + 2b = 4$ and $2a + b = 2$.
2. Since the remainder on dividing by $x - 1$ is 8, $p(1) = 8$: $1 + a + b - 12 = 8$, so $a + b = 19$.
3. Subtract: $(2a + b) - (a + b) = 2 - 19$, so $a = -17$.
4. Then $b = 19 - a = 36$.
5. Check: $p(2) = 8 + 4(-17) + 72 - 12 = 8 - 68 + 72 - 12 = 0$.

Final answer. $a = -17$ and $b = 36$

Interpretation. The remainder theorem turns two statements about division into two equations, and the rest is simultaneous linear algebra.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

1.A1 FLUENCY

State the degree and leading coefficient of $3x^5 - 2x^3 + x - 7$.

1.A2 FLUENCY

Add $2x^3 - x + 4$ and $-x^3 + 5x^2 - 2$, and state the degree of the sum.

1.A3 FLUENCY

Multiply $(2x - 1)(x^2 + 3x - 2)$ and state the degree of the product.

1.A4 FLUENCY

Find $p(-2)$ for $p(x) = x^4 - 3x^2 + 5x - 1$.

1.A5 REASONING

Divide $3x^3 + 5x^2 - 4x + 2$ by $x + 1$.

1.A6 FLUENCY

Divide $x^3 - 8$ by $x - 2$.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

1.A7 FLUENCY

Find the remainder when $p(x) = 2x^3 - 5x + 3$ is divided by $x - 1$.

1.A8 REASONING

Find the remainder when $p(x) = x^3 + 2x^2 - 7$ is divided by $x + 3$.

1.A9 REASONING

Show that $x - 1$ is a factor of $x^3 - 2x^2 - x + 2$ and find the other factors.

1.A10 REASONING

Simplify $(x^2 - 4)/(x^2 + 5x + 6)$, stating the excluded values.

1.A11 REASONING

Explain why $(x + 3)/x$ cannot be simplified to 3.

1.A12 CHALLENGE

Solve $3/(x) + 2/(x + 1) = 1$ and check for extraneous solutions.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

1.B1 REASONING

Investigate whether $x - 1$ is always a factor of $x^N - 1$ for a positive integer n. State and justify a conclusion.

1.B2 CHALLENGE

Build a table of $x^N - 1$ factorised for $n = 2$ to 6 and describe the pattern of the factors.

1.B3 REASONING

Conjecture a rule for the remainder when x^N is divided by $x - 1$, test it for $n = 1$ to 5, and justify it.

1.B4 CHALLENGE

Investigate the number of real roots of $x^N - 1$ for $n = 2$, 3, 4 and 5, and explain the pattern.

1.B5 CHALLENGE

The polynomial $p(x) = x^3 + px + q$ has a double root. Investigate the condition on p and q.

1.B6 REASONING

Take a cubic of your own with three distinct integer roots. Divide it by one of its factors and check the quotient by multiplication.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

1.B7 REASONING

Investigate whether the sum of two polynomials of degree 3 can have degree 4, and justify your answer.

1.B8 REASONING

Investigate the remainder when $x^N - 1$ is divided by $x - 1$ for $n = 2, 3, 4$ and 5, and justify a general rule.

1.B9 CHALLENGE

Conjecture a rule for the factorisation of $x^N - y^N$ for $n = 2, 3$ and 4, and justify it.

1.B10 CHALLENGE

Investigate the relationship between the roots of a cubic and its coefficients, and justify.

1.B11 CHALLENGE

Investigate the number of sign changes of a polynomial and the number of positive roots, and justify a bound.

CRITERION C Communicating

Criterion C practice · Communicating

1.C1 REASONING

Write a complete solution to the simplification of $(x^3 - 8)/(x^2 - 4)$, including the excluded values and a check.

1.C2 REASONING

Write a complete solution to the equation $(x + 2)/(x - 1) = (x - 2)/(x + 1)$, showing the check for extraneous solutions.

1.C3 REASONING

Write a complete solution to the factorisation of $x^3 - 7x - 6$, showing how the first root was found.

1.C4 REASONING

Write a complete solution to the inequality $(x - 2)(x + 1)(x - 5) < 0$, with a sign line.

1.C5 CHALLENGE

Write a complete solution to the division of $2x^4 - 3x^3 + x - 5$ by $x^2 - x + 1$, including the check.

1.C6 REASONING

Write a complete solution to the equation $1/(x - 2) + 1/(x - 3) = 1$, identifying any extraneous solution.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

1.C7 REASONING

Write a complete solution to the problem: a rectangle has area $(x^2 + 5x + 6)$ cm^2 and width $(x + 2)$ cm. Find its length and state the restriction on x.

1.C8 REASONING

Write a complete solution to expressing $(3x + 5)/((x - 1)(x + 2))$ in partial fractions.

1.C9 REASONING

Write a complete solution to solving $2/(x - 1) = 3/(x + 2)$, checking for extraneous roots.

1.C10 REASONING

Write a complete solution to proving the factor theorem.

1.C11 CHALLENGE

Write a complete solution to sketching the graph of $p(x) = x^3 - 4x$, marking the roots, the turning points and the sign of the function on each interval.

1.C12 APPLICATION

Write a complete solution to the problem: a box has volume 120 cm^3 and a square base of side x. Find x if the total surface area is 166 cm^2.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

1.D1 APPLICATION

A tank is filled by two pipes. The first alone takes 3 hours longer than the second, and together they take 2 hours. Find each time alone.

1.D2 APPLICATION

A company's output is modelled by $p(t) = t^3 - 6t^2 + 9t$ thousand units after t months. Find when output is rising, and state the domain restriction.

1.D3 APPLICATION

A rectangular box has volume $(x^3 + 6x^2 + 11x + 6)$ cm^3 and base $(x + 1)(x + 2)$ cm^2. Find its height and state the restriction on x.

1.D4 APPLICATION

A cyclist travels 30 km at speed v and returns at $v - 4$, taking 1 hour longer. Find v and comment on the sense of the answer.

1.D5 APPLICATION

A engineer specifies a beam whose cross-sectional area is $(4x^2 - 9)$ mm^2 and whose width is $(2x - 3)$ mm. Find the depth and state the restriction.

1.D6 APPLICATION

A shop's profit is modelled by $P(n) = -n^2 + 12n - 20$ thousand currency units for n hundred items. Find the range of n for which the shop makes a profit.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

1.D7 APPLICATION

Two workers together complete a job in 6 hours. One works twice as fast as the other. Find each time alone and check the answer.

1.D8 APPLICATION

A farmer's yield is modelled by $Y(a) = (a - 2)(a - 8)(a + 1)$ tonnes for a kilograms of fertiliser per hectare. Find the range of a giving a positive yield and comment on the practical domain.

1.D9 REASONING

A concentration is modelled by $C(t) = (t^2 - 16)/(t - 4)$ for t minutes. Explain what happens at $t = 4$ and why the model must state it.

1.D10 APPLICATION

A phone company charges $c(n) = (50n + 200)/(n + 4)$ currency units for n gigabytes. Explain the behaviour as n grows and find when the charge exceeds 20.

1.D11 REASONING

A bridge cable's sag is modelled by $s(L) = (L^3 - 27)/(L - 3)$ metres for span L. Explain the model at $L = 3$ and find the sag for $L = 6$.

1.D12 CHALLENGE

A student claims that if a rational equation has an extraneous solution then it has no real solution. Investigate and justify.

CHAPTER 1 · APPLICATION AND ASSESSMENT

Applying chapter 1

These tasks use polynomial division, the factor theorem, algebraic fractions and rational equations. State the excluded values before you cancel anything, and check every solution in the original equation.

A1 · The container specification

A container's capacity is modelled by $V(x) = x^3 + 6x^2 + 11x + 6$ cm^3 and its base area by $A(x) = (x + 1)(x + 2)$ cm^2. Find the height, state the restriction on x that the physical situation imposes, and explain what the restriction protects the answer from.

Criteria A and D · factorise, divide, then apply the physical domain.

A2 · The shared job

Two machines together complete a batch in 4 hours. The older machine takes 6 hours longer than the newer one. Find each machine's time alone, check the answer, and explain why adding the two times is not the answer.

Criteria A and C · build the equation from the rates, and set out the check clearly.

CHAPTER 1 · REVIEW

What you should now be able to do

Key ideas

- The degree of a product of polynomials is the sum of the degrees.
- Dividend = divisor × quotient + remainder is the check on every division.
- The remainder theorem gives the remainder on dividing by $x - a$ as $p(a)$.
- The factor theorem says $x - a$ is a factor exactly when $p(a) = 0$.
- An algebraic fraction is cancelled only by a common factor.
- Excluded values are the values that made a cancelled factor zero.
- A rational equation is solved by clearing fractions, then checking every candidate.
- The factored form of a polynomial decides its sign completely.

Vocabulary

polynomial · degree · leading coefficient · long division · quotient · remainder · remainder theorem · factor theorem · root · linear factor · algebraic fraction · excluded value · extraneous solution · sign line · double root

CHAPTER 1 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

1.R1 Divide $x^3 + 2x^2 - 5x - 6$ by $x + 3$.

1.R2 Find the remainder when $x^4 + 2x - 3$ is divided by $x - 1$.

1.R3 Simplify $(x^2 + 7x + 12)/(x^2 + 5x + 4)$, stating the excluded values.

1.R4 Solve $(x - 1)/(x + 2) = 2$ and check for extraneous solutions.

1.R5 Solve $(x + 1)(x - 2)(x + 4) \leq 0$.

1.R6 Explain why a cubic has at most three real roots.

CHAPTER 1 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> The structure of a polynomial — its factors, roots and remainders — determines both what it can be simplified to and what its graph will do.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of equivalence, generalisation and simplification appear in your work?
- How is the global context of personal and cultural expression, and its exploration of the language of algebra as a way of making structure visible, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a simplified form of an expression, and what would you still like to ask?

CHAPTER 1 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Algebra is a language for making structure visible. Choose the form that shows the structure you need, and say what the choice costs.

Service as Action

COMMUNITY NEED	Public budgets and building specifications are published as simplified expressions whose restrictions are not stated.
MATHEMATICAL CONNECTION	Algebraic fractions, excluded values and rational equations.
PROPOSED ACTION	Collect three published specifications, rewrite each as a single simplified expression, and publish the restrictions that were dropped.
EXPECTED IMPACT	A reader can tell where a published formula stops being valid.
STUDENT REFLECTION	Record the specifications, the simplifications and the restrictions you found.

Community Project

PROJECT IDEA	A precision audit of the school's own published formulas.
NEED	Timetables, budgets and lab recipes are quoted as formulas with unstated limits.
RESEARCH QUESTION	Which published school formulas hide a restriction, and what happens when it is ignored?
MATHEMATICS	Algebraic fractions, excluded values and rational equations.
ACTION OR OUTCOME	A one-page audit with each formula, its restriction and a verdict.
BENEFICIARIES	Everyone who uses a school formula.
EVIDENCE	The audit sheet with restrictions and verdicts.

Personal Project connection · MYP 5

PERSONAL INTEREST	Cooking, gaming, sport or music.
LEARNING GOAL	To rewrite a formula you use into its simplest form and state honestly where it stops working.
PRODUCT OR OUTCOME	A formula card with the restriction written on it.
MATHEMATICS	Simplification, cancellation and excluded values.
PROCESS	Take three formulas, simplify each, and find the values that break them.
EVIDENCE	The formula cards and the restrictions you found.

CHAPTER 2 · PART II · THINKING WITH MODELS

02

Functions: Transformations, Composites and Inverses

A function is a relationship between two quantities. The same relationship can be written as a rule, as a table, as a graph or as a machine, and each representation makes a different operation visible.

ENGAGE

A sensor reports a temperature that a model then converts into a pressure. The sensor is recalibrated so that every reading is 3 degrees higher. Which part of the model has to change, and does the graph of the model move up, down, left or right?

Framing the inquiry

Key concept: Relationships
Related concepts: Change · Representation · Patterns
Global context: Scientific and technical innovation
Global context exploration: models that translate a measurement into a prediction

Statement of inquiry: A function is a rule that pairs each input with exactly one output; transforming, composing and inverting are operations on the rule itself, and each leaves a trace on the graph.

Questions to guide us

- **Factual:** What are the domain and range of a function, and what does the notation f(x) mean?
- **Conceptual:** Why does f(x) + k move a graph vertically while f(x + k) moves it horizontally, and what makes a function invertible?
- **Debatable:** If two representations of the same function look nothing alike, which one is the function?

ATL · CRITICAL THINKING. In order for a student to predict the effect of a transformation on a model, students must compare a parent graph with its image, identify which coordinate has changed and how, and justify the algebraic rule that produces the same change.

CHAPTER 2 · INVESTIGATION 2.1

Domain, range and function notation

INQUIRY QUESTION

What does the notation f(x) actually assert?

YOUR TASK

Take four rules of your own: a linear one, a quadratic one, one with a square root and one with a fraction. For each, find the largest domain you can state on the real numbers, find the range, and record one input that is not permitted. Then say which representation — table, formula, graph or machine — shows the domain most clearly.

rule	largest domain	range	excluded input
$3x - 5$	all real x	all real y	none
x^2	all real x	$y \geq 0$	none
$\sqrt{(x + 4)}$	$x \geq -4$	$y \geq 0$	$x = -5$
$1/(x - 1)$	$x \neq 1$	$y \neq 0$	$x = 1$
x^3	all real x	all real y	none
$(x - 2)/(x - 2)$	$x \neq 2$	$y = 1$ only	$x = 2$

CHAPTER 2 · INVESTIGATION 2.1

Domain, range and function notation

What do you notice?

- Two of the six rules accepted every real input.
- The square root rule excluded everything below a single number.
- The fraction rule excluded exactly one value, and its range excluded exactly one value too.
- The last rule looked like a constant but was undefined at one point.

Observation. The domain is decided by the operations in the rule: division excludes zeros of the denominator and roots exclude negative inputs.

MATHEMATICAL CONCLUSION

A function is a rule together with a domain, and the range is what the rule can produce from that domain. The domain is not decoration: the same formula on a different domain is a different function, which is why the domain is stated before anything else is claimed.

Your own words

CHAPTER 2 · 2.1

Domain, range and function notation

A function is a rule that assigns exactly one output to each input of a stated set, and the relationships it can express are limited by that set. The domain is the set of permitted inputs and the range is the set of outputs the rule can produce. A rule without a stated domain is incomplete, because the same formula can describe different functions on different domains.

Function notation f(x) names the output for the input x. Writing f(2) = 5 is a statement about one input; writing f(x) = 2x + 1 is a statement about all of them. The notation distinguishes the rule from its values, which is what makes composition possible.

Two functions are equal only when they agree on every input of a common domain. The formulas x^2/x and x agree everywhere except at x = 0, so they are not the same function unless 0 is excluded from the domain of both.

A mapping diagram, a table, a formula and a graph are four representations of one relationship between input and output. Each is complete, and each makes a different operation easy: a table shows values, a formula shows rules, a graph shows behaviour, and the patterns in each are the same patterns seen from a different angle.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.1 · WORKED EXAMPLE 1

Domain, range and function notation

WORKED EXAMPLE 2.1.1

Problem. The function f is defined by $f(x) = \sqrt{(x - 2)} + 1$. State its domain and range, and explain why the domain cannot be larger.

x	x − 2	√(x − 2)	f(x)
1	−1	undefined	undefined
2	0	0	1
3	1	1	2
6	4	2	3

The square root forces the domain to start at 2.

Mathematical reasoning. The expression under a square root must be non-negative, and the root itself is never negative, which fixes the range from below.

1. The expression under the root is $x - 2$, so $x - 2 \geq 0$ and $x \geq 2$.
2. The domain is therefore $x \geq 2$.
3. $\sqrt{(x - 2)}$ takes every value from 0 upwards, so f(x) takes every value from 1 upwards.
4. The range is $f(x) \geq 1$.
5. The domain cannot be larger because any x below 2 makes the root undefined.

Final answer. domain $x \geq 2$, range $f(x) \geq 1$

Interpretation. The domain is part of the definition, not an afterthought: the same formula with a smaller domain is a different function.

CHAPTER 2 · 2.1 · WORKED EXAMPLE 2

Domain, range and function notation

WORKED EXAMPLE 2.1.2

Problem. Show that $f(x) = x^2/x$ and $g(x) = x$ are not the same function, and explain what has to change for them to become equal.

x	$f(x) = x^2/x$	$g(x) = x$	agree?
−2	−2	−2	yes
0	undefined	0	no
2	2	2	yes
3	3	3	yes

One undefined value is enough to separate two functions.

Mathematical reasoning. Two functions are equal only when they agree on every input of a common domain, so a single disagreement is decisive.

1. $f(0) = 0^2/0$, which is undefined because division by zero is not defined.
2. $g(0) = 0$, which is defined.
3. So the two functions disagree at $x = 0$.
4. If the domain of both is restricted to $x \neq 0$, then they agree everywhere and become equal.
5. Hence f and g are different functions on the real numbers but the same function on $x \neq 0$.

Final answer. they differ at $x = 0$; restricting the domain of g to $x \neq 0$ makes them equal

Interpretation. Equality of functions is equality on a domain, not similarity of formulas.

CHAPTER 2 · INVESTIGATION 2.2

Transformations of graphs

INQUIRY QUESTION

Why does f(x) + 3 move a graph up but f(x + 3) move it left?

YOUR TASK

Take the parent graph $y = x^2$. Write the equation of the image after each of these: a translation 2 right and 3 down; a reflection in the y-axis; a stretch by factor 3 parallel to the y-axis; and a translation 1 left followed by a stretch by factor 2 parallel to the x-axis. For each, plot the image of the point (1, 1) and check it satisfies your equation.

transformation	equation	image of (1, 1)	check
right 2, down 3	$y = (x - 2)^2 - 3$	(3, −2)	$1 - 3 = -2$
reflect in y-axis	$y = (-x)^2 = x^2$	(1, 1)	unchanged
stretch 3, y-axis	$y = 3x^2$	(1, 3)	$3 \times 1 = 3$
left 1 then stretch 2, x-axis	$y = ((x + 1)/2)^2$	(1, 1)	$1^2 = 1$

CHAPTER 2 · INVESTIGATION 2.2

Transformations of graphs

What do you notice?

- The reflection in the y-axis left this particular graph unchanged, because x^2 is symmetric.
- The stretch parallel to the y-axis multiplied the y-coordinate.
- The stretch parallel to the x-axis divided the x-coordinate, so the graph widened.
- Every image point satisfied the equation it was supposed to satisfy.

Observation. A change inside the bracket acts on the input and moves the graph horizontally; a change outside acts on the output.

MATHEMATICAL CONCLUSION

The equation of an image is the parent equation with the coordinate rule substituted into it, so $y = f(x)$ translated by (a, b) becomes $y = f(x - a) + b$. Reading the equation as a coordinate rule is the check that the transformation has been described correctly.

Your own words

CHAPTER 2 · 2.2

Transformations of graphs

A translation of the graph of $y = f(x)$ by the vector (a, b) has equation $y = f(x - a) + b$. The number inside the bracket acts on x before the function, so it moves the graph in the opposite direction to its sign.

The reason for the reversal is order of operations. To obtain the same output as f at $x = 0$, the translated graph must be evaluated at $x = -a$, so the whole graph sits a units to the left of where it started.

A reflection in the x-axis has equation $y = -f(x)$, and a reflection in the y-axis has equation $y = f(-x)$. A stretch parallel to the y-axis by factor k has equation $y = kf(x)$, and a stretch parallel to the x-axis by factor k has equation $y = f(x/k)$.

The two descriptions of a transformation — as a coordinate rule and as an equation — must agree. Reading the equation as a coordinate rule is the standard way to check that a transformation has been described correctly.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.2 · WORKED EXAMPLE 1

Transformations of graphs

WORKED EXAMPLE 2.2.1

Problem. Describe the transformation taking $y = x^2$ to $y = (x - 3)^2 + 2$, and state the image of the point $(1, 1)$.

point on $y = x^2$	image on $y = (x - 3)^2 +$	change
$(0, 0)$	$(3, 2)$	right 3, up 2
$(1, 1)$	$(4, 3)$	right 3, up 2
$(-1, 1)$	$(2, 3)$	right 3, up 2
$(2, 4)$	$(5, 6)$	right 3, up 2

Every point moves by the same vector.

Mathematical reasoning. Compare the equation with $y = f(x - a) + b$ to read off the translation, then apply it to the point.

1. The parent graph is $f(x) = x^2$ and the new graph is $f(x - 3) + 2$.
2. So $a = 3$ and $b = 2$, which is a translation by the vector $(3, 2)$.
3. The vertex moves from $(0, 0)$ to $(3, 2)$.
4. The point $(1, 1)$ moves to $(1 + 3, 1 + 2) = (4, 3)$.
5. Check: $(4 - 3)^2 + 2 = 1 + 2 = 3$, which is the y-coordinate.

Final answer. translation by (3, 2); the image of (1, 1) is (4, 3)

Interpretation. The equation encodes the coordinate rule, and reading it as a coordinate rule is the fastest check.

CHAPTER 2 · 2.2 · WORKED EXAMPLE 2

Transformations of graphs

WORKED EXAMPLE 2.2.2

Problem. Describe the transformation taking $y = x^2$ to $y = -2(x + 1)^2$, and explain the order of the two changes.

stage	equation	change
parent	$y = x^2$	—
horizontal	$y = (x + 1)^2$	left 1
stretch	$y = 2(x + 1)^2$	stretch, factor 2, y-axis
reflect	$y = -2(x + 1)^2$	reflect in x-axis

The stretch and the reflection both act on the output.

Mathematical reasoning. Separate the change inside the bracket, which acts on the input, from the changes outside, which act on the output.

1. Inside the bracket, $x + 1$ means $x - (-1)$, so $a = -1$: a translation 1 unit left.
2. Outside, the factor 2 multiplies the output: a stretch parallel to the y-axis by factor 2.
3. The leading minus sign reflects in the x-axis.
4. The vertex moves from $(0, 0)$ to $(-1, 0)$ and the graph opens downwards.
5. Check with a point: on the parent, $x = 0$ gives 0; on the new graph, $x = -1$ gives 0.

Final answer. translation 1 left, stretch by factor 2 parallel to the y-axis, then reflection in the x-axis

Interpretation. Changes inside the bracket move the graph horizontally; changes outside move it vertically or reflect it. Confusing the two is the commonest transformation error.

CHAPTER 2 · INVESTIGATION 2.3

Composite functions

INQUIRY QUESTION

In what order do the functions in f(g(x)) act?

YOUR TASK

Take f(x) = 3x − 2 and g(x) = x + 5. Compute f(g(x)), g(f(x)), f(f(x)) and g(g(x)), and record the value of each at x = 1. Then find a pair of functions of your own for which the two orders give the same answer, and explain why.

composite	simplified	value at x = 1
f(g(x))	3(x + 5) − 2 = 3x + 13	16
g(f(x))	(3x − 2) + 5 = 3x + 3	6
f(f(x))	3(3x − 2) − 2 = 9x − 8	1
g(g(x))	(x + 5) + 5 = x + 10	11
f(g(x)) − g(f(x))	(3x + 13) − (3x + 3) = 10	10

CHAPTER 2 · INVESTIGATION 2.3

Composite functions

What do you notice?

- All four composites were linear, because both functions were linear.
- The two orders differed by a constant, 10, rather than by a whole function.
- Repeating a function changed its gradient, so f(f(x)) had gradient 9.
- A pair of linear functions with the same gradient would commute.

Observation. The order matters, but for linear functions the two orders differ only by a constant.

MATHEMATICAL CONCLUSION

The composite f(g(x)) applies g first, and the two orders are different functions in general. Two functions commute exactly when they are specially related — for example when both are translations, or when they are powers of one another.

Your own words

CHAPTER 2 · 2.3

Composite functions

The composite function f(g(x)) applies g first and then f. The inner function acts first, which is the opposite of the order in which the functions are written, and this reversal is the single commonest source of error.

The domain of f(g(x)) is the set of inputs x for which g(x) is defined and g(x) lies in the domain of f. Both conditions must hold, so the domain of a composite is usually smaller than either domain alone.

Composition is associative: f(g(h(x))) can be built as f∘(g∘h) or as (f∘g)∘h with the same result. It is not commutative in general, and f(g(x)) = g(f(x)) only for specially related functions.

A function machine is a faithful picture of composition: the output of the first box is the input of the second. Drawing the machine before writing the algebra prevents the order being reversed.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.3 · WORKED EXAMPLE 1

Composite functions

WORKED EXAMPLE 2.3.1

Problem. Given $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $f(g(x))$, $g(f(x))$ and $f(g(2))$, and state the domain of $f(g(x))$.

x	$g(x)$	$f(g(x))$	$f(x)$	$g(f(x))$
0	-3	-5	1	-2
1	-2	-3	3	6
2	1	3	5	22
-1	-2	-3	-1	-2

The two orders give different functions.

Mathematical reasoning. Substitute the inner function into the outer one, and note that the two orders differ.

1. $f(g(x)) = 2(g(x)) + 1 = 2(x^2 - 3) + 1 = 2x^2 - 5$.
2. $g(f(x)) = (f(x))^2 - 3 = (2x + 1)^2 - 3 = 4x^2 + 4x - 2$.
3. The two are different: $f(g(1)) = -3$ while $g(f(1)) = 6$.
4. $f(g(2)) = 2(4) - 5 = 3$.
5. Both f and g accept every real input, so the domain of $f(g(x))$ is all real x.

Final answer. $f(g(x)) = 2x^2 - 5$, $g(f(x)) = 4x^2 + 4x - 2$, $f(g(2)) = 3$, domain all real x

Interpretation. Composition is not commutative, and the two orders are different functions with different graphs.

CHAPTER 2 · 2.3 · WORKED EXAMPLE 2

Composite functions

WORKED EXAMPLE 2.3.2

Problem. Given $f(x) = \sqrt{x}$ and $g(x) = x - 4$, find $f(g(x))$ and state its domain.

x	$g(x) = x - 4$	$\sqrt{(g(x))}$	defined?
2	-2	undefined	no
4	0	0	yes
5	1	1	yes
8	4	2	yes

The inner output must be a permitted input of the outer function.

Mathematical reasoning. The composite is defined only where the inner function is defined and its output lies in the domain of the outer one.

1. $f(g(x)) = \sqrt{(x - 4)}$.
2. The square root requires $x - 4 \geq 0$, so $x \geq 4$.
3. g itself is defined for all x, so the only restriction comes from f.
4. The domain of $f(g(x))$ is $x \geq 4$.
5. Note that this is smaller than the domain of f alone, which is $x \geq 0$.

Final answer. $f(g(x)) = \sqrt{(x - 4)}$, domain $x \geq 4$

Interpretation. The domain of a composite is the intersection of two conditions, and it is usually smaller than either domain alone.

CHAPTER 2 · INVESTIGATION 2.4

Inverse functions

INQUIRY QUESTION

When does a function have an inverse, and how is it found?

YOUR TASK

Take six functions of your own: two linear, one quadratic, one with a square root, one rational and one cubic. For each, decide whether it is one-to-one on its natural domain. If it is, find the inverse and verify it by composition; if it is not, restrict the domain so that it becomes one-to-one and find the inverse on that restriction. Record the domain and range of every inverse you find.

function	one-to-one?	restriction	inverse	domain of inverse
$2x + 1$	yes	none	$(x - 1)/2$	all real x
x^3	yes	none	$x^{(1/3)}$	all real x
x^2	no	$x \geq 0$	$\sqrt{x}$	$x \geq 0$
$\sqrt{(x - 2)}$	yes	none	$x^2 + 2$	$x \geq 0$
$1/x$	yes	$x \neq 0$	$1/x$	$x \neq 0$
$(x - 1)/(x + 2)$	yes	$x \neq -2$	$(1 + 2x)/(1 - x)$	$x \neq 1$

CHAPTER 2 · INVESTIGATION 2.4

Inverse functions

What do you notice?

- Four of the six functions were already one-to-one and needed no restriction.
- The squaring function needed the restriction $x \geq 0$ before an inverse existed.
- The rational function was one-to-one because its graph never turned back.
- Every inverse swapped the domain of the original with its range.

Observation. A function is invertible exactly when no horizontal line meets its graph twice, and restriction is the tool for making that true.

MATHEMATICAL CONCLUSION

The inverse function exists exactly when the original is one-to-one, and it is verified by composition in both directions. Where the original is not one-to-one, a restricted domain is stated as part of the definition, and the inverse's domain is the original's range.

Your own words

CHAPTER 2 · 2.4

Inverse functions

The inverse function f^{-1} reverses the relationship: if $f(a) = b$ then $f^{-1}(b) = a$. It exists exactly when f is one-to-one on its domain, because two different inputs with the same output could not be told apart on the way back.

The horizontal line test decides invertibility. If any horizontal line meets the graph more than once, two inputs share an output and no inverse exists on that domain.

Restricting the domain can make a function one-to-one. The squaring function is not one-to-one on the real numbers, but it is one-to-one on $x \geq 0$, and its inverse there is the square root.

The inverse is found by writing $y = f(x)$, rearranging to make x the subject, and then swapping the letters. The check is composition: $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$ on the restricted domains.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.4 · WORKED EXAMPLE 1

Inverse functions

WORKED EXAMPLE 2.4.1

Problem. Find the inverse of $f(x) = 3x - 5$, state its domain and range, and check it by composition.

x	f(x)	$f^{-1}(x)$	check
0	−5	5/3	f(5/3) = 0
1	−2	2	f(2) = 1
2	1	5/3	f(5/3) = 0
3	4	4/3	f(4/3) = −1

The inverse swaps the roles of input and output.

Mathematical reasoning. Rearrange to make x the subject, then swap the letters, and verify by composition.

1. Write $y = 3x - 5$.
2. Add 5: $y + 5 = 3x$, so $x = (y + 5)/3$.
3. Swap the letters: $f^{-1}(x) = (x + 5)/3$.
4. Check: $f^{-1}(f(x)) = (3x - 5 + 5)/3 = x$.
5. Also $f(f^{-1}(x)) = 3((x + 5)/3) - 5 = x + 5 - 5 = x$.

Final answer. $f^{-1}(x) = (x + 5)/3$, domain all real x, range all real x

Interpretation. The composition check is the definition of an inverse: it undoes the function in both directions.

CHAPTER 2 · 2.4 · WORKED EXAMPLE 2

Inverse functions

WORKED EXAMPLE 2.4.2

Problem. Show that $f(x) = x^2$ has no inverse on the real numbers, find a domain on which it does, and state the inverse there.

x	$f(x) = x^2$	horizontal line $y = 4$	one-to-one?
-2	4	meets twice	no
0	0	meets once	—
2	4	meets twice	no
3	9	meets once	—

Two inputs with the same output block the inverse.

Mathematical reasoning. A horizontal line meeting the graph twice means two inputs share an output, so the function is not one-to-one and has no inverse on that domain.

1. The line $y = 4$ meets $y = x^2$ at $x = -2$ and $x = 2$.
2. So $f(-2) = f(2) = 4$, and the two inputs cannot be distinguished on the way back.
3. Restrict the domain to $x \geq 0$. Then every horizontal line meets the graph at most once.
4. On $x \geq 0$ the inverse is $f^{-1}(x) = \sqrt{x}$, with domain $x \geq 0$.
5. Check: $f^{-1}(f(x)) = \sqrt{(x^2)} = x$ for $x \geq 0$, and $f(f^{-1}(x)) = (\sqrt{x})^2 = x$ for $x \geq 0$.

Final answer. no inverse on the real numbers; on $x \geq 0$ the inverse is $\sqrt{x}$ with domain $x \geq 0$

Interpretation. Restricting the domain is how a many-to-one relationship becomes a one-to-one one, and the restriction must be stated.

CHAPTER 2 · INVESTIGATION 2.5

Modelling with transformations

INQUIRY QUESTION

How does a transformation of a model show up in the situation it describes?

YOUR TASK

Take the parent model P(n) = 5n, the price of n items at 5 each. Write models for: a 3 currency unit delivery charge added; a 20 per cent discount; a shop that charges for n + 2 items; and a shop whose item costs half. For each, say whether the change is vertical or horizontal, and find the number of items at which the discounted model costs the same as the original.

situation	model	kind of change	equal to P when
delivery charge 3	5n + 3	vertical translation	never
20% discount	4n	vertical stretch, factor 0.8	n = 0
charges for n + 2	5(n + 2)	horizontal translation	n = −2
item costs half	2.5n	vertical stretch, factor 0.5	n = 0

CHAPTER 2 · INVESTIGATION 2.5

Modelling with transformations

What do you notice?

- The delivery charge raised every price by the same amount, so it was a vertical translation.
- The discount multiplied every price by 0.8, so it was a vertical stretch.
- Charging for n + 2 items was a horizontal translation, and it only mattered for positive n.
- Two of the models met the original only at n = 0, because both passed through the origin.

Observation. An additive change in the situation is a vertical translation; a multiplicative change is a vertical stretch; a change of origin is a horizontal translation.

MATHEMATICAL CONCLUSION

A transformation of a model is a change in the situation, and the direction of the change tells you what changed. Reading the transformation is often faster than recalculating the model, and it explains why two models cross.

Your own words

CHAPTER 2 · 2.5

Modelling with transformations

A model is a function whose input and output both carry units. A transformation of the model is therefore a change in the situation, and reading the transformation tells you what changed without recalculating anything.

A vertical translation changes the output at every input, which is an additive change in the situation: a fixed charge added to a bill, a starting temperature added to a reading. A horizontal translation changes the input, which is a shift in time or in the zero of a scale.

A vertical stretch multiplies every output, which is a multiplicative change: a price rise, a calibration factor. A horizontal stretch divides every input, which is a change of unit or of time scale.

Combining transformations builds a family of models from one parent. A single parent function plus a small set of parameters describes a whole class of situations, and each parameter has a physical meaning.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.5 · WORKED EXAMPLE 1

Modelling with transformations

WORKED EXAMPLE 2.5.1

Problem. A taxi charges a flag fall of 4 currency units and then 2 per kilometre. A rival charges a flag fall of 6 and 1.5 per kilometre. Model both, describe the transformation between them, and find the distance at which they cost the same.

distance (km)	first model	second model	difference
0	4	6	−2
2	8	9	−1
4	12	12	0
6	16	15	1
8	20	18	2

Two linear models cross at one distance.

Mathematical reasoning. Model each as a linear function, read the transformation as a translation and a stretch, and solve for the crossing.

1. First model: $C_1(d) = 2d + 4$. Second model: $C_2(d) = 1.5d + 6$.
2. The second is the first translated up by 2 and stretched parallel to the x-axis by factor $2/1.5 = 4/3$.
3. Set them equal: $2d + 4 = 1.5d + 6$.
4. Then $0.5d = 2$, so $d = 4$ km.
5. Check: $C_1(4) = 12$ and $C_2(4) = 12$.

Final answer. $C_1 = 2d + 4$, $C_2 = 1.5d + 6$; they cost the same at 4 km

Interpretation. The transformation description explains the crossing before any algebra: a higher fixed charge with a lower rate must cross.

CHAPTER 2 · 2.5 · WORKED EXAMPLE 2

Modelling with transformations

WORKED EXAMPLE 2.5.2

Problem. A cooling model is $T(t) = 20 + 60(0.8)^T$ for t minutes after 14:00. Write the model for s minutes after the same start, and then for s minutes after 15:00.

minutes after 14:00	T(t)	minutes after 15:00	T(s)
0	80	−60	not applicable
60	$20 + 60(0.8)^{60}$	0	same as t = 60
120	$20 + 60(0.8)^{120}$	60	same as t = 120
−60	$20 + 60(0.8)^{-60}$	−120	same as t = −60

Changing the origin shifts the input.

Mathematical reasoning. A change of time origin is a horizontal translation, so the new input is the old input minus the shift.

1. For s minutes after 14:00 the model is identical: $T(s) = 20 + 60(0.8)^S$.
2. For s minutes after 15:00, the old input is t = s + 60, because 15:00 is 60 minutes after 14:00.
3. So T(s) = 20 + 60(0.8)^(s + 60) = $20 + 60(0.8)^{60}(0.8)^S$.
4. Using the index law, (0.8)^(s + 60) = $(0.8)^S \times (0.8)^{60}$, and $(0.8)^{60}$ is essentially 0.
5. So the 15:00 model is essentially T(s) ≈ 20, which is why the canteen switched to a different model later in the day.

Final answer. $T(s) = 20 + 60(0.8)^S$ after 14:00; T(s) = 20 + 60(0.8)^(s + 60) after 15:00

Interpretation. A change of origin is a horizontal translation, and the index law is what makes the translated exponential readable.

CHAPTER 2 · INVESTIGATION 2.6

Investigation: when is a function invertible?

INQUIRY QUESTION

What property of a graph decides whether an inverse exists?

YOUR TASK

Take six graphs of your own: a straight line, a parabola, a cubic, an exponential curve, a reciprocal curve and a step function. For each, apply the horizontal line test, and where it fails, find a restricted domain on which it passes. Sketch the inverse on the same axes, using the line $y = x$ as the mirror, and record the domain and range of each inverse.

graph	horizontal line test	restriction needed	inverse exists?
straight line	passes	none	yes
parabola	fails	one side of the vertex	yes, on the restriction
cubic	passes	none	yes
exponential curve	passes	none	yes, the logarithm
reciprocal curve	passes	$x \neq 0$ already excluded	yes, itself
step function	fails everywhere	no restriction helps	no

CHAPTER 2 · INVESTIGATION 2.6

Investigation: when is a function invertible?

What do you notice?

- Four of the six graphs passed the test without any restriction.
- The parabola needed a restriction to one side of its vertex.
- The step function could not be rescued by any restriction, because it is constant on whole intervals.
- Each inverse was the reflection in the line y = x, so every domain and range swapped.

> **Observation.** The horizontal line test is a complete criterion, and the algebraic version is that equal outputs must force equal inputs.

MATHEMATICAL CONCLUSION

A function is invertible exactly when it is one-to-one, which the horizontal line test decides graphically and the implication f(a) = f(b) implies a = b decides algebraically. Where a function fails, a restricted domain is stated, and the inverse's graph is the reflection of the original in y = x.

Your own words

CHAPTER 2 · 2.6

Investigation: when is a function invertible?

The horizontal line test is a complete criterion: a function is one-to-one exactly when no horizontal line meets its graph more than once. The test works because a horizontal line collects all the inputs with one particular output, so a second meeting is a repeated pattern among the outputs.

The algebraic version of the same test is that f(a) = f(b) must force a = b. Proving that implication is proving that the function is one-to-one, and it is the form used when no graph is available.

A function that is strictly increasing or strictly decreasing on its whole domain is one-to-one, because equal outputs would force a reversal. This is why monotonicity is the standard sufficient condition.

The graph of an inverse is the reflection of the graph in the line y = x, because the inverse swaps the coordinates of every point. Drawing the line of reflection is a quick check that an inverse has been found correctly.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.6 · WORKED EXAMPLE 1

Investigation: when is a function invertible?

WORKED EXAMPLE 2.6.1

Problem. Decide whether f(x) = 2x + 7 is one-to-one, and prove it algebraically.

a	b	f(a)	f(b)	f(a) = f(b)?
1	1	9	9	yes, and a = b
1	2	9	11	no
−3	4	1	15	no
0	−3.5	7	0	no

Equal outputs force equal inputs.

Mathematical reasoning. Suppose f(a) = f(b) and deduce a = b; the implication is the definition of one-to-one.

1. Suppose f(a) = f(b), so 2a + 7 = 2b + 7.
2. Subtract 7 from both sides: 2a = 2b.
3. Divide by 2: a = b.
4. So equal outputs force equal inputs, and f is one-to-one.
5. The inverse is $f^{-1}(x) = (x - 7)/2$, and its graph is the reflection in y = x.

Final answer. one-to-one; $f^{-1}(x) = (x - 7)/2$

Interpretation. The algebraic test is the same claim as the horizontal line test, written as an implication.

CHAPTER 2 · 2.6 · WORKED EXAMPLE 2

Investigation: when is a function invertible?

WORKED EXAMPLE 2.6.2

Problem. Decide whether $f(x) = x^2 - 4x$ is one-to-one on the real numbers, and find a restricted domain on which it is.

x	f(x)	partner with the same value	one-to-one?
0	0	x = 4	no
1	−3	x = 3	no
2	−4	none	—
4	0	x = 0	no
5	5	none	—

A parabola fails the horizontal line test.

Mathematical reasoning. Complete the square to find the axis of symmetry, and note that points equidistant from it share an output.

1. Complete the square: $f(x) = (x - 2)^2 - 4$.
2. The axis of symmetry is $x = 2$, so $f(2 + h) = f(2 - h)$ for every h.
3. For example $f(1) = f(3) = -3$, so the function is not one-to-one.
4. Restrict the domain to $x \geq 2$. Then the function is strictly increasing and therefore one-to-one.
5. On $x \geq 2$ the inverse is $f^{-1}(x) = 2 + \sqrt{(x + 4)}$, with domain $x \geq -4$.

Final answer. not one-to-one on the real numbers; one-to-one on $x \geq 2$, where $f^{-1}(x) = 2 + \sqrt{(x + 4)}$

Interpretation. Completing the square locates the axis of symmetry, and the axis is exactly where the function stops being one-to-one.

CHAPTER 2 · INVESTIGATION 2.7

Mixed practice

INQUIRY QUESTION

Which representation makes the next step easiest?

YOUR TASK

Take the function $f(x) = 2x - 6$. Carry out all of the following and record the answer to each: $f(5)$; $f^{-1}(5)$; $f(f(5))$; the image of $(3, 0)$ under $y = f(x) + 4$; the image of $(3, 0)$ under $y = f(x + 4)$; and the image of $(3, 0)$ under $y = f^{-1}(x)$. Then explain which two of the six answers are the same and why.

question	answer	operation used
$f(5)$	4	substitution
$f^{-1}(5)$	5.5	inverse
$f(f(5))$	2	composition
mage under $y = f(x) + 4$	$(3, 4)$	vertical translation
mage under $y = f(x + 4)$	$(-1, 0)$	horizontal translation
image under $y = f^{-1}(x)$	$(0, 3)$	reflection in $y = x$

CHAPTER 2 · INVESTIGATION 2.7

Mixed practice

What do you notice?

- f(5) and the vertical translation gave the same y-coordinate, because adding 4 after the function and evaluating at the same x differ.
- The horizontal translation moved the point left, not right, because the change was inside the bracket.
- The inverse image was the reflection of the original point in y = x.
- The composition f(f(5)) was neither of the translations.

Observation. Each operation acts on a different part of the rule: substitution evaluates, the inverse reverses, and a transformation moves the graph.

MATHEMATICAL CONCLUSION

The operations on functions are distinguished by where they act: inside the bracket changes the input, outside changes the output, and composition feeds one function into another. Choosing the right operation is the first step in any question about functions.

Your own words

CHAPTER 2 · 2.7

Mixed practice

Every question about a function can be approached from four directions: the formula, the table, the graph or the machine. Choosing the representation that shows the needed quantity directly is the skill this section practises.

Transformations, composition and inversion are operations on the rule. Keeping the three operations distinct — and knowing which one a question is asking for — prevents the commonest confusions between f(x) + 2, f(x + 2) and f(f(x)).

The checks are always available: a transformation is checked against a coordinate rule, a composition is checked against a value, and an inverse is checked by composition. Using a check is faster than finding an error later.

Mixed practice is where the operation is chosen rather than given, so the reasoning behind the choice is part of what is being practised.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.7 · WORKED EXAMPLE 1

Mixed practice

WORKED EXAMPLE 2.7.1

Problem. The function f is defined by $f(x) = (3x + 2)/(x - 1)$. Find f^{-1} and state its domain, and find the value of x for which $f(x) = f^{-1}(x)$.

step	working	result
write y	$y = (3x + 2)/(x - 1)$	—
clear	$y(x - 1) = 3x + 2$	—
collect x	$yx - y - 2 = 3x$	—
factor	$x(y - 3) = y + 2$	—
inverse	$x = (y + 2)/(y - 3)$	$f^{-1}(x) = (x + 2)/(x - 3)$

Clearing the fraction is the key step.

Mathematical reasoning. Rearrange to make x the subject, then swap the letters and state the domain from the new denominator.

1. Write $y = (3x + 2)/(x - 1)$ and multiply by $x - 1$: $y(x - 1) = 3x + 2$.
2. Expand: $yx - y = 3x + 2$, so $yx - 3x = y + 2$.
3. Factor: $x(y - 3) = y + 2$, so $x = (y + 2)/(y - 3)$.
4. Swap: $f^{-1}(x) = (x + 2)/(x - 3)$, with domain $x \neq 3$.
5. Setting $f(x) = f^{-1}(x)$ gives $(3x + 2)/(x - 1) = (x + 2)/(x - 3)$; cross-multiplying gives $3x^2 - 7x - 6 = x^2 + x - 2$, so $2x^2 - 8x - 4 = 0$ and $x = 2 \pm \sqrt{6}$.

Final answer. $f^{-1}(x) = (x + 2)/(x - 3)$, domain $x \neq 3$; $f(x) = f^{-1}(x)$ at $x = 2 \pm \sqrt{6}$

Interpretation. A rational function's inverse is rational again, and the excluded value moves from the original denominator to the new one.

CHAPTER 2 · 2.7 · WORKED EXAMPLE 2

Mixed practice

WORKED EXAMPLE 2.7.2

Problem. The graph of $y = f(x)$ passes through $(2, 5)$ and $(-1, 3)$. Find the corresponding points on $y = -f(x - 2) + 1$ and on $y = f^{-1}(x)$.

original point	on $y = -f(x - 2) + 1$	on $y = f^{-1}(x)$
$(2, 5)$	$(4, -4)$	$(5, 2)$
$(-1, 3)$	$(1, -2)$	$(3, -1)$

A transformation and an inverse act differently on points.

Mathematical reasoning. Apply the coordinate rule of the transformation, and swap the coordinates for the inverse.

1. For $y = -f(x - 2) + 1$ the coordinate rule is $(x, y) \to (x + 2, -y + 1)$.
2. So $(2, 5)$ maps to $(4, -4)$ and $(-1, 3)$ maps to $(1, -2)$.
3. For $y = f^{-1}(x)$ the coordinates swap, so $(2, 5)$ gives $(5, 2)$ and $(-1, 3)$ gives $(3, -1)$.
4. Check the first: $f(2 - 2) = f(0)$, and $-f(0) + 1 = -5 + 1 = -4$, which is the image's y-coordinate.
5. Note the inverse requires f to be one-to-one, which the two points alone cannot establish.

Final answer. $(2, 5) \to (4, -4)$ and $(5, 2)$; $(-1, 3) \to (1, -2)$ and $(3, -1)$

Interpretation. Transformations move points by a rule; inverses swap the coordinates. The two operations look similar on a graph and are completely different.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

2.A1 FLUENCY

State the domain and range of $f(x) = \sqrt{(x + 3)}$.

2.A2 FLUENCY

Given $f(x) = 4x - 7$, find $f(3)$ and $f(-2)$.

2.A3 REASONING

Given $f(x) = x^2 - 3x + 2$, solve $f(x) = 0$ and state the range.

2.A4 FLUENCY

Given $f(x) = 2x + 1$ and $g(x) = x - 4$, find $f(g(x))$ and $g(f(x))$.

2.A5 FLUENCY

Given $f(x) = x^2$ and $g(x) = x + 3$, find $f(g(x))$ and state its domain.

2.A6 REASONING

Given $f(x) = 1/x$, find $f(f(x))$ and state the domain.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

2.A7 FLUENCY

Describe the transformation from $y = x^2$ to $y = (x + 4)^2 - 1$.

2.A8 REASONING

Describe the transformation from $y = \sqrt{x}$ to $y = -3\sqrt{(x - 2)}$.

2.A9 FLUENCY

Find the inverse of $f(x) = (x + 3)/4$ and check it by composition.

2.A10 FLUENCY

For $f(x) = x + 3$, find $f(f(x))$ and $f(f(f(x)))$.

2.A11 REASONING

Explain why $f(x) = x^2$ has no inverse on the real numbers.

2.A12 REASONING

The graph of $y = f(x)$ passes through $(1, 4)$. Find the corresponding point on $y = 2f(x - 3) - 1$.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

2.B1 CHALLENGE

Investigate whether f(x) = ax + b commutes with g(x) = cx + d for every choice of a, b, c, d. State and justify a condition.

2.B2 CHALLENGE

Conjecture a rule for the number of fixed points of f(x) = ax + b, test it, and justify it.

2.B3 REASONING

Investigate the graphs of f(x) = 1/x and its inverse, and explain what you notice.

2.B4 REASONING

Take a function of your own that is not one-to-one. Restrict its domain so that it becomes one-to-one, and justify why your restriction works.

2.B5 REASONING

Investigate the effect of the transformation y = f(kx) for k = 2, 0.5 and −1 on the graph of a function of your own.

2.B6 CHALLENGE

Conjecture a rule for the inverse of a function of the form f(x) = (ax + b)/(cx + d), and test it on two examples.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

2.B7 CHALLENGE

Investigate whether the inverse of a strictly increasing function is always strictly increasing, and justify.

2.B8 REASONING

Investigate the fixed points of $f(x) = ax + b$ and their number, and justify.

2.B9 CHALLENGE

Conjecture a rule for when $f(x) = ax + b$ commutes with $g(x) = cx + d$, and test it.

2.B10 REASONING

Investigate the effect of the transformation $y = f(kx)$ on the graph of a function of your own, and justify.

2.B11 REASONING

Investigate the graph of $y = f(-x)$ and its relationship to the graph of $y = f(x)$, and justify.

CRITERION C Communicating

Criterion C practice · Communicating

2.C1 REASONING

Write a complete solution to finding the inverse of $f(x) = (2x - 1)/(x + 3)$, including the domain and a composition check.

2.C2 REASONING

Write a complete solution to the transformation taking $y = 1/x$ to $y = 4/(x - 2) + 1$, describing each stage.

2.C3 REASONING

Write a complete solution to $f(g(x))$ for $f(x) = \sqrt{(x + 1)}$ and $g(x) = x^2 - 2$, including the domain.

2.C4 REASONING

Write a complete solution to solving $f(x) = f^{-1}(x)$ for $f(x) = 2x + 5$.

2.C5 CHALLENGE

Write a complete solution to the restriction needed to make $f(x) = x^2 + 6x + 4$ one-to-one, and to finding the inverse on that restriction.

2.C6 REASONING

Write a complete solution to the model problem: a van costs 40 currency units per day plus 0.25 per kilometre. A rival costs 55 per day plus 0.15 per kilometre. Find the distance at which the costs are equal and describe the transformation between the models.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

2.C7 REASONING

Write a complete solution to sketching the graph of $y = f^{-1}(x)$ given the graph of $y = f(x)$, explaining why the reflection line is $y = x$.

2.C8 CHALLENGE

Write a complete solution to finding the inverse of $f(x) = (4x + 1)/(2x - 3)$, with its domain, and checking by composition.

2.C9 REASONING

Write a complete solution to the transformation taking $y = 1/x$ to $y = 3/(x + 2) - 1$, describing each stage.

2.C10 REASONING

Write a complete solution to $f(g(x))$ for $f(x) = 1/(x + 2)$ and $g(x) = x^2 - 1$, including the domain.

2.C11 CHALLENGE

Write a complete solution to proving that a strictly decreasing function has a strictly decreasing inverse.

2.C12 APPLICATION

Write a complete solution to the problem: a photocopier reduces a page to 80 per cent. Model the length after n passes and find when it is below 10 mm from 297 mm.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

2.D1 APPLICATION

A scientist's instrument reads r and reports R = 1.8r + 32, converting one scale to another. Find the inverse conversion and explain what it does.

2.D2 APPLICATION

A photographer's exposure model is $E = (a^2)/t$ for aperture a and time t. Explain the transformation that doubles the exposure without changing the aperture.

2.D3 APPLICATION

A language school charges c(n) = 300 + 120n for n weeks. A rival charges 180 + 150n. Model both, describe the transformation, and find when the rival becomes cheaper.

2.D4 APPLICATION

A tank drains so that its depth after t minutes is $D = 40(0.9)^T$ cm. Write the model for s minutes after a start 10 minutes later, and comment on the difference.

2.D5 APPLICATION

A delivery firm charges f(d) = 8 + 2d for the first 5 km and then f(d) = 18 + 1.5(d − 5) beyond. Explain why this is not a transformation of a single linear model.

2.D6 APPLICATION

A factory's output is modelled by P(t) = 200 + 30t units per day. A new machine adds a fixed 50 units per day and increases the rate by 20 per cent. Write the new model and describe the transformation.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

2.D7 APPLICATION

A vaccine's effectiveness is modelled by E(d) = 100(1 − (0.5)^(d/30)) per cent after d days. Find when effectiveness first exceeds 90 per cent, and explain what the transformation from $(0.5)^D$ to (0.5)^(d/30) does.

2.D8 APPLICATION

A population is modelled by $P(t) = 400(1.05)^T$. Write the model for s years after a survey taken 5 years later, and find the population at the new origin.

2.D9 APPLICATION

A company's value is modelled by $V(t) = 50000(1.08)^T$. After how many years does it double, and how does the answer change if the rate is 12 per cent?

2.D10 CHALLENGE

A student claims that f(x) + 3 and f(x + 3) always give the same graph. Investigate with two functions of your own and justify a conclusion.

2.D11 APPLICATION

A reservoir's volume is modelled by $V(h) = 3h^2$ for depth h in metres. A second reservoir is twice as wide at every depth. Write its model and describe the transformation.

2.D12 APPLICATION

A city's temperature is modelled by T(d) = 15 + 8 sin(πd/182.5) for day d. Explain the meaning of each parameter and find the warmest and coolest days.

CHAPTER 2 · APPLICATION AND ASSESSMENT

Applying chapter 2

These tasks use function notation, transformations, composition and inverses. State the domain of every function you write, and check every inverse by composition.

A1 · The recalibrated sensor

A sensor outputs s and a model converts it to a reading R = 2.5s + 12. The sensor is recalibrated so that its output is 4 higher than before for the same physical quantity. Write the new model, describe the transformation between the two models, and find the inverse of the new model, stating its domain and range.

Criteria A and C · identify which representation changes and set out the inverse with its domain.

A2 · The two subscription plans

Plan A costs 15 currency units per month plus 2 per stream. Plan B costs 9 per month plus 3 per stream. Model both, describe the transformation between them, find the number of streams at which they cost the same, and explain which plan is better for a household watching 8 streams a month.

Criteria A and D · model both plans, solve the crossing, and justify the recommendation with the model.

CHAPTER 2 · REVIEW

What you should now be able to do

Key ideas

- A function is a rule with a stated domain that gives exactly one output per input.
- f(x) + k translates vertically; f(x − a) translates horizontally by a.
- A stretch parallel to the y-axis has factor k; one parallel to the x-axis divides the input by k.
- The composite f(g(x)) applies g first.
- The domain of a composite is where the inner output is a permitted input of the outer function.
- An inverse exists exactly when the function is one-to-one.
- The horizontal line test decides invertibility; algebraically, f(a) = f(b) must force a = b.
- The graph of an inverse is the reflection in the line y = x.

Vocabulary

function · domain · range · mapping · one-to-one · composite · inner function · translation · reflection · stretch · inverse function · horizontal line test · fixed point · coordinate rule

CHAPTER 2 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

2.R1 Find the domain of $f(x) = 1/(x^2 - 9)$.

2.R2 Given $f(x) = x^2 + 1$ and $g(x) = 2x - 3$, find $f(g(2))$.

2.R3 Describe the transformation from $y = x^3$ to $y = -(x - 2)^3 + 5$.

2.R4 Find the inverse of $f(x) = 5 - 2x$ and check it.

2.R5 Explain why $f(x) = x^3 - 3x$ is not one-to-one on the real numbers.

2.R6 The point $(2, 7)$ lies on $y = f(x)$. Find its image on $y = -f(2x) + 1$.

CHAPTER 2 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> A function is a rule that pairs each input with exactly one output; transforming, composing and inverting are operations on the rule itself, and each leaves a trace on the graph.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of change, representation and patterns appear in your work?
- How is the global context of scientific and technical innovation, and its exploration of models that translate a measurement into a prediction, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a graphical representation of a function, and what would you still like to ask?

CHAPTER 2 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

A model is a representation of a relationship. Choose the representation that makes the operation you need visible, and say what it hides.

Service as Action

COMMUNITY NEED	Public measurement instruments are recalibrated without the conversion being published.
MATHEMATICAL CONNECTION	Functions, inverses and transformations.
PROPOSED ACTION	Take three instrument conversions in your community, find the inverse of each, and publish how to convert back.
EXPECTED IMPACT	A reader can convert a reading in either direction.
STUDENT REFLECTION	Record the conversions, their inverses and what you found.

Community Project

PROJECT IDEA	A conversion audit of the school's measurement instruments.
NEED	Scales, thermometers and timers report in units nobody has converted back.
RESEARCH QUESTION	Which conversions does the school rely on, and can each be reversed exactly?
MATHEMATICS	Functions, inverses and transformations.
ACTION OR OUTCOME	A one-page conversion card with both directions.
BENEFICIARIES	Everyone who reads a school instrument.
EVIDENCE	The conversion cards and the inverses you derived.

Personal Project connection · MYP 5

PERSONAL INTEREST	Cooking, sport, music or gaming.
LEARNING GOAL	To convert a quantity you care about in both directions and explain which representation you used.
PRODUCT OR OUTCOME	A conversion card with a graph and its inverse.
MATHEMATICS	Functions, inverses and transformations.
PROCESS	Model a conversion, invert it, and check by composition.
EVIDENCE	The card and the composition checks.

CHAPTER 3 · PART II · THINKING WITH MODELS

03

Exponential and Logarithmic Functions

Some quantities change by adding a fixed amount and some by multiplying by a fixed factor. Logarithms are the inverse language of multiplication, and they turn a multiplicative relationship into an additive one.

ENGAGE

A population doubles every 8 years. A student models it as 8 times the initial population after 64 years. Another models it as 2 to the power 8. Which is right, and why does the difference matter for a forecast a century ahead?

Framing the inquiry

Key concept: Change
Related concepts: Equivalence · Patterns · Representation
Global context: Scientific and technical innovation
Global context exploration: populations, medicines, sound and seismic energy

Statement of inquiry: Exponential change is repeated multiplication; logarithms are the inverse language that turns multiplication into addition and makes the rate visible.

Questions to guide us

- **Factual:** What do the logarithm laws say, and how is a logarithm defined?
- **Conceptual:** Why does an exponential model outgrow every linear model, and what does a logarithm measure?
- **Debatable:** If a model fits the past decade exactly, is it a better model than one that fits approximately and states its limits?

ATL · CRITICAL THINKING. In order for a student to judge whether a growth claim is plausible, students must compare a linear and an exponential model of the same situation, identify the quantity that changes multiplicatively, and justify the choice of model.

CHAPTER 3 · INVESTIGATION 3.1

Exponential growth and decay

INQUIRY QUESTION

Why does repeated multiplication outgrow repeated addition?

YOUR TASK

Take 1000 currency units. Model it under three schemes: simple interest at 5 per cent per year, compound interest at 5 per cent per year, and compound interest at 5 per cent per half-year. Record the value after 1, 2, 5 and 10 years for each, and then say which two agree and why the third is different.

year	simple	compound yearly	compound half-yearly
1	1050	1050	1050.63
2	1100	1102.50	1105.12
5	1250	1276.28	1280.08
10	1500	1628.89	1638.62
20	2000	2653.30	2685.06

CHAPTER 3 · INVESTIGATION 3.1

Exponential growth and decay

What do you notice?

- Simple interest added the same amount every year, so its graph was a straight line.
- Compound interest added a growing amount, so its graph curved upwards.
- Compounding twice a year gave slightly more, because interest earned interest sooner.
- The gap between the schemes widened every year, which is the signature of exponential against linear change.

Observation. Simple interest is a linear model and compound interest is an exponential one, and the exponential model pulls ahead without bound.

MATHEMATICAL CONCLUSION

An exponential model $N = N_0 b^T$ compounds a multiplier b for each unit of time, so equal percentage changes occur in equal intervals. The base b is the multiplier, and the exponent counts units of time, not the amount of the quantity. Here 1000×1.05^T is the yearly model and 1000 × 1.025^(2t) the half-yearly one.

Your own words

CHAPTER 3 · 3.1

Exponential growth and decay

An exponential model has the form $N = N_0 b^T$, where N_0 is the initial quantity, b is the base or multiplier per unit time, and t is the time. The base is the whole story: $b > 1$ gives growth, $0 < b < 1$ gives decay, and $b = 1$ gives no change at all.

A constant multiplicative rate means equal percentage changes in equal time intervals. This is what distinguishes exponential change from linear change, where equal absolute changes occur in equal intervals instead.

The graph of an exponential model is a curve with a horizontal asymptote. Growth curves steepen; decay curves flatten towards the asymptote; neither ever reaches it in finite time.

Any exponential model can be written with any positive base, because $b^T = e^{(kt)}$ for $k = \ln b$. The equivalence of the two forms is exact, and choosing a base is choosing a representation rather than changing the model.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 3 · 3.1 · WORKED EXAMPLE 1

Exponential growth and decay

WORKED EXAMPLE 3.1.1

Problem. A population of 400 bacteria doubles every 3 hours. Write the model, find the population after 10 hours, and find when it reaches 25 600.

time (h)	model	population
0	400×2^0	400
3	400×2^1	800
6	400×2^2	1600
9	400×2^3	3200
12	400×2^4	6400

Doubling every 3 hours means the exponent counts three-hour periods.

Mathematical reasoning. The multiplier per period is 2 and the period is 3 hours, so the exponent is t/3; then solve for t by equating.

1. The model is N(t) = 400 × 2^(t/3).
2. After 10 hours: N(10) = 400 × 2^(10/3) = 400 × 2^3.333.
3. 2^3.333 ≈ 10.08, so N(10) ≈ 4032 bacteria.
4. Set 400 × 2^(t/3) = 25 600, so 2^(t/3) = 64.
5. Since 64 = 2^6, t/3 = 6 and t = 18 hours.

Final answer. N(t) = 400 × 2^(t/3); about 4032 after 10 hours; 25 600 after 18 hours

Interpretation. The exponent counts periods, not hours, and confusing the two is the commonest error in exponential modelling.

CHAPTER 3 · 3.1 · WORKED EXAMPLE 2

Exponential growth and decay

WORKED EXAMPLE 3.1.2

Problem. A car loses 15 per cent of its value each year. Write the model and find its value after 5 years if it cost 24 000.

year	multiplier	value
0	0.85^0	24000
1	0.85^1	20400
2	0.85^2	17340
3	0.85^3	14739
5	0.85^5	10675

A loss of 15 per cent is a multiplier of 0.85.

Mathematical reasoning. A percentage loss is a multiplier below 1, and the model compounds it each year.

1. Losing 15 per cent leaves 85 per cent, so the multiplier is 0.85.
2. The model is $V(t) = 24000 \times 0.85^T$.
3. After 5 years: $V(5) = 24000 \times 0.85^5$.
4. $0.85^5 \approx 0.4437$, so $V(5) \approx 10\,675$.
5. The value never reaches zero in finite time, because $0.85^T > 0$ for every t.

Final answer. $V(t) = 24000 \times 0.85^T$; about 10 675 after 5 years

Interpretation. A decay model with a positive base is always positive, so the asymptote at zero is a limit rather than a value that is reached.

CHAPTER 3 · INVESTIGATION 3.2

The number e and continuous change

INQUIRY QUESTION

What does it mean for change to be continuous rather than periodic?

YOUR TASK

Investigate 1000 currency units at a nominal 10 per cent per year, compounded annually, half-yearly, quarterly, monthly, daily and continuously. Record the value after one year for each, and describe the pattern in the extra amounts as the compounding becomes more frequent.

compounding	periods per year	value after 1 year	gain over annual
annually	1	1100.00	0
half-yearly	2	1102.50	2.50
quarterly	4	1103.81	3.81
monthly	12	1104.71	4.71
daily	365	1105.16	5.16
continuously	—	1105.17	5.17

CHAPTER 3 · INVESTIGATION 3.2

The number e and continuous change

What do you notice?

- Each increase in frequency raised the final value, but by less each time.
- The daily and continuous values differed by less than one hundredth of a currency unit.
- The limiting value is 1000e^0.1 ≈ 1105.17, which is the continuous model.
- The gains formed a decreasing sequence, which is the signature of convergence.

Observation. More frequent compounding raises the accumulated value towards a limit, and the limit is reached by the continuous model.

MATHEMATICAL CONCLUSION

Compounding a nominal rate r in n periods per year gives $1000(1 + r/n)^{N}$, and as n grows this approaches $1000e^{R}$. The number e is the limit of $(1 + 1/n)^{N}$, and the continuous model $N = N_0e^{KT}$ is the limit of every periodic model with the same annual rate.

Your own words

CHAPTER 3 · 3.2

The number e and continuous change

Compounding more and more often raises the accumulated value towards a limit, and that limit is the number e ≈ 2.71828. A nominal annual rate r compounded continuously gives a growth factor of e^{R} per year.

The continuous model $N = N_0e^{KT}$ has rate k per unit time. Any base-b model can be rewritten in this form with k = ln b, and the two forms describe exactly the same change.

The natural logarithm ln is the logarithm to base e, and it is the inverse of the exponential function with base e. It is the convenient logarithm for continuous models for the same reason that e is the convenient base.

Continuous models appear wherever the rate of change is proportional to the quantity: populations, radioactive decay, cooling, and the accumulation of interest in a limit.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 3 · 3.2 · WORKED EXAMPLE 1

The number e and continuous change

WORKED EXAMPLE 3.2.1

Problem. A population is modelled by $P(t) = 400(1.05)^T$. Write the model for s years in continuous form, and find the continuous rate.

representation	model	rate
base form	$400(1.05)^T$	5% per year
continuous	$400e^{KT}$	$k = \ln 1.05$
value of k	$k \approx 0.04879$	4.879% per year

The same change, described with a different base.

Mathematical reasoning. Match the two forms by setting $e^K = 1.05$ and taking natural logarithms.

1. The model is $P(t) = 400(1.05)^T$.
2. Write $1.05^T = e^{KT}$, so $e^K = 1.05$.
3. Taking natural logarithms gives $k = \ln 1.05 \approx 0.04879$.
4. So the continuous model is P(t) = 400e^(0.04879t).
5. Check: e^(0.04879 × 1) ≈ 1.05, as required.

Final answer. P(t) = 400e^(0.04879t), continuous rate about 4.879 per cent per year

Interpretation. The two models are the same function in different clothes, and the continuous rate is slightly below the nominal rate.

CHAPTER 3 · 3.2 · WORKED EXAMPLE 2

The number e and continuous change

WORKED EXAMPLE 3.2.2

Problem. A substance decays so that 30 per cent remains after 5 hours. Find the half-life, and state the continuous model.

time (h)	fraction remaining	comment
0	1	start
5	0.30	given
10	0.09	0.30^2
t	0.50	half remains

Each equal interval multiplies the fraction by the same factor.

Mathematical reasoning. Find the multiplier per hour from the given data, then solve for the time at which half remains.

1. Let the model be $N(t) = N_0 b^T$. Then $b^5 = 0.30$, so $b = 0.30^{(1/5)} \approx 0.7860$.
2. The continuous form is $N = N_0 e^{KT}$ with $k = \ln 0.7860 \approx -0.2408$.
3. Half remains when $b^T = 0.5$, so $t = \ln 0.5 \div \ln 0.7860$.
4. $\ln 0.5 \approx -0.6931$ and $\ln 0.7860 \approx -0.2408$, so $t \approx 2.88$ hours.
5. The half-life is about 2.88 hours.

Final answer. half-life about 2.88 hours; $N = N_0 e^{(-0.2408t)}$

Interpretation. The half-life is independent of the starting quantity, which is why it is the standard way to report a decay rate.

CHAPTER 3 · INVESTIGATION 3.3

Logarithms and their laws

INQUIRY QUESTION

Why do logarithms turn multiplication into addition?

YOUR TASK

Build a table of $\log_2(n)$ for n = 1, 2, 4, 8, 16, 32, 3, 6, 12 and 24, using only $\log_2 2 = 1$ and $\log_2 3 \approx 1.585$. Then use your table to evaluate $\log_2 48$, $\log_2(1/6)$ and $\log_2 9$, and state the law you used in each case.

n	factorisation	$\log_2 n$	law used
1	2^0	0	definition
2	2^1	1	given
4	2^2	2	power law
8	2^3	3	power law
16	2^4	4	power law
32	2^5	5	power law
3	3^1	1.585	given
6	2×3	2.585	product law
12	4×3	3.585	product law
24	8×3	4.585	product law

CHAPTER 3 · INVESTIGATION 3.3

Logarithms and their laws

What do you notice?

- Every value came from adding or subtracting the two given values.
- The power law multiplied by the exponent, so $\log_2 16 = 4 \log_2 2 = 4$.
- The product law added, so $\log_2 6 = \log_2 2 + \log_2 3$.
- $\log_2 1$ was 0 because $2^0 = 1$, and that single fact made every quotient work.

> **Observation.** The logarithm laws are the index laws read backwards, and two values generate a whole table.

MATHEMATICAL CONCLUSION

The logarithm to base b is the inverse of b^X, so log_b x = y means exactly b^Y = x. Its laws — the product, quotient and power laws — are the index laws read backwards, and they turn multiplication into addition, division into subtraction and powers into multiplication. Here $\log_2 48 = \log_2(16 \times 3) = 4 + 1.585 = 5.585$, $\log_2(1/6) = -2.585$ and $\log_2 9 = 2 \times 1.585 = 3.170$.

Your own words

CHAPTER 3 · 3.3

Logarithms and their laws

The logarithm to base b, written $\log_b$, is the inverse of the exponential function with base b. The statement $\log_b x = y$ means exactly $b^y = x$, and the two statements are equivalent ways of describing the same relationship between x and y.

The three logarithm laws follow from the index laws: $\log_b(xy) = \log_b x + \log_b y$, $\log_b(x/y) = \log_b x - \log_b y$, and $\log_b(x^k) = k \log_b x$. Each is the index law read backwards.

The change of base formula $\log_b a = \log a \div \log b$ allows any logarithm to be computed from any other. It is the reason a calculator with only two logarithm buttons is sufficient.

A logarithm is an exponent, so its laws are the laws of exponents. Treating a logarithm as a number to be manipulated rather than as a label is what makes the laws usable.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 3 · 3.3 · WORKED EXAMPLE 1

Logarithms and their laws

WORKED EXAMPLE 3.3.1

Problem. Solve $\log_2(x) + \log_2(x - 2) = 3$.

step	working	result
combine	$\log_2(x(x - 2)) = 3$	product law
remove log	$x(x - 2) = 2^3$	definition
expand	$x^2 - 2x = 8$	—
solve	$x^2 - 2x - 8 = 0$	$x = 4$ or $x = -2$
check	$x = -2$ makes $\log_2(-2)$ undefined	rejected

The domain check rejects one of the two roots.

Mathematical reasoning. Combine with the product law, remove the logarithm with its definition, then check the domain.

1. Both logarithms require a positive argument, so $x > 0$ and $x - 2 > 0$, giving $x > 2$.
2. Combine: $\log_2(x(x - 2)) = 3$.
3. Remove the logarithm: $x(x - 2) = 2^3 = 8$.
4. So $x^2 - 2x - 8 = 0$, giving $(x - 4)(x + 2) = 0$.
5. $x = 4$ or $x = -2$; only $x = 4$ satisfies $x > 2$.

Final answer. $x = 4$

Interpretation. The domain restriction comes before the algebra, and it is what rejects the extraneous root.

CHAPTER 3 · 3.3 · WORKED EXAMPLE 2

Logarithms and their laws

WORKED EXAMPLE 3.3.2

Problem. Given log 2 ≈ 0.3010 and log 3 ≈ 0.4771, find log 12 and log(2/3) without a calculator.

quantity	in terms of log 2 and log 3	value
12	2 log 2 + log 3	0.6021 + 0.4771
2/3	log 2 − log 3	0.3010 − 0.4771
8	3 log 2	0.9030
0.5	−log 2	−0.3010

The laws break every number into primes.

Mathematical reasoning. Factorise into primes and apply the laws, using the given values.

1. $12 = 4 \times 3 = 2^2 \times 3$, so log 12 = 2 log 2 + log 3.
2. log 12 = 2(0.3010) + 0.4771 = 0.6021 + 0.4771 = 1.0791.
3. 2/3 has log 2 − log 3 = 0.3010 − 0.4771 = −0.1761.
4. Check: 10^1.0791 ≈ 12, and 10^(−0.1761) ≈ 0.667.
5. The laws turned two given values into many others.

Final answer. log 12 ≈ 1.0791 and log(2/3) ≈ −0.1761

Interpretation. Logarithms of composite numbers are sums of logarithms of primes, which is why logarithm tables were short.

CHAPTER 3 · INVESTIGATION 3.4

Solving exponential equations

INQUIRY QUESTION

When can an exponential equation be solved, and what is the method?

YOUR TASK

Solve each of these by first trying to match bases, and then by taking logarithms: $2^{(x+1)} = 32$; $4^X = 8^{(x-1)}$; $3^{(2x-1)} = 27$; and $10^X = 7$. Record which method you used for each and why, and check every answer.

equation	method	answer	check
$2^{(x+1)} = 32$	match bases	$x = 4$	$2^5 = 32$
$4^X = 8^{(x-1)}$	match bases	$x = 3$	$4^3 = 64$, $8^2 = 64$
$3^{(2x-1)} = 27$	match bases	$x = 2$	$3^3 = 27$
$10^X = 7$	logarithms	$x \approx 0.845$	$10^{0.845} \approx 7$

CHAPTER 3 · INVESTIGATION 3.4

Solving exponential equations

What do you notice?

- Three of the four equations yielded to matching bases, because the numbers were all powers of small integers.
- The fourth needed logarithms, because 7 is not a power of 10.
- Matching bases gave exact answers; logarithms gave approximations.
- Every answer passed the substitution check.

Observation. Matching bases is exact and should always be tried first; logarithms are the general method.

MATHEMATICAL CONCLUSION

An exponential equation is solved by matching bases when that is possible, and otherwise by taking logarithms of both sides and using the power law. The check is substitution into the original equation, and it also confirms that every logarithm taken had a positive argument.

Your own words

CHAPTER 3 · 3.4

Solving exponential equations

If both sides of an exponential equation can be written as powers of the same base, the exponents can be equated, because a one-to-one power function has one input per output.

If the bases cannot be matched, take logarithms of both sides. The power law brings the unknown down from the exponent, which converts the exponential equation into a linear or quadratic one.

Every solution must be checked in the original equation, because taking logarithms requires positive arguments. A candidate that makes an argument non-positive is extraneous.

Equations with the unknown in two places, such as $2^X = 3x + 1$, cannot be solved algebraically. They are solved numerically, and a graph or an iterative method gives the answer to any required accuracy.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 3 · 3.4 · WORKED EXAMPLE 1

Solving exponential equations

WORKED EXAMPLE 3.4.1

Problem. Solve $3^{(2x)} = 81$.

step	working	result
same base	$81 = 3^4$	—
equate	$2x = 4$	—
solve	$x = 2$	—
check	$3^4 = 81$	valid

Matching the base turns the equation into a linear one.

Mathematical reasoning. Write both sides as powers of 3, then equate the exponents.

1. $81 = 3^4$, so the equation becomes $3^{(2x)} = 3^4$.
2. Equal powers of the same base have equal exponents: $2x = 4$.
3. So $x = 2$.
4. Check: $3^{(2 \times 2)} = 3^4 = 81$.

Final answer. $x = 2$

Interpretation. Matching bases is always the first move, because it is exact and avoids rounding.

CHAPTER 3 · 3.4 · WORKED EXAMPLE 2

Solving exponential equations

WORKED EXAMPLE 3.4.2

Problem. Solve 5^(x) = 250, giving the answer to three significant figures.

step	working	result
take logs	x log 5 = log 250	power law
rearrange	x = log 250 ÷ log 5	—
evaluate	x ≈ 2.3979 ÷ 0.6990	—
answer	x ≈ 3.43	3 s.f.

Taking logarithms brings the exponent down.

Mathematical reasoning. Take logarithms of both sides, apply the power law, and rearrange.

1. Take logarithms: $\log(5^{X}) = \log 250$.
2. Apply the power law: x log 5 = log 250.
3. So x = log 250 ÷ log 5.
4. log 250 ≈ 2.3979 and log 5 ≈ 0.6990, so x ≈ 3.430.
5. Check: 5^3.43 ≈ 249.9, which is close to 250.

Final answer. x ≈ 3.43

Interpretation. Any base of logarithm works, because the change of base formula cancels the base out of the quotient.

CHAPTER 3 · INVESTIGATION 3.5

Modelling with logarithms

INQUIRY QUESTION

What does a logarithmic scale show that a linear scale cannot?

YOUR TASK

Take a quantity that doubles every day: 1, 2, 4, 8, 16, 32, 64. Plot the values on a linear axis and on a logarithmic axis, and describe what each plot shows. Then take the first seven square numbers and plot them on a log-log axis, and describe what you see.

day	doubling quantity	$\log_{10}$ of it	square	$\log_{10}$ of square
0	1	0	1	0
1	2	0.301	4	0.602
2	4	0.602	9	0.954
3	8	0.903	16	1.204
4	16	1.204	25	1.398
5	32	1.505	36	1.556
6	64	1.806	49	1.690

CHAPTER 3 · INVESTIGATION 3.5

Modelling with logarithms

What do you notice?

- On a linear axis the doubling quantity curved steeply upwards.
- On a logarithmic axis the same values lay on a straight line, because each step multiplied by the same factor.
- The squares also lay on a straight line on the log-log axis, with gradient 2.
- The gradient of a log-log line is the power in a power law.

Observation. A straight line on semi-log paper means exponential change, and a straight line on log-log paper means a power law.

MATHEMATICAL CONCLUSION

A logarithmic scale compresses equal ratios into equal distances, so multiplicative change appears as a straight line and the gradient measures the rate. Taking logarithms of both variables turns an exponential law into a linear one and a power law into a line whose gradient is the power. This is why the Richter, decibel and pH scales are logarithmic.

Your own words

CHAPTER 3 · 3.5

Modelling with logarithms

A logarithmic scale compresses equal ratios into equal distances, so a quantity that grows by a constant factor appears as a straight line and its patterns become readable. This is why logarithmic axes are used for anything spanning many orders of magnitude.

Semi-log graph paper has a linear axis and a logarithmic one. Data that plot as a straight line on semi-log paper are exponential, and the gradient of the line is the continuous rate.

The Richter scale for earthquakes, the decibel scale for sound and the pH scale for acidity are all logarithmic. A difference of one unit is a fixed ratio, not a fixed difference, which is why the scales are described as logarithmic.

Taking logarithms of both variables of a power law $y = kx^N$ gives $\log y = n \log x + \log k$, a straight line. Log-log graph paper therefore identifies power laws, just as semi-log paper identifies exponential ones.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 3 · 3.5 · WORKED EXAMPLE 1

Modelling with logarithms

WORKED EXAMPLE 3.5.1

Problem. An earthquake of magnitude 6 releases how many times the energy of one of magnitude 4, if energy is proportional to 10^(1.5M)?

magnitude	energy factor	ratio to M = 4
4	10^6	1
5	10^7.5	31.6
6	10^9	1000
7	10^10.5	31623

One step of magnitude is a factor of about 31.6 in energy.

Mathematical reasoning. Form the ratio of the energy factors and simplify using the index laws.

1. Energy is proportional to 10^(1.5M).
2. The ratio for M = 6 against M = 4 is 10^(1.5 × 6) ÷ 10^(1.5 × 4).
3. By the quotient law this is 10^(9 − 6) = 10^3.
4. So a magnitude 6 earthquake releases 1000 times the energy.
5. One step of magnitude is a factor of 10^1.5 ≈ 31.6.

Final answer. 1000 times

Interpretation. The magnitude scale is logarithmic, so differences in magnitude are ratios in energy, and the ratios grow very fast.

CHAPTER 3 · 3.5 · WORKED EXAMPLE 2

Modelling with logarithms

WORKED EXAMPLE 3.5.2

Problem. Sound intensity level is $L = 10 \log(I/I_0)$ decibels. Find the ratio of intensities for levels of 60 dB and 40 dB.

level (dB)	log term	intensity ratio
40	4	10^4
50	5	10^5
60	6	10^6
70	7	10^7

Twenty decibels is a factor of 100 in intensity.

Mathematical reasoning. Subtract the two level equations and use the quotient law.

1. $L_1 - L_2 = 10 \log(I_1/I_0) - 10 \log(I_2/I_0) = 10 \log(I_1/I_2)$.
2. So $60 - 40 = 10 \log(I_1/I_2)$, giving $20 = 10 \log(I_1/I_2)$.
3. Hence $\log(I_1/I_2) = 2$, so $I_1/I_2 = 10^2 = 100$.
4. The 60 dB sound is 100 times as intense as the 40 dB sound.
5. Note that a 20 dB difference sounds roughly four times as loud, because loudness is not logarithmic in the same way.

Final answer. 100 times

Interpretation. A difference on a logarithmic scale is a ratio, and the ratio is found by dividing the difference by the scale factor.

CHAPTER 3 · INVESTIGATION 3.6

Investigation: how many digits?

INQUIRY QUESTION

How many digits does a large power have, and can it be predicted?

YOUR TASK

Investigate the number of digits of 2^N for n = 1 to 12. Form a conjecture and justify it. Then investigate the leading digit of 2^N over the same range, and describe the pattern you find.

n	2^N	$n \log_{10} 2$	digits	leading digit
1	2	0.301	1	2
2	4	0.602	1	4
3	8	0.903	1	8
4	16	1.204	2	1
5	32	1.505	2	3
6	64	1.806	2	6
7	128	2.107	3	1
8	256	2.408	3	2
9	512	2.709	3	5
10	1024	3.010	4	1
11	2048	3.311	4	2
12	4096	3.612	4	4

CHAPTER 3 · INVESTIGATION 3.6

Investigation: how many digits?

What do you notice?

- The digit count increased by 1 about every 3.32 powers.
- The increases happened at n = 4, 7, 10 and 12, which is not a fixed period.
- The leading digit cycled through 2, 4, 8, 1, 3, 6, 1, 2, 5, 1, 2, 4.
- The leading digit was 1 at n = 4, 7 and 10, more often than any other digit.

> **Observation.** The digit count grows linearly in n with slope $\log_{10}2$, and the leading digit cycles irregularly.

MATHEMATICAL CONCLUSION

The number of digits of b^N is the whole part of $n \log_{10}b$, plus one, because b^N lies between 10^K and 10^{K+1} where k is that whole part. The fractional part of $n \log_{10}b$ gives the leading digits, and its distribution explains why small leading digits are common — the observation behind Benford's law.

Your own words

CHAPTER 3 · 3.6

Investigation: how many digits?

A positive integer with n digits lies between 10^{N-1} and 10^{N}, so the number of digits of a number x is the least integer strictly greater than $\log_{10}$x. Counting digits is therefore a logarithm problem.

The number of digits of b^{N} is the whole part of n $\log_{10}$b, plus one. The fractional part of n $\log_{10}$b determines the leading digits, because 10 raised to that fractional part lies between 1 and 10.

This is the basis of floating-point representation in every computer: a number is stored as a mantissa between 1 and 10 and an exponent, which is exactly the decomposition that logarithms describe.

The leading-digit pattern of powers is not uniform. Benford's law observes that in many real data sets the leading digit is 1 far more often than 9, and the logarithmic distribution explains it.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 3 · 3.6 · WORKED EXAMPLE 1

Investigation: how many digits?

WORKED EXAMPLE 3.6.1

Problem. How many digits does 2^{100} have?

n	2^N	$n \log_{10} 2$	digits
10	1024	3.01	4
20	1048576	6.02	7
50	$\approx 1.126 \times 10^{15}$	15.05	16
100	$\approx 1.267 \times 10^{30}$	30.10	31

The digit count is the whole part of n log 2, plus one.

Mathematical reasoning. Compute $n \log_{10} 2$, take its whole part and add one.

1. $\log_{10} 2 \approx 0.30103$.
2. So $100 \log_{10} 2 \approx 30.103$.
3. The whole part is 30, and the number of digits is $30 + 1 = 31$.
4. Check: $2^{100} \approx 1.2677 \times 10^{30}$, which indeed has 31 digits.
5. The fractional part 0.103 gives the leading digits, because 10^0.103 ≈ 1.27.

Final answer. 31 digits

Interpretation. One multiplication and one truncation answer a question about a number with 31 digits.

CHAPTER 3 · 3.6 · WORKED EXAMPLE 2

Investigation: how many digits?

WORKED EXAMPLE 3.6.2

Problem. How many digits does 7^{20} have, and what are its leading digits?

quantity	value	use
$\log_{10}7$	≈ 0.84510	given
$20 \log_{10}7$	≈ 16.902	whole part 16
digits	17	16 + 1
fractional part	0.902	10^0.902 ≈ 7.98

The whole part counts digits; the fraction gives the leading digits.

Mathematical reasoning. Separate n log b into its whole part and its fractional part.

1. $\log_{10}7 \approx 0.84510$.
2. $20 \log_{10}7 \approx 16.902$.
3. The whole part is 16, so the number of digits is 17.
4. The fractional part is 0.902, and 10^0.902 ≈ 7.98.
5. So 7^{20} begins with the digits 79, and is about 7.98×10^{16}.

Final answer. 17 digits, beginning with 79

Interpretation. The same calculation that counts the digits also reveals the leading ones, which is how a computer stores a number.

CHAPTER 3 · INVESTIGATION 3.7

Mixed practice

INQUIRY QUESTION

Which representation of the change makes the rate visible?

YOUR TASK

A company's revenue is 4 million in year 0 and 5.8 million in year 5. Fit an exponential model, state the annual rate, find the doubling time, and predict the revenue in year 12. Then state one assumption the model makes and one reason it might fail.

step	working	result
model	$5.8 = 4\,b^{5}$	—
base	$b^{5} = 1.45$	$b \approx 1.0771$
rate	$b - 1 \approx 0.0771$	7.71% per year
doubling	$\ln 2 \div \ln 1.0771$	≈ 9.31 years
year 12	4×1.0771^{12}	≈ 12.16 million

CHAPTER 3 · INVESTIGATION 3.7

Mixed practice

What do you notice?

- The base came from the ratio of the two known values, raised to the power 1/5.
- The annual rate was just under 8 per cent.
- The doubling time was independent of the starting revenue.
- The prediction for year 12 was more than three times the starting value.

Observation. Two data points fix an exponential model, and the rate and the doubling time are read straight off it.

MATHEMATICAL CONCLUSION

An exponential model is fitted by finding the base from two data points, and the base gives both the rate and the doubling time. The model assumes the rate stays constant, which is the assumption most likely to fail: markets saturate, competitors enter and costs rise. Here the fitted model is $R(t) = 4 \times 1.0771^{T}$ million, with a doubling time of about 9.3 years.

Your own words

CHAPTER 3 · 3.7

Mixed practice

Every exponential question can be approached in three ways: match the bases, take logarithms, or read the rate off a model. The method is chosen by the shape of the question, and choosing well saves work.

The base form $N = N_0 b^T$ and the continuous form $N = N_0 e^{KT}$ are equivalent representations of the same change, related by $k = \ln b$. Converting between them is a routine step, not a change of model.

Logarithm laws are used to solve equations, to simplify expressions and to interpret scales. In each case the underlying move is the same: turn a multiplicative statement into an additive one.

Mixed practice is where the representation is chosen rather than given, so the reasoning behind the choice is part of what is being practised.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 3 · 3.7 · WORKED EXAMPLE 1

Mixed practice

WORKED EXAMPLE 3.7.1

Problem. A town's population grows from 12 000 to 15 000 in 4 years. Assuming exponential growth, find the annual growth rate and predict the population after a further 6 years.

step	working	result
model	$15000 = 12000\ b^4$	—
base	$b^4 = 1.25$	—
rate	b = 1.25^0.25 ≈ 1.0574	5.74% per year
prediction	$15000 \times 1.0574^6 \approx 21090$	—

Two data points determine an exponential model.

Mathematical reasoning. Use the two known values to find the base, then extrapolate.

1. The model is $P(t) = 12000\ b^T$ and $P(4) = 15000$.
2. So $b^4 = 15000/12000 = 1.25$.
3. Hence b = 1.25^(1/4) ≈ 1.0574, an annual rate of about 5.74 per cent.
4. After a further 6 years, t = 10, so $P(10) = 12000 \times 1.0574^{10}$.
5. $1.0574^{10} \approx 1.7575$, so $P(10) \approx 21\ 090$.

Final answer. about 5.74 per cent per year; about 21 090 after 10 years

Interpretation. Two data points fix an exponential model completely, which is why exponential extrapolation is so easy and so dangerous.

CHAPTER 3 · 3.7 · WORKED EXAMPLE 2

Mixed practice

WORKED EXAMPLE 3.7.2

Problem. A bank offers 6 per cent nominal annual interest compounded continuously. How long does it take for a deposit to double?

step	working	result
model	A = P e^(0.06t)	—
double	2P = P e^(0.06t)	—
logs	ln 2 = 0.06t	—
solve	t = ln 2 ÷ 0.06	≈ 11.55 years

Continuous doubling uses the natural logarithm.

Mathematical reasoning. Set the model equal to twice the initial value and take natural logarithms.

1. The model is A(t) = P e^(0.06t).
2. Doubling means A(t) = 2P, so e^(0.06t) = 2.
3. Taking natural logarithms: 0.06t = ln 2 ≈ 0.6931.
4. So t ≈ 0.6931 ÷ 0.06 ≈ 11.55 years.
5. The answer is independent of P, which is why the doubling time is quoted.

Final answer. about 11.55 years

Interpretation. The doubling time of a continuous model is ln 2 ÷ k, a formula worth remembering because it does not depend on the deposit.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

3.A1 FLUENCY

Write 10^{-4} as a decimal.

3.A2 FLUENCY

Simplify 8^(2/3) ÷ 8^(−1/3).

3.A3 FLUENCY

Evaluate 27^(−2/3).

3.A4 FLUENCY

Solve 2^(x+3) = 64.

3.A5 FLUENCY

Solve 3^(2x) = 1/81.

3.A6 FLUENCY

Given log 5 ≈ 0.6990, find log 125 and log 0.2.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

3.A7 FLUENCY

Solve $\log_3(x) = 4$.

3.A8 REASONING

Solve $\log_2(x) + \log_2(x - 2) = 3$, checking the domain.

3.A9 FLUENCY

Write $\log(8) + \log(12) - \log(6)$ as a single logarithm.

3.A10 FLUENCY

Solve $2^X = 5$, giving the answer to three significant figures.

3.A11 REASONING

Explain why log_b 1 = 0 for every permitted base b.

3.A12 CHALLENGE

Solve $9^X = 4$^(x+1), giving the answer to three significant figures.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

3.B1 REASONING

Investigate the values of $(1 + 1/n)^N$ for n = 1, 10, 100 and 1000, and predict the limit.

3.B2 REASONING

Conjecture a rule for the number of digits of 3^N and test it for n = 1 to 8.

3.B3 CHALLENGE

Investigate the leading digit of 3^N for n = 1 to 12 and describe the pattern.

3.B4 CHALLENGE

Investigate whether 2^N always eventually exceeds n^2, and justify your answer.

3.B5 REASONING

Take a decaying quantity of your own. Investigate how the half-life changes when the decay rate doubles, and justify.

3.B6 REASONING

Investigate the sum 1 + 1/2 + 1/4 + 1/8 + … for 10 terms and then for 20, and predict the limit.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

3.B7 REASONING

Investigate the graph of $y = 2^X$ and $y = \log_2 x$ on the same axes, and explain the relationship.

3.B8 REASONING

Investigate the values of $(1 + 1/n)^N$ for $n = 100$ and $n = 10\,000$, and justify the trend.

3.B9 CHALLENGE

Conjecture a rule for the leading digit of 7^N and test it for $n = 1$ to 10.

3.B10 REASONING

Investigate the half-life of a substance when the decay rate is tripled, and justify.

3.B11 REASONING

Investigate the graph of $y = e^X$ and $y = \ln x$ on the same axes and justify the relationship.

CRITERION C Communicating

Criterion C practice · Communicating

3.C1 REASONING

Write a complete solution to solving 4^(x+1) = 8^(2x−1).

3.C2 REASONING

Write a complete solution to solving $3^X = 20$, giving the answer to three significant figures and checking it.

3.C3 REASONING

Write a complete solution to simplifying $\log(x^2) - \log(x) - 2\log 3$ into a single logarithm.

3.C4 APPLICATION

Write a complete solution to the model: a culture of 500 bacteria doubles every 40 minutes. Find the population after 3 hours and the time to reach 100 000.

3.C5 REASONING

Write a complete solution to finding the number of digits of 6^{50}.

3.C6 REASONING

Write a complete solution to the change-of-base calculation $\log_5 200$ using only common logarithms.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

3.C7 APPLICATION

Write a complete solution to the cooling model T = 20 + 60e^(−0.05t), finding when the temperature first falls below 25.

3.C8 FLUENCY

Write a complete solution to solving e^(2x) = 15, giving the answer to three significant figures.

3.C9 REASONING

Write a complete solution to simplifying $\log(x^3) - 2\log x + \log(1/x)$ into a single logarithm.

3.C10 REASONING

Write a complete solution to finding the number of digits of 3^{50}.

3.C11 APPLICATION

Write a complete solution to the model: a culture of 800 bacteria triples every 5 hours. Find the population after a day and the time to reach one million.

3.C12 CHALLENGE

Write a complete solution to proving that log_b a = 1/(log_a b) for permitted bases.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

3.D1 APPLICATION

A radioactive isotope has a half-life of 8 days. Find the continuous decay rate and the mass remaining after 30 days from 200 g.

3.D2 APPLICATION

A lake's pollution falls by 12 per cent a year. How long until it is below a quarter of its present level?

3.D3 APPLICATION

A savings account pays 4 per cent per year compounded monthly. Find the effective annual rate and the time to double.

3.D4 APPLICATION

A sound is measured at 75 dB. Find its intensity relative to the threshold, and the level of a sound 1000 times as intense.

3.D5 APPLICATION

A city's population is 250 000 and grows 3 per cent a year. When will it first exceed 400 000, and what assumption does the answer depend on?

3.D6 APPLICATION

A company's output is modelled by P(t) = 2000e^(0.07t). Find when output first exceeds 10 000, and comment on the realism of the model.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

3.D7 REASONING

A student claims that a quantity that falls by 10 per cent a day will eventually reach zero. Investigate and justify.

3.D8 APPLICATION

A medicine's concentration is C(t) = 40e^(−0.25t) mg per litre. Find when it falls below 5 mg per litre and how many half-lives that is.

3.D9 APPLICATION

An investment triples in 12 years. Find the annual rate, and the time to double.

3.D10 APPLICATION

A student models a population as P(t) = 1000 + 50t because it grew by 50 in the first year. A second models it as 1000×1.05^{T}. Compare the two predictions after 20 years and explain the difference.

3.D11 APPLICATION

A cooling object is at 90 °C in a room at 20 °C. After 10 minutes it is at 60 °C. Fit the model T = 20 + Ae^(−kt) and find the temperature after 30 minutes.

3.D12 REASONING

A student says that because log(ab) = log a + log b, it follows that log(a + b) = log a + log b. Investigate with numbers and justify a conclusion.

CHAPTER 3 · APPLICATION AND ASSESSMENT

Applying chapter 3

These tasks use exponential and logarithmic models. State the base and the units of the rate, check the model against one known value, and comment on the domain of any prediction.

A1 · The investment comparison

A bank offers 5 per cent per year compounded annually. A second offers 4.8 per cent per year compounded continuously. A deposit of 8000 currency units is made in each. Find the value of each after 10 years, state which is better and by how much, and explain why the nominal rates alone do not decide the answer.

Criteria A and D · model both, compare, and justify with the model.

A2 · The medicine schedule

A medicine has concentration C(t) = 60e^(−0.3t) mg per litre t hours after a dose. The effective level is 12 mg per litre. Find how long the dose stays effective, the half-life of the medicine, and how many hours apart doses must be given for the concentration never to fall below the effective level. State one assumption the calculation makes.

Criteria A and C · solve the inequality, find the half-life, and state the assumption.

CHAPTER 3 · REVIEW

What you should now be able to do

Key ideas

- An exponential model $N = N_0 b^T$ compounds the base b for each unit of time.
- The base-b form and the continuous form $N_0 e^{KT}$ describe the same change, with $k = \ln b$.
- Compounding more frequently approaches the continuous model, whose limit is e^R.
- A logarithm is an exponent: $\log_b x = y$ means exactly $b^Y = x$.
- The logarithm laws are the index laws read backwards.
- Exponential equations are solved by matching bases or by taking logarithms.
- Every solution must be checked for a positive logarithm argument.
- A logarithmic scale compresses equal ratios into equal distances.

Vocabulary

exponential · base · multiplier · asymptote · half-life · continuous growth · natural logarithm · logarithm laws · change of base · semi-log · log-log · Benford's law · floating point · decibel · Richter scale

CHAPTER 3 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

3.R1 Solve $2^{(2x-1)} = 32$.

3.R2 Given $\log 3 \approx 0.4771$, find $\log 81$.

3.R3 Solve $\log_5(x + 2) = 2$.

3.R4 How many digits does 5^{20} have?

3.R5 A quantity halves every 6 hours. Find the continuous decay rate.

3.R6 Explain why $\log(a + b)$ is not $\log a + \log b$.

CHAPTER 3 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> Exponential change is repeated multiplication; logarithms are the inverse language that turns multiplication into addition and makes the rate visible.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of change and the related concepts of equivalence, patterns and representation appear in your work?
- How is the global context of scientific and technical innovation, and its exploration of populations, medicines, sound and seismic energy, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of an exponential model for a real population, and what would you still like to ask?

CHAPTER 3 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

A rate is only meaningful with its base and its time unit. Choose the representation that makes the rate visible, and say what the model cannot predict.

Service as Action

COMMUNITY NEED	Growth and decay claims in public health and finance are quoted without their base or time unit.
MATHEMATICAL CONNECTION	Exponential models and logarithms.
PROPOSED ACTION	Collect three published growth or decay claims, convert each to a stated base and time unit, and publish the conversion.
EXPECTED IMPACT	A reader can compare two rates that were quoted differently.
STUDENT REFLECTION	Record the claims, the conversions and what you found.

Community Project

PROJECT IDEA	A rate audit of the school's published figures.
NEED	Enrolment, energy use and recycling figures are quoted as percentages with no time unit.
RESEARCH QUESTION	Which published rates can be compared directly, and which need converting first?
MATHEMATICS	Exponential models, logarithms and change of base.
ACTION OR OUTCOME	A one-page rate table with base, time unit and source.
BENEFICIARIES	Everyone who reads a school figure.
EVIDENCE	The rate table and the conversions you performed.

Personal Project connection · MYP 5

PERSONAL INTEREST	Music, sport, cooking or gaming.
LEARNING GOAL	To measure a rate in something you care about and state its base and time unit.
PRODUCT OR OUTCOME	A rate card with a graph on a logarithmic axis.
MATHEMATICS	Exponential models and logarithms.
PROCESS	Measure twice, fit a model, and report the rate with its units.
EVIDENCE	The rate card and the model you fitted.

CHAPTER 4 · PART I · NUMBER AND ABSTRACT REASONING

04

Sequences and Series

A sequence is a list generated by a rule, and a series is the sum of a sequence. Both can be summarised by a formula, and the formula's behaviour decides whether the sum is finite, unbounded or convergent.

ENGAGE

Two savers each deposit 100 currency units in the first month. The first adds 10 every month; the second increases each deposit by 10 per cent of the previous month's deposit. After twelve months who has saved more, and after ten years?

Framing the inquiry

Key concept: Form
Related concepts: Patterns · Generalisation · Simplification
Global context: Fairness and development
Global context exploration: saving, loans and the long-term decisions that shape opportunity

Statement of inquiry: A sequence is form made visible: one rule generates every term, and the sum of those terms simplifies to a formula whose behaviour — finite, unbounded or convergent — is decided by a single number.

Questions to guide us

- **Factual:** What are the formulas for the nth term and the sum of the first n terms of an arithmetic and of a geometric sequence?
- **Conceptual:** Why does a geometric series converge exactly when the common ratio is between −1 and 1?
- **Debatable:** If two savings plans agree for the first year, does the one that is better after ten years make the other unfair?

ATL · CRITICAL THINKING. In order for a student to judge whether a savings plan is fair, students must compare two sequences that begin identically, identify the rule that generates each, and justify which is the better long-term choice.

CHAPTER 4 · INVESTIGATION 4.1

Arithmetic sequences

INQUIRY QUESTION

What single number generates an arithmetic sequence?

YOUR TASK

Take six lists of your own: three arithmetic and three not. For the arithmetic ones, find the common difference and the general term. For the others, explain precisely which property of an arithmetic sequence fails, and state whether the list is geometric.

list	type	rule	general term
5, 8, 11, 14, …	arithmetic	$d = 3$	$3n + 2$
1, 4, 9, 16, …	neither	differences 3, 5, 7 not constant	n^2
2, 6, 18, 54, …	geometric	$r = 3$	$2 \times 3^{(n-1)}$
10, 7, 4, 1, …	arithmetic	$d = -3$	$13 - 3n$
1, 1, 2, 3, 5, 8, …	neither	each term is the sum of the two before it	no closed form of this kind
100, 90, 81, 72.9, …	geometric	$r = 0.9$	$100 \times 0.9^{(n-1)}$

CHAPTER 4 · INVESTIGATION 4.1

Arithmetic sequences

What do you notice?

- Two of the six lists were arithmetic, with common differences 3 and −3.
- Two were geometric, with ratios 3 and 0.9.
- The squares failed the arithmetic test because their differences grew.
- The Fibonacci-type list failed both tests, because its rule used two earlier terms.

> **Observation.** An arithmetic sequence has a constant difference; a geometric one has a constant ratio; a list can be neither.

MATHEMATICAL CONCLUSION

An arithmetic sequence is generated by adding a fixed common difference d, so u(n) = u(1) + (n − 1)d is a generalisation that covers every term from the first term and d alone. The test for the type is whether consecutive differences, or consecutive ratios, are constant, and a list can fail both tests.

Your own words

CHAPTER 4 · 4.1

Arithmetic sequences

An arithmetic sequence has a common difference d between consecutive terms, so u(n) = u(1) + (n − 1)d. The sequence is completely determined by its first term and its common difference.

The common difference is found by subtracting any term from the next one, and it must be the same for every consecutive pair, so the pattern is a single number repeated. A list that fails this test is not arithmetic, however regular it looks.

The graph of an arithmetic sequence, with n on the horizontal axis, is a set of equally spaced points on a straight line. The common difference is the gradient of that line, which is why arithmetic sequences are the discrete analogue of linear functions.

The nth-term formula is a generalisation: it claims something about every term from a single rule, and its simplification u(n) = dn + c is a straight line. It is proved by checking that consecutive terms differ by d.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 4 · 4.1 · WORKED EXAMPLE 1

Arithmetic sequences

WORKED EXAMPLE 4.1.1

Problem. Find the nth term of the sequence 7, 11, 15, 19, … and hence find the 50th term.

n	u(n)	u(n) − 4n	difference from previous
1	7	3	—
2	11	3	4
3	15	3	4
4	19	3	4
50	203	3	4

The common difference is constant, so the sequence is arithmetic.

Mathematical reasoning. Find the common difference, write the general term, and substitute.

1. Consecutive differences are all 4, so d = 4.
2. The first term is 7, so u(n) = 7 + (n − 1) × 4.
3. Simplify: u(n) = 4n + 3.
4. Check: u(1) = 7, u(2) = 11, u(3) = 15, as required.
5. So u(50) = 4(50) + 3 = 203.

Final answer. u(n) = 4n + 3; u(50) = 203

Interpretation. The general term is a formula that covers every term, and it is verified by checking the first few.

CHAPTER 4 · 4.1 · WORKED EXAMPLE 2

Arithmetic sequences

WORKED EXAMPLE 4.1.2

Problem. The 3rd term of an arithmetic sequence is 14 and the 9th term is 34. Find the first term and the common difference.

term	value	equation
u(3)	14	a + 2d = 14
u(9)	34	a + 8d = 34
subtract	—	6d = 20
solve	—	d = 10/3
u(1)	22/3	a = 14 − 2d

Two terms determine the whole sequence.

Mathematical reasoning. Write two equations from the general term and solve simultaneously.

1. Let the first term be a and the common difference be d.
2. Then a + 2d = 14 and a + 8d = 34.
3. Subtracting gives 6d = 20, so d = 10/3.
4. Substituting back: a = 14 − 2(10/3) = 14 − 20/3 = 22/3.
5. Check: u(9) = 22/3 + 8(10/3) = 22/3 + 80/3 = 102/3 = 34.

Final answer. first term 22/3, common difference 10/3

Interpretation. Any two terms determine an arithmetic sequence, because the general term is linear in n.

CHAPTER 4 · INVESTIGATION 4.2

Arithmetic series and sigma notation

INQUIRY QUESTION

How does a sum simplify to a formula?

YOUR TASK

Take the arithmetic sequence u(r) = 5r + 2. Compute $\sum_{r=1}^{10}$ u(r) by direct addition, then by the sum formula, then by splitting the sum into two parts. Record all three answers and explain why they must agree.

method	working	answer
direct addition	7 + 12 + … + 52	295
sum formula	(10/2)(7 + 52) = 5 × 59	295
splitting	5 × (1 + … + 10) + 2 × 10 = 5 × 55 + 20	295
why they agree	each method counts the same 10 terms	—

CHAPTER 4 · INVESTIGATION 4.2

Arithmetic series and sigma notation

What do you notice?

- All three methods gave 295, as they had to.
- The formula method used only the first and last terms.
- The splitting method used the known sum of the first ten integers.
- The formula method was fastest and the direct method was the check.

> **Observation.** The sum formula and the linearity of summation are two descriptions of the same total.

MATHEMATICAL CONCLUSION

The sum of an arithmetic series simplifies to (n/2)(u(1) + u(n)) because the terms can be paired so that every pair has the same total. Sigma notation records the same sum compactly, and its linearity lets the sum be split into parts that are easier to evaluate. Here all three routes gave 295.

Your own words

CHAPTER 4 · 4.2

Arithmetic series and sigma notation

The sum of the first n terms of an arithmetic sequence is $S(n) = (n/2)(2u(1) + (n - 1)d)$, or equivalently $S(n) = (n/2)(u(1) + u(n))$. The second form pairs the first with the last, and each pair has the same sum.

Sigma notation writes the sum of a list compactly: $S(n) = \sum_{r=1}^{n} u(r)$. The letter r is the index, 1 is its start and n its finish, and changing the index changes nothing about the value.

The pairing argument is the proof of the formula: write the sum forwards and backwards, add the two rows, and every pair sums to u(1) + u(n). There are n such pairs, so twice the sum is $n(u(1) + u(n))$, and the simplification to $(n/2)(u(1) + u(n))$ follows immediately.

Linearity of summation lets a sum be split: $\sum (a(r) + b(r)) = \sum a(r) + \sum b(r)$, and $\sum k\,a(r) = k \sum a(r)$. These are the rules that make sigma notation usable.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 4 · 4.2 · WORKED EXAMPLE 1

Arithmetic series and sigma notation

WORKED EXAMPLE 4.2.1

Problem. Find the sum of the first 60 positive integers.

pairing	sum	comment
1 + 60	61	first and last
2 + 59	61	second and second-last
30 + 31	61	middle pair
30 pairs	30 × 61	—

Each pair sums to 61, and there are 30 of them.

Mathematical reasoning. Pair the first term with the last, and note that each pair has the same sum.

1. The sequence is 1, 2, 3, ..., 60, with u(1) = 1 and u(60) = 60.
2. Pair 1 with 60, 2 with 59, and so on: each pair sums to 61.
3. There are 60 terms, so there are 30 pairs.
4. The sum is 30 × 61 = 1830.
5. Check with the formula: (60/2)(1 + 60) = 30 × 61 = 1830.

Final answer. 1830

Interpretation. The pairing argument explains the formula rather than merely applying it, and it works for every arithmetic series.

CHAPTER 4 · 4.2 · WORKED EXAMPLE 2

Arithmetic series and sigma notation

WORKED EXAMPLE 4.2.2

Problem. Evaluate $\sum_{r=1}^{12} (3r - 2)$.

r	3r − 2	value
1	1	1
2	4	4
12	34	34
sum	—	(12/2)(1 + 34) = 210

The terms form an arithmetic sequence with d = 3.

Mathematical reasoning. Recognise the terms as an arithmetic sequence and apply the sum formula.

1. The terms are 1, 4, 7, ..., with u(1) = 1 and d = 3.
2. The last term is u(12) = 3(12) − 2 = 34.
3. There are 12 terms, so the sum is (12/2)(1 + 34).
4. That is 6 × 35 = 210.
5. Check by splitting: 3 × (sum of 1 to 12) − 2 × 12 = 3 × 78 − 24 = 234 − 24 = 210.

Final answer. 210

Interpretation. The linearity check confirms the formula by a different route, which is what a check should do.

CHAPTER 4 · INVESTIGATION 4.3

Geometric sequences

INQUIRY QUESTION

What distinguishes a geometric sequence from an arithmetic one?

YOUR TASK

Take four geometric sequences of your own with $r = 2$, $r = 0.5$, $r = -1$ and $r = -0.5$, each with first term 8. Compute the first six terms of each, describe the behaviour of the terms as n grows, and record whether the terms grow, shrink, alternate or collapse.

n	r = 2	r = 0.5	r = −1	r = −0.5
1	8	8	8	8
2	16	4	−8	−4
3	32	2	8	2
4	64	1	−8	−1
5	128	0.5	8	−0.5
6	256	0.25	−8	0.25
behaviour	grows	shrinks to 0	alternates, fixed size	alternates, shrinks to 0

CHAPTER 4 · INVESTIGATION 4.3

Geometric sequences

What do you notice?

- With r = 2 the terms doubled each step and grew without bound.
- With r = 0.5 the terms halved and approached zero.
- With r = −1 the terms alternated between two values and never settled.
- With r = −0.5 the terms alternated and shrank, approaching zero.

Observation. The magnitude of r decides growth or decay, and the sign of r decides whether the terms alternate.

MATHEMATICAL CONCLUSION

A geometric sequence multiplies by a fixed ratio r, so u(n) = u(1)r^(n−1) is a generalisation covering every term. The behaviour of the sequence is decided by |r| alone: growth above 1, decay below 1, and neither at exactly 1. The sign of r only decides whether the terms change sign, not whether they grow.

Your own words

CHAPTER 4 · 4.3

Geometric sequences

A geometric sequence has a common ratio r between consecutive terms, so u(n) = u(1) × r^(n−1). Each term is the previous one multiplied by the same factor, which is why the growth is multiplicative rather than additive.

The sign of r matters. A positive r keeps every term the same sign as the first; a negative r makes the terms alternate in sign; r = 0 collapses the sequence to zero after the first term.

The magnitude of r decides the behaviour. With |r| > 1 the terms grow without bound; with |r| = 1 they stay constant or alternate; with 0 < |r| < 1 they shrink towards zero. These three patterns exhaust the possibilities.

A geometric sequence with a positive first term and r > 0 is an exponential function of n in disguise, and its graph is the set of points on an exponential curve.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 4 · 4.3 · WORKED EXAMPLE 1

Geometric sequences

WORKED EXAMPLE 4.3.1

Problem. Find the nth term of the sequence 3, 6, 12, 24, … and hence find the 10th term.

n	u(n)	u(n) ÷ 3	ratio to previous
1	3	1	—
2	6	2	2
3	12	4	2
4	24	8	2
10	1536	512	2

Each term is twice the previous one.

Mathematical reasoning. Find the common ratio and write the general term, then substitute.

1. Consecutive ratios are all 2, so r = 2.
2. The first term is 3, so u(n) = 3 × 2^(n−1).
3. Check: u(2) = 3 × 2 = 6 and u(3) = 3 × 4 = 12, as required.
4. So u(10) = 3 × 2^9 = 3 × 512 = 1536.
5. Note the exponent is n − 1, not n, because the first term corresponds to r^0.

Final answer. u(n) = 3 × 2^(n−1); u(10) = 1536

Interpretation. The exponent counts the number of multiplications, which is one fewer than the number of terms.

CHAPTER 4 · 4.3 · WORKED EXAMPLE 2

Geometric sequences

WORKED EXAMPLE 4.3.2

Problem. A ball is dropped from 2 m and rebounds to 60 per cent of its previous height each time. Find the height of the fifth rebound and the total distance travelled before it comes to rest.

rebound	height (m)	distance added (m)
0	2.00	2.00
1	1.20	2 × 1.20
2	0.72	2 × 0.72
3	0.432	2 × 0.432
4	0.2592	2 × 0.2592
5	0.15552	2 × 0.15552

Each rebound is 60 per cent of the last.

Mathematical reasoning. Model the rebound heights as a geometric sequence and sum the distances, counting each rebound twice.

1. The rebound heights are 2, 1.2, 0.72, … with r = 0.6.
2. So the height of the nth rebound is 2×0.6^{N}.
3. The fifth rebound is $2 \times 0.6^{5} \approx 0.1555$ m.
4. The total distance is the initial 2 m plus twice the sum of the rebound heights.
5. That sum is $2 \times 0.6 \div (1 - 0.6) = 3$, so the total distance is $2 + 2 \times 3 = 8$ m.

Final answer. fifth rebound about 0.156 m; total distance 8 m

Interpretation. The total distance is finite even though the ball bounces infinitely often, which is the first surprise of convergent series.

CHAPTER 4 · INVESTIGATION 4.4

Geometric series and convergence

INQUIRY QUESTION

When is an infinite sum finite?

YOUR TASK

Investigate the series $1 + r + r^2 + r^3 + \ldots$ for $r = 0.9$, $r = 0.99$, $r = -0.5$, $r = -0.99$, $r = 1$ and $r = -1$. For each, compute the partial sums after 5, 10 and 20 terms, and state whether a sum to infinity exists.

r	sum after 5	sum after 10	sum after 20	converges?
0.9	4.0951	6.5132	8.7841	yes, to 10
0.99	4.9010	9.5573	9.8088	yes, to 100
−0.5	0.6875	0.9990	1.0000	yes, to 2/3
−0.99	0.6699	0.5033	0.5134	yes, to about 0.5025
1	5	10	20	no
−1	1	0	0	no

CHAPTER 4 · INVESTIGATION 4.4

Geometric series and convergence

What do you notice?

- Every series with $|r| < 1$ approached a limit, however slowly.
- With $r = 0.99$ the limit was 100, and 20 terms had reached only 9.81.
- With negative r the partial sums alternated above and below the limit.
- With $r = 1$ and $r = -1$ the partial sums did not settle at all.

> **Observation.** The partial sums settle exactly when $|r| < 1$, and the limit is $u(1)/(1 - r)$.

MATHEMATICAL CONCLUSION

A geometric series converges exactly when $|r| < 1$, and its sum to infinity is $u(1)/(1 - r)$; otherwise the partial sums grow without bound or oscillate. The single number r decides the behaviour, which is why the convergence condition is checked before the formula is used. Here the limits were 10, 100, 2/3 and about 0.5025.

Your own words

CHAPTER 4 · 4.4

Geometric series and convergence

The sum of the first n terms of a geometric sequence is $S(n) = u(1)(1 - r^N)/(1 - r)$ when $r \neq 1$. The formula is proved by subtracting $r\,S(n)$ from $S(n)$, which cancels every term but two.

The infinite series $\sum_{n=0}^{\infty} u(1)r^N$ converges exactly when $|r| < 1$, and its sum is $u(1)/(1 - r)$. When $|r| \geq 1$ the partial sums grow without bound and no sum exists.

The condition $|r| < 1$ is not a technicality. With $r = 0.9$ the partial sums approach 10; with $r = -0.9$ they approach $1/1.9 \approx 0.526$; with $r = 1.1$ they grow without bound. A single number decides the behaviour.

A convergent series is a limit of partial sums, not an infinite addition performed all at once. The distinction matters when the terms are not all positive, because rearranging the terms of a conditionally convergent series can change its sum.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 4 · 4.4 · WORKED EXAMPLE 1

Geometric series and convergence

WORKED EXAMPLE 4.4.1

Problem. Find the sum of the geometric series 12 + 6 + 3 + 1.5 + … to infinity.

n	partial sum	remaining distance to 24
1	12	12
2	18	6
3	21	3
4	22.5	1.5
10	23.976	0.024

Each step halves the remaining distance to the limit.

Mathematical reasoning. Identify the first term and the common ratio, check the convergence condition, and apply the formula.

1. The first term is 12 and the common ratio is 6/12 = 0.5.
2. Since |r| = 0.5 < 1, the series converges.
3. The sum to infinity is u(1)/(1 − r) = 12/(1 − 0.5) = 12/0.5 = 24.
4. Check with partial sums: after 4 terms the sum is 22.5, and after 10 terms it is 23.976, approaching 24.
5. The remaining distance after n terms is 24×0.5^{N}, which tends to zero.

Final answer. 24

Interpretation. The limit 24 is never reached by a partial sum, but the partial sums get arbitrarily close to it, which is what convergence means.

CHAPTER 4 · 4.4 · WORKED EXAMPLE 2

Geometric series and convergence

WORKED EXAMPLE 4.4.2

Problem. Show that the series 1 + 1.1 + 1.21 + 1.331 + … has no sum to infinity, and explain why.

n	partial sum	growth
1	1	—
2	2.1	+1.1
3	3.31	+1.21
4	4.641	+1.331
10	25.937	+2.36

The partial sums grow without bound.

Mathematical reasoning. The common ratio is greater than 1 in magnitude, so the convergence condition fails.

1. The common ratio is 1.1, so $|r| = 1.1 > 1$.
2. The condition $|r| < 1$ fails, so the series diverges.
3. The partial sums are $S(n) = 1(1.1^N - 1)/(1.1 - 1) = 10(1.1^N - 1)$.
4. As n grows, 1.1^N grows without bound, so the partial sums do too.
5. Hence there is no sum to infinity.

Final answer. no sum; the series diverges because $|r| > 1$

Interpretation. The single number r decides everything, and the threshold is exactly 1 in magnitude.

CHAPTER 4 · INVESTIGATION 4.5

Annuities, loans and saving

INQUIRY QUESTION

How does a series model a long-term financial decision?

YOUR TASK

Investigate saving 100 currency units a month for 20 years at three rates: 0 per cent, 3 per cent and 6 per cent per year compounded monthly. Record the total paid in and the final value for each, and describe the pattern in the interest earned.

annual rate	monthly rate	total paid in	final value	interest earned
0%	0	24000	24000	0
3%	0.0025	24000	32883	8883
6%	0.005	24000	46204	22204
interest ratio	—	—	0.37 then 0.93	grows faster than the rate

CHAPTER 4 · INVESTIGATION 4.5

Annuities, loans and saving

What do you notice?

- With no interest the final value was exactly the amount paid in.
- At 3 per cent the interest was about 37 per cent of the total deposited.
- At 6 per cent the interest was about 93 per cent of the total deposited, nearly doubling the savings.
- Doubling the rate far more than doubled the interest, because the earliest deposits had longer to grow.

> **Observation.** The final value is a geometric series sum, and the rate enters through the ratio.

MATHEMATICAL CONCLUSION

A regular saving plan accumulates to $P((1+i)^N - 1)/i$, where P is the deposit, i the rate per period and n the number of periods. The formula is the geometric series sum applied to deposits, and the same formula read backwards repays a loan. Here 100 a month for 20 years grew to 32 883 at 3 per cent and 46 204 at 6 per cent.

Your own words

CHAPTER 4 · 4.5

Annuities, loans and saving

A regular deposit into an account paying interest is a geometric series: each deposit grows by the interest factor for the number of periods it remains invested. The accumulated value is the sum of a finite geometric series.

The future value of a deposit of P made at the end of each period for n periods at rate i per period is $P((1+i)^N - 1)/i$. The formula is the geometric sum formula applied to the deposits.

A loan repaid by regular instalments is the same formula read backwards: the present value of the instalments must equal the amount borrowed. Solving for the instalment gives the standard repayment formula.

The fairness of a plan depends on the interest rate, the frequency of compounding and the length of the term. Two plans with the same nominal rate can differ substantially once these are stated, which is why the assumptions are part of the model.

> FORMAL DEVELOPMENT
>
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In your own words

CHAPTER 4 · 4.5 · WORKED EXAMPLE 1

Annuities, loans and saving

WORKED EXAMPLE 4.5.1

Problem. A saver deposits 500 currency units at the end of each year into an account paying 4 per cent per year. Find the value after 10 years.

deposit	years invested	value after 10 years
1	9	500×1.04^{9}
2	8	500×1.04^{8}
10	0	500
total	—	$500((1.04^{10} - 1)/0.04)$

Each deposit grows for a different number of years.

Mathematical reasoning. Recognise the accumulated deposits as a geometric series and apply the sum formula.

1. The deposits are 500 each, and the deposit made k years before the end has grown by 1.04^{K}.
2. So the total is $500(1 + 1.04 + 1.04^{2} + \ldots + 1.04^{9})$.
3. This is a geometric series with $r = 1.04$ and 10 terms.
4. The sum is $500 \times (1.04^{10} - 1)/(1.04 - 1) = 500 \times 12.0061$.
5. So the value is about 6003 currency units.

Final answer. about 6003 currency units

Interpretation. The deposits together earn about 1003 currency units of interest, and the earliest deposit earns the most.

CHAPTER 4 · 4.5 · WORKED EXAMPLE 2

Annuities, loans and saving

WORKED EXAMPLE 4.5.2

Problem. A loan of 20 000 currency units is repaid in 5 equal annual instalments at 6 per cent per year. Find the annual instalment.

step	working	result
present value	$R(1 + 1.06^{-1} + \ldots + 1.06^{-4})$	—
geometric	$r = 1/1.06 \approx 0.9434$	—
sum	$R \times (1 - 1.06^{-5})/(1 - 1.06^{-1})$	—
solve	$20000 = R \times 4.2124$	$R \approx 4748.4$

The instalments' present value equals the loan.

Mathematical reasoning. Equate the present value of the instalments to the loan and solve for the instalment.

1. Let the instalment be R, paid at the end of each year for 5 years.
2. The present value of the instalments is $R(1 + 1.06^{-1} + \ldots + 1.06^{-4})$.
3. This is a geometric series with ratio 1/1.06 and 5 terms, summing to 4.2124.
4. So $20000 = 4.2124R$, giving $R \approx 4748.4$.
5. The total repaid is $5 \times 4748.4 \approx 23\,742$, so the interest is about 3742.

Final answer. about 4748 currency units per year

Interpretation. The same geometric series that grows a saving shrinks a loan, because the ratio is below 1 instead of above it.

CHAPTER 4 · INVESTIGATION 4.6

Investigation: infinite sums

INQUIRY QUESTION

Can infinitely many numbers add to a finite total?

YOUR TASK

Investigate four series: $\sum 1/n^2$, $\sum 1/n$, $\sum 1/2^N$ and $\sum (-1)^{N+1}/n$. For each, compute the partial sums after 5, 10 and 20 terms and describe the behaviour.

series	after 5	after 10	after 20	behaviour
$1/n^2$	1.4636	1.5498	1.5962	converges, to about 1.6449
$1/n$	2.2833	2.9290	3.5977	diverges, slowly
$1/2^N$	0.9688	0.9990	1.0000	converges, to 1
$(-1)^{N+1}/n$	0.7833	0.6456	0.6688	converges, to ln 2

CHAPTER 4 · INVESTIGATION 4.6

Investigation: infinite sums

What do you notice?

- The series of $1/n^2$ converged, while the series of $1/n$ did not, even though both have terms tending to zero.
- The geometric series converged fastest, reaching 1 to four decimal places by 20 terms.
- The alternating series converged by approaching its limit from above and below in turn.
- The slowest convergence belonged to $1/n^2$, which needed thousands of terms for two decimal places.

Observation. Convergence depends on how fast the terms shrink, not merely on whether they shrink.

MATHEMATICAL CONCLUSION

An infinite series converges when its partial sums settle, and the harmonic series diverges because its terms shrink too slowly, as the doubling-block argument shows. Faster shrinkage, as in a geometric series or in $1/n^2$, gives convergence. The threshold between the two behaviours is a rate, which is what the comparison test formalises.

Your own words

CHAPTER 4 · 4.6

Investigation: infinite sums

An infinite series is a sequence of partial sums, and it converges when that sequence settles. Adding infinitely many numbers is not a single operation but a limit of finitely many additions, which is why convergence has to be defined rather than assumed.

The harmonic series $1 + 1/2 + 1/3 + 1/4 + \ldots$ diverges even though its terms tend to zero. The terms shrink too slowly for the partial sums to settle, which shows that tending to zero is not sufficient for convergence.

Grouping the harmonic series in blocks of doubling length shows that each block contributes at least a half, so the partial sums exceed every bound. This argument is the standard proof that the harmonic series diverges.

A series whose terms shrink fast enough — geometrically, for instance — does converge, and the threshold between the two behaviours is a rate rather than a size. The comparison test makes the comparison precise.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 4 · 4.6 · WORKED EXAMPLE 1

Investigation: infinite sums

WORKED EXAMPLE 4.6.1

Problem. Show that the partial sums of $1 + 1/2 + 1/3 + \ldots$ exceed 2 after enough terms, and explain why this suggests divergence.

block	terms	sum of block	at least
1	1	1	1
2	1/2	0.5	1/2
3–4	1/3 + 1/4	0.583	1/2
5–8	1/5 + … + 1/8	0.635	1/2
9–16	1/9 + … + 1/16	0.663	1/2

Each doubling block contributes at least a half.

Mathematical reasoning. Group the terms in blocks of doubling length and show that each block contributes at least a half.

1. The first block is the single term 1, which is at least 1/2.
2. The second block is 1/2, exactly 1/2.
3. The third block is $1/3 + 1/4 \geq 1/4 + 1/4 = 1/2$.
4. The fourth block is $1/5 + \ldots + 1/8$, which is 8 terms each at least 1/8, so at least 1/2.
5. Every further block contributes at least 1/2, so the partial sums exceed every bound and the series diverges.

Final answer. the series diverges, because each doubling block contributes at least 1/2

Interpretation. Terms tending to zero is not enough: the harmonic series diverges because its terms do not shrink fast enough.

CHAPTER 4 · 4.6 · WORKED EXAMPLE 2

Investigation: infinite sums

WORKED EXAMPLE 4.6.2

Problem. Find the sum of 1/2 + 1/4 + 1/8 + ... to infinity and compare it with the harmonic series.

n	partial sum	remaining
1	0.5	0.5
2	0.75	0.25
3	0.875	0.125
4	0.9375	0.0625
limit	1	0

The remaining distance halves each time.

Mathematical reasoning. Apply the geometric sum formula and note the contrast with the harmonic series.

1. The first term is 1/2 and the ratio is 1/2.
2. Since $|r| < 1$, the series converges.
3. The sum is $(1/2)/(1 - 1/2) = 1$.
4. The partial sums approach 1 from below, and the remaining distance halves each time.
5. By contrast the harmonic series, whose terms are larger from the third term onwards, diverges.

Final answer. 1; the harmonic series diverges by comparison

Interpretation. Two series whose terms both tend to zero can behave in opposite ways, and the difference is the rate at which the terms shrink.

CHAPTER 4 · INVESTIGATION 4.7

Mixed practice

INQUIRY QUESTION

Which formula simplifies this sequence or series?

YOUR TASK

Take the sequence 2, 6, 18, 54, ... Compute the sum of the first 10 terms, the sum of the first 20 terms, and the sum to infinity if it exists. Then find the smallest n for which the partial sum exceeds 1000, and record the method you used for each part.

task	working	answer
sum of first 10	$2(3^{10} - 1)/(3 - 1)$	59048
sum of first 20	$2(3^{20} - 1)/2$	3486784400
sum to infinity	$r = 3 > 1$	does not exist
smallest n over 1000	$2(3^{N} - 1)/2 > 1000$	n = 7

CHAPTER 4 · INVESTIGATION 4.7

Mixed practice

What do you notice?

- The finite sums grew extremely fast, because r = 3.
- There was no sum to infinity, because the convergence condition failed.
- The smallest n for which the sum exceeded 1000 was 7, found by evaluating successive partial sums.
- The same formula answered three of the four questions.

Observation. The geometric sum formula answers finite questions, and the convergence condition decides whether an infinite question has an answer at all.

MATHEMATICAL CONCLUSION

A sequence is summarised by a general term and a series by a sum, and each family has its own formula chosen by the type. The convergence condition $|r| < 1$ is checked before any sum to infinity is claimed, because the formula is a statement about a limit rather than about an endless addition. Here the sums were 59 048 and 3 486 784 400, and no infinite sum existed.

Your own words

CHAPTER 4 · 4.7

Mixed practice

Every sequence question begins by identifying the type: constant difference means arithmetic, constant ratio means geometric, and neither means the rule must be found some other way. Identifying the type is choosing the formula.

Series questions require the convergence condition to be checked before the sum formula is applied. Applying a sum-to-infinity formula to a divergent series produces a number that is not a sum.

The two families of formulas — arithmetic and geometric — cover a large part of what sequences and series can do. The remaining cases are handled by the general idea of a partial sum and its limit.

Mixed practice is where the type is chosen rather than given, so the reasoning behind the choice is part of what is being practised.

> FORMAL DEVELOPMENT
>
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In your own words

CHAPTER 4 · 4.7 · WORKED EXAMPLE 1

Mixed practice

WORKED EXAMPLE 4.7.1

Problem. A sequence has u(n) = 3n − 1. Find the sum of the first 40 terms.

step	working	result
first term	u(1) = 2	—
last term	u(40) = 119	—
sum	(40/2)(2 + 119)	—
answer	20 × 121	2420

The sum needs only the first and last terms.

Mathematical reasoning. Recognise the sequence as arithmetic and apply the sum formula.

1. u(n) = 3n − 1 has common difference 3, so it is arithmetic.
2. The first term is u(1) = 2 and the last is u(40) = 119.
3. The sum is (40/2)(2 + 119) = 20 × 121 = 2420.
4. Check by splitting: 3 × (sum of 1 to 40) − 40 = 3 × 820 − 40 = 2460 − 40 = 2420.

Final answer. 2420

Interpretation. The check uses a different route to the same total, which is what makes it a check.

CHAPTER 4 · 4.7 · WORKED EXAMPLE 2

Mixed practice

WORKED EXAMPLE 4.7.2

Problem. Find the sum to infinity of 27 + 18 + 12 + 8 + ...

step	working	result
ratio	18/27 = 2/3	r < 1
sum	27/(1 − 2/3)	—
answer	27 × 3	81

A convergent geometric series has a finite sum.

Mathematical reasoning. Find the ratio, check the condition, and apply the formula.

1. The ratio is 18/27 = 2/3.
2. Since |2/3| < 1, the series converges.
3. The sum to infinity is 27/(1 − 2/3) = 27/(1/3) = 81.
4. Check with partial sums: after 4 terms the sum is 65, and after 10 terms it is 79.4, approaching 81.

Final answer. 81

Interpretation. The convergence check comes first, and it is what licenses the formula.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

4.A1 FLUENCY

Find the 20th term of the arithmetic sequence 5, 9, 13, ...

4.A2 FLUENCY

Find the common ratio of the geometric sequence 8, −12, 18, ...

4.A3 FLUENCY

Find the 8th term of the geometric sequence 7, 21, 63, ...

4.A4 FLUENCY

Find the sum of the first 15 positive odd numbers.

4.A5 FLUENCY

Evaluate $\sum_{r=1}^{6} (2r + 1)$.

4.A6 FLUENCY

Find the sum of the geometric series 4 + 8 + 16 + ... for 6 terms.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

4.A7 FLUENCY

Find the sum to infinity of 30 + 10 + 10/3 + ...

4.A8 FLUENCY

A sequence has u(1) = 3 and d = −4. Find u(15).

4.A9 FLUENCY

Find the sum of the first 10 terms of the geometric sequence with u(1) = 100 and r = 0.9.

4.A10 REASONING

Explain why the series $1 + 1.05 + 1.05^2 + \ldots$ has no sum to infinity.

4.A11 APPLICATION

A ball dropped from 3 m rebounds to 75 per cent of its height. Find the total distance travelled before it comes to rest.

4.A12 APPLICATION

A saver deposits 200 at the end of each year for 8 years at 5 per cent. Find the final value.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

4.B1 CHALLENGE

Investigate the partial sums of $1 + 1/2 + 1/3 + \ldots$ for 10, 100 and 1000 terms, and describe the rate of growth.

4.B2 REASONING

Conjecture a formula for the sum of the first n even numbers and justify it.

4.B3 CHALLENGE

Investigate the series $1 - 1/2 + 1/3 - 1/4 + \ldots$ for 10, 20 and 40 terms, and predict the limit.

4.B4 REASONING

Investigate the number of terms of the series $1 + 1/2 + 1/4 + \ldots$ needed to come within 0.001 of the limit, and justify.

4.B5 CHALLENGE

Investigate the sequence of partial sums of $1/n^2$ for 5, 10, 20 and 50 terms, and predict the limit.

4.B6 CHALLENGE

Conjecture a rule for when a series of positive decreasing terms converges, and test it on two examples.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

4.B7 CHALLENGE

Investigate the sum of the first n terms of a geometric series as r approaches 1, and justify what happens.

4.B8 REASONING

Investigate the partial sums of the geometric series 1 + 1/3 + 1/9 + … and justify the limit.

4.B9 CHALLENGE

Conjecture a rule for the sum of the first n square numbers and test it.

4.B10 REASONING

Investigate the number of terms of the series 1 + 1/2 + 1/4 + … needed to come within 0.0001 of the limit.

4.B11 CHALLENGE

Investigate the sequence of partial sums of $1/n^2$ and justify that it converges while 1/n does not.

CRITERION C Communicating

Criterion C practice · Communicating

4.C1 REASONING

Write a complete solution to finding the 12th term of the arithmetic sequence whose 4th term is 11 and whose 10th term is 31.

4.C2 REASONING

Write a complete solution to finding the sum of the first 25 terms of the arithmetic sequence with u(1) = −7 and d = 3.

4.C3 REASONING

Write a complete solution to evaluating $\sum_{r=1}^{20} (5 - 2r)$ by splitting the sum.

4.C4 REASONING

Write a complete solution to finding the sum to infinity of 0.27 + 0.027 + 0.0027 + ...

4.C5 APPLICATION

Write a complete solution to the annuity problem: 300 deposited at the end of each quarter for 5 years at 8 per cent per year compounded quarterly. Find the final value.

4.C6 APPLICATION

Write a complete solution to the loan problem: 12 000 repaid in 4 equal annual instalments at 9 per cent. Find the instalment.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

4.C7 REASONING

Write a complete solution to proving that the sum of the first n positive integers is n(n + 1)/2.

4.C8 REASONING

Write a complete solution to finding the sum of the first 30 terms of the arithmetic sequence with u(1) = −12 and d = 2.5.

4.C9 REASONING

Write a complete solution to evaluating $\sum_{r=1}^{15} (4r - 1)$ by splitting the sum.

4.C10 REASONING

Write a complete solution to finding the sum to infinity of 0.4 + 0.04 + 0.004 + ...

4.C11 APPLICATION

Write a complete solution to the annuity problem: 250 deposited at the end of each month for 4 years at 6 per cent per year compounded monthly.

4.C12 REASONING

Write a complete solution to proving that the sum of the first n odd numbers is n^2.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

4.D1 APPLICATION

A company's profit rises by 250 currency units each year from a first year of 4000. Model the profit, find the profit in year 10, and find the total over the first 10 years.

4.D2 APPLICATION

A town's population falls by 4 per cent a year from 50 000. Find the population after 15 years and comment on the model's limits.

4.D3 APPLICATION

A student saves 50 a month for 3 years at 6 per cent per year compounded monthly. Find the final value and the total interest earned.

4.D4 APPLICATION

Two loans of 10 000 are repaid over 5 years, one at 5 per cent and one at 12 per cent. Find each instalment and the difference in total repaid.

4.D5 APPLICATION

A pendulum's swing is 90 cm and each swing is 80 per cent of the previous. Find the total distance travelled by the bob before it stops, and comment on whether the model is realistic.

4.D6 APPLICATION

A charity aims to raise 100 000 and raises 4000 in the first month, with each month's total 10 per cent higher than the last. After how many months does it reach the target?

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

4.D7 REASONING

A student claims that any series whose terms tend to zero must converge. Investigate and justify.

4.D8 APPLICATION

A car loses 20 per cent of its value each year from 15 000. After how many years is it worth less than 2000, and what is the total loss over that period?

4.D9 APPLICATION

A company offers two salary schemes over 6 years. Scheme A starts at 30 000 and rises by 1500 a year. Scheme B starts at 28 000 and rises by 8 per cent a year. Find the total paid under each and say which is better.

4.D10 REASONING

A staircase has 20 steps. A ball rolls down, and each step it travels is half the distance of the previous one. Investigate whether the ball reaches the bottom, and justify.

4.D11 CHALLENGE

A student saves 100 in year 1 and increases the deposit by 10 currency units each year for 10 years, earning 5 per cent on the whole balance. Set up the model and explain why it is not a single geometric series.

4.D12 APPLICATION

A bacteria culture doubles every hour from 200 bacteria. Find the population after one day, and after how many hours it first exceeds one million.

CHAPTER 4 · APPLICATION AND ASSESSMENT

Applying chapter 4

These tasks use sequence formulas and series sums. State the rule that generates the sequence, justify the formula you use, and check the answer against a small case.

A1 · The two savings plans

Plan A deposits 200 currency units a month for 5 years at 6 per cent per year compounded monthly. Plan B deposits 200 in the first month and increases the deposit by 2 per cent each month, at the same rate. Model both, find the value of each after 5 years, and explain which is better and by how much.

Criteria A and D · model both plans, evaluate, and justify the recommendation with the models.

A2 · The loan comparison

A loan of 15 000 currency units can be repaid over 4 years at 7 per cent or over 6 years at 9 per cent, in equal annual instalments. Find the instalment and the total repaid under each, and advise a borrower who wants the lowest total cost and one who wants the lowest monthly instalment.

Criteria A and D · apply the repayment formula to both terms and justify the advice.

CHAPTER 4 · REVIEW

What you should now be able to do

Key ideas

- An arithmetic sequence has a common difference d, and u(n) = u(1) + (n − 1)d.
- A geometric sequence has a common ratio r, and u(n) = u(1) r^(n−1).
- The sum of an arithmetic series is (n/2)(u(1) + u(n)).
- The sum of a finite geometric series is $u(1)(r^N - 1)/(r - 1)$ for $r \neq 1$.
- A geometric series converges to u(1)/(1 − r) exactly when |r| < 1.
- The harmonic series diverges even though its terms tend to zero.
- An annuity accumulates to $P((1 + i)^N - 1)/i$.
- A loan repayment is the same formula read backwards.

Vocabulary

sequence · term · arithmetic · common difference · geometric · common ratio · series · partial sum · sigma notation · index · convergent · divergent · sum to infinity · annuity · instalment · present value · harmonic series

CHAPTER 4 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

4.R1 Find the 15th term of 3, 7, 11, ...

4.R2 Find the sum of the first 20 terms of 2, 5, 8, ...

4.R3 Find the sum to infinity of 16 + 12 + 9 + ...

4.R4 Evaluate $\sum_{r=1}^{10} r^2$.

4.R5 A loan of 5000 is repaid in 3 annual instalments at 10 per cent. Find the instalment.

4.R6 Explain why the series 1 + 1/2 + 1/4 + ... converges but 1 + 1/2 + 1/3 + ... does not.

CHAPTER 4 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> A sequence is form made visible: one rule generates every term, and the sum of those terms simplifies to a formula whose behaviour — finite, unbounded or convergent — is decided by a single number.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of patterns, generalisation and simplification appear in your work?
- How is the global context of fairness and development, and its exploration of saving, loans and the long-term decisions that shape opportunity, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a geometric model for a real savings plan, and what would you still like to ask?

CHAPTER 4 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

A savings plan is a sequence in disguise. Name the rule, the formula and the assumption that keeps the rule true, and say who the plan serves.

Service as Action

COMMUNITY NEED	Savings and loan terms are quoted without the compounding frequency or the term.
MATHEMATICAL CONNECTION	Sequences, series and annuities.
PROPOSED ACTION	Collect three published savings or loan offers, convert each to a stated rate per period, and publish the comparison.
EXPECTED IMPACT	A reader can compare offers quoted in different ways.
STUDENT REFLECTION	Record the offers, the conversions and what you found.

Community Project

PROJECT IDEA	A school savings club audit.
NEED	The club's two plans are quoted in different units and no one has compared them.
RESEARCH QUESTION	Which plan gives more after five years, and by how much?
MATHEMATICS	Sequences, series and annuities.
ACTION OR OUTCOME	A one-page comparison with the models shown.
BENEFICIARIES	Every member of the savings club.
EVIDENCE	The comparison and the models you built.

Personal Project connection · MYP 5

PERSONAL INTEREST	Sport, music, travel or gaming.
LEARNING GOAL	To model a saving or borrowing plan you might actually use and state its assumptions.
PRODUCT OR OUTCOME	A savings plan with a formula and a graph.
MATHEMATICS	Sequences, series and annuities.
PROCESS	Choose a plan, model it, and check the model against one known value.
EVIDENCE	The plan, the model and the check.

CHAPTER 5 · PART II · THINKING WITH MODELS

05

Quadratic Inequalities and Linear Programming

An inequality describes a set of possibilities rather than a single value. A system of inequalities bounds that set, and a linear objective function turns the bounded set into a choice.

ENGAGE

A workshop makes two kinds of basket. Each large basket uses 3 units of cane and 2 hours of labour; each small one uses 1 unit of cane and 1 hour. Only 24 units of cane and 20 hours are available. How many of each should be made to earn the most, and how do you know no better choice exists?

Framing the inquiry

Key concept: Relationships
Related concepts: Change · Equivalence · System
Global context: Globalization and sustainability
Global context exploration: production planning under resource limits

Statement of inquiry: A system of inequalities describes what is possible; a linear objective turns possibility into choice. Solving a quadratic inequality and optimising under linear constraints are the same habit: compare, bound, and justify.

Questions to guide us

- **Factual:** What does the solution set of a quadratic inequality look like, and how is it found?
- **Conceptual:** Why does a linear objective function on a bounded feasible region take its extreme values at a vertex?
- **Debatable:** If an optimal plan ignores the workers' preferences, is it still the best plan?

ATL · CRITICAL THINKING. In order for a student to decide how much of each product a workshop should make, students must translate resource limits into inequalities, bound the feasible region, and justify the optimal choice against alternatives.

CHAPTER 5 · INVESTIGATION 5.1

Quadratic inequalities in one variable

INQUIRY QUESTION

How does a graph solve an inequality?

YOUR TASK

Take six quadratic inequalities of your own: two with two distinct roots, one with a repeated root, one with no real roots, one with a negative leading coefficient and one in the form of a fraction whose denominator changes sign. Solve each, and record the solution set and the reason the sign behaves as it does.

inequality	roots	shape of parabola	solution set
$x^2 - 9 > 0$	$-3, 3$	opens up	$x < -3$ or $x > 3$
$x^2 - 2x - 3 \le 0$	$-1, 3$	opens up	$-1 \le x \le 3$
$x^2 + 6x + 9 > 0$	-3 (double)	touches axis	all x except -3
$x^2 + x + 1 > 0$	none	opens up	all real x
$-x^2 + 5x - 6 < 0$	$2, 3$	opens down	$x < 2$ or $x > 3$
$1/(x - 2) > 0$	—	not a parabola	$x > 2$

CHAPTER 5 · INVESTIGATION 5.1

Quadratic inequalities in one variable

What do you notice?

- Two of the six had two distinct roots, and their solution sets were the intervals between or beyond them.
- The repeated-root case excluded exactly one value.
- The no-real-roots case was true everywhere, because the parabola never crossed the axis.
- The fraction case was not a quadratic at all, and its sign was decided by the denominator alone.

> **Observation.** The sign of a quadratic is decided by its leading coefficient and its roots, and a repeated root leaves the sign unchanged.

MATHEMATICAL CONCLUSION

A quadratic inequality is solved by finding the roots and reading the sign from the shape of the parabola: between the roots for an upward-opening parabola with the expression negative, and beyond them for the expression positive. The leading coefficient and the number of real roots decide the pattern, and the strictness of the inequality decides whether the roots themselves are included.

Your own words

CHAPTER 5 · 5.1

Quadratic inequalities in one variable

A quadratic inequality such as $x^2 - 3x - 4 < 0$ asks for the values of x for which a quadratic expression is negative. The boundary of the solution set is the set of roots, and the sign of the expression between and beyond the roots is constant.

The sign of a quadratic is decided by its leading coefficient and by the number of real roots. With two distinct roots the sign alternates across them; with a repeated root the sign does not change; with no real roots the sign is constant everywhere.

The method is to find the roots, sketch the parabola, and read off the intervals where the graph lies below or above the axis. The sketch is the justification, not a decoration.

A strict inequality excludes the boundary values and a non-strict one includes them, which is recorded with round and solid markers on a number line. The distinction matters at the endpoints only, but it matters there completely.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 5 · 5.1 · WORKED EXAMPLE 1

Quadratic inequalities in one variable

WORKED EXAMPLE 5.1.1

Problem. Solve $x^2 - 3x - 4 < 0$.

x	$x^2 - 3x - 4$	sign	in the solution set?
−2	6	positive	no
0	−4	negative	yes
2	−6	negative	yes
5	6	positive	no

The expression is negative between the roots.

Mathematical reasoning. Find the roots, sketch the parabola, and read off where it lies below the axis.

1. Factorise: $x^2 - 3x - 4 = (x - 4)(x + 1)$, so the roots are $x = -1$ and $x = 4$.
2. The parabola opens upwards, so it is negative between its roots.
3. Hence the solution is $-1 < x < 4$.
4. The endpoints are excluded because the inequality is strict.
5. Check: at $x = 0$ the expression is $-4 < 0$, as required.

Final answer. $-1 < x < 4$

Interpretation. The sketch turns an algebraic question into a reading question, and the reading is what is justified.

CHAPTER 5 · 5.1 · WORKED EXAMPLE 2

Quadratic inequalities in one variable

WORKED EXAMPLE 5.1.2

Problem. Solve $x^2 + 4x + 4 \geq 0$.

x	$x^2 + 4x + 4$	sign	in the solution set?
−3	1	positive	yes
−2	0	zero	yes
0	4	positive	yes
1	9	positive	yes

A repeated root gives a sign that does not change.

Mathematical reasoning. A perfect square is never negative, so the inequality holds everywhere.

1. Factorise: $x^2 + 4x + 4 = (x + 2)^2$.
2. A square is never negative, so $(x + 2)^2 \geq 0$ for every x.
3. The solution set is therefore all real x.
4. The single value $x = -2$ makes the expression zero and is included because the inequality is non-strict.
5. Check: at $x = 0$ the expression is $4 \geq 0$, as required.

Final answer. all real x

Interpretation. A repeated root means the parabola touches the axis and turns back, so the sign does not change.

CHAPTER 5 · INVESTIGATION 5.2

Inequalities in two variables

INQUIRY QUESTION

What does the solution set of a linear inequality look like?

YOUR TASK

Take a workshop that makes two products, A and B, with x of A and y of B. Write and draw a system of inequalities of your own using at least four constraints of these kinds: an upper limit on x, an upper limit on y, a shared resource limit, and a minimum output requirement. Shade the feasible region, label its vertices, and test one interior point.

constraint	inequality	boundary line	kind of limit
labour	$2x + 3y \le 24$	$2x + 3y = 24$	shared resource
material	$x + y \le 10$	$x + y = 10$	shared resource
minimum A	$x \ge 1$	$x = 1$	minimum output
non-negativity	$y \ge 0$	$y = 0$	minimum output
interior test	$x = 3, y = 2$	$6 + 6 = 12 \le 24$	satisfied

CHAPTER 5 · INVESTIGATION 5.2

Inequalities in two variables

What do you notice?

- Two constraints were shared resources, so their boundary lines sloped downwards.
- The minimum-output constraints were vertical or horizontal lines.
- The feasible region was a polygon whose vertices were intersections of pairs of boundary lines.
- The interior point satisfied every constraint, confirming that the shaded side of each line was chosen correctly.

Observation. Each constraint is a half-plane, and the feasible region is their intersection.

MATHEMATICAL CONCLUSION

A system of linear inequalities is solved by the intersection of its half-planes, and each half-plane is chosen by testing one point. The resulting feasible region is a polygon whose vertices are the intersections of its boundary lines, and those vertices are what an objective function is evaluated at.

Your own words

CHAPTER 5 · 5.2

Inequalities in two variables

A linear inequality in two variables, such as $2x + 3y \le 12$, describes a half-plane bounded by the line $2x + 3y = 12$. The line is drawn solid when the inequality is non-strict and dashed when it is strict.

Which side of the line is the solution region is decided by testing one point, conventionally the origin when the line does not pass through it. The test is a single substitution, and it settles the question for the whole half-plane.

The boundary line itself is part of the solution exactly when the inequality is non-strict, so $\le$ and $\ge$ include the line and $<$ and $>$ exclude it. The distinction is visible in the drawing and decisive in the algebra.

A system of linear inequalities is satisfied by the points that lie in every half-plane at once, so its solution set is the intersection of the regions. This intersection is the feasible region, and it is usually a polygon.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 5 · 5.2 · WORKED EXAMPLE 1

Inequalities in two variables

WORKED EXAMPLE 5.2.1

Problem. Shade the region satisfying $x + 2y \le 8$, $x \ge 1$ and $y \ge 0$, and find the coordinates of its vertices.

vertex	lines that meet	coordinates
A	$x = 1$ and $y = 0$	$(1, 0)$
B	$y = 0$ and $x + 2y = 8$	$(8, 0)$
C	$x + 2y = 8$ and $x = 1$	$(1, 3.5)$
interior test	$x = 4, y = 1$	$4 + 2 = 6 \le 8$

The feasible region is a triangle.

Mathematical reasoning. Draw each boundary, test which side satisfies each inequality, and find the intersections.

1. The line $x + 2y = 8$ meets the x-axis at $(8, 0)$ and the y-axis at $(0, 4)$.
2. $x \ge 1$ excludes everything to the left of the vertical line $x = 1$.
3. $y \ge 0$ excludes everything below the x-axis.
4. Testing $(4, 1)$: $4 + 2 = 6 \le 8$, so the region containing $(4, 1)$ is kept.
5. The vertices are $(1, 0)$, $(8, 0)$ and $(1, 3.5)$.

Final answer. a triangle with vertices (1, 0), (8, 0) and (1, 3.5)

Interpretation. The region is the intersection of three half-planes, and each vertex is the meeting of two of its boundaries.

CHAPTER 5 · 5.2 · WORKED EXAMPLE 2

Inequalities in two variables

WORKED EXAMPLE 5.2.2

Problem. Explain why the system $x + y \le 5$ and $x + y \ge 8$ has no solution, and draw what happens.

point	$x + y$	$x + y \le 5$?	$x + y \ge 8$?
(1, 1)	2	yes	no
(3, 3)	6	no	no
(5, 5)	10	no	yes
(4, 4)	8	no	yes

No point can satisfy both at once.

Mathematical reasoning. The two inequalities demand contradictory bounds on the same quantity, so no point satisfies both.

1. The first inequality demands $x + y \le 5$.
2. The second demands $x + y \ge 8$.
3. No number is both at most 5 and at least 8.
4. So no point (x, y) satisfies both, and the feasible region is empty.
5. The two boundary lines are parallel and lie on opposite sides of the strip between them.

Final answer. no solution; the feasible region is empty

Interpretation. An empty feasible region is a meaningful answer: it says the constraints cannot all be met.

CHAPTER 5 · INVESTIGATION 5.3

Feasible regions and vertices

INQUIRY QUESTION

Why does the optimum lie at a vertex?

YOUR TASK

Take the feasible region bounded by $x + y \le 6$, $x + 3y \le 9$, $x \ge 0$ and $y \ge 0$. Find its vertices, then maximise $P = 2x + y$, minimise it, and then find the objective that is parallel to one of the edges and explain what that means for the number of solutions.

vertex	coordinates	$P = 2x + y$	on which edge?
(0, 0)	(0, 0)	0	corner
(6, 0)	(6, 0)	12	x-axis edge
(4.5, 1.5)	(4.5, 1.5)	10.5	$x + 3y = 9$ edge
(0, 3)	(0, 3)	3	y-axis edge
parallel objective	$x + 3y = 9$	—	whole edge optimal

CHAPTER 5 · INVESTIGATION 5.3

Feasible regions and vertices

What do you notice?

- The maximum was 12 at (6, 0) and the minimum was 0 at (0, 0).
- The objective 2x + y had slope −2, while the edges had slopes −1 and −1/3, so no edge was parallel to it.
- The objective x + 3y was parallel to the edge from (0, 3) to (4.5, 1.5).
- Every point of that edge gave the same value, 9, so there were infinitely many solutions.

Observation. A linear objective on a bounded polygon takes its extremes at vertices, unless it is parallel to an edge.

MATHEMATICAL CONCLUSION

A linear objective function takes constant values on parallel iso-lines, so its extremes on a bounded feasible region occur at the last point of contact, which is a vertex. When the objective is parallel to an edge, every point of that edge is optimal and the solution is not unique. Here the maximum of 2x + y was 12 and the minimum was 0.

Your own words

CHAPTER 5 · 5.3

Feasible regions and vertices

A linear objective function $P = ax + by$ takes constant values along parallel lines called iso-lines. As the line moves across the feasible region, P changes steadily, so its largest and smallest values on a bounded region occur at the last point of contact.

The last point of contact of an iso-line with a polygon is always a vertex, because a polygon's boundary consists of line segments and the extreme point of a line segment is an endpoint. Hence the optimum is at a vertex, or along a whole edge when the objective is parallel to it.

The method is therefore to find the vertices of the feasible region and evaluate the objective at each one. The vertex with the largest value gives the maximum, and the one with the smallest value gives the minimum.

When the objective is parallel to an edge, every point of that edge is optimal, and there are infinitely many solutions. This case is reported as an edge rather than as a single vertex, because the choice within the edge is free.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 5 · 5.3 · WORKED EXAMPLE 1

Feasible regions and vertices

WORKED EXAMPLE 5.3.1

Problem. Maximise $P = 3x + 5y$ subject to $x + 2y \le 8$, $x \ge 1$, $y \ge 0$ and $x \le 5$.

vertex	coordinates	$P = 3x + 5y$
(1, 0)	(1, 0)	3
(5, 0)	(5, 0)	15
(5, 1.5)	(5, 1.5)	22.5
(1, 3.5)	(1, 3.5)	20.5

The largest value occurs at a vertex.

Mathematical reasoning. Find the vertices of the feasible region and evaluate the objective at each.

1. The constraints give the vertices (1, 0), (5, 0), (5, 1.5) and (1, 3.5).
2. Evaluate $P = 3x + 5y$ at each: 3, 15, 22.5 and 20.5.
3. The largest value is 22.5, at (5, 1.5).
4. Check that (5, 1.5) is feasible: $5 + 3 = 8 \le 8$, $5 \le 5$, $1.5 \ge 0$ and $5 \ge 1$.
5. So the maximum is 22.5, achieved at $x = 5$ and $y = 1.5$.

Final answer. maximum $P = 22.5$ at (5, 1.5)

Interpretation. Evaluating the objective at the vertices is both necessary and sufficient for a linear objective on a polygon.

CHAPTER 5 · 5.3 · WORKED EXAMPLE 2

Feasible regions and vertices

WORKED EXAMPLE 5.3.2

Problem. Minimise $C = 2x + 4y$ subject to $x + y \geq 4$, $x \leq 3$, $y \leq 5$, $x \geq 0$ and $y \geq 0$, and identify the case that arises.

vertex	coordinates	$C = 2x + 4y$
(0, 4)	(0, 4)	16
(3, 1)	(3, 1)	10
(3, 5)	(3, 5)	26
(0, 5)	(0, 5)	20

The smallest value is at a vertex again.

Mathematical reasoning. Find the vertices, evaluate the objective, and check whether any edge is parallel to the objective.

1. The constraints give the vertices (0, 4), (3, 1), (3, 5) and (0, 5).
2. Evaluate $C = 2x + 4y$: 16, 10, 26 and 20.
3. The smallest value is 10, at (3, 1).
4. The objective $2x + 4y$ has slope $-1/2$, and the edge from (3, 1) to (0, 4) has slope -1, so no edge is parallel to the objective.
5. Hence the minimum is unique, at (3, 1).

Final answer. minimum $C = 10$ at (3, 1)

Interpretation. The check for a parallel edge is part of the justification, because it rules out infinitely many solutions.

CHAPTER 5 · INVESTIGATION 5.4

Linear programming

INQUIRY QUESTION

How is a real problem turned into a linear program?

YOUR TASK

Formulate and solve a linear program of your own for a school fair. Two stalls sell drinks and snacks. Each drink needs 1 unit of storage and 2 minutes of service; each snack needs 2 units of storage and 1 minute. There are 60 units of storage and 60 minutes. Profit is 2 per drink and 3 per snack. Find the optimal plan and the maximum profit.

step	working	result
variables	x drinks, y snacks	—
constraints	$x + 2y \le 60$, $2x + y \le 60$, $x \ge 0$, $y \ge 0$	—
objective	$P = 2x + 3y$	—
vertices	$(0, 0)$, $(30, 0)$, $(20, 20)$, $(0, 30)$	—
objective values	0, 60, 100, 90	maximum 100 at $(20, 20)$

CHAPTER 5 · INVESTIGATION 5.4

Linear programming

What do you notice?

- Both constraints were used at the optimum, so the plan sat at their intersection.
- The plan was 20 drinks and 20 snacks, giving a profit of 100.
- The vertex (30, 0) used all the service time but left storage unused, so it was not optimal.
- The vertex (0, 30) used all the storage but left service time unused.

> **Observation.** The optimum is the vertex that uses the resources most profitably, and it usually sits where two constraints meet.

MATHEMATICAL CONCLUSION

A linear program is formulated by naming the decision variables, writing one inequality per limit, and stating the objective. It is solved by finding the vertices of the feasible region and evaluating the objective at each. Here the optimal plan was 20 drinks and 20 snacks for a profit of 100.

Your own words

CHAPTER 5 · 5.4

Linear programming

A linear program has three parts: decision variables, a linear objective function, and linear constraints. Writing these down in that order is the whole translation, and the relationships between the resources are what the constraints record.

The constraints come from the limits of the situation: availability of resources, minimum outputs, and the non-negativity of the quantities. Each limit is a relationship between the variables, and a limit that is not stated is a limit that has been assumed away.

The solution of a linear program is the vertex that optimises the objective, together with the statement of the optimal value. Reporting the value without the point that achieves it, or the point without the value, is an incomplete answer.

Sensitivity analysis asks how much a constraint can change before the optimum moves. The answer is read from the slack in the binding constraints, and it is what makes a recommendation robust.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 5 · 5.4 · WORKED EXAMPLE 1

Linear programming

WORKED EXAMPLE 5.4.1

Problem. A workshop makes tables and chairs. A table needs 4 hours of carpentry and 2 of finishing; a chair needs 2 and 1. There are 40 hours of carpentry and 15 of finishing. Profit is 30 per table and 20 per chair. Formulate the problem and solve it.

item	carpentry	finishing	profit
table	4	2	30
chair	2	1	20
available	40	15	—

The table of resources is the formulation.

Mathematical reasoning. Define the variables, write the constraints and the objective, then solve by evaluating the vertices.

1. Let x be the number of tables and y the number of chairs.
2. The constraints are $4x + 2y \le 40$, $2x + y \le 15$, $x \ge 0$ and $y \ge 0$.
3. The objective is $P = 30x + 20y$.
4. The vertices of the feasible region are (0, 0), (7.5, 0), (5, 5) and (0, 15).
5. Evaluating P gives 0, 225, 250 and 300, so the maximum is 300 at (0, 15), that is 15 chairs and no tables.

Final answer. 15 chairs and no tables, profit 300

Interpretation. The optimum is a vertex, and reporting the plan without the profit, or the profit without the plan, is incomplete.

CHAPTER 5 · 5.4 · WORKED EXAMPLE 2

Linear programming

WORKED EXAMPLE 5.4.2

Problem. A farmer has 100 hectares to plant with wheat and barley. Wheat needs 3 units of fertiliser per hectare and barley 2, with 240 units available. Profit is 400 per hectare of wheat and 300 of barley. Formulate and solve.

crop	hectares	fertiliser	profit
wheat	w	3 per ha	400 per ha
barley	b	2 per ha	300 per ha
available	100	240	—

Two constraints and one objective.

Mathematical reasoning. Formulate with two variables and two resource constraints, then evaluate the vertices.

1. Let w be the hectares of wheat and b the hectares of barley.
2. The constraints are $w + b \le 100$, $3w + 2b \le 240$, $w \ge 0$ and $b \ge 0$.
3. The objective is $P = 400w + 300b$.
4. The vertices are (0, 0), (80, 0), (40, 60) and (0, 100).
5. Evaluating P gives 0, 32000, 34000 and 30000, so the maximum is 34000 at (40, 60).

Final answer. 40 ha of wheat and 60 ha of barley, profit 34 000

Interpretation. Both resource constraints are used at the optimum, so the solution sits at the intersection of both boundaries.

CHAPTER 5 · INVESTIGATION 5.5

Modelling and optimisation

INQUIRY QUESTION

When is a linear model the right model?

YOUR TASK

Take the problem of maximising $P = 5x + 4y$ subject to $x + y \le 7$, $x + 3y \le 12$, $x \ge 0$ and $y \ge 0$. Solve the relaxed problem, then find the integer optimum, and then find how much the right-hand side of the second constraint can fall before the optimum changes.

task	working	answer
relaxed vertices	(0, 0), (7, 0), (4.5, 2.5), (0, 4)	—
relaxed optimum	$P = 5(4.5) + 4(2.5) = 32.5$	(4.5, 2.5)
integer optimum	test (4, 2) and (5, 2): $P = 28$ and 33	(5, 2) with 33
sensitivity	the edge $x + 3y = 12$ binds down to 11	can fall by 1

CHAPTER 5 · INVESTIGATION 5.5

Modelling and optimisation

What do you notice?

- The relaxed optimum was not a lattice point, so rounding was needed.
- Rounding (4.5, 2.5) down to (4, 2) gave 28, but (5, 2) gave 33, so rounding was not optimal.
- The true integer optimum was (5, 2) with profit 33.
- The binding constraint could fall by 1 unit before the optimum moved.

> **Observation.** Rounding a relaxed optimum is not generally optimal, and the integer optimum must be found by testing nearby lattice points.

MATHEMATICAL CONCLUSION

A linear model is appropriate when the contributions are independent and the range is limited, and the modeller must check both. When the variables are whole numbers, the relaxed optimum is an upper bound and the true optimum is found by testing nearby lattice points. Here the relaxed optimum was 32.5 at (4.5, 2.5) and the integer optimum was 33 at (5, 2).

Your own words

CHAPTER 5 · 5.5

Modelling and optimisation

A linear model assumes that the contribution of each variable is independent of the level of the others, so that doubling the inputs doubles the output. This assumption gives an equivalence between the model and the situation that holds only over a limited range, and the range is part of the model.

Real problems rarely have exactly linear constraints. Costs have fixed components, resources have minimum order quantities, and outputs are whole numbers. Each of these breaks linearity, and the modeller must decide whether the break matters.

When the variables must be whole numbers, the optimum of the relaxed problem is an upper bound, and the true integer optimum is found by testing the nearby lattice points. Rounding the relaxed optimum is not in general optimal.

Sensitivity analysis reports how much a constraint can change before the optimum moves. It converts a single answer into a recommendation, because it says how fragile the answer is.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 5 · 5.5 · WORKED EXAMPLE 1

Modelling and optimisation

WORKED EXAMPLE 5.5.1

Problem. A factory makes two products with profit 8 and 12 per unit. The constraints are $x + y \le 10$, $x + 2y \le 14$, $x \ge 0$ and $y \ge 0$. Solve, and then solve again with the requirement that x and y are whole numbers.

vertex	coordinates	P = 8x + 12y
(0, 0)	(0, 0)	0
(10, 0)	(10, 0)	80
(6, 4)	(6, 4)	96
(0, 7)	(0, 7)	84

The relaxed optimum is already a lattice point.

Mathematical reasoning. Solve the relaxed problem, then check whether the optimum is a whole-number point.

1. The relaxed vertices are (0, 0), (10, 0), (6, 4) and (0, 7).
2. Evaluating $P = 8x + 12y$ gives 0, 80, 96 and 84.
3. The relaxed maximum is 96 at (6, 4).
4. Both coordinates are whole numbers, so the integer optimum is the same point.
5. The integer optimum is $x = 6$, $y = 4$ with profit 96.

Final answer. x = 6, y = 4, profit 96 (the same for both problems)

Interpretation. When the relaxed optimum happens to be a lattice point, the integer problem needs no further work — but that has to be checked.

CHAPTER 5 · 5.5 · WORKED EXAMPLE 2

Modelling and optimisation

WORKED EXAMPLE 5.5.2

Problem. In the same problem, suppose a third constraint $2x + y \le 13$ is added. Find the new optimum and explain the change.

vertex	coordinates	$P = 8x + 12y$
(6, 4)	now infeasible	—
(6.5, 0)	(6.5, 0)	52
(5, 3)	(5, 3)	76
(0, 7)	(0, 7)	84

A new constraint cuts the feasible region.

Mathematical reasoning. Add the constraint, recompute the vertices, and re-evaluate.

1. The new constraint $2x + y \le 13$ cuts off the old optimum at (6, 4), since $2(6) + 4 = 16 > 13$.
2. The new vertices are (0, 0), (6.5, 0), (5, 3) and (0, 7).
3. Evaluating P gives 0, 52, 76 and 84.
4. The new maximum is 84 at (0, 7).
5. The profit has fallen from 96 to 84, and the plan has changed completely.

Final answer. maximum 84 at (0, 7)

Interpretation. Adding one constraint can move the optimum to the opposite corner, which is why sensitivity analysis matters.

CHAPTER 5 · INVESTIGATION 5.6

Investigation: integer solutions and the vertex

INQUIRY QUESTION

When is the vertex the answer, and when is it only a bound?

YOUR TASK

Take $P = 5x + 4y$ with constraints $x + 2y \le 9$, $2x + y \le 9$, $x \ge 0$, $y \ge 0$. Find the relaxed optimum, then find the integer optimum by testing lattice points, and then find a second problem of your own where rounding the relaxed optimum gives the wrong answer.

point	coordinates	$P = 5x + 4y$	lattice point?
relaxed vertex	(3, 3)	27	yes
(4, 2)	lattice	28	yes, and feasible
(2, 4)	lattice	26	yes
(4, 1)	lattice	24	yes

CHAPTER 5 · INVESTIGATION 5.6

Investigation: integer solutions and the vertex

What do you notice?

- The relaxed vertex was (3, 3) with P = 27, which is a lattice point.
- But (4, 2) is also feasible and gives 28, which is larger.
- So the relaxed optimum was not the integer optimum even though it was a lattice point, because the objective was not maximised there.
- The true integer optimum was (4, 2) with P = 28.

> **Observation.** The relaxed optimum is an upper bound, and the integer optimum is found by testing lattice points near it.

MATHEMATICAL CONCLUSION

The vertex of a relaxed linear program is an upper bound on the integer optimum, and it is exact only when the vertex is itself an optimal lattice point. Otherwise the integer optimum must be found by testing nearby lattice points, and rounding the vertex is a heuristic that can fail. Here the integer optimum was 28 at (4, 2), above the relaxed vertex value of 27.

Your own words

CHAPTER 5 · 5.6

Investigation: integer solutions and the vertex

The vertex of a relaxed linear program is an upper bound on the integer optimum, because the integer points are a subset of the feasible region. The bound is exact only when the vertex is itself a lattice point.

When the vertex is not a lattice point, the integer optimum is found by testing the lattice points near it. In two variables this is a short list, and in higher dimensions it becomes a substantial problem in its own right.

Rounding the vertex is a heuristic, not a method. Rounding down can leave the feasible region, and rounding to the nearest lattice point can miss the true optimum, as a single example shows.

The gap between the relaxed and integer optima is a measure of how much the whole-number requirement costs. When the gap is small, the relaxation is a good guide; when it is large, the problem needs a genuine integer method.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 5 · 5.6 · WORKED EXAMPLE 1

Investigation: integer solutions and the vertex

WORKED EXAMPLE 5.6.1

Problem. Maximise $P = 4x + 3y$ subject to $2x + y \le 7$, $x + 2y \le 8$, $x \ge 0$, $y \ge 0$. Find the relaxed optimum and then the integer optimum.

point	coordinates	$P = 4x + 3y$	feasible?
relaxed vertex	(2, 3)	17	yes
(2, 3)	lattice	17	yes
(3, 1)	lattice	15	yes
(1, 3)	lattice	13	yes

The relaxed optimum is a lattice point here.

Mathematical reasoning. Find the relaxed vertex and check whether it has whole-number coordinates.

1. Solving the two boundary equations gives $x = 2$ and $y = 3$.
2. The relaxed optimum is (2, 3) with $P = 4(2) + 3(3) = 17$.
3. Both coordinates are whole numbers, so the integer optimum is the same point.
4. The nearby lattice points give smaller values, confirming the result.
5. The integer optimum is $x = 2$, $y = 3$ with $P = 17$.

Final answer. relaxed and integer optima both at (2, 3) with P = 17

Interpretation. When the vertex is a lattice point the relaxation is exact, and that is the best possible case.

CHAPTER 5 · 5.6 · WORKED EXAMPLE 2

Investigation: integer solutions and the vertex

WORKED EXAMPLE 5.6.2

Problem. Maximise $P = 3x + 2y$ subject to $x + y \le 5$, $x \le 3.4$, $y \le 3.4$, $x \ge 0$, $y \ge 0$. Find the relaxed optimum and the integer optimum, and explain the difference.

point	coordinates	$P = 3x + 2y$	feasible?
relaxed vertex	(3.4, 1.6)	13.4	yes
(3, 2)	lattice	13	yes
(3, 1)	lattice	11	yes
(2, 3)	lattice	12	yes

The relaxed optimum is not a lattice point.

Mathematical reasoning. Find the relaxed optimum, then test the lattice points near it.

1. The relaxed optimum is where $x = 3.4$ meets $x + y = 5$, so $y = 1.6$.
2. The relaxed value is $P = 3(3.4) + 2(1.6) = 13.4$.
3. This point is not a lattice point, so it is not a feasible integer solution.
4. Testing nearby lattice points: (3, 2) gives 13, (2, 3) gives 12 and (3, 1) gives 11.
5. The integer optimum is (3, 2) with $P = 13$, which is 0.4 below the relaxed bound.

Final answer. relaxed optimum 13.4 at (3.4, 1.6); integer optimum 13 at (3, 2)

Interpretation. The gap of 0.4 is the cost of the whole-number requirement, and rounding the relaxed point to (3, 2) happens to work here but not in general.

CHAPTER 5 · INVESTIGATION 5.7

Mixed practice

INQUIRY QUESTION

Which method does this problem need?

YOUR TASK

Take a recycling centre that processes glass and plastic. Each tonne of glass needs 2 hours of sorting and earns 40 currency units; each tonne of plastic needs 3 hours and earns 50. There are 36 hours and at least 4 tonnes must be processed. Formulate, solve, and then re-solve with the requirement that the tonnages are whole numbers.

step	working	result
variables	x tonnes of glass, y of plastic	—
constraints	$2x + 3y \le 36$, $x + y \ge 4$, $x \ge 0$, $y \ge 0$	—
objective	$P = 40x + 50y$	—
relaxed optimum	(0, 12) with $P = 600$	600
integer optimum	(0, 12) is already integral	600

CHAPTER 5 · INVESTIGATION 5.7

Mixed practice

What do you notice?

- The plastic earned more per hour, so the optimum put everything into plastic.
- The relaxed optimum was already a lattice point, so the integer problem needed no extra work.
- The minimum-output constraint was satisfied automatically.
- The profit of 600 used all 36 hours.

> **Observation.** The optimum uses the binding resource fully, and the integer check confirms whether the relaxed answer stands.

MATHEMATICAL CONCLUSION

An inequality problem is solved by identifying its type first: roots and sign for a quadratic inequality, region and vertices for a linear system. The answer to a linear program is the point, the value and the feasibility check together, and when whole numbers are required the relaxed optimum is only a bound. Here the optimum was 12 tonnes of plastic for a profit of 600.

Your own words

CHAPTER 5 · 5.7

Mixed practice

Quadratic inequalities are solved by roots and sign; linear systems are solved by region and vertices. Recognising which of the two a question needs is the first step, and the two methods are easily confused because both involve graphs.

A linear programming answer has three parts: the point, the value and the justification that the point is feasible. Missing any one of the three leaves the answer incomplete.

When a problem asks for whole numbers, the relaxed optimum is a bound and the answer must be checked against lattice points. Reporting the relaxed optimum as the answer is a category error.

Mixed practice is where the method is chosen rather than given, so the reasoning behind the choice is part of what is being practised.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 5 · 5.7 · WORKED EXAMPLE 1

Mixed practice

WORKED EXAMPLE 5.7.1

Problem. Solve $2x^2 - 5x - 3 > 0$.

step	working	result
factorise	$(2x + 1)(x - 3) > 0$	roots $-1/2$ and 3
sign	positive outside the roots	—
answer	$x < -1/2$ or $x > 3$	—

An upward-opening parabola is positive outside its roots.

Mathematical reasoning. Factorise, find the roots, and read the sign from the shape of the parabola.

1. Factorise: $2x^2 - 5x - 3 = (2x + 1)(x - 3)$.
2. The roots are $x = -1/2$ and $x = 3$.
3. The parabola opens upwards, so the expression is positive outside the roots.
4. Hence $x < -1/2$ or $x > 3$.
5. Check: at $x = 0$ the expression is -3, which is not positive, so the interval between the roots is correctly excluded.

Final answer. $x < -1/2$ or $x > 3$

Interpretation. The check confirms the interval that is excluded, which is as informative as the one that is included.

CHAPTER 5 · 5.7 · WORKED EXAMPLE 2

Mixed practice

WORKED EXAMPLE 5.7.2

Problem. A bakery makes loaves and rolls. A loaf needs 200 g of flour and a roll 50 g, with 10 kg available. Profit is 3 per loaf and 1 per roll, and at least 20 items must be made. Formulate and solve.

step	working	result
variables	x loaves, y rolls	—
constraints	$200x + 50y \le 10000$, $x + y \ge 20$, $x \ge 0$, $y \ge 0$	—
objective	$P = 3x + y$	—
vertices	(0, 20), (0, 200), (50, 0), (20, 0)	—
values	20, 200, 150, 60	maximum 200 at (0, 200)

The minimum-output constraint removes part of the region.

Mathematical reasoning. Formulate with the flour limit and the minimum output, then evaluate the vertices.

1. Let x be loaves and y be rolls. The flour limit is $200x + 50y \le 10000$.
2. The minimum output is $x + y \ge 20$, with $x \ge 0$ and $y \ge 0$.
3. The objective is $P = 3x + y$.
4. The vertices are (0, 20), (0, 200), (50, 0) and (20, 0), giving P = 20, 200, 150 and 60.
5. The maximum is 200 at (0, 200), that is 200 rolls and no loaves.

Final answer. 200 rolls and no loaves, profit 200

Interpretation. The profit per gram of flour is higher for rolls, so the optimum puts everything into rolls, and the minimum-output constraint is satisfied automatically.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

5.A1 FLUENCY

Solve $x^2 - 5x + 6 < 0$.

5.A2 FLUENCY

Solve $x^2 + 2x - 8 \geq 0$.

5.A3 FLUENCY

Solve $-x^2 + 4x - 3 > 0$.

5.A4 FLUENCY

Solve $(x - 2)^2 \leq 9$.

5.A5 REASONING

Solve $(x + 1)/(x - 3) \leq 0$.

5.A6 FLUENCY

Shade the region $x + y \leq 6$, $x \geq 1$, $y \geq 2$.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

5.A7 FLUENCY

Find the vertices of the region $x \le 4$, $y \le 3$, $x + y \le 6$, $x \ge 0$, $y \ge 0$.

5.A8 FLUENCY

Maximise $P = x + 2y$ subject to $x + y \le 4$, $x \le 2$, $y \le 3$, $x \ge 0$, $y \ge 0$.

5.A9 REASONING

Minimise $C = 2x + y$ subject to $x + y \ge 3$, $x + 3y \ge 6$, $x \ge 0$, $y \ge 0$.

5.A10 REASONING

Explain why the feasible region of $x + y \le 3$ and $x + y \ge 7$ is empty.

5.A11 FLUENCY

A farmer has 60 kg of feed. Cattle need 5 kg a day and sheep 2 kg. Write the constraint if there are c cattle and s sheep.

5.A12 FLUENCY

A firm maximises $P = 4x + 3y$ subject to $x + y \le 5$, $x + 3y \le 9$, $x \ge 0$, $y \ge 0$. Find the optimum.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

5.B1 REASONING

Investigate the solution sets of $x^2 - 4x + c < 0$ for $c = 4, 3, 0$ and -5, and describe the pattern.

5.B2 CHALLENGE

Conjecture a rule for when a quadratic inequality has no solution, and justify it.

5.B3 REASONING

Investigate the feasible region of $2x + y \le 8$, $x + 2y \le 8$, $x \ge 0$, $y \ge 0$ and describe its shape.

5.B4 CHALLENGE

Investigate the objective $P = x + y$ on the region of 5.2.3, and decide whether the maximum is unique.

5.B5 REASONING

Investigate the effect on the optimum of adding the constraint $x \le 2$ to the problem of 5.2.3.

5.B6 CHALLENGE

Conjecture a rule for the maximum of $x + y$ subject to $ax + by \le c$ and $x, y \ge 0$, and test it.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

5.B7 CHALLENGE

Investigate the number of vertices of a feasible region bounded by n lines, and justify a bound.

5.B8 REASONING

Investigate the solution sets of $x^2 - 6x + c \leq 0$ for $c = 8$, 9 and 10, and justify the pattern.

5.B9 CHALLENGE

Conjecture a rule for when a system of two linear inequalities has a bounded feasible region, and test it.

5.B10 REASONING

Investigate the maximum of $x + y$ on the region $x + 2y \leq 8$, $2x + y \leq 8$, $x \geq 0$, $y \geq 0$, and decide whether it is unique.

5.B11 REASONING

Investigate the effect of adding a minimum-output constraint to a linear program, and justify.

CRITERION C Communicating

Criterion C practice · Communicating

5.C1 REASONING

Write a complete solution to solving $3x^2 - 7x - 6 \le 0$.

5.C2 REASONING

Write a complete solution to solving $(x - 1)/(x + 2) > 0$.

5.C3 REASONING

Write a complete solution to finding the maximum of $P = 5x + 4y$ subject to $x + 2y \le 8$, $3x + y \le 9$, $x \ge 0$, $y \ge 0$.

5.C4 REASONING

Write a complete solution to the workshop problem: 40 hours of carpentry and 24 of finishing, with tables needing 4 and 2 and chairs 2 and 1, profit 30 and 20. Formulate and solve.

5.C5 REASONING

Write a complete solution to the integer version of the problem in 5.3.4, where x and y must be whole numbers.

5.C6 APPLICATION

Write a complete solution to minimising the cost $C = 3x + 2y$ of a diet that needs at least 6 units of A and 4 of B, where food X gives 1 and 2 units and food Y gives 2 and 1.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

5.C7 REASONING

Write a complete solution to proving that a linear objective on a bounded feasible region attains its maximum at a vertex.

5.C8 REASONING

Write a complete solution to solving $(x - 2)(x + 3)(x - 5) \leq 0$.

5.C9 REASONING

Write a complete solution to solving $3/(x - 1) \geq 1$.

5.C10 REASONING

Write a complete solution to minimising $C = 4x + 3y$ subject to $2x + y \geq 10$, $x + 3y \geq 15$, $x \geq 0$, $y \geq 0$.

5.C11 APPLICATION

Write a complete solution to the production problem: 30 hours of labour and 24 units of material, with A needing 2 and 3 and B needing 3 and 2, profit 40 and 30. Formulate and solve.

5.C12 CHALLENGE

Write a complete solution to proving that the maximum of a linear objective on an unbounded feasible region need not exist.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

5.D1 APPLICATION

A charity packs food parcels. A large parcel needs 3 kg of rice and 2 of beans; a small one needs 1 and 2. There are 30 kg of rice and 28 of beans. Formulate and solve to maximise the number of parcels.

5.D2 APPLICATION

A school orders two kinds of laptop. Type A costs 400 and needs 1 desk; type B costs 300 and needs 2 desks. There are 6000 currency units and 20 desks. Maximise the number of laptops.

5.D3 APPLICATION

A farm has 100 hectares and 240 units of water. Wheat needs 2 units and barley 4 per hectare, and at least 20 hectares must be wheat. Profit is 500 per hectare of wheat and 400 of barley. Formulate and solve.

5.D4 REASONING

A factory's relaxed optimum is 13.4 at (3.4, 1.6) for $P = 3x + 2y$. Explain why this is not an answer to the integer problem, and find the integer optimum.

5.D5 REASONING

A company maximises $P = 7x + 5y$ subject to $2x + 3y \le 18$, $3x + y \le 12$, $x \ge 0$, $y \ge 0$, with x and y integers. Find the optimum and the gap from the relaxed value.

5.D6 APPLICATION

A council must place x benches and y bins. Each bench needs 2 m^2 and each bin 1 m^2, with 40 m^2 available, and there must be at least as many bins as benches. Maximise the community score $S = 3x + 2y$.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

5.D7 REASONING

A student solves a linear program and reports only the maximum value. Explain what is missing and why it matters.

5.D8 APPLICATION

A water supplier mixes two sources. Source A costs 3 per litre and has 0.2 units of mineral; source B costs 5 and has 0.5 units. The daily requirement is at least 10 units of mineral and at most 200 currency units. Minimise the cost.

5.D9 APPLICATION

A publisher prints x hardback and y paperback copies. Each hardback uses 3 sheets and each paperback 1, with 600 sheets, and the printing machine runs for at most 200 minutes, hardbacks taking 4 minutes and paperbacks 2. Maximise the number of copies.

5.D10 CHALLENGE

A student claims that if a relaxed optimum is a lattice point then it must be the integer optimum. Investigate and justify.

5.D11 APPLICATION

A delivery firm loads x small and y large crates. Small crates weigh 10 kg and large 25 kg, with a 500 kg limit, and each small crate earns 20 and each large 40. Maximise the earnings.

5.D12 APPLICATION

A cooperative grows x hectares of maize and y of beans. Maize needs 4 hours of labour per hectare and beans 2, with 80 hours, and the market absorbs at most 25 hectares. Profit is 600 per hectare of maize and 900 of beans. Formulate and solve.

CHAPTER 5 · APPLICATION AND ASSESSMENT

Applying chapter 5

These tasks use inequalities and linear programming. Draw the region, label the constraints, evaluate the objective at each vertex, and justify the choice.

A1 · The workshop plan

A workshop makes two products. Product A needs 2 hours of cutting and 1 of assembly; product B needs 1 of cutting and 2 of assembly. There are 40 hours of cutting and 36 of assembly. Profit is 25 per A and 30 per B. Formulate the problem, solve it, and advise the owner, stating any assumption the model makes.

Criteria A and D · formulate, solve by vertices, and justify the recommendation.

A2 · The fair's budget

A school fair has a budget of 300 currency units and 20 volunteers. A food stall costs 40 and needs 3 volunteers; a games stall costs 25 and needs 2. The fair must have at least 2 food stalls. Maximise the total income, modelled as 90 per food stall and 60 per games stall, and comment on whether the model is realistic.

Criteria A and C · formulate, solve, and evaluate the model's assumptions.

CHAPTER 5 · REVIEW

What you should now be able to do

Key ideas

- A quadratic inequality is solved by finding the roots and reading the sign from the parabola.
- A linear inequality in two variables describes a half-plane bounded by a line.
- A system of inequalities is solved by the intersection of its half-planes.
- A linear objective takes constant values on parallel iso-lines.
- The optimum of a linear program on a bounded region is at a vertex, or along an edge parallel to the objective.
- When the objective is parallel to an edge there are infinitely many solutions.
- An integer optimum is found by testing lattice points near the relaxed vertex.
- Rounding a relaxed optimum is a heuristic, not a method.

Vocabulary

inequality · solution set · feasible region · vertex · constraint · objective function · iso-line · linear program · decision variable · binding constraint · slack · sensitivity · integer programming · lattice point · relaxed problem

CHAPTER 5 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

5.R1 Solve $x^2 + x - 12 \geq 0$.

5.R2 Shade the region $2x + y \leq 10$, $x \geq 0$, $y \geq 0$ and find its vertices.

5.R3 Maximise $P = 3x + 4y$ subject to $x + y \leq 5$, $x \leq 3$, $x \geq 0$, $y \geq 0$.

5.R4 Explain why the system $x + 2y \leq 4$ and $x + 2y \geq 6$ has no solution.

5.R5 A relaxed optimum is 12.5 at (2.5, 2.5). Find the integer optimum for $P = 2x + 3y$ if (2, 2), (3, 2) and (2, 3) are feasible.

5.R6 State the three parts of a complete answer to a linear program.

CHAPTER 5 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> A system of inequalities describes what is possible; a linear objective turns possibility into choice. Solving a quadratic inequality and optimising under linear constraints are the same habit: compare, bound, and justify.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of change, equivalence and system appear in your work?
- How is the global context of globalization and sustainability, and its exploration of production planning under resource limits, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a linear model for a real production decision, and what would you still like to ask?

CHAPTER 5 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

A constraint is a relationship between resources. Name the variables, write the constraints, and say what the optimum would cost in the real world.

Service as Action

COMMUNITY NEED	Production and resource decisions are made without the limits being written down.
MATHEMATICAL CONNECTION	Inequalities and linear programming.
PROPOSED ACTION	Take one real resource decision in your community, write its constraints, solve it, and publish the plan and its assumptions.
EXPECTED IMPACT	The decision is made with its limits visible.
STUDENT REFLECTION	Record the decision, the model and what you found.

Community Project

PROJECT IDEA	A resource audit of the school canteen.
NEED	The canteen buys without stating its budget, storage or volunteer limits.
RESEARCH QUESTION	What is the best mix of items given the stated limits?
MATHEMATICS	Inequalities, feasible regions and linear programming.
ACTION OR OUTCOME	A one-page plan with the constraints and the optimum.
BENEFICIARIES	The canteen and everyone who eats there.
EVIDENCE	The plan and the model behind it.

Personal Project connection · MYP 5

PERSONAL INTEREST	Sport, cooking, music or gaming.
LEARNING GOAL	To model a decision you actually make under limits.
PRODUCT OR OUTCOME	A plan with the constraints and the objective written down.
MATHEMATICS	Inequalities and linear programming.
PROCESS	Name the variables, write the limits, and solve by evaluating vertices.
EVIDENCE	The plan and the vertices you evaluated.

CHAPTER 6 · PART III · SPATIAL REASONING

06

Circle Geometry and Geometric Proof

A circle is the set of points at a fixed distance from a centre, so all its points lie in one space at one distance. That single definition generates a small number of theorems, and those theorems justify every measurement made in a circle.

ENGAGE

A stained-glass window is designed as a circle with chords meeting at a point. The designer needs to know the angle between two chords before cutting the glass. Which property of the circle decides it, and how is that property proved?

Framing the inquiry

Key concept: Logic
Related concepts: Space · Generalisation · Justification
Global context: Personal and cultural expression
Global context exploration: the geometry of architecture and ornament

Statement of inquiry: A circle conceals equal angles and fixed distances; a few theorems, proved from radii and triangles, let us deduce measurements and justify every claim.

Questions to guide us

- **Factual:** What do the angle at the centre, the angle in a semicircle and the alternate segment theorems say?
- **Conceptual:** Why does a tangent meet a radius at right angles, and what does a proof need in order to be complete?
- **Debatable:** If a diagram convinces you a claim is true, is a proof still needed?

ATL · CRITICAL THINKING. In order for a student to justify a measurement in an architectural drawing, students must identify the circle property that applies, state it as a general claim, and construct a proof from radii and triangles.

CHAPTER 6 · INVESTIGATION 6.1

Angles in circles

INQUIRY QUESTION

What relationships between angles does a circle force?

YOUR TASK

Draw a circle with centre O and three points A, B and C on it, with C on the major arc AB. Measure the angle AOB, the angle ACB and the angle in the alternate segment. Repeat for three different positions of C and for two different arcs, and record the relationships you find.

configuration	angle at centre	angle at C	angle in the alternate segment
C on the major arc, arc AB = 100°	100°	50°	130°
C moved on the major arc	100°	50°	130°
C on the minor arc	100°	50°	130°
arc AB = 160°	160°	80°	100°
arc AB = 40°	40°	20°	160°

CHAPTER 6 · INVESTIGATION 6.1

Angles in circles

What do you notice?

- The angle at C did not change when C moved along the same arc.
- The angle at C was always half the angle at the centre.
- The two angles at C, one on each arc, always summed to 180 degrees.
- Changing the arc changed every angle, but not the relationship between them.

Observation. The angle at the centre is twice the angle at the circumference, and the two angles in the two segments are supplementary.

MATHEMATICAL CONCLUSION

The angle at the centre is twice the angle at the circumference on the same arc, and this one theorem explains angles in the same segment, the right angle in a semicircle, and the supplementary opposite angles of a cyclic quadrilateral. The measurements confirmed it in every configuration, with the angle at C always half the centre angle.

Your own words

CHAPTER 6 · 6.1

Angles in circles

The angle at the centre is twice the angle at the circumference standing on the same arc. This is the fundamental circle theorem, and the others are consequences of it, so one generalisation covers many cases.

The proof splits into cases according to whether the centre lies on a side of the inscribed angle. Each case reduces to an isosceles triangle and the exterior angle theorem, and together the cases cover every configuration.

Angles in the same segment are equal, because each is half the same angle at the centre. The angle in a semicircle is a right angle, because the angle at the centre is 180 degrees.

A cyclic quadrilateral has opposite angles summing to 180 degrees, because the two opposite angles stand on arcs that together make the whole circumference. This is the theorem that makes cyclic quadrilaterals recognisable.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 6 · 6.1 · WORKED EXAMPLE 1

Angles in circles

WORKED EXAMPLE 6.1.1

Problem. An angle at the centre of a circle is 118 degrees and stands on the same arc as an angle at the circumference. Find the angle at the circumference.

angle	position	measure
at the centre	standing on arc AB	118°
at the circumference	same arc AB	59°
check	twice 59°	118°
other segment	major arc AB	121°

The centre angle is twice the circumference angle.

Mathematical reasoning. Apply the angle at the centre theorem directly.

1. The angle at the centre is 118 degrees and the angle at the circumference stands on the same arc.
2. By the theorem, the centre angle is twice the circumference angle.
3. So the circumference angle is $118 \div 2 = 59$ degrees.
4. Check: $2 \times 59 = 118$, as required.
5. A point on the other arc would give $180 - 59 = 121$ degrees, because opposite angles of a cyclic quadrilateral sum to 180.

Final answer. 59 degrees

Interpretation. The theorem relates two angles through the arc they share, which is why naming the arc is part of the justification.

CHAPTER 6 · 6.1 · WORKED EXAMPLE 2

Angles in circles

WORKED EXAMPLE 6.1.2

Problem. Prove that the angle in a semicircle is a right angle.

step	claim	reason
1	AB is a diameter, so the arc AB is 180°	definition of a diameter
2	the angle at the centre on arc AB is 180°	straight line
3	the angle at C is half of 180°	angle at the centre theorem
4	the angle at C is 90°	arithmetic

The proof is one application of the main theorem.

Mathematical reasoning. Treat the diameter as an arc of 180 degrees and apply the angle at the centre theorem.

1. Let AB be a diameter and C a point on the circle distinct from A and B.
2. The arc AB is a semicircle, so the angle AOB at the centre is 180 degrees.
3. The angle ACB stands on the same arc AB.
4. By the angle at the centre theorem, angle ACB is half of 180 degrees.
5. Hence angle ACB is 90 degrees, as required.

Final answer. the angle in a semicircle is 90 degrees

Interpretation. The proof names the theorem it uses and the arc it applies to, which is what makes it a proof rather than a description.

CHAPTER 6 · INVESTIGATION 6.2

Cyclic quadrilaterals

INQUIRY QUESTION

What does a quadrilateral being cyclic tell you?

YOUR TASK

Draw four cyclic quadrilaterals of your own, each with a different shape, and measure all four interior angles of each. Check that opposite angles are supplementary in every case. Then draw one quadrilateral that is not cyclic and show that the property fails.

quadrilateral	angles	opposite pair 1	opposite pair 2	cyclic?
square in a circle	90, 90, 90, 90	180	180	yes
rectangle in a circle	70, 110, 70, 110	180	180	yes
kite in a circle	60, 120, 60, 120	180	180	yes
irregular cyclic	100, 80, 75, 105	180	180	yes
non-cyclic	90, 90, 80, 100	180	170	no

CHAPTER 6 · INVESTIGATION 6.2

Cyclic quadrilaterals

What do you notice?

- Every cyclic quadrilateral had opposite angles summing to 180 degrees.
- The non-cyclic quadrilateral failed one of the two pairs, which was enough to rule it out.
- The test is an if and only if, so failing it proves non-cyclicity.
- The irregular case confirmed the property is not restricted to symmetric shapes.

Observation. Cyclicity is decided by one pair of supplementary opposite angles.

MATHEMATICAL CONCLUSION

A quadrilateral is cyclic exactly when a pair of opposite angles is supplementary, and then both pairs are. The exterior angle equals the interior opposite angle, which is the quickest way to find an unknown angle. The measurements confirmed the property in every cyclic case and its failure in the non-cyclic one.

Your own words

CHAPTER 6 · 6.2

Cyclic quadrilaterals

A cyclic quadrilateral is one whose four vertices lie on a circle, so its four points share one space of fixed radius. Its defining property is that opposite angles sum to 180 degrees, and the converse is also true: a quadrilateral with supplementary opposite angles is cyclic.

The proof of the defining property uses the angle at the centre theorem twice. Each opposite pair of angles stands on arcs that together form the whole circumference, so their halves sum to 180 degrees.

The converse gives a test for cyclicity that is used constantly: if one pair of opposite angles is supplementary, the fourth vertex lies on the circle through the other three. This is how a circle is constructed to pass through given points.

An exterior angle of a cyclic quadrilateral equals the interior opposite angle, because the interior angle is supplementary to its opposite and the exterior angle is supplementary to the interior. This is the property used to find an unknown angle quickly.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 6 · 6.2 · WORKED EXAMPLE 1

Cyclic quadrilaterals

WORKED EXAMPLE 6.2.1

Problem. A cyclic quadrilateral has angles 3x, 5x, 2x and 4x in order. Find each angle.

angle	expression	value
first	3x	54°
second	5x	90°
third	2x	36°
fourth	4x	72°

Opposite pairs sum to 180 degrees.

Mathematical reasoning. Use the supplementary property on both pairs of opposite angles.

1. Opposite angles sum to 180, so 3x + 2x = 180, giving x = 36.
2. Check the other pair: 5x + 4x = 9x = 9 × 36 = 324, which is not 180, so the ordering must be reconsidered.
3. The angles are in order around the quadrilateral, so opposite pairs are (3x, 2x) and (5x, 4x).
4. Both pairs must sum to 180, which requires 5x = 180, so x = 36, and then 5x = 180 is impossible for a genuine quadrilateral angle.
5. The only consistent reading is 3x + 5x = 180 and 2x + 4x = 180, both giving x = 22.5, so the angles are 67.5, 112.5, 45 and 90 degrees.

Final answer. 67.5°, 112.5°, 45° and 90°

Interpretation. Naming which pairs are opposite is part of the justification, and the two conditions must be checked together.

CHAPTER 6 · 6.2 · WORKED EXAMPLE 2

Cyclic quadrilaterals

WORKED EXAMPLE 6.2.2

Problem. Prove that an exterior angle of a cyclic quadrilateral equals the interior opposite angle.

step	claim	reason
1	angle DAB + angle BCD = 180°	cyclic quadrilateral
2	angle DAE + angle DAB = 180°	angles on a straight line
3	angle DAE = angle BCD	both are 180° − angle DAB

Two supplementary claims give the result.

Mathematical reasoning. Use the cyclic property and the straight-line property together.

1. Let ABCD be cyclic and let AE extend the side DA beyond A.
2. Then angle DAB + angle BCD = 180 degrees, because ABCD is cyclic.
3. Also angle DAE + angle DAB = 180 degrees, because they lie on a straight line.
4. Both angle DAE and angle BCD equal 180 degrees minus angle DAB.
5. Hence angle DAE = angle BCD, as required.

Final answer. the exterior angle equals the interior opposite angle

Interpretation. The proof chains two known facts, and each step cites the fact it uses.

CHAPTER 6 · INVESTIGATION 6.3

Tangents and radii

INQUIRY QUESTION

Why is a tangent perpendicular to the radius at the point of contact?

YOUR TASK

Draw a circle and an external point P. Draw the two tangents from P, touching at A and B. Measure PA, PB, the angles at P and the angles at A and B. Repeat for three different positions of P and record what stays the same and what changes.

position of P	PA	PB	angle APB	angle OAP
P close to the circle	4.1	4.1	112°	90°
P further away	8.7	8.7	64°	90°
P far away	19.4	19.4	30°	90°
P on the other side	8.7	8.7	64°	90°

CHAPTER 6 · INVESTIGATION 6.3

Tangents and radii

What do you notice?

- The two tangent lengths were equal in every case.
- The angles at the points of contact were always 90 degrees.
- The angle at P shrank as P moved away from the circle.
- Moving P to the other side changed nothing except the orientation.

Observation. The tangent lengths from a point are equal, and the tangent is always perpendicular to the radius.

MATHEMATICAL CONCLUSION

A tangent is perpendicular to the radius at the point of contact, which is proved by contradiction from the definition of a tangent. Two consequences follow immediately: the two tangents from an external point are equal, and the line from the centre to that point bisects the angle between them. The measurements confirmed both, with PA = PB in every case.

Your own words

CHAPTER 6 · 6.3

Tangents and radii

A tangent to a circle is a line that meets the circle at exactly one point, called the point of contact. The tangent is perpendicular to the radius drawn to that point, and this is a theorem rather than a definition.

The proof is by contradiction. If the tangent were not perpendicular to the radius, a foot of the perpendicular from the centre to the line would give a second point of the circle on the line, contradicting the definition of a tangent.

Two tangents drawn from an external point to a circle are equal in length, and the line from the centre to that point bisects the angle between them. Both facts follow from congruent right-angled triangles.

The angle between a tangent and a chord equals the angle in the alternate segment, which is the alternate segment theorem. It is the theorem that makes tangents useful for computing angles, and it is proved from the tangent-radius property.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 6 · 6.3 · WORKED EXAMPLE 1

Tangents and radii

WORKED EXAMPLE 6.3.1

Problem. A point P is 13 cm from the centre O of a circle of radius 5 cm. A tangent from P touches the circle at T. Find the length PT.

quantity	value	reason
OP	13 cm	given
OT	5 cm	radius
angle OTP	90°	tangent perpendicular to radius
PT	12 cm	Pythagoras: $13^2 - 5^2 = 144$

The tangent forms a right-angled triangle with the radius.

Mathematical reasoning. Use the tangent-radius property to get a right angle, then apply Pythagoras.

1. The tangent PT is perpendicular to the radius OT, so triangle OTP is right-angled at T.
2. By Pythagoras, $OP^2 = OT^2 + PT^2$.
3. So $13^2 = 5^2 + PT^2$, giving $169 = 25 + PT^2$.
4. Hence $PT^2 = 144$ and $PT = 12$ cm.
5. The two tangents from P are both 12 cm, by the equal-tangents theorem.

Final answer. PT = 12 cm

Interpretation. The tangent-radius property converts a tangent problem into a Pythagoras problem, which is why it is used so often.

CHAPTER 6 · 6.3 · WORKED EXAMPLE 2

Tangents and radii

WORKED EXAMPLE 6.3.2

Problem. Prove that the two tangents drawn to a circle from an external point are equal in length.

step	claim	reason
1	OA = OB	radii of the same circle
2	OP is common	shared side
3	angle OAP = angle OBP = 90°	tangent perpendicular to radius
4	triangles OAP and OBP are congruent	RHS
5	PA = PB	corresponding parts

Two right-angled triangles share the hypotenuse.

Mathematical reasoning. Show the two right-angled triangles are congruent and read off the equal hypotenuses.

1. Let the tangents from P touch the circle at A and B, and let O be the centre.
2. The radii OA and OB are equal, and OP is common to both triangles.
3. Both triangles are right-angled, because a tangent is perpendicular to the radius at the point of contact.
4. So triangles OAP and OBP are congruent by the right angle, hypotenuse, side criterion.
5. Hence PA = PB, as required.

Final answer. the two tangents are equal in length

Interpretation. The proof uses the tangent-radius property as one of its congruency conditions, which is why that property is proved first.

CHAPTER 6 · INVESTIGATION 6.4

Chords, secants and the alternate segment

INQUIRY QUESTION

How do the lengths of intersecting chords relate?

YOUR TASK

Draw a circle and two chords intersecting inside it. Measure the four segments and check the intersecting chords theorem. Then draw a tangent and a chord from the point of contact, and check the alternate segment theorem against an angle in the alternate segment. Repeat for a second circle.

measurement	circle 1	circle 2	relationship
AX × XB	2 × 6 = 12	3 × 4 = 12	—
CX × XD	4 × 3 = 12	6 × 2 = 12	equal to AX × XB
tangent-chord angle	38°	52°	—
angle in alternate segment	38°	52°	equal to the tangent-chord angle

CHAPTER 6 · INVESTIGATION 6.4

Chords, secants and the alternate segment

What do you notice?

- The two products were equal in both circles.
- The tangent-chord angle equalled the angle in the alternate segment in both cases.
- The second circle had the angle pairs swapped, which showed the relationship is not a coincidence of one diagram.
- Both theorems held regardless of where the chords crossed.

> **Observation.** Intersecting chords give equal products, and a tangent-chord angle equals the alternate segment angle.

MATHEMATICAL CONCLUSION

The intersecting chords theorem follows from the similarity of the triangles formed by the chords, which in turn follows from equal angles. The alternate segment theorem follows from the tangent-radius property and the angle at the centre theorem. The measurements confirmed both, with the products equal and the angle pairs equal in each circle.

Your own words

CHAPTER 6 · 6.4

Chords, secants and the alternate segment

When two chords AB and CD intersect inside a circle at X, the products AX × XB and CX × XD are equal. This is the intersecting chords theorem, and it follows from the similarity of the triangles formed.

The similarity comes from the equal angles: the angles at X are vertically opposite, and the angles subtended by the same arc are equal. Equal angles give similar triangles, and similar triangles give the equal ratios.

When two secants meet outside the circle, the product of the whole secant and its external part is the same for both. The tangent case is the limit of this, giving the tangent-secant theorem.

The alternate segment theorem states that the angle between a tangent and a chord equals the angle in the alternate segment. It is proved by drawing the radius to the point of contact and using the tangent-radius property together with the angle at the centre theorem.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 6 · 6.4 · WORKED EXAMPLE 1

Chords, secants and the alternate segment

WORKED EXAMPLE 6.4.1

Problem. Two chords AB and CD intersect at X inside a circle, with AX = 3, XB = 4, CX = 2. Find XD.

segment	length	product
AX	3	—
XB	4	—
AX × XB	12	—
CX × XD	2 × XD	= 12

The two products are equal.

Mathematical reasoning. Apply the intersecting chords theorem and solve for the unknown length.

1. By the intersecting chords theorem, AX × XB = CX × XD.
2. So 3 × 4 = 2 × XD, giving 12 = 2 × XD.
3. Hence XD = 6.
4. Check: 2 × 6 = 12, which equals 3 × 4.
5. The result is independent of where X lies along CD, so long as the products are formed correctly.

Final answer. XD = 6

Interpretation. The theorem converts a length problem into a product equation, and the check is the equality of the two products.

CHAPTER 6 · 6.4 · WORKED EXAMPLE 2

Chords, secants and the alternate segment

WORKED EXAMPLE 6.4.2

Problem. Prove the alternate segment theorem: the angle between a tangent at A and a chord AB equals the angle in the alternate segment.

step	claim	reason
1	OA is perpendicular to the tangent	tangent-radius property
2	angle OAB = angle OBA	isosceles triangle OAB
3	angle between tangent and AB = 90° − angle OAB	straight angle
4	angle at the centre = 2 × angle ACB	angle at the centre theorem
5	angle at the centre = 180° − 2 angle OAB	angle sum of triangle OAB
6	angle ACB = 90° − angle OAB	combining steps 4 and 5

The proof chains three circle theorems.

Mathematical reasoning. Express both angles in terms of the base angle of the isosceles triangle and compare.

1. Let the tangent at A meet the chord AB, and let C be a point on the circle in the alternate segment.
2. The radius OA is perpendicular to the tangent, so the angle between the tangent and AB is 90 degrees minus angle OAB.
3. Triangle OAB is isosceles, so angle OAB = angle OBA.
4. The angle at the centre AOB is 180 degrees minus twice angle OAB, and it is twice angle ACB.
5. So angle ACB is 90 degrees minus angle OAB, which equals the angle between the tangent and the chord.

Final answer. the angle between the tangent and the chord equals the angle in the alternate segment

Interpretation. The proof is a chain of three theorems, and each link is named.

CHAPTER 6 · INVESTIGATION 6.5

Constructing and writing proofs

INQUIRY QUESTION

What makes a geometric argument a proof?

YOUR TASK

Take four circle theorems from this chapter: the angle at the centre theorem, the angle in a semicircle, the tangent-radius property and the perpendicular from the centre to a chord. For each, write out a complete proof with a reason beside every step, and then state the converse and decide whether it is true.

theorem	proof method	converse	converse true?
angle at the centre	isosceles triangle plus exterior angle	if a centre angle is twice a circumference angle on the same arc then the four points are concyclic	yes
angle in a semicircle	special case of the above	if a right angle is subtended by a segment then the segment is a diameter	yes
tangent-radius	contradiction from the definition of a tangent	if a line through a point of a circle is perpendicular to the radius then it is a tangent	yes
perpendicular bisects chord	congruent triangles	the line from the centre to the midpoint of a chord is perpendicular to it	yes

CHAPTER 6 · INVESTIGATION 6.5

Constructing and writing proofs

What do you notice?

- Two proofs used congruent triangles and two used a contradiction or a reduction to an earlier result.
- Every converse turned out to be true, but each needed its own proof.
- The angle in a semicircle was the quickest, because it was a special case.
- Writing the reasons beside the steps exposed the assumptions immediately.

> **Observation.** A proof is a chain of justified statements, and the converse of a true statement is a separate claim.

MATHEMATICAL CONCLUSION

A geometric proof is a sequence of statements each justified by a definition, a given, or a previously proved result, and the justification is what distinguishes it from a description. The converse of a proved result is a separate claim requiring its own proof, and all four converses here were true.

Your own words

CHAPTER 6 · 6.5

Constructing and writing proofs

A proof is a sequence of statements, each justified by a definition, a previously proved result, or a given fact. The logic of the chain is the proof: a list of true statements without reasons is a description.

Proofs in circle geometry are usually written in two columns, or as a numbered chain with the reason beside each step. Either form is acceptable provided every step is justified and the conclusion follows by logic alone.

A proof by contradiction assumes the opposite of what is to be proved and derives a statement that contradicts a known fact. It is the natural form for the tangent-radius property, because the definition of a tangent is a statement about the number of intersection points.

The converse of a proved result is a separate claim that needs its own proof. The angle-in-a-semicircle theorem and its converse are both true, but neither follows from the other.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 6 · 6.5 · WORKED EXAMPLE 1

Constructing and writing proofs

WORKED EXAMPLE 6.5.1

Problem. Prove that if a pair of opposite angles of a quadrilateral is supplementary then the quadrilateral is cyclic.

step	claim	reason
1	suppose D is not on the circle through A, B, C	assumption for contradiction
2	let the circle meet BD again at E	construction
3	angle AEB = 180° − angle BCD	cyclic quadrilateral ABCE
4	angle AEB = angle ADB	E lies on BD
5	angle ADB = 180° − angle BCD	combining steps 3 and 4
6	D and E coincide	both satisfy the same condition on the line BD

A contradiction proof constructs a second point.

Mathematical reasoning. Assume the fourth vertex is off the circle, construct the point on the circle, and show the two must coincide.

1. Let ABCD have angle DAB + angle BCD = 180 degrees.
2. Suppose D is not on the circle through A, B and C.
3. Let that circle meet the line BD again at E, so ABCE is cyclic.
4. Then angle AEB = 180 degrees minus angle BCD, which equals angle ADB.
5. So D and E make the same angle with AB while lying on the same line, and hence coincide.
6. This contradicts the assumption, so D lies on the circle and ABCD is cyclic.

Final answer. the quadrilateral is cyclic

Interpretation. The proof is by contradiction, and the contradiction is that two points with the same position must be distinct.

CHAPTER 6 · 6.5 · WORKED EXAMPLE 2

Constructing and writing proofs

WORKED EXAMPLE 6.5.2

Problem. Write a proof that the perpendicular from the centre of a circle to a chord bisects the chord.

step	claim	reason
1	OM is perpendicular to chord AB	construction
2	OA = OB	radii
3	OM is common	shared side
4	triangles OMA and OMB are congruent	RHS
5	AM = MB	corresponding parts

Two right-angled triangles with a shared hypotenuse.

Mathematical reasoning. Form two congruent right-angled triangles and read off the equal segments.

1. Let M be the foot of the perpendicular from the centre O to the chord AB.
2. The radii OA and OB are equal, and OM is common.
3. Both triangles OMA and OMB are right-angled at M.
4. So they are congruent by the right angle, hypotenuse, side criterion.
5. Hence AM = MB, so M bisects the chord.

Final answer. the perpendicular from the centre bisects the chord

Interpretation. The proof is short because the congruency is immediate, and every step cites its reason.

CHAPTER 6 · INVESTIGATION 6.6

Investigation: circle theorems under transformation

INQUIRY QUESTION

Which circle properties survive a transformation?

YOUR TASK

Draw a circle with a chord and an inscribed angle. Apply a translation, a rotation, a reflection and an enlargement to the whole figure, and measure the inscribed angle in each image. Then repeat with the alternate segment theorem, and record which quantities change and which stay the same.

transformation	inscribed angle	arc length	radius	chord length
none	42°	84°	5	7.2
translation	42°	84°	5	7.2
rotation	42°	84°	5	7.2
reflection	42°	84°	5	7.2
enlargement, factor 3	42°	252°	15	21.6

CHAPTER 6 · INVESTIGATION 6.6

Investigation: circle theorems under transformation

What do you notice?

- The inscribed angle survived every transformation.
- The isometries left every measurement unchanged.
- The enlargement multiplied the lengths by 3 but left the angles alone.
- The alternate segment angle was likewise unchanged in every case.

Observation. Angles are preserved by all four transformations, and lengths only by the isometries.

MATHEMATICAL CONCLUSION

Translations, rotations and reflections preserve both angles and lengths, and enlargements preserve angles while scaling lengths. Every circle theorem is therefore invariant under all of them, which is why a single diagram is enough to establish a circle theorem: any other configuration is obtained by a transformation, and the transformation preserves the quantities the theorem relates.

Your own words

CHAPTER 6 · 6.6

Investigation: circle theorems under transformation

A circle is defined by a centre and a radius, so any transformation that preserves distances preserves circles. Translations, rotations and reflections all send circles to circles, and enlargements send circles to circles with a scaled radius.

An enlargement with centre on the circle sends the circle to a line, which is the tangent at that centre in the limit. This is the transformation view of a tangent: it is the image of a circle under an enlargement about a point of the circle.

Angles are preserved by all of these transformations, so every circle theorem is invariant under them. A theorem proved for one configuration is therefore proved for every configuration obtained by a transformation, which is why one diagram is enough.

Inversion in a circle is a transformation that sends circles to circles and lines to circles, and it is the transformation in which the intersecting chords theorem becomes most transparent. It is the tool behind many advanced results in circle geometry.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 6 · 6.6 · WORKED EXAMPLE 1

Investigation: circle theorems under transformation

WORKED EXAMPLE 6.6.1

Problem. A circle of radius 3 with centre (2, 5) is translated by the vector (−4, 1) and then enlarged by factor 2 about the origin. Find the final centre and radius.

stage	centre	radius
original	(2, 5)	3
after translation	(−2, 6)	3
after enlargement	(−4, 12)	6

Distances are scaled, so the radius scales too.

Mathematical reasoning. Apply each transformation to the centre and scale the radius by the enlargement factor.

1. The original centre is (2, 5) and the radius is 3.
2. Translation by (−4, 1) moves the centre to (−2, 6).
3. An enlargement of factor 2 about the origin multiplies every distance from the origin by 2.
4. So the centre moves to (−4, 12) and the radius becomes 6.
5. The image is a circle, because translations and enlargements preserve circles.

Final answer. centre (−4, 12), radius 6

Interpretation. Tracking the centre and the radius is enough, because the shape of a circle is determined by those two quantities.

CHAPTER 6 · 6.6 · WORKED EXAMPLE 2

Investigation: circle theorems under transformation

WORKED EXAMPLE 6.6.2

Problem. Explain why a tangent can be regarded as the limit of circles under an enlargement about a point of the circle.

factor	image of the circle about a point on it	behaviour
1	the circle itself	unchanged
2	a larger circle through the centre	curvature halves
10	a much larger circle	locally nearly straight
limit	a straight line	the tangent

Larger circles look straighter near the point.

Mathematical reasoning. An enlargement about a point of the circle increases the radius without bound, and a circle of infinite radius is a line.

1. An enlargement of factor k about a point P on the circle keeps P fixed and multiplies the radius by k.
2. The curvature of a circle is the reciprocal of its radius, so the curvature is divided by k.
3. As k grows, the curvature tends to zero and the circle becomes locally straight.
4. The limiting line passes through P and is perpendicular to the radius OP.
5. That line is the tangent, which is why the tangent is perpendicular to the radius.

Final answer. the tangent is the limit of the images, and it is perpendicular to the radius

Interpretation. The transformation view explains the tangent-radius property rather than merely restating it.

CHAPTER 6 · INVESTIGATION 6.7

Mixed practice

INQUIRY QUESTION

Which theorem does this figure need?

YOUR TASK

Take a circle of radius 5 with a point P at distance 13 from the centre. Draw the two tangents from P and the chord of contact AB. Find the angle APB, the length of AB, the angle AOB and the area of the quadrilateral OAPB, and name the theorem used at each step.

quantity	method	answer
length of a tangent	Pythagoras: $\sqrt{(13^2 - 5^2)}$	12
angle APB	2 arctan(5/12)	45.2°
angle AOB	180° − angle APB	134.8°
chord AB	2 × 5 × sin(angle AOB ÷ 2)	9.23
area of OAPB	2 × (1/2 × 5 × 12)	60

CHAPTER 6 · INVESTIGATION 6.7

Mixed practice

What do you notice?

- Every quantity came from the tangent-radius property plus one further theorem.
- The area of the quadrilateral was twice the area of one right-angled triangle.
- The chord of contact was found from the isosceles triangle OAB.
- Naming the theorem at each step made the whole solution checkable.

Observation. The tangent-radius property converts every tangent question into a right-angled triangle question.

MATHEMATICAL CONCLUSION

A circle question is solved by naming the theorem that applies, stating it in full, and applying it to the named parts. The tangent-radius property, the equal-tangents theorem and the perpendicular-bisector property between them handle most tangent and chord questions, and each answer cites the property it uses. Here the tangent was 12, the angle APB was about 45.2 degrees and the chord of contact was about 9.23.

Your own words

CHAPTER 6 · 6.7

Mixed practice

Every circle question begins by naming the theorem that applies. Naming it forces the relevant arc, chord or tangent to be identified, and the identification is most of the work.

A complete answer states the theorem in full, applies it to the named parts, and derives the required quantity. Omitting the statement of the theorem leaves the answer a calculation without a justification.

Proofs in this chapter are chains of at most five steps, each citing a definition, a given, or an earlier theorem. Writing the reason beside each step is the discipline that makes the chain checkable.

Mixed practice is where the theorem is chosen rather than given, so the reasoning behind the choice is part of what is being practised.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 6 · 6.7 · WORKED EXAMPLE 1

Mixed practice

WORKED EXAMPLE 6.7.1

Problem. A chord of length 8 cm is 3 cm from the centre of a circle. Find the radius.

step	working	result
half-chord	$8 \div 2 = 4$	—
right triangle	$r^2 = 4^2 + 3^2$	—
radius	$r^2 = 25$	$r = 5$

The perpendicular from the centre bisects the chord.

Mathematical reasoning. Use the perpendicular-bisector property to form a right-angled triangle, then apply Pythagoras.

1. The perpendicular from the centre to the chord bisects it, so half the chord is 4 cm.
2. The distance from the centre to the chord is 3 cm.
3. The radius, the half-chord and the distance form a right-angled triangle.
4. So $r^2 = 4^2 + 3^2 = 16 + 9 = 25$.
5. Hence $r = 5$ cm.

Final answer. radius 5 cm

Interpretation. The perpendicular-bisector property is what makes the right-angled triangle available, and it must be named.

CHAPTER 6 · 6.7 · WORKED EXAMPLE 2

Mixed practice

WORKED EXAMPLE 6.7.2

Problem. Two tangents from P touch a circle of radius 6 at A and B, and the angle APB is 70 degrees. Find the angle AOB and the length PA.

step	working	result
right angles	angle OAP = 90°	—
quadrilateral	angle AOB = 180° − 70°	110°
Pythagoras	$PA^2 = OP^2 - 36$	—
half-angle	angle APO = 35°	bisected

The two tangents and the two radii form a quadrilateral.

Mathematical reasoning. Use the tangent-radius property and the angle sum of the quadrilateral, then Pythagoras.

1. The radii OA and OB are perpendicular to the tangents at A and B.
2. So the quadrilateral OAPB has angles 90, 70, 90 and angle AOB.
3. The angle sum is 360, so angle AOB = 360 − 250 = 110 degrees.
4. The line OP bisects the angle APB, so angle APO is 35 degrees.
5. In the right-angled triangle OAP, tan 35° = 6/PA, so PA = 6 ÷ tan 35° ≈ 8.56 cm.

Final answer. angle AOB = 110 degrees; PA ≈ 8.56 cm

Interpretation. The two properties used are the tangent-radius property and the equal-tangents theorem, and both must be named.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

6.A1 FLUENCY

An angle at the centre is 96 degrees and stands on the same arc as an angle at the circumference. Find the angle at the circumference.

6.A2 REASONING

An angle in a semicircle is a right angle. Explain why.

6.A3 FLUENCY

A cyclic quadrilateral has angles 70, 110 and 85 degrees. Find the fourth.

6.A4 FLUENCY

A chord is 6 cm from the centre of a circle of radius 10 cm. Find the chord length.

6.A5 FLUENCY

A tangent from P touches a circle of radius 5 at T, and OP = 13. Find PT.

6.A6 FLUENCY

Two chords intersect with segments 3 and 4 on one chord and 2 and x on the other. Find x.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

6.A7 FLUENCY

The angle between a tangent and a chord is 35 degrees. Find the angle in the alternate segment.

6.A8 FLUENCY

A cyclic quadrilateral has an exterior angle of 62 degrees at one vertex. Find the interior opposite angle.

6.A9 FLUENCY

Two tangents from P touch a circle at A and B, and PA = 9. Find PB.

6.A10 FLUENCY

An angle at the circumference standing on a major arc is 40 degrees. Find the angle at the centre on the same arc.

6.A11 REASONING

A circle has radius 7 and a chord of length 10. Find the distance from the centre to the chord.

6.A12 REASONING

Prove that the perpendicular from the centre of a circle to a chord bisects the chord.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

6.B1 REASONING

Investigate the angle at the centre theorem by drawing three circles with different arcs and measuring the pairs of angles, and justify the relationship.

6.B2 REASONING

Conjecture a rule for the angle between two tangents from an external point and the angle at the centre between the radii, and test it.

6.B3 CHALLENGE

Investigate the locus of points from which the two tangents to a circle are perpendicular, and justify.

6.B4 REASONING

Investigate whether a rectangle is always cyclic, and justify.

6.B5 CHALLENGE

Investigate the length of the chord of contact AB in terms of the radius r and the distance d from the centre to the external point.

6.B6 REASONING

Conjecture a rule for when two circles have exactly two common tangents, and test it on three pairs.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

6.B7 REASONING

Investigate the angle in the alternate segment for a tangent and a diameter, and justify the result.

6.B8 REASONING

Investigate the locus of the centre of a circle tangent to two intersecting lines, and justify.

6.B9 CHALLENGE

Conjecture a rule for the angle between two tangents from a point at distance d from the centre of a circle of radius r, and test it.

6.B10 REASONING

Investigate whether every rhombus is cyclic, and justify.

6.B11 CHALLENGE

Investigate the common chord of two intersecting circles and justify that it is perpendicular to the line of centres.

CRITERION C Communicating

Criterion C practice · Communicating

6.C1 REASONING

Write a complete solution to finding the radius of a circle in which a chord of length 24 is 5 cm from the centre.

6.C2 REASONING

Write a complete solution to proving that the angle in a semicircle is a right angle.

6.C3 REASONING

Write a complete solution to finding the angle between two tangents from P to a circle of radius 6 when OP = 10.

6.C4 REASONING

Write a complete solution to proving that the two tangents from an external point are equal.

6.C5 CHALLENGE

Write a complete solution to finding the length of the chord of contact when the tangents from P meet at 60 degrees and the radius is 8.

6.C6 CHALLENGE

Write a complete solution to proving the intersecting chords theorem for chords meeting inside a circle.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

6.C7 FLUENCY

Write a complete solution to finding the angle between a tangent and a chord that subtends an angle of 44 degrees in the alternate segment.

6.C8 REASONING

Write a complete solution to finding the radius of a circle in which a chord of length 30 is 8 cm from the centre.

6.C9 REASONING

Write a complete solution to proving that an exterior angle of a cyclic quadrilateral equals the interior opposite angle.

6.C10 REASONING

Write a complete solution to finding the angle at the centre subtended by a chord of length 9 in a circle of radius 7.5.

6.C11 CHALLENGE

Write a complete solution to finding the area of the region between a tangent from P and a circle of radius 6 when OP = 10.

6.C12 CHALLENGE

Write a complete solution to proving the converse of the angle in a semicircle.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

6.D1 APPLICATION

A surveyor stands 40 m from the foot of a tower and measures the angle of elevation to the top as 28 degrees. Find the height of the tower.

6.D2 APPLICATION

A circular garden of radius 12 m is enclosed by a fence tangent to it at three points, each pair of fence sections meeting at 90 degrees. Find the length of each fence section.

6.D3 APPLICATION

A wheel of radius 30 cm rolls along a straight track. Describe the path of a point on the rim and explain why the track is tangent to the wheel at every instant.

6.D4 APPLICATION

An architect designs a window as a semicircle on top of a rectangle. The rectangle is 1.6 m wide and the semicircle has radius 0.8 m. Find the total height of the window if the rectangle is 1.2 m tall.

6.D5 APPLICATION

A roundabout of radius 8 m is surrounded by a path 2 m wide. Find the length of the outer edge of the path.

6.D6 APPLICATION

A lighthouse beam sweeps a circle of radius 20 km. A ship is 15 km from the lighthouse. Find the angle through which the beam must turn for the ship to pass from the edge of the beam to the closest point.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

6.D7 REASONING

A student claims that any quadrilateral with one pair of opposite angles supplementary is cyclic. Investigate and justify.

6.D8 APPLICATION

A circular medallion of radius 4 cm is to be framed by a square frame touching it on all four sides. Find the side of the square and the area of frame not covered by the medallion.

6.D9 APPLICATION

A pulley of radius 10 cm drives a belt that also passes round a pulley of radius 4 cm with centres 30 cm apart. Find the length of the belt.

6.D10 CHALLENGE

A student draws a circle, marks a point outside it and draws two lines through the point that each cut the circle twice, claiming they are secants. Investigate the products of the segments and justify.

6.D11 REASONING

An ornamental pattern is built from a circle with four equal chords meeting at a point inside. Investigate the products of the segments and relate them to the intersecting chords theorem.

6.D12 REASONING

A student proves the tangent-radius property by contradiction. Write out the proof and explain each step.

CHAPTER 6 · APPLICATION AND ASSESSMENT

Applying chapter 6

These tasks use circle theorems and geometric proof. Name the theorem you use, state it in full, and set out the deduction in order.

A1 · The window design

A stained-glass window is a circle of radius 60 cm with three chords meeting at a point 25 cm from the centre. One chord is divided into segments of 15 cm and 35 cm. Find the segments of a second chord passing through the same point, and the angle at the centre subtended by the shorter segment of the first chord.

Criteria C and D · apply the intersecting chords theorem and the angle at the centre theorem, justifying each step.

A2 · The circular garden

A circular garden of radius 10 m has a straight path tangent to it at one point and a second path meeting the first at 50 degrees, both tangent to the garden. Find the length of each path from the corner to its point of contact, the angle at the centre between the two radii to the points of contact, and the area of the region enclosed by the two paths and the minor arc between the contact points.

Criteria C and D · use the tangent-radius property and the equal-tangents theorem, then compute the area.

CHAPTER 6 · REVIEW

What you should now be able to do

Key ideas

- The angle at the centre is twice the angle at the circumference on the same arc.
- Angles in the same segment are equal.
- The angle in a semicircle is a right angle.
- Opposite angles of a cyclic quadrilateral are supplementary.
- An exterior angle of a cyclic quadrilateral equals the interior opposite angle.
- A tangent is perpendicular to the radius at the point of contact.
- The two tangents from an external point are equal in length.
- The angle between a tangent and a chord equals the angle in the alternate segment.
- For intersecting chords, $AX \times XB = CX \times XD$.
- The perpendicular from the centre to a chord bisects the chord.

Vocabulary

circle · centre · radius · diameter · chord · arc · tangent · secant · point of contact · cyclic quadrilateral · alternate segment · angle at the centre · angle at the circumference · concyclic · proof by contradiction · congruent · similar

CHAPTER 6 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

6.R1 An angle at the circumference is 33 degrees. Find the angle at the centre on the same arc.

6.R2 A chord is 9 cm from the centre of a circle of radius 15 cm. Find the chord length.

6.R3 A cyclic quadrilateral has angles 3x, 4x, 2x and 5x in order. Find each angle.

6.R4 A tangent from P touches a circle of radius 7, with OP = 25. Find the tangent length.

6.R5 Two chords meet with segments 5 and 6 on one and 4 and x on the other. Find x.

6.R6 Prove that the angle between a tangent and a diameter is 90 degrees.

CHAPTER 6 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> A circle conceals equal angles and fixed distances; a few theorems, proved from radii and triangles, let us deduce measurements and justify every claim.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of logic and the related concepts of space, generalisation and justification appear in your work?
- How is the global context of personal and cultural expression, and its exploration of the geometry of architecture and ornament, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a diagram as a justification, and what would you still like to ask?

CHAPTER 6 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

An ornament is a theorem made visible. Name the property the design depends on, prove it, and say what the design would lose without it.

Service as Action

COMMUNITY NEED	Architectural and ornamental designs are drawn without the geometry that makes them accurate.
MATHEMATICAL CONNECTION	Circle geometry and geometric proof.
PROPOSED ACTION	Take one ornamental design in your community, identify the circle theorems it depends on, and publish a worked proof of one of them.
EXPECTED IMPACT	The design can be reproduced accurately by anyone.
STUDENT REFLECTION	Record the design, the theorem and the proof.

Community Project

PROJECT IDEA	A geometry audit of a local building or monument.
NEED	The proportions are admired but not explained.
RESEARCH QUESTION	Which circle theorems does the design use, and can each be proved from radii and triangles?
MATHEMATICS	Circle geometry and geometric proof.
ACTION OR OUTCOME	A one-page explanation with a proof.
BENEFICIARIES	Anyone who visits or studies the building.
EVIDENCE	The explanation and the proofs you wrote.

Personal Project connection · MYP 5

PERSONAL INTEREST	Art, design, architecture or craft.
LEARNING GOAL	To prove a circle theorem and use it to make something accurate.
PRODUCT OR OUTCOME	A drawing or model that depends on one circle theorem.
MATHEMATICS	Circle geometry and geometric proof.
PROCESS	Choose a theorem, prove it, and apply it to a design.
EVIDENCE	The design and the proof behind it.

CHAPTER 7 · PART III · SPATIAL REASONING

07

Sine and Cosine Rules; Trigonometry in Three Dimensions

A triangle is determined by three independent measurements. The sine rule and the cosine rule are the two relationships that connect them, and together they solve every triangle and extend trigonometry into three dimensions.

ENGAGE

A surveyor measures two angles and one side of a triangular plot. A colleague measures two sides and a non-included angle of a different plot and finds two possible positions for the third corner. Why does one set of measurements give one triangle and the other give two?

Framing the inquiry

Key concept: Relationships
Related concepts: Space · Change · Measurement
Global context: Orientation in space and time
Global context exploration: navigation, surveying and the geometry of place

Statement of inquiry: Triangles that are not right-angled can still be solved: the sine and cosine rules turn measurements of angle and distance into positions, areas and three-dimensional structure.

Questions to guide us

- **Factual:** What do the sine rule and the cosine rule state, and when is each used?
- **Conceptual:** Why can two sides and a non-included angle determine two triangles, and when does it determine none?
- **Debatable:** If a navigation calculation gives two possible positions, which one should be acted on, and on what grounds?

ATL · CRITICAL THINKING. In order for a student to decide how many triangles a set of measurements determines, students must compare the given side with the perpendicular height, identify the zero, one or two-triangle cases, and justify the conclusion.

CHAPTER 7 · INVESTIGATION 7.1

The sine rule

INQUIRY QUESTION

How do the sine rule and the ambiguous case fit together?

YOUR TASK

Take a fixed angle A = 40° and a fixed side b = 12. Vary a over the values 6, 8, 7.7, 7.71, 12 and 14. For each, compute b sin A, decide how many triangles exist, and where two exist, find both values of B. Record the case boundaries.

a	b sin A	comparison	number of triangles	value(s) of B
6	7.713	a < b sin A	0	none
7.7	7.713	a < b sin A	0	none
7.71	7.713	a < b sin A	0	none
7.713	7.713	b sin A = a	1 (right-angled)	90°
8	7.713	b sin A < a < b	2	73.9° and 106.1°
12	7.713	a = b	1	40°
14	7.713	a > b	1	33.7°

CHAPTER 7 · INVESTIGATION 7.1

The sine rule

What do you notice?

- Below the height there were no triangles at all, however close the values were.
- At the height there was exactly one triangle, and it was right-angled.
- Between the height and b there were exactly two triangles, with supplementary values of B.
- At and above b there was exactly one triangle again.

Observation. The number of triangles is decided by comparing a with b sin A and b, and the two-triangle case gives supplementary values of B.

MATHEMATICAL CONCLUSION

Two sides and a non-included angle determine a triangle only when the given side is at least the perpendicular height b sin A. When b sin A = a there is exactly one triangle and it is right-angled; when b sin A < a < b there are two, with the two values of B supplementary; when a < b sin A there is none. Here the height was 7.713, and the cases were 0, 1 and 2 triangles around it.

Your own words

CHAPTER 7 · 7.1

The sine rule

The sine rule states that a/sin A = b/sin B = c/sin C, where a is the side opposite the angle A. It is proved by dropping a perpendicular from one vertex and expressing its length in two ways.

The rule is used when a triangle has a side and its opposite angle, together with one further angle or side. It converts a known pair into an unknown one through a single ratio, which is why it is the rule for angle-side problems.

The ambiguous case arises when two sides and a non-included angle are given, written SSA. The given side opposite the given angle may be shorter than, equal to, or longer than the perpendicular height, and the three cases give two, one or no triangle.

The condition is exact: with A given and sides a and b given, the perpendicular height is b sin A. Two triangles exist when b sin A < a < b; exactly one right triangle exists when b sin A = a; no triangle exists when a < b sin A; and exactly one triangle exists when a ≥ b.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 7 · 7.1 · WORKED EXAMPLE 1

The sine rule

WORKED EXAMPLE 7.1.1

Problem. In triangle ABC, A = 38°, B = 64° and a = 9.4. Find b and c.

quantity	working	value
C	180° − 38° − 64°	78°
b	9.4 sin 64° ÷ sin 38°	13.72
c	9.4 sin 78° ÷ sin 38°	14.93
check	largest side opposite largest angle	c > b > a

Each side follows from one ratio.

Mathematical reasoning. Find the third angle, then apply the sine rule twice.

1. The third angle is C = 180° − 38° − 64° = 78°.
2. By the sine rule, b = a sin B ÷ sin A.
3. So b = 9.4 × sin 64° ÷ sin 38° = 9.4 × 0.8988 ÷ 0.6157 ≈ 13.72.
4. Similarly c = 9.4 × sin 78° ÷ sin 38° = 9.4 × 0.9781 ÷ 0.6157 ≈ 14.93.
5. Check: C is the largest angle, so c is the largest side, as found.

Final answer. b ≈ 13.72 and c ≈ 14.93

Interpretation. Two angles and one side determine exactly one triangle, so there is no ambiguity here.

CHAPTER 7 · 7.1 · WORKED EXAMPLE 2

The sine rule

WORKED EXAMPLE 7.1.2

Problem. In triangle ABC, A = 30°, a = 5 and b = 10. Determine how many triangles exist and find the remaining measurements.

case	condition	holds?
no triangle	a < b sin A	5 < 5, no
one right triangle	b sin A = a	10 × 0.5 = 5, yes
two triangles	b sin A < a < b	5 < 5, no
one triangle	a ≥ b	5 ≥ 10, no

The height b sin A equals the given side.

Mathematical reasoning. Compute the perpendicular height b sin A and compare it with the given side a to decide the number of triangles.

1. The perpendicular height from C to the side AB extended is b sin A = 10 × sin 30° = 10 × 0.5 = 5.
2. This equals the given side a = 5, so the height condition b sin A = a holds.
3. Hence there is exactly one triangle, and it is right-angled at C.
4. The remaining angles are B = 90° and C = 60°.
5. Check: a/sin A = 5/0.5 = 10 and b/sin B = 10/1 = 10, so the sine rule is satisfied.

Final answer. exactly one triangle, right-angled at C, with B = 90° and C = 60°

Interpretation. The ambiguous case is decided before any angle is computed, and the check confirms the single solution.

CHAPTER 7 · INVESTIGATION 7.2

The cosine rule

INQUIRY QUESTION

When is the cosine rule the right rule, and what does it generalise?

YOUR TASK

Take a triangle with sides 8 and 11 and included angle 55°. Compute the third side by the cosine rule, then all three angles, and check the angle sum. Then repeat with the included angle 125° and compare the two results.

included angle	third side	largest angle	smallest angle	angle sum
55°	9.22	62.4°	42.6°	180.0°
90°	13.60	90.0°	35.8°	180.0°
125°	16.87	125.0°	19.9°	180.0°
155°	18.19	155.0°	9.7°	180.0°

CHAPTER 7 · INVESTIGATION 7.2

The cosine rule

What do you notice?

- A larger included angle gave a much larger third side, from 9.22 to 16.87.
- In both cases the largest angle was the given one, because it was opposite the largest side.
- The angle sum was 180 degrees in both cases, confirming the calculations.
- The change in the third side was far from proportional to the change in the angle, because the cosine is not linear.

Observation. The third side grows with the included angle, and fastest when the cosine is changing fastest.

MATHEMATICAL CONCLUSION

The cosine rule $a^2 = b^2 + c^2 - 2bc \cos A$ determines the third side from two sides and the included angle, and rearranged determines an angle from three sides. It is a generalisation of Pythagoras, to which it reduces when the included angle is a right angle. Here the third side was 9.22 for an included angle of 55 degrees and 16.87 for 125 degrees.

Your own words

CHAPTER 7 · 7.2

The cosine rule

The cosine rule states that $a^2 = b^2 + c^2 - 2bc\cos A$, and rearranged gives $\cos A = (b^2 + c^2 - a^2)/(2bc)$. It is proved by placing the triangle in a coordinate system and applying Pythagoras to two right-angled triangles.

The rule is used when three sides are given, or two sides and the included angle. It is the only rule that handles the included angle directly, and it reduces to Pythagoras when the included angle is 90 degrees.

The cosine rule is a generalisation of Pythagoras: the term $2bc\cos A$ vanishes when A is a right angle, and the formula becomes $a^2 = b^2 + c^2$. Every right-angled calculation is a special case of it.

When the cosine rule gives a quadratic in an unknown side, both roots must be considered and rejected if they are not positive. This happens when two sides and a non-included angle are given, and it is the algebraic form of the ambiguous case.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 7 · 7.2 · WORKED EXAMPLE 1

The cosine rule

WORKED EXAMPLE 7.2.1

Problem. In triangle ABC, $b = 7$, $c = 10$ and $A = 47°$. Find a and then find B.

quantity	working	value
a^2	$7^2 + 10^2 - 2(7)(10)\cos 47°$	95.59
a	$\sqrt{95.59}$	9.78
$\cos B$	$(49 + 95.59 - 100) \div (2 \times 7 \times 9.78)$	0.3253
B	$\arccos 0.3253$	71.0°

The included angle fixes the third side.

Mathematical reasoning. Apply the cosine rule for the side, then again for the angle.

1. The cosine rule gives $a^2 = b^2 + c^2 - 2bc\cos A$.
2. So $a^2 = 49 + 100 - 140\cos 47° = 149 - 140(0.6820) \approx 95.59$.
3. Hence $a \approx 9.78$.
4. For the angle, $\cos B = (a^2 + c^2 - b^2)/(2ac) = (95.59 + 100 - 49)/(2 \times 9.78 \times 10) \approx 0.7493$.
5. So $B \approx 41.5°$, and $C \approx 180 - 47 - 41.5 = 91.5°$.

Final answer. $a \approx 9.78$ and $B \approx 41.5°$

Interpretation. Two sides and the included angle determine exactly one triangle, so there is no ambiguity.

CHAPTER 7 · 7.2 · WORKED EXAMPLE 2

The cosine rule

WORKED EXAMPLE 7.2.2

Problem. Find the angles of the triangle with sides 5, 7 and 9.

angle	cosine	value
A (opposite 9)	(25 + 49 − 81)/70 = −0.1	95.7°
B (opposite 7)	(25 + 81 − 49)/90 = 0.6333	50.7°
C (opposite 5)	(49 + 81 − 25)/126 = 0.8333	33.6°
check	95.7 + 50.7 + 33.6	180.0°

Three sides determine the angles uniquely.

Mathematical reasoning. Apply the cosine rule three times, or twice and use the angle sum.

1. For the angle opposite the side 9, $\cos A = (5^2 + 7^2 - 9^2)/(2 \times 5 \times 7) = (25 + 49 - 81)/70 = -7/70 = -0.1$.
2. So $A \approx 95.7°$, which is obtuse as the negative cosine indicates.
3. For the angle opposite 7, $\cos B = (25 + 81 - 49)/(2 \times 5 \times 9) = 57/90 \approx 0.6333$, so $B \approx 50.7°$.
4. For the angle opposite 5, $\cos C = (49 + 81 - 25)/(2 \times 7 \times 9) = 105/126 \approx 0.8333$, so $C \approx 33.6°$.
5. The three angles sum to 180°, as required.

Final answer. approximately 95.7°, 50.7° and 33.6°

Interpretation. Three sides determine exactly one triangle, so the answer is unique and the angle sum is the check.

CHAPTER 7 · INVESTIGATION 7.3

Area and the included angle

INQUIRY QUESTION

How is the area of a triangle found without a height?

YOUR TASK

Take a triangle with sides 6, 8 and 11. Compute its area using each of the three forms of the area formula, and check that all three agree. Then find the radius of its circumcircle using $a = 2R \sin A$.

form	angle needed	working	area
$\frac{1}{2}ab \sin C$	C opposite 11	$\frac{1}{2} \times 6 \times 8 \times \sin C$	22.4
$\frac{1}{2}bc \sin A$	A opposite 6	$\frac{1}{2} \times 8 \times 11 \times \sin A$	22.4
$\frac{1}{2}ca \sin B$	B opposite 8	$\frac{1}{2} \times 11 \times 6 \times \sin B$	22.4
circumradius	A opposite 6	$6 \div (2 \sin A)$	5.49

CHAPTER 7 · INVESTIGATION 7.3

Area and the included angle

What do you notice?

- All three forms gave the same area, as they had to.
- The angles were found first by the cosine rule, since only the sides were given.
- The circumradius came from a = 2R sin A, which is the sine rule rearranged.
- The area was about 22.4 square units, and the circumradius about 5.49.

Observation. The area formula is symmetric in the three pairs of sides, and the sine rule extends to the circumradius.

MATHEMATICAL CONCLUSION

The area of a triangle is ½bc sin A, where A is the angle included between the sides b and c, and the formula is symmetric in the three pairs. The same relationship rearranged gives a = 2R sin A, where R is the circumradius. Here every form gave an area of about 22.4 square units and a circumradius of about 5.49.

Your own words

CHAPTER 7 · 7.3

Area and the included angle

The area of a triangle is half the base times the perpendicular height. When the height is not given but two sides and the included angle are, the height is one side times the sine of the included angle, giving area = ½bc sin A.

The formula is symmetric: area = ½ab sin C = ½bc sin A = ½ca sin B. Any pair of sides with the angle between them gives the same area, which is a check on the calculation.

The formula is derived by dropping the perpendicular and using the definition of sine in the resulting right-angled triangle. The derivation is short, and it explains why the included angle is the one that appears.

The same formula applied to a sector or a segment gives the area of a circular sector as $\frac{1}{2}r^2\theta$ in radians, and the segment as the sector minus the triangle. The radian measure is what makes the sector formula simple.

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In your own words

CHAPTER 7 · 7.3 · WORKED EXAMPLE 1

Area and the included angle

WORKED EXAMPLE 7.3.1

Problem. Find the area of a triangle with sides 9 and 13 and included angle 52°.

step	working	value
height	$9 \sin 52°$	7.09
area	$\frac{1}{2} \times 13 \times 7.09$	46.1
check	$\frac{1}{2} \times 9 \times 13 \times \sin 52°$	46.1
units	square units	—

The height is one side times the sine of the included angle.

Mathematical reasoning. Drop the perpendicular and express the height with the sine of the included angle.

1. Take the sides 9 and 13 with the included angle 52°.
2. The height on the base 13 is $9 \sin 52° \approx 9 \times 0.7880 \approx 7.09$.
3. The area is $\frac{1}{2} \times 13 \times 7.09 \approx 46.1$ square units.
4. Check with the formula: $\frac{1}{2} \times 9 \times 13 \times \sin 52° \approx 46.1$.
5. The two calculations agree, as they must.

Final answer. about 46.1 square units

Interpretation. The direct formula is faster, and the height calculation explains where it comes from.

CHAPTER 7 · 7.3 · WORKED EXAMPLE 2

Area and the included angle

WORKED EXAMPLE 7.3.2

Problem. A sector of a circle of radius 8 has angle 0.7 radians. Find its area and the area of the segment cut off by the chord.

region	working	area
sector	$\frac{1}{2} \times 8^2 \times 0.7$	22.4
triangle	$\frac{1}{2} \times 8 \times 8 \times \sin 0.7$	20.5
segment	$22.4 - 20.5$	1.9

The segment is the sector minus the triangle.

Mathematical reasoning. Use the sector formula with radians and subtract the triangle.

1. The sector area is $\frac{1}{2}r^2\theta = \frac{1}{2} \times 64 \times 0.7 = 22.4$.
2. The triangle formed by the two radii and the chord has area $\frac{1}{2}r^2 \sin\theta = \frac{1}{2} \times 64 \times \sin 0.7$.
3. $\sin 0.7 \approx 0.6442$, so the triangle area is $32 \times 0.6442 \approx 20.5$.
4. The segment is the sector minus the triangle: $22.4 - 20.5 \approx 1.9$.
5. The segment is much smaller than the sector because the angle is small.

Final answer. sector 22.4, segment about 1.9

Interpretation. The radian measure makes the sector formula $\frac{1}{2}r^2\theta$, and the same triangle formula gives the correction.

CHAPTER 7 · INVESTIGATION 7.4

Bearings and navigation

INQUIRY QUESTION

How do bearings turn a triangle into a position?

YOUR TASK

Set up a navigation problem of your own with two legs and a turn, solve it by the cosine rule, and then solve the same problem by resolving each leg into north and east components. Compare the two answers and record which method you found more reliable.

leg	distance	bearing	north component	east component
first	25	020°	23.49	8.55
second	18	130°	−11.57	13.79
total	—	—	11.92	22.34
resultant	—	—	distance 25.31	bearing 118.1°

CHAPTER 7 · INVESTIGATION 7.4

Bearings and navigation

What do you notice?

- Both methods gave the same distance and the same bearing.
- The component method made the sign of the northward distance visible, which the cosine rule hid.
- The cosine rule was fewer steps once the included angle was found.
- The component method needed the angle of each leg measured from north, which is what a bearing gives directly.

> **Observation.** Bearings are angles from north, so they suit the component method; the cosine rule suits a triangle with a known included angle.

MATHEMATICAL CONCLUSION

A bearing is a direction measured clockwise from north, and a navigation problem is a triangle whose sides are distances and whose angles come from bearings. The interior angles must be extracted carefully, and the two solution methods — the cosine rule and resolving into components — must agree. Here both gave a distance of 34.7 on a bearing of 137 degrees.

Your own words

CHAPTER 7 · 7.4

Bearings and navigation

A bearing is a direction measured clockwise from north, written as a three-figure angle. Bearings are the language in which positions are described, and they convert a diagram into a set of numbers that can be transmitted.

A navigation problem is a triangle in which the sides are distances and the angles are bearings. The sine and cosine rules solve it exactly as they solve any triangle, once the interior angles have been extracted from the bearings.

The interior angle between two bearings is the difference of the bearings, or 360 degrees minus that difference, depending on which way round the triangle goes. Getting this wrong is the commonest navigation error, and drawing the diagram prevents it.

Speed, distance and time enter as a scale on the sides: a ship travelling at 12 knots for 3 hours covers 36 nautical miles. The bearing gives the direction and the distance the magnitude, and together they give a position vector.

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In your own words

CHAPTER 7 · 7.4 · WORKED EXAMPLE 1

Bearings and navigation

WORKED EXAMPLE 7.4.1

Problem. A ship sails 40 km on a bearing of 060° and then 30 km on a bearing of 150°. Find its distance and bearing from the start.

leg	distance	bearing	interior angle at the turn
first	40	060°	—
second	30	150°	90°
resultant	—	—	—

The turn is a right angle.

Mathematical reasoning. Extract the interior angle from the bearings, then apply Pythagoras and trigonometry.

1. The bearings differ by 150 − 60 = 90 degrees, so the turn is a right angle.
2. The two legs are perpendicular, so the distance is $\sqrt{(40^2 + 30^2)} = \sqrt{2500} = 50$ km.
3. The bearing of the resultant is 060° + arctan(30/40) = 060° + 36.87° = 096.87°.
4. So the ship is 50 km away on a bearing of approximately 097°.
5. Check: the eastward distance is 40 sin 60° + 30 sin 150° = 34.64 + 15 = 49.64, and the northward is 40 cos 60° + 30 cos 150° = 20 − 25.98 = −5.98, which gives the same distance.

Final answer. 50 km on a bearing of about 097°

Interpretation. The bearing difference is the interior angle, and the components check is a different route to the same position.

CHAPTER 7 · 7.4 · WORKED EXAMPLE 2

Bearings and navigation

WORKED EXAMPLE 7.4.2

Problem. Two ports A and B are 80 km apart, with B on a bearing of 035° from A. A ship leaves A and a second leaves B at the same time, the first on a bearing of 080° at 20 km/h and the second on a bearing of 200° at 15 km/h. Find the distance between them after 2 hours.

quantity	working	value
first ship's distance	20 × 2	40 km
second ship's distance	15 × 2	30 km
angle at A between AB and the first leg	080° − 035°	45°
angle between the two legs	—	140°

The included angle is not the difference of the bearings.

Mathematical reasoning. Extract the two distances and the included angle from the bearings, then apply the cosine rule.

1. After 2 hours the first ship has travelled 40 km and the second 30 km.
2. The angle at A between AB and the first ship's course is 080° − 035° = 45°.
3. The second ship's course from B is on a bearing of 200°, so the angle between the two courses at the triangle's vertices is 180° − 45° − (200° − 180°) = 140°.
4. By the cosine rule, the distance squared is $40^2 + 30^2 - 2(40)(30)\cos 140°$.
5. That is 1600 + 900 + 1838.5 = 4338.5, so the distance is about 65.9 km.

Final answer. about 65.9 km

Interpretation. The included angle must be extracted from the bearings carefully, and drawing the diagram is what prevents the error.

CHAPTER 7 · INVESTIGATION 7.5

Trigonometry in three dimensions

INQUIRY QUESTION

How does a three-dimensional problem become a two-dimensional one?

YOUR TASK

Take a cone of radius 5 and vertical height 12. Find the slant height, the angle between the slant height and the base, the angle between the slant height and the axis, and the area of the curved surface. Then repeat for a cone of radius 5 and height 5, and compare.

quantity	cone 1 (r = 5, h = 12)	cone 2 (r = 5, h = 5)
slant height	13	$5\sqrt{2} \approx 7.07$
angle with the base	67.4°	45°
angle with the axis	22.6°	45°
curved surface area	$\pi \times 5 \times 13 \approx 204.2$	$\pi \times 5 \times 7.07 \approx 111.1$

CHAPTER 7 · INVESTIGATION 7.5

Trigonometry in three dimensions

What do you notice?

- The two angles were complementary in each cone, as they had to be in a right-angled triangle.
- Halving the height halved the slant height only partly, from 13 to 7.07.
- The curved surface area fell by nearly half, because it is proportional to the slant height.
- The second cone had equal base radius and height, so its angles were both 45 degrees.

Observation. Every angle in a cone is read from the right-angled triangle formed by the radius, the height and the slant height.

MATHEMATICAL CONCLUSION

A three-dimensional problem is solved by choosing a plane that contains the required lengths and solving the two-dimensional triangle in it. In a cone that plane gives the radius, the vertical height and the slant height, and from them every angle and the curved surface area. Here the slant heights were 13 and about 7.07, with curved surface areas of about 204.2 and 111.1.

Your own words

CHAPTER 7 · 7.5

Trigonometry in three dimensions

A three-dimensional problem is solved by finding a plane that contains the required lengths and angles, solving the two-dimensional triangle in that plane, and then relating it back to the solid. The choice of plane is the whole skill.

In a cuboid with edges a, b and c, the space diagonal is $\sqrt{(a^2 + b^2 + c^2)}$ and the angle between the diagonal and an edge is found from a right-angled triangle containing both. The face diagonal is an intermediate step.

The angle between a line and a plane is the angle between the line and its projection onto the plane, and it is found in the right-angled triangle formed by the line, its projection and the perpendicular. It is never the angle the line makes with the plane's normal.

In a pyramid or cone, the slant height, the vertical height and the radius of the base form a right-angled triangle, and every angle in the solid is found from that triangle or from a triangle in a face.

> FORMAL DEVELOPMENT
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In your own words

CHAPTER 7 · 7.5 · WORKED EXAMPLE 1

Trigonometry in three dimensions

WORKED EXAMPLE 7.5.1

Problem. A cuboid has edges 6, 8 and 24. Find the space diagonal and the angle it makes with the longest edge.

step	working	value
face diagonal	$\sqrt{(6^2 + 8^2)}$	10
space diagonal	$\sqrt{(10^2 + 24^2)}$	26
cos of the angle	24/26	0.9231
the angle	arccos 0.9231	22.6°

The space diagonal closes a right-angled triangle.

Mathematical reasoning. Find the face diagonal, then use it to find the space diagonal and the angle.

1. The face diagonal on the 6 by 8 face is $\sqrt{(36 + 64)} = 10$.
2. The space diagonal is $\sqrt{(10^2 + 24^2)} = \sqrt{676} = 26$.
3. The angle with the longest edge has cosine 24/26, because the edge and the diagonal are in the same right-angled triangle.
4. So the angle is $\arccos(24/26) \approx 22.6°$.
5. Check: $6^2 + 8^2 + 24^2 = 36 + 64 + 576 = 676 = 26^2$.

Final answer. space diagonal 26; angle about 22.6°

Interpretation. The two-step Pythagoras is the standard way through a cuboid, and the angle is read from the final triangle.

CHAPTER 7 · 7.5 · WORKED EXAMPLE 2

Trigonometry in three dimensions

WORKED EXAMPLE 7.5.2

Problem. A square pyramid has a base of side 10 and a vertical height of 12. Find the slant height and the angle between a sloping face and the base.

step	working	value
half the base side	$10 \div 2$	5
slant height	$\sqrt{(5^2 + 12^2)}$	13
tan of the angle	12/5	2.4
the angle	arctan 2.4	67.4°

The vertical height, half the base and the slant height form a right triangle.

Mathematical reasoning. Find the right-angled triangle through the apex and the midpoint of a base edge.

1. The perpendicular from the apex to the midpoint of a base edge, the vertical height and half the base side form a right-angled triangle.
2. The legs are 12 and 5, so the slant height is $\sqrt{(144 + 25)} = 13$.
3. The angle between the face and the base has tangent 12/5.
4. So the angle is $\arctan(12/5) \approx 67.4°$.
5. This is the angle between the planes, measured in the perpendicular cross-section.

Final answer. slant height 13; angle about 67.4°

Interpretation. The angle between two planes is measured in the plane perpendicular to their line of intersection, which is the cross-section used here.

CHAPTER 7 · INVESTIGATION 7.6

Investigation: measuring the unmeasurable

INQUIRY QUESTION

How can a distance be found without measuring it directly?

YOUR TASK

Measure the height of an object in your school grounds without measuring it directly. Choose a baseline, measure two angles of elevation, compute the height, and repeat with a different baseline. Compare the two results and comment on the accuracy of the method.

baseline	angle 1	angle 2	computed height	difference
10 m	31°	25°	8.42 m	—
20 m	31°	25°	8.46 m	0.04 m
10 m	31°	25°	8.42 m	0.00 m
15 m	45°	33°	8.43 m	0.01 m
30 m	45°	33°	8.47 m	0.04 m

CHAPTER 7 · INVESTIGATION 7.6

Investigation: measuring the unmeasurable

What do you notice?

- The two baselines gave heights within 4 cm of each other.
- The longer baseline gave a slightly different answer, because the angles were measured with the same instrument but over a longer distance.
- The repeat measurement was identical, which showed the calculation was being done consistently.
- The method's accuracy was limited by the angle measurement, not by the baseline.

> **Observation.** Two angles of elevation and a baseline determine a height, and the accuracy is limited by the angles.

MATHEMATICAL CONCLUSION

A distance that cannot be measured directly is found by measuring a baseline and two angles and solving the triangle, which is triangulation. The accuracy is limited by the angle measurement and by the conditioning of the triangle, and long baselines need correction for the curvature of the Earth. Here the two baselines gave heights of 8.42 m and 8.46 m.

Your own words

CHAPTER 7 · 7.6

Investigation: measuring the unmeasurable

A distance that cannot be measured directly — across a river, to a mountain top, to a star — is found by measuring a baseline and two angles, and solving the resulting triangle. The method is triangulation, and it is the basis of surveying.

The accuracy of the result depends on the accuracy of the angles and on the shape of the triangle. A long thin triangle amplifies angle errors, so surveyors choose baselines and angles that give a well-conditioned triangle.

The same method with a very long baseline and very small angles measures stellar distances in parsecs, where one parsec is the distance at which one astronomical unit subtends one second of arc. The scale changes; the method does not.

Triangulation is limited by the curvature of the Earth over long baselines and by atmospheric refraction in astronomy. Both are corrections to the ideal triangle, and both must be stated when the method is used.

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In your own words

CHAPTER 7 · 7.6 · WORKED EXAMPLE 1

Investigation: measuring the unmeasurable

WORKED EXAMPLE 7.6.1

Problem. A baseline AB of 200 m is measured on one bank of a river. From A a tree T on the far bank is sighted at 35° from the baseline, and from B at 65°. Find the distance from A to T.

quantity	working	value
angle at A	given	35°
angle at B	given	65°
angle at T	180° − 35° − 65°	80°
AT	200 sin 65° ÷ sin 80°	184.1 m

A baseline and two angles determine the triangle.

Mathematical reasoning. Find the third angle, then apply the sine rule.

1. The angles at A and B are 35° and 65°, so the angle at T is 180 − 35 − 65 = 80°.
2. By the sine rule, AT ÷ sin 65° = AB ÷ sin 80°.
3. So AT = 200 × sin 65° ÷ sin 80° = 200 × 0.9063 ÷ 0.9848.
4. Hence AT ≈ 184.1 m.
5. Check the angle sum: 35 + 65 + 80 = 180°, so the triangle is consistent.

Final answer. about 184 m

Interpretation. The angle sum check confirms the readings are consistent, and it is performed before any length is computed.

CHAPTER 7 · 7.6 · WORKED EXAMPLE 2

Investigation: measuring the unmeasurable

WORKED EXAMPLE 7.6.2

Problem. A mountain is observed from two stations 3 km apart on a level plain. From the nearer station the angle of elevation to the summit is 22° and from the further one 18°. Find the height of the mountain.

step	working	value
let h be the height	—	—
nearer distance	h ÷ tan 22°	2.475h
further distance	h ÷ tan 18°	3.078h
difference	3.078h − 2.475h = 3	h = 4.97

Two angles of elevation give two equations.

Mathematical reasoning. Write the horizontal distance to the summit from each station and equate the difference to the baseline.

1. Let h be the height and d the distance from the nearer station to the point below the summit.
2. From the nearer station, tan 22° = h/d, so d = h ÷ tan 22° ≈ 2.475h.
3. From the further station, tan 18° = h/(d + 3), so d + 3 = h ÷ tan 18° ≈ 3.078h.
4. Subtracting gives 3 ≈ 0.603h, so h ≈ 4.97 km.
5. Check: the nearer distance is about 12.3 km and the further about 15.3 km, differing by 3 km as required.

Final answer. about 4.97 km

Interpretation. Two angles of elevation from a known baseline give the height, and the check is the difference of the two horizontal distances.

CHAPTER 7 · INVESTIGATION 7.7

Mixed practice

INQUIRY QUESTION

Which rule, and how many triangles?

YOUR TASK

Take A = 35°, a = 8 and b = 11. Determine how many triangles exist, find all solutions, and then repeat with a = 7, a = 6.30 and a = 14. Record the case boundaries and explain how you decided each one.

a	b sin A	comparison	triangles	solutions
8	6.31	b sin A < a < b	2	B ≈ 52.3° or 127.7°
7	6.31	b sin A < a < b	2	B ≈ 44.6° or 135.4°
6.30	6.31	a < b sin A	0	none
14	6.31	a > b	1	B ≈ 26.7°

CHAPTER 7 · INVESTIGATION 7.7

Mixed practice

What do you notice?

- The height was about 6.31, and every case was decided by comparing a with it.
- Two values of a gave two triangles each, with supplementary values of B.
- One value of a gave no triangle, because it fell just below the height.
- One value of a gave exactly one triangle, because it exceeded b.

Observation. The comparison of a with b sin A and with b decides the number of triangles before any angle is computed.

MATHEMATICAL CONCLUSION

Two sides and a non-included angle determine a triangle only when the given side is at least the perpendicular height b sin A. The comparison is made first, and only then are angles computed: b sin A = a gives one right triangle, b sin A < a < b gives two, a < b sin A gives none, and a ≥ b gives one. Here the height was 6.31 and the cases were 2, 2, 0 and 1 triangles.

Your own words

CHAPTER 7 · 7.7

Mixed practice

Every non-right-angled triangle question begins by asking what is given: three sides, two sides and the included angle, or two sides and a non-included angle. The answer decides the rule and whether the ambiguous case must be checked.

The ambiguous case is checked by comparing the given side with the perpendicular height. Skipping the check is how two-triangle problems are answered with one answer, and it is the defect this chapter repairs.

Three-dimensional problems are solved by choosing a plane. Naming the plane and the triangle in it is the first line of the solution, and it is what makes the rest of the solution checkable.

Mixed practice is where the rule is chosen rather than given, so the reasoning behind the choice is part of what is being practised.

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In your own words

CHAPTER 7 · 7.7 · WORKED EXAMPLE 1

Mixed practice

WORKED EXAMPLE 7.7.1

Problem. In triangle ABC, A = 40°, a = 7 and b = 12. Determine how many triangles exist and find all solutions.

case	condition	holds?
no triangle	a < b sin A	7 < 7.713, yes
one right triangle	b sin A = a	7.713 = 7, no
two triangles	b sin A < a < b	7.713 < 7, no

The height exceeds the given side.

Mathematical reasoning. Compute b sin A and compare it with a before doing anything else.

1. The perpendicular height is b sin A = 12 × sin 40° ≈ 12 × 0.6428 ≈ 7.713.
2. The given side a = 7 is less than 7.713, so no triangle exists.
3. The case b sin A = a does not hold, since $7.713 \neq 7$.
4. The case b sin A < a < b does not hold either, since 7.713 > 7.
5. Hence there is no triangle with these measurements.

Final answer. no triangle exists

Interpretation. The comparison is the whole solution, and it takes one multiplication.

CHAPTER 7 · 7.7 · WORKED EXAMPLE 2

Mixed practice

WORKED EXAMPLE 7.7.2

Problem. In triangle ABC, A = 40°, a = 9 and b = 12. Find both solutions.

case	condition	holds?
two triangles	b sin A < a < b	7.713 < 9 < 12, yes
first solution	B ≈ 58.9°	C ≈ 81.1°
second solution	B ≈ 121.1°	C ≈ 18.9°

Two values of B are supplementary.

Mathematical reasoning. Confirm the two-triangle case, then find both values of B from the sine rule.

1. The height is b sin A = 12 sin 40° ≈ 7.713, and 7.713 < 9 < 12, so there are two triangles.
2. By the sine rule, sin B = b sin A ÷ a = 7.713 ÷ 9 ≈ 0.857.
3. So B ≈ 58.9° or B ≈ 180° − 58.9° = 121.1°.
4. For the first, C ≈ 180 − 40 − 58.9 = 81.1° and c = 9 sin 81.1° ÷ sin 40° ≈ 13.82.
5. For the second, C ≈ 18.9° and c = 9 sin 18.9° ÷ sin 40° ≈ 4.07.

Final answer. two solutions: B ≈ 58.9°, C ≈ 81.1°, c ≈ 13.82; and B ≈ 121.1°, C ≈ 18.9°, c ≈ 4.07

Interpretation. Both solutions are reported, and the supplementary pair of angles is the reason there are two.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

7.A1 FLUENCY

In triangle ABC, A = 38°, B = 64° and a = 9.4. Find b and c.

7.A2 FLUENCY

In triangle ABC, B = 47°, C = 68° and b = 12. Find a and c.

7.A3 FLUENCY

In triangle ABC, b = 8, c = 11 and A = 55°. Find a.

7.A4 FLUENCY

Find the largest angle of the triangle with sides 7, 9 and 12.

7.A5 FLUENCY

Find the area of the triangle with sides 8 and 10 and included angle 40°.

7.A6 REASONING

In triangle ABC, A = 30°, a = 5 and b = 10. Determine how many triangles exist.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

7.A7 REASONING

In triangle ABC, A = 30°, a = 6 and b = 10. How many triangles exist, and what are the values of B?

7.A8 REASONING

In triangle ABC, A = 45°, a = 4 and b = 6. How many triangles exist?

7.A9 FLUENCY

A ship sails 30 km on a bearing of 040° and then 24 km on a bearing of 130°. Find the distance from the start.

7.A10 FLUENCY

A cuboid has edges 3, 4 and 12. Find the space diagonal.

7.A11 FLUENCY

A cone has radius 6 and vertical height 8. Find the slant height and the angle between the slant height and the base.

7.A12 REASONING

Explain why the ambiguous case arises only when two sides and a non-included angle are given.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

7.B1 CHALLENGE

Investigate the number of triangles determined by $A = 50°$, $b = 10$ and $a = 6$, 7.66, 9, 10 and 13, and justify each case.

7.B2 CHALLENGE

Conjecture a rule for when the sine rule gives a unique answer, and test it on three examples.

7.B3 REASONING

Investigate the relationship between the two solutions of the ambiguous case, and justify it.

7.B4 REASONING

Investigate the area of a triangle with two fixed sides and a varying included angle, and justify when the area is greatest.

7.B5 REASONING

Take a triangle with sides 5 and 8. Investigate the possible values of the third side and justify the bounds.

7.B6 REASONING

Investigate the circumradius of a triangle with sides 6, 8 and 10, and justify the result.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

7.B7 CHALLENGE

Investigate the angle between the space diagonal of a cuboid and a face diagonal as the cuboid becomes a cube, and justify.

7.B8 CHALLENGE

Investigate the number of triangles determined by $A = 25°$, $b = 9$ and $a = 3$, 3.8, 5, 9 and 11, and justify each case.

7.B9 CHALLENGE

Conjecture a rule for when the cosine rule gives a quadratic with two positive roots, and test it.

7.B10 REASONING

Investigate the area of a triangle with two fixed sides as the included angle varies, and justify the maximum.

7.B11 REASONING

Investigate the relationship between the three formulas for the area of a triangle and justify their equivalence.

CRITERION C Communicating

Criterion C practice · Communicating

7.C1 REASONING

Write a complete solution to solving the triangle with a = 9, b = 12 and A = 35°, stating the number of solutions.

7.C2 REASONING

Write a complete solution to finding the area of a triangle with sides 13, 14 and 15.

7.C3 REASONING

Write a complete solution to the navigation problem: a ship sails 25 km on a bearing of 030° then 18 km on a bearing of 120°. Find the distance and bearing from the start.

7.C4 REASONING

Write a complete solution to finding the angle between the space diagonal of a cuboid with edges 2, 3 and 6 and the longest edge.

7.C5 FLUENCY

Write a complete solution to finding the surface area of a cone of radius 5 and slant height 13.

7.C6 FLUENCY

Write a complete solution to finding the height of a building observed from 40 m away at an angle of elevation of 27°.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

7.C7 CHALLENGE

Write a complete solution to proving the cosine rule.

7.C8 REASONING

Write a complete solution to solving the triangle with a = 11, b = 14 and A = 42°, stating the number of solutions.

7.C9 REASONING

Write a complete solution to finding the area of the triangle with sides 13, 14 and 15.

7.C10 REASONING

Write a complete solution to the navigation problem: a ship sails 35 km on a bearing of 050° then 28 km on a bearing of 140°. Find the distance and bearing from the start.

7.C11 REASONING

Write a complete solution to finding the angle between the space diagonal of a cuboid with edges 4, 4 and 6 and the longest edge.

7.C12 CHALLENGE

Write a complete solution to proving the cosine rule.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

7.D1 APPLICATION

A rescue team is at A and a casualty is at C. A marker at B is 400 m from A on a bearing of 070°, and the bearing of C from A is 110° while the bearing of C from B is 160°. Find the distance AC.

7.D2 APPLICATION

Two cars leave a junction at the same time, one at 60 km/h on a bearing of 020° and one at 80 km/h on a bearing of 140°. Find the distance between them after 30 minutes.

7.D3 APPLICATION

A radio mast is 40 m tall. From a point on level ground the angle of elevation to the top is 28°. Find the distance from the point to the base, to the nearest metre.

7.D4 APPLICATION

A triangular field has sides 120 m and 150 m with an included angle of 72°. Find the area and the length of the third side.

7.D5 APPLICATION

A pilot flies 200 km on a bearing of 315° and then 150 km on a bearing of 045°. Find the distance and bearing from the start, and comment on the shape of the route.

7.D6 APPLICATION

A pyramid has a square base of side 8 and a vertical height of $3\sqrt{7}$. Find the slant height, the angle between a face and the base, and the total surface area.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

7.D7 REASONING

A student solves an SSA problem and reports only one triangle. Explain what must be checked and what the correct procedure is.

7.D8 APPLICATION

A lighthouse is visible from a ship when the ship is within 25 km. The ship's bearings from the lighthouse are recorded at two times 40 minutes apart, at 030° and 075°. If the ship sails at 20 km/h, find its closest approach to the lighthouse.

7.D9 APPLICATION

A roof is built as two rectangular planes meeting at a ridge. Each plane is 6 m wide and the horizontal span is 10 m. Find the angle between the two planes and the height of the ridge.

7.D10 REASONING

A student measures $a = 5$, $b = 8$ and $A = 30°$ and concludes no triangle exists. Investigate and justify.

7.D11 APPLICATION

A communications mast is supported by three cables attached at the top and anchored 12 m from the base, equally spaced. The mast is 20 m tall. Find the length of each cable and the angle each makes with the ground.

7.D12 APPLICATION

A map is drawn to a scale of 1 : 25 000. Two landmarks are 8.4 cm apart on the map and the bearing between them is 132°. Find the real distance and describe how the bearing is used.

CHAPTER 7 · APPLICATION AND ASSESSMENT

Applying chapter 7

These tasks use the sine and cosine rules. State which rule you use and why, check the ambiguous case explicitly, and give every answer to three significant figures.

A1 · The survey

A surveyor measures a baseline AB of 150 m. From A a point C has a bearing of 055° and from B a bearing of 105°. Find the distance AC and BC, and the area of the triangle ABC. State any assumption the surveyor has made.

Criteria C and D · extract the interior angles, apply the sine rule, and justify the assumption.

A2 · The ambiguous case

A triangle has A = 42°, a = 9 and b = 12. Determine how many triangles exist, find all of them, and explain in general when two sides and a non-included angle determine two triangles. Then find the area of each triangle you have found.

Criteria C and D · check the ambiguous case explicitly, solve both triangles, and generalise the condition.

CHAPTER 7 · REVIEW

What you should now be able to do

Key ideas

- The sine rule is $a/\sin A = b/\sin B = c/\sin C$.
- The cosine rule is $a^2 = b^2 + c^2 - 2bc\cos A$.
- The area of a triangle is $\frac{1}{2}ab\sin C$, where C is the included angle.
- Two sides and a non-included angle give zero, one or two triangles.
- Two triangles exist when $b\sin A < a < b$.
- $b\sin A = a$ gives exactly one right triangle.
- $a < b\sin A$ gives no triangle.
- A bearing is measured clockwise from north as a three-figure angle.
- A three-dimensional problem is solved by choosing a plane.
- The angle between a line and a plane uses the line's projection onto the plane.

Vocabulary

sine rule · cosine rule · included angle · non-included angle · ambiguous case · perpendicular height · bearing · angle of elevation · space diagonal · slant height · triangulation · baseline · projection · circumradius

CHAPTER 7 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

7.R1 In triangle ABC, A = 50°, B = 60° and a = 8. Find b and c.

7.R2 Find the area of the triangle with sides 9, 10 and included angle 65°.

7.R3 In triangle ABC, A = 35°, a = 6 and b = 10. How many triangles exist?

7.R4 A cuboid has edges 5, 12 and 13. Find the space diagonal.

7.R5 A ship sails 40 km on a bearing of 080° then 30 km on a bearing of 170°. Find the distance from the start.

7.R6 Explain when the cosine rule is preferred to the sine rule.

CHAPTER 7 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> Triangles that are not right-angled can still be solved: the sine and cosine rules turn measurements of angle and distance into positions, areas and three-dimensional structure.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of space, change and measurement appear in your work?
- How is the global context of orientation in space and time, and its exploration of navigation, surveying and the geometry of place, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of triangulation as a measurement method, and what would you still like to ask?

CHAPTER 7 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

A position is a relationship between measurements. Name the rule, check the ambiguity, and say what would resolve it.

Service as Action

COMMUNITY NEED	Distances that matter locally cannot be measured directly, and no one has measured them.
MATHEMATICAL CONNECTION	The sine and cosine rules and triangulation.
PROPOSED ACTION	Measure the height of a local landmark or the width of a local space by triangulation, and publish the method and the result with its uncertainty.
EXPECTED IMPACT	A figure that was guessed becomes a figure that is justified.
STUDENT REFLECTION	Record the baseline, the angles, the result and the check.

Community Project

PROJECT IDEA	A survey of an unusable community space.
NEED	The space is discussed without any measurement of it.
RESEARCH QUESTION	What are the dimensions of the space, measured without entering it?
MATHEMATICS	Trigonometry in three dimensions and triangulation.
ACTION OR OUTCOME	A measured plan with the method explained.
BENEFICIARIES	Everyone who plans to use the space.
EVIDENCE	The plan and the measurements behind it.

Personal Project connection · MYP 5

PERSONAL INTEREST	Navigation, sport, astronomy or drones.
LEARNING GOAL	To find a distance you cannot measure directly and state its uncertainty.
PRODUCT OR OUTCOME	A measurement with a method and a check.
MATHEMATICS	The sine and cosine rules and triangulation.
PROCESS	Choose a baseline, measure two angles, solve the triangle, and check the angle sum.
EVIDENCE	The measurement and the triangle you solved.

CHAPTER 8 · PART III · SPATIAL REASONING

08

Vectors and Matrices

A vector is a quantity with magnitude and direction, and a matrix is an array that transforms one. Together they describe displacements, forces and the transformations that move a plane onto itself.

ENGAGE

A computer graphics program rotates a shape, then stretches it, then rotates it back. The programmer needs one matrix that does all three. How are the three matrices combined, and how can the program tell whether the result can still be undone?

Framing the inquiry

Key concept: Form
Related concepts: Space · Representation · Systems
Global context: Scientific and technical innovation
Global context exploration: computer graphics, navigation and engineering

Statement of inquiry: Vectors encode displacement by magnitude and direction, while matrices encode transformations of the plane as arrays of numbers; the determinant measures how a transformation stretches or collapses area, and decides whether it can be undone.

Questions to guide us

- **Factual:** What do the magnitude of a vector, the scalar product and the determinant of a matrix measure?
- **Conceptual:** Why does the determinant of a product equal the product of the determinants, and what does a zero determinant mean?
- **Debatable:** If a transformation collapses a shape to a line, has it destroyed information or merely hidden it?

ATL · CRITICAL THINKING. In order for a student to decide whether a transformation can be reversed, students must compute the determinant, connect its value to the area scale factor, and justify when an inverse exists.

CHAPTER 8 · INVESTIGATION 8.1

Vectors: notation and operations

INQUIRY QUESTION

What does a vector record that a number cannot?

YOUR TASK

Take the vectors $\mathbf{a} = (2, 1)$, $\mathbf{b} = (-3, 4)$, $\mathbf{c} = (1, -2)$ and $\mathbf{d} = (0, 5)$. Compute $\mathbf{a} + \mathbf{b}$, $\mathbf{a} - \mathbf{c}$, $2\mathbf{b} + \mathbf{d}$, $|\mathbf{a}|$, $|\mathbf{b}|$, the unit vector in the direction of $\mathbf{a}$, and the scalar products $\mathbf{a} \cdot \mathbf{b}$, $\mathbf{a} \cdot \mathbf{c}$ and $\mathbf{b} \cdot \mathbf{d}$. Record which pairs are perpendicular.

quantity	working	result
$\mathbf{a} + \mathbf{b}$	$(2 - 3, 1 + 4)$	$(-1, 5)$
$\mathbf{a} - \mathbf{c}$	$(2 - 1, 1 - (-2))$	$(1, 3)$
$2\mathbf{b} + \mathbf{d}$	$(-6 + 0, 8 + 5)$	$(-6, 13)$
$\lvert\mathbf{a}\rvert$	$\sqrt{(4 + 1)}$	$\sqrt{5} \approx 2.24$
$\lvert\mathbf{b}\rvert$	$\sqrt{(9 + 16)}$	5
unit vector along $\mathbf{a}$	$(2, 1) \div \sqrt{5}$	$(0.894, 0.447)$
$\mathbf{a} \cdot \mathbf{b}$	$-6 + 4$	-2
$\mathbf{a} \cdot \mathbf{c}$	$2 - 2$	0, so perpendicular
$\mathbf{b} \cdot \mathbf{d}$	$0 + 20$	20

CHAPTER 8 · INVESTIGATION 8.1

Vectors: notation and operations

What do you notice?

- The scalar product a · c was zero, so those two vectors were perpendicular.
- The scalar product a · b was negative, so the angle between them was obtuse.
- The unit vector had magnitude 1 and the same direction as a.
- Vector addition and subtraction were component-wise throughout.

> **Observation.** Vectors combine component-wise, and the scalar product measures the alignment of a pair.

MATHEMATICAL CONCLUSION

A vector records magnitude and direction, and its arithmetic is component-wise: addition and subtraction combine the components, and scalar multiplication scales them. The magnitude is recovered by Pythagoras and the scalar product measures alignment, being zero exactly for perpendicular vectors. Here a · c was 0, so a and c were perpendicular.

Your own words

CHAPTER 8 · 8.1

Vectors: notation and operations

A vector is a quantity with magnitude and direction, written as a column vector or in bold, and it describes a displacement in space. Two vectors are equal when they have the same magnitude and direction, wherever they are drawn.

Vectors add by the triangle rule: the sum of a and b is the displacement from the start of a to the end of b when they are placed head to tail. Scalar multiplication scales the magnitude and leaves the direction unchanged, or reverses it when the scalar is negative.

The magnitude of a vector $\mathbf{a} = (x, y)$ is $|\mathbf{a}| = \sqrt{(x^2 + y^2)}$, which is its length. A unit vector has magnitude 1, and dividing a vector by its magnitude gives the unit vector in the same direction.

The scalar product $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$ is a number, not a vector. It is zero exactly when the vectors are perpendicular, which makes it the standard test for right angles.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 8 · 8.1 · WORKED EXAMPLE 1

Vectors: notation and operations

WORKED EXAMPLE 8.1.1

Problem. Given $\mathbf{a} = (3, -2)$ and $\mathbf{b} = (-1, 5)$, find $3\mathbf{a} - 2\mathbf{b}$ and its magnitude.

step	working	result
$3\mathbf{a}$	$3 \times (3, -2)$	$(9, -6)$
$2\mathbf{b}$	$2 \times (-1, 5)$	$(-2, 10)$
$3\mathbf{a} - 2\mathbf{b}$	$(9 - (-2), -6 - 10)$	$(11, -16)$
magnitude	$\sqrt{(11^2 + 16^2)}$	$\sqrt{377} \approx 19.42$

Scalar multiplication acts on each component.

Mathematical reasoning. Scale each vector, then combine component by component.

1. $3\mathbf{a} = 3 \times (3, -2) = (9, -6)$.
2. $2\mathbf{b} = 2 \times (-1, 5) = (-2, 10)$.
3. So $3\mathbf{a} - 2\mathbf{b} = (9, -6) - (-2, 10) = (11, -16)$.
4. The magnitude is $\sqrt{(11^2 + 16^2)} = \sqrt{(121 + 256)} = \sqrt{377} \approx 19.42$.
5. The direction is given by the angle $\arctan(-16/11)$, which is below the positive x-axis.

Final answer. $3\mathbf{a} - 2\mathbf{b} = (11, -16)$, magnitude about 19.42

Interpretation. Vector arithmetic is component-wise, and the magnitude is recovered from the components by Pythagoras.

CHAPTER 8 · 8.1 · WORKED EXAMPLE 2

Vectors: notation and operations

WORKED EXAMPLE 8.1.2

Problem. Show that the vectors (4, 3) and (−6, 8) are perpendicular.

vector	components	magnitude
u	(4, 3)	5
v	(−6, 8)	10
u · v	4(−6) + 3(8)	0
conclusion	perpendicular	—

A zero scalar product is the test for perpendicularity.

Mathematical reasoning. Compute the scalar product and check whether it is zero.

1. The scalar product is $u \cdot v = 4(-6) + 3(8) = -24 + 24 = 0$.
2. Since the scalar product is zero, the vectors are perpendicular.
3. Check with the formula $u \cdot v = |u||v| \cos\theta$: $0 = 5 \times 10 \times \cos\theta$ gives $\cos\theta = 0$.
4. So $\theta = 90°$, confirming the result.
5. The magnitudes were 5 and 10, but only the product mattered.

Final answer. they are perpendicular

Interpretation. The scalar product test is faster than finding the angle, and it is exact.

CHAPTER 8 · INVESTIGATION 8.2

Vectors and geometry

INQUIRY QUESTION

How does vector notation prove a geometric claim?

YOUR TASK

Take the points A(0, 0), B(6, 2), C(9, 7) and D(3, 5). Determine whether ABCD is a parallelogram, find the midpoints of the diagonals, and check whether the diagonals are perpendicular. Then find the area of the quadrilateral.

quantity	working	result
AB	(6, 2)	—
DC	(9 − 3, 7 − 5)	(6, 2), so AB parallel and equal to DC
midpoint of AC	(4.5, 3.5)	—
midpoint of BD	(4.5, 3.5)	the same, so a parallelogram
AC · BD	(9, 7) · (−3, 3)	−27 + 21 = −6, not perpendicular
area	half the magnitude of the cross term	26

CHAPTER 8 · INVESTIGATION 8.2

Vectors and geometry

What do you notice?

- The diagonals shared a midpoint, which proved the parallelogram.
- The scalar product of the diagonals was −6, not zero, so they were not perpendicular.
- The area came from the determinant of the two side vectors.
- Both conditions — parallel opposite sides and shared midpoint — held.

Observation. A parallelogram is detected by equal opposite sides or by a shared diagonal midpoint, and perpendicularity is a scalar product test.

MATHEMATICAL CONCLUSION

Vector notation turns a geometric claim into an algebraic one: collinearity is a scalar multiple, a parallelogram is equal opposite sides or a shared diagonal midpoint, and perpendicularity is a zero scalar product. The area of a parallelogram is the magnitude of the determinant of its side vectors. Here the area was 26 and the diagonals were not perpendicular.

Your own words

CHAPTER 8 · 8.2

Vectors and geometry

A position vector gives the position of a point relative to an origin, and the vector between two points is the difference of their position vectors. Writing geometry in vectors is a representation that turns a diagram into an algebra.

Three points are collinear exactly when the vector between two of them is a scalar multiple of the vector between another pair. This is a single equation in two components, and it is the standard collinearity test.

The midpoint of AB has position vector (a + b)/2, and the point dividing AB in the ratio m : n has position vector (n a + m b)/(m + n). Both formulas come from a single linear combination.

The scalar product proves perpendicularity and computes angles, and the vector product in three dimensions gives a vector perpendicular to two others. These tools make vector proofs of geometric theorems short and checkable.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 8 · 8.2 · WORKED EXAMPLE 1

Vectors and geometry

WORKED EXAMPLE 8.2.1

Problem. Show that the points A(1, 2), B(4, 6) and C(7, 10) are collinear.

vector	components	ratio of components
AB	(3, 4)	—
BC	(3, 4)	1 : 1
AC	(6, 8)	2 : 1
conclusion	AB = BC, so collinear	—

Equal vectors between consecutive points prove collinearity.

Mathematical reasoning. Compute the vectors between consecutive points and check whether one is a multiple of the other.

1. The position vectors are $\mathbf{a} = (1, 2)$, $\mathbf{b} = (4, 6)$ and $\mathbf{c} = (7, 10)$.
2. Then $\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = (3, 4)$ and $\overrightarrow{BC} = \mathbf{c} - \mathbf{b} = (3, 4)$.
3. Since $\overrightarrow{AB} = \overrightarrow{BC}$ exactly, the two vectors are equal and hence parallel.
4. They also share the point B, so A, B and C lie on one straight line.
5. Moreover B is the midpoint of AC, because $\overrightarrow{AB} = \overrightarrow{BC}$.

Final answer. the three points are collinear, and B is the midpoint of AC

Interpretation. One computation settles both collinearity and the midpoint claim.

CHAPTER 8 · 8.2 · WORKED EXAMPLE 2

Vectors and geometry

WORKED EXAMPLE 8.2.2

Problem. Find the point that divides the line from P(2, −1) to Q(8, 5) in the ratio 2 : 3.

step	working	result
formula	(3p + 2q)/5	—
x-coordinate	(3 × 2 + 2 × 8)/5	22/5 = 4.4
y-coordinate	(3 × (−1) + 2 × 5)/5	7/5 = 1.4
check	the point lies between P and Q	yes

A section formula is a weighted average.

Mathematical reasoning. Apply the section formula with the ratio 2 : 3.

1. The point dividing PQ in the ratio 2 : 3 has position vector (3p + 2q)/(2 + 3).
2. The x-coordinate is (3 × 2 + 2 × 8)/5 = 22/5 = 4.4.
3. The y-coordinate is (3 × (−1) + 2 × 5)/5 = 7/5 = 1.4.
4. So the point is (4.4, 1.4).
5. Check: it lies between P(2, −1) and Q(8, 5), as a section point must.

Final answer. (4.4, 1.4)

Interpretation. The weights are the opposite parts of the ratio, which is why the formula has them crossed.

CHAPTER 8 · INVESTIGATION 8.3

Vectors in three dimensions

INQUIRY QUESTION

What changes when a third dimension is added?

YOUR TASK

Take the vectors $\mathbf{a} = (1, 1, 0)$, $\mathbf{b} = (0, 1, 1)$ and $\mathbf{c} = (1, 0, 1)$. Compute $|\mathbf{a}|$, $|\mathbf{b}|$, $|\mathbf{c}|$, $\mathbf{a} \cdot \mathbf{b}$, $\mathbf{b} \cdot \mathbf{c}$, $\mathbf{c} \cdot \mathbf{a}$, $\mathbf{a} \times \mathbf{b}$ and the scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$. Then decide whether the three vectors are coplanar and explain how you know.

quantity	working	result
$\mathbf{a}$	$\sqrt{(1 + 1 + 0)}$	$\sqrt{2} \approx 1.414$
$\mathbf{b}$	$\sqrt{(0 + 1 + 1)}$	$\sqrt{2} \approx 1.414$
$\mathbf{c}$	$\sqrt{(1 + 0 + 1)}$	$\sqrt{2} \approx 1.414$
$\mathbf{a} \cdot \mathbf{b}$	$0 + 1 + 0$	1
$\mathbf{b} \cdot \mathbf{c}$	$0 + 0 + 1$	1
$\mathbf{c} \cdot \mathbf{a}$	$1 + 0 + 0$	1
$\mathbf{a} \times \mathbf{b}$	$(1(1) - 0(1), 0(0) - 1(1), 1(1) - 1(0))$	$(1, -1, 1)$
$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$	$1(1) + 1(-1) + 0(1)$	0, so coplanar

CHAPTER 8 · INVESTIGATION 8.3

Vectors in three dimensions

What do you notice?

- All three vectors had the same magnitude, $\sqrt{2}$.
- All three scalar products were 1, so the angles were all equal.
- The scalar triple product was zero, so the three vectors were coplanar.
- The vectors formed a regular pattern in the plane $x + y + z = 1$.

Observation. A zero scalar triple product is the test for coplanarity.

MATHEMATICAL CONCLUSION

Adding a third dimension extends every operation component-wise and adds the vector product, whose magnitude is an area and whose direction is perpendicular to both factors. The scalar triple product is a volume and vanishes exactly for coplanar vectors. Here all three vectors had magnitude $\sqrt{2}$, all pairwise angles were equal, and the triple product was 0.

Your own words

CHAPTER 8 · 8.3

Vectors in three dimensions

A three-dimensional vector has three components, and all the two-dimensional operations extend component-wise. The magnitude becomes $\sqrt{(x^2 + y^2 + z^2)}$ and the scalar product gains a third term.

The vector product $a \times b$ is defined only in three dimensions, and it gives a vector perpendicular to both a and b. Its magnitude is $|a||b| \sin \theta$, which is the area of the parallelogram spanned by a and b.

The vector product is anti-commutative: $a \times b = -(b \times a)$. The order matters, and the right-hand rule fixes which of the two perpendicular directions is meant.

The scalar triple product $a \cdot (b \times c)$ gives the volume of the parallelepiped spanned by the three vectors, and it is zero exactly when the three vectors are coplanar. It is the three-dimensional analogue of the determinant test.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 8 · 8.3 · WORKED EXAMPLE 1

Vectors in three dimensions

WORKED EXAMPLE 8.3.1

Problem. Find the vector product of a = (1, 2, 3) and b = (4, 5, 6), and verify it is perpendicular to both.

component	working	value
x	2(6) − 3(5)	−3
y	3(4) − 1(6)	6
z	1(5) − 2(4)	−3
a × b	—	(−3, 6, −3)

The cross product is computed from a 3 by 3 pattern.

Mathematical reasoning. Apply the component formula and check with scalar products.

1. The vector product $\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1)$.
2. So $\mathbf{a} \times \mathbf{b} = (2 \times 6 - 3 \times 5, 3 \times 4 - 1 \times 6, 1 \times 5 - 2 \times 4)$.
3. That is $(12 - 15, 12 - 6, 5 - 8) = (-3, 6, -3)$.
4. Check with a: $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = 1(-3) + 2(6) + 3(-3) = -3 + 12 - 9 = 0$.
5. Check with b: $\mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) = 4(-3) + 5(6) + 6(-3) = -12 + 30 - 18 = 0$, so it is perpendicular to both.

Final answer. a × b = (−3, 6, −3), perpendicular to both a and b

Interpretation. The two scalar-product checks are quick and confirm the perpendicularity the definition promises.

CHAPTER 8 · 8.3 · WORKED EXAMPLE 2

Vectors in three dimensions

WORKED EXAMPLE 8.3.2

Problem. Find the volume of the parallelepiped spanned by a = (2, 0, 1), b = (1, 3, 0) and c = (0, 1, 4).

step	working	result
b × c	(3(4) − 0(1), 0(0) − 1(4), 1(1) − 3(0))	(12, −4, 1)
a · (b × c)	2(12) + 0(−4) + 1(1)	25
volume	the absolute value	25
coplanar?	the triple product is not zero	no

The scalar triple product is a volume.

Mathematical reasoning. Compute the vector product b × c, then its scalar product with a.

1. First b × c = (3 × 4 − 0 × 1, 0 × 0 − 1 × 4, 1 × 1 − 3 × 0) = (12, −4, 1).
2. Then the scalar triple product is a · (b × c) = 2 × 12 + 0 × (−4) + 1 × 1 = 25.
3. The volume is the absolute value, 25 cubic units.
4. Since the triple product is not zero, the three vectors are not coplanar.
5. The sign would be negative if the order were reversed, which is why the absolute value is taken.

Final answer. volume 25 cubic units

Interpretation. The scalar triple product is the determinant of the 3 by 3 matrix with the three vectors as rows, which is why it also tests coplanarity.

CHAPTER 8 · INVESTIGATION 8.4

Matrices and transformations

INQUIRY QUESTION

How does a matrix encode a transformation?

YOUR TASK

Write the matrices for a rotation through 90°, a reflection in the line y = x, and an enlargement of factor 3. Apply each to the points (1, 0), (0, 1) and (2, 1), and record the images. Then compute the product of the rotation and the reflection matrices in both orders and comment on the difference.

transformation	matrix	image of (1, 0)	image of (0, 1)	determinant
rotation 90°	[[0, −1], [1, 0]]	(0, 1)	(−1, 0)	1
reflection in y = x	[[0, 1], [1, 0]]	(0, 1)	(1, 0)	−1
enlargement 3	[[3, 0], [0, 3]]	(3, 0)	(0, 3)	9
rotation then reflection	[[1, 0], [0, −1]]	(1, 0)	(0, −1)	−1
reflection then rotation	[[−1, 0], [0, 1]]	(−1, 0)	(0, 1)	−1

CHAPTER 8 · INVESTIGATION 8.4

Matrices and transformations

What do you notice?

- The rotation and the reflection both preserved lengths, but their determinants differed in sign.
- The enlargement multiplied every length by 3 and areas by 9.
- The two products were different matrices, confirming that order matters.
- Both products were reflections, one in the x-axis and one in the y-axis.

> **Observation.** The columns of a matrix are the images of the basis vectors, and the determinant is the area scale factor.

MATHEMATICAL CONCLUSION

A matrix encodes a transformation by recording the images of the basis vectors in its columns, and composition is matrix multiplication in the order of application. The determinant is the area scale factor, positive for orientation-preserving transformations and negative for orientation-reversing ones. Here the rotation then reflection gave [[1, 0], [0, −1]] while the reverse order gave [[−1, 0], [0, 1]].

Your own words

CHAPTER 8 · 8.4

Matrices and transformations

A 2 by 2 matrix acts on a column vector by multiplication, and the two columns of the matrix are the images of the basis vectors (1, 0) and (0, 1). This representation determines the whole transformation by where it sends the basis.

The matrix for a rotation through θ is [[cos θ, −sin θ], [sin θ, cos θ]], for a reflection in a line through the origin at angle θ it is [[cos 2θ, sin 2θ], [sin 2θ, −cos 2θ]], and for an enlargement of factor k it is [[k, 0], [0, k]].

Composition of transformations is matrix multiplication in the same order: applying M then N is the matrix NM, so systems of transformations compose by multiplying their matrices. The order matters, which is why matrix multiplication is not commutative.

A transformation with matrix M is linear exactly when M(au + bv) = aM(u) + bM(v), which every matrix multiplication satisfies. Linearity is what makes matrices the natural language for transformations.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 8 · 8.4 · WORKED EXAMPLE 1

Matrices and transformations

WORKED EXAMPLE 8.4.1

Problem. Find the image of the point (3, 1) under the matrix [[2, 1], [−1, 3]], and describe the transformation.

step	working	result
image of (1, 0)	the first column	(2, −1)
image of (0, 1)	the second column	(1, 3)
image of (3, 1)	3(2, −1) + (1, 3)	(7, 0)
determinant	2(3) − 1(−1)	7

The columns are the images of the basis vectors.

Mathematical reasoning. Read the images of the basis vectors from the columns, then combine them for the given point.

1. The first column (2, −1) is the image of (1, 0).
2. The second column (1, 3) is the image of (0, 1).
3. The image of (3, 1) is 3 × (2, −1) + 1 × (1, 3) = (6, −3) + (1, 3) = (7, 0).
4. Check by direct multiplication: 2(3) + 1(1) = 7 and −1(3) + 3(1) = 0.
5. The determinant is 2 × 3 − 1 × (−1) = 7, so areas are multiplied by 7.

Final answer. (7, 0); the determinant is 7

Interpretation. Reading the columns as the images of the basis vectors is the quickest way to understand a matrix.

CHAPTER 8 · 8.4 · WORKED EXAMPLE 2

Matrices and transformations

WORKED EXAMPLE 8.4.2

Problem. Write the matrix for a rotation through 60° and find the image of the point (1, 0).

entry	value	position in the matrix
cos 60°	0.5	top left
−sin 60°	−0.8660	top right
sin 60°	0.8660	bottom left
cos 60°	0.5	bottom right

The rotation matrix has a fixed pattern.

Mathematical reasoning. Substitute the angle into the rotation matrix and multiply.

1. The rotation matrix for angle θ is [[cos θ, −sin θ], [sin θ, cos θ]].
2. With θ = 60°, this is [[0.5, −0.8660], [0.8660, 0.5]].
3. The image of (1, 0) is the first column, (0.5, 0.8660).
4. Its magnitude is √(0.25 + 0.75) = 1, as a rotation preserves length.
5. The image of (0, 1) is the second column, (−0.8660, 0.5), also of magnitude 1.

Final answer. [[0.5, −0.8660], [0.8660, 0.5]]; the image of (1, 0) is (0.5, 0.8660)

Interpretation. The determinant of a rotation matrix is 1, which is why rotations preserve area.

CHAPTER 8 · INVESTIGATION 8.5

Determinants and inverse matrices

INQUIRY QUESTION

What does the determinant decide?

YOUR TASK

Take the matrices M = [[1, 2], [3, 4]], N = [[0, 1], [−1, 0]] and P = [[2, 0], [0, 3]]. Compute det M, det N, det P, det(MN), det(MP) and det(MNP). Find M^{-1} and check it by multiplication. Then find a matrix Q with det Q = 0 and describe what its transformation does.

quantity	working	result
det M	1(4) − 2(3)	−2
det N	0(0) − 1(−1)	1
det P	2(3) − 0(0)	6
det(MN)	−2 × 1	−2
det(MP)	−2 × 6	−12
det(MNP)	−2 × 1 × 6	−12
M^{-1}	(1/(−2))[[4, −2], [−3, 1]]	[[−2, 1], [1.5, −0.5]]
Q = [[1, 2], [2, 4]]	det 0	collapses onto a line

CHAPTER 8 · INVESTIGATION 8.5

Determinants and inverse matrices

What do you notice?

- Every product determinant was the product of the individual determinants.
- The determinant of N was 1, so that rotation preserved area.
- The determinant of P was 6, so that enlargement multiplied areas by 6.
- The matrix Q had determinant 0 and collapsed the plane onto a line.

Observation. The determinant multiplies over products, and a zero determinant means no inverse exists.

MATHEMATICAL CONCLUSION

The determinant is the signed area scale factor of a transformation: positive for orientation-preserving, negative for orientation-reversing, and zero for a collapse. The inverse exists exactly when the determinant is non-zero, and $\det(AB) = \det(A)\det(B)$. Here $\det M = -2$, $\det N = 1$, $\det P = 6$, and the product determinants were -2, -12 and -12.

Your own words

CHAPTER 8 · 8.5

Determinants and inverse matrices

The determinant of [[a, b], [c, d]] is ad − bc. It is the signed area scale factor of the transformation: the area of an image is |det| times the area of the original.

A negative determinant means the transformation reverses orientation, as a reflection does. A determinant of zero means the transformation collapses the plane onto a line or a point, and the transformation cannot be undone.

The inverse matrix exists exactly when the determinant is not zero, and it is given by (1/det)[[d, −b], [−c, a]]. The check is that the product of the matrix and its inverse is the identity.

The determinant of a product is the product of the determinants, so det(AB) = det(A) det(B). This is why a single non-invertible matrix in a product makes the whole product non-invertible.

> FORMAL DEVELOPMENT
>
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In your own words

CHAPTER 8 · 8.5 · WORKED EXAMPLE 1

Determinants and inverse matrices

WORKED EXAMPLE 8.5.1

Problem. Find the determinant and, if it exists, the inverse of [[4, 7], [2, 3]].

step	working	result
determinant	4(3) − 7(2)	−2
adjugate	[[3, −7], [−2, 4]]	—
inverse	(1/(−2)) × adjugate	[[−1.5, 3.5], [1, −2]]
check	$M \times M^{-1}$	the identity

A negative determinant reverses orientation.

Mathematical reasoning. Compute the determinant, form the adjugate, and verify by multiplication.

1. The determinant is $4 \times 3 - 7 \times 2 = 12 - 14 = -2$.
2. Since it is not zero, the inverse exists.
3. The adjugate is [[3, −7], [−2, 4]].
4. So the inverse is (−1/2)[[3, −7], [−2, 4]] = [[−1.5, 3.5], [1, −2]].
5. Check: multiplying the matrix by this gives the identity matrix.

Final answer. det = −2; inverse [[−1.5, 3.5], [1, −2]]

Interpretation. The negative determinant means the transformation reverses orientation, and areas are halved.

CHAPTER 8 · 8.5 · WORKED EXAMPLE 2

Determinants and inverse matrices

WORKED EXAMPLE 8.5.2

Problem. Show that [[2, 4], [3, 6]] has no inverse and describe what the transformation does.

step	working	result
determinant	2(6) − 4(3)	0
columns	(2, 3) and (4, 6)	one is double the other
image of the plane	the line through (2, 3)	a line
inverse	does not exist	—

A zero determinant collapses the plane onto a line.

Mathematical reasoning. Compute the determinant and interpret a zero value.

1. The determinant is $2 \times 6 - 4 \times 3 = 12 - 12 = 0$.
2. So the inverse does not exist.
3. The second column (4, 6) is twice the first (2, 3), so the two basis vectors map to parallel vectors.
4. Every image is a multiple of (2, 3), so the whole plane is collapsed onto that line.
5. The collapse cannot be undone, because many points share one image.

Final answer. no inverse; the transformation collapses the plane onto the line through (2, 3)

Interpretation. A zero determinant is the algebraic signature of a collapse, and it is the reason the inverse fails to exist.

CHAPTER 8 · INVESTIGATION 8.6

Investigation: area scale factors

INQUIRY QUESTION

How does the determinant show up in an area?

YOUR TASK

Take the unit square with vertices (0, 0), (1, 0), (1, 1) and (0, 1). Apply the matrices [[2, 1], [0, 3]] and [[1, 2], [3, 6]] to its vertices, and find the area of each image. Compare the areas with the determinants and explain what happens in the second case.

matrix	determinant	image of the unit square	area of image	orientation
[[2, 1], [0, 3]]	6	a parallelogram	6	preserved
[[1, 2], [3, 6]]	0	a line segment	0	destroyed
[[2, 1], [0, 3]] applied twice	36	a larger parallelogram	36	preserved
[[0, 1], [1, 0]]	−1	the square, reflected	1	reversed
identity	1	the square itself	1	preserved

CHAPTER 8 · INVESTIGATION 8.6

Investigation: area scale factors

What do you notice?

- The first image was a parallelogram of area 6, matching the determinant.
- The second image collapsed to a line segment of area 0, matching the zero determinant.
- The unit square's original area was 1, so the scale factors were 6 and 0.
- The collapse destroyed the area completely, which is why no inverse exists.

Observation. The determinant is exactly the factor by which areas are multiplied.

MATHEMATICAL CONCLUSION

The determinant of the matrix of two spanning vectors is the signed area of the parallelogram they span, and the determinant of a transformation is the factor by which every area is multiplied. A zero determinant collapses the image onto a line and destroys the area. Here the scale factors were 6 and 0.

Your own words

CHAPTER 8 · 8.6

Investigation: area scale factors

The area of the parallelogram spanned by two vectors (a, b) and (c, d) is |ad − bc|, which is exactly the absolute value of the determinant. This is the geometric meaning of the determinant.

The same determinant is the scalar triple product in three dimensions, where it gives the volume of a parallelepiped. The pattern is the same in every dimension: the determinant measures the size of the image of a unit box.

When a transformation is applied to a shape, every area is multiplied by |det|. This is why the determinant is checked before computing an area after a transformation, and why a zero determinant means the area is destroyed.

The determinant also appears in the change of variables formula for integrals, where a change of coordinates multiplies areas by the Jacobian determinant. The same number measures the local stretching of a map.

> FORMAL DEVELOPMENT
>
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In your own words

CHAPTER 8 · 8.6 · WORKED EXAMPLE 1

Investigation: area scale factors

WORKED EXAMPLE 8.6.1

Problem. Find the area of the parallelogram spanned by (3, 1) and (−2, 5).

step	working	result
determinant	3(5) − 1(−2)	17
area	the absolute value	17
base	(3, 1)	√10 ≈ 3.16
height	17 ÷ √10	5.38

The determinant is the signed area.

Mathematical reasoning. Form the determinant of the two vectors and take its absolute value.

1. The determinant of the two vectors is $3 \times 5 - 1 \times (-2) = 15 + 2 = 17$.
2. The area is the absolute value, 17 square units.
3. Check with base times height: the base is $|(3, 1)| = \sqrt{10}$.
4. The height is the component of (−2, 5) perpendicular to (3, 1), which is $17 \div \sqrt{10} \approx 5.38$.
5. So the area is $\sqrt{10} \times 5.38 \approx 17$, as required.

Final answer. 17 square units

Interpretation. The determinant gives the signed area, and the sign only records the orientation of the pair.

CHAPTER 8 · 8.6 · WORKED EXAMPLE 2

Investigation: area scale factors

WORKED EXAMPLE 8.6.2

Problem. A triangle has vertices (0, 0), (4, 1) and (1, 5). Find its area using a determinant.

step	working	result
side vectors	(4, 1) and (1, 5)	—
determinant	4(5) − 1(1)	19
parallelogram area	19	—
triangle area	19 ÷ 2	9.5

A triangle is half the parallelogram.

Mathematical reasoning. Take the two side vectors from one vertex, form the determinant, and halve it.

1. The side vectors from the origin are (4, 1) and (1, 5).
2. Their determinant is $4 \times 5 - 1 \times 1 = 19$.
3. The parallelogram spanned by them has area 19.
4. The triangle is half of that, so its area is 9.5.
5. Check with the shoelace formula, which gives the same value.

Final answer. 9.5 square units

Interpretation. The determinant formula for a triangle's area is faster than finding a base and a height.

CHAPTER 8 · INVESTIGATION 8.7

Mixed practice

INQUIRY QUESTION

Which operation does this problem need?

YOUR TASK

Take the vectors $\mathbf{a} = (2, -1)$, $\mathbf{b} = (4, 3)$ and the matrices $M = [[1, -1], [2, 0]]$ and $N = [[0, 2], [1, 1]]$. Compute $\mathbf{a} + \mathbf{b}$, $2\mathbf{a} - 3\mathbf{b}$, $|\mathbf{a} + \mathbf{b}|$, $\mathbf{a} \cdot \mathbf{b}$, $\det M$, $\det N$, $\det(MN)$, M^{-1}, and the area of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$. Record the geometric meaning of each determinant.

quantity	working	result
$\mathbf{a} + \mathbf{b}$	$(6, 2)$	—
$2\mathbf{a} - 3\mathbf{b}$	$(4 - 12, -2 - 9)$	$(-8, -11)$
$\lvert\mathbf{a} + \mathbf{b}\rvert$	$\sqrt{(36 + 4)}$	$\sqrt{40} \approx 6.32$
$\mathbf{a} \cdot \mathbf{b}$	$8 - 3$	5
$\det M$	$1(0) - (-1)(2)$	2
$\det N$	$0(1) - 2(1)$	-2
$\det(MN)$	$2 \times (-2)$	-4
M^{-1}	$(1/2)[[0, 1], [-2, 1]]$	$[[0, 0.5], [-1, 0.5]]$
area of parallelogram	$2(3) - (-1)(4)$	10

CHAPTER 8 · INVESTIGATION 8.7

Mixed practice

What do you notice?

- The determinant of M was positive, so that transformation preserved orientation.
- The determinant of N was negative, so that transformation reversed it.
- The product determinant was the product, −4, reversing orientation and scaling areas by 4.
- The parallelogram area came from the determinant of the two side vectors.

Observation. The determinant links the algebra of a matrix to the geometry of area and orientation.

MATHEMATICAL CONCLUSION

Vectors combine component-wise and matrices encode transformations, with composition read in the order of application. The determinant is the signed area scale factor and decides invertibility: it multiplies over products, is positive for orientation-preserving transformations and negative for orientation-reversing ones, and zero exactly when no inverse exists. Here det M = 2, det N = −2 and det(MN) = −4.

Your own words

CHAPTER 8 · 8.7

Mixed practice

Vector problems are recognised by the presence of directions: displacement, force, velocity and position are all vectors. Scalar problems have magnitude only, and confusing the two is the commonest error.

Matrix problems are recognised by the presence of transformations. The order of composition is the order of application, read from right to left, and reversing it changes the answer.

The determinant is the bridge between the algebra and the geometry: it tells you the area scale factor and whether an inverse exists. Computing it early is a good habit, because it settles both questions at once.

Mixed practice is where the operation is chosen rather than given, so the reasoning behind the choice is part of what is being practised.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 8 · 8.7 · WORKED EXAMPLE 1

Mixed practice

WORKED EXAMPLE 8.7.1

Problem. Given $a = (3, -2)$ and $b = (-1, 5)$, find $3a - 2b$.

step	working	result
$3a$	$(9, -6)$	—
$2b$	$(-2, 10)$	—
$3a - 2b$	$(9 - (-2), -6 - 10)$	$(11, -16)$

Component-wise arithmetic.

Mathematical reasoning. Scale each vector and combine component by component.

1. $3a = (9, -6)$ and $2b = (-2, 10)$.
2. Subtracting gives $(9 - (-2), -6 - 10) = (11, -16)$.
3. The magnitude is $\sqrt{(121 + 256)} = \sqrt{377} \approx 19.42$.
4. The direction is $\arctan(-16/11)$, below the positive x-axis.
5. There is no ambiguity, because vector arithmetic is component-wise.

Final answer. (11, −16)

Interpretation. The components are combined separately, and the magnitude is recovered only if it is needed.

CHAPTER 8 · 8.7 · WORKED EXAMPLE 2

Mixed practice

WORKED EXAMPLE 8.7.2

Problem. Find the determinant of [[3, 5], [6, 10]] and explain why the matrix has no inverse.

step	working	result
determinant	3(10) − 5(6)	0
columns	(3, 6) and (5, 10)	one is 5/3 of the other
image	the line through (3, 6)	a line
inverse	does not exist	—

Proportional columns give a zero determinant.

Mathematical reasoning. Compute the determinant and interpret a zero value.

1. The determinant is $3 \times 10 - 5 \times 6 = 30 - 30 = 0$.
2. The two columns are proportional, since (5, 10) is 5/3 of (3, 6).
3. So the two basis vectors map to parallel vectors and the plane collapses onto a line.
4. The collapse cannot be reversed, so no inverse exists.
5. The area scale factor is zero, which is the geometric statement of the same fact.

Final answer. determinant 0; no inverse exists

Interpretation. A zero determinant, proportional columns and a collapsed image are three descriptions of one situation.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

8.A1 FLUENCY

Given $a = (3, -2)$ and $b = (-1, 5)$, find $3a - 2b$.

8.A2 FLUENCY

Find the magnitude of the vector $(6, 8)$.

8.A3 FLUENCY

Find the unit vector in the direction of $(3, 4)$.

8.A4 FLUENCY

Given $a = (2, 5)$ and $b = (-3, 1)$, find $a \cdot b$.

8.A5 FLUENCY

Show that $(1, 2)$ and $(4, -2)$ are perpendicular.

8.A6 FLUENCY

Given $a = (1, 2, 3)$ and $b = (4, 0, -1)$, find $a \cdot b$.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

8.A7 FLUENCY

Find the determinant of [[5, 2], [3, 7]].

8.A8 FLUENCY

Find the inverse of [[2, 0], [0, 4]], if it exists.

8.A9 FLUENCY

Write the matrix for a rotation through 90°.

8.A10 FLUENCY

The point (2, 1) is transformed by [[1, 3], [0, 1]]. Find its image.

8.A11 REASONING

Find the area of the parallelogram spanned by (4, 1) and (2, 5).

8.A12 REASONING

Explain why [[1, 2], [2, 4]] has no inverse.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

8.B1 REASONING

Investigate the effect of the matrix [[k, 0], [0, 1/k]] for k = 2, 3 and 0.5, and justify what it does to areas.

8.B2 CHALLENGE

Conjecture a rule for when a 2 by 2 matrix represents a rotation, and test it on three matrices.

8.B3 REASONING

Investigate the images of the unit square under [[1, 1], [0, 1]] and [[1, 0], [1, 1]], and describe the transformations.

8.B4 REASONING

Investigate the determinant of the product of two reflections, and justify the result.

8.B5 REASONING

Take three vectors of your own in three dimensions and determine whether they are coplanar, justifying your answer.

8.B6 REASONING

Investigate the relationship between the determinant of a matrix and the determinant of its transpose, and justify.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

8.B7 REASONING

Investigate the vector product of two parallel vectors, and justify the result.

8.B8 REASONING

Investigate the determinant of the matrix [[cos θ, −sin θ], [sin θ, cos θ]] and justify why rotations preserve area.

8.B9 CHALLENGE

Conjecture a rule for the determinant of the inverse of a matrix, and test it.

8.B10 REASONING

Investigate the image of a straight line under a matrix transformation and justify the result.

8.B11 REASONING

Investigate the determinant of a matrix representing a shear, and justify.

CRITERION C Communicating

Criterion C practice · Communicating

8.C1 REASONING

Write a complete solution to finding the image of (3, −1) under the matrix [[2, −1], [4, 3]] and the determinant of the matrix.

8.C2 REASONING

Write a complete solution to showing that the points A(2, 1), B(5, 4) and C(−1, −5) are collinear.

8.C3 REASONING

Write a complete solution to finding the point dividing AB in the ratio 3 : 2 where A = (−2, 4) and B = (6, −2).

8.C4 CHALLENGE

Write a complete solution to finding the vector product of (2, −1, 3) and (1, 4, −2), and verifying it is perpendicular to both.

8.C5 CHALLENGE

Write a complete solution to finding the volume of the parallelepiped spanned by (1, 2, 3), (4, 5, 6) and (7, 8, 10).

8.C6 CHALLENGE

Write a complete solution to proving that the determinant of a product equals the product of the determinants, for 2 by 2 matrices.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

8.C7 REASONING

Write a complete solution to finding the area of the triangle with vertices (1, 1), (6, 2) and (3, 7).

8.C8 REASONING

Write a complete solution to finding the image of (5, −2) under [[1, 2], [-1, 3]] and the determinant of the matrix.

8.C9 REASONING

Write a complete solution to showing that A(−1, 2), B(3, 5) and C(9, 11) are collinear.

8.C10 REASONING

Write a complete solution to finding the point dividing AB in the ratio 1 : 4 where A = (2, 7) and B = (18, −3).

8.C11 CHALLENGE

Write a complete solution to finding the vector product of (3, −2, 1) and (1, 4, −2), and verifying it is perpendicular to both.

8.C12 REASONING

Write a complete solution to finding the area of the triangle with vertices (1, 2), (5, 3) and (2, 7).

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

8.D1 APPLICATION

A drone's displacement is modelled by the vector 3a + 2b where a = (40, 30) metres and b = (−20, 50). Find the displacement and its distance from the start.

8.D2 APPLICATION

A force of (3, 4) newtons acts on a body that moves through the displacement (5, 0) metres. Find the work done.

8.D3 APPLICATION

A graphics program applies the matrix [[0, −1], [1, 0]] to a shape. Describe the transformation and find the image of the point (2, 5).

8.D4 APPLICATION

A bridge support is modelled by two vectors (4, 0, 3) and (0, 5, 12). Find the angle between them and the area of the parallelogram they span.

8.D5 APPLICATION

A machine part is transformed by [[1, 0], [0, −1]] and then by [[−1, 0], [0, 1]]. Find the combined matrix and describe the single transformation.

8.D6 APPLICATION

A student applies the matrix [[2, 0], [0, 2]] to a triangle of area 7. Find the area of the image and explain the calculation.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

8.D7 REASONING

A navigation system stores positions as vectors and applies a correction matrix. Explain how a zero determinant would affect the system.

8.D8 APPLICATION

A parallelogram has diagonals with vectors (6, 2) and (2, 6). Find its area.

8.D9 REASONING

A transformation has matrix [[3, 1], [5, 2]]. A student claims it cannot be a rotation. Investigate and justify.

8.D10 APPLICATION

A crane's arm is modelled by the vector 8(cos 35°, sin 35°) and its counterweight by −5(cos 35°, sin 35°). Find the resultant and its magnitude.

8.D11 APPLICATION

A school's coordinate grid is transformed by [[1, 2], [3, 4]]. Find the image of the unit square, its area, and whether the transformation can be undone.

8.D12 REASONING

A student computes the vector product of two vectors in two dimensions and gets a number. Investigate and justify.

CHAPTER 8 · APPLICATION AND ASSESSMENT

Applying chapter 8

These tasks use vectors and matrices. State the vector or matrix you compute, give the geometric meaning of the determinant, and check every inverse by multiplication.

A1 · The graphic transformation

A graphics program applies the matrix [[1, 1], [−1, 1]] to a shape and then the matrix [[2, 0], [0, 0.5]]. Describe each transformation, find the combined matrix, find its determinant, and say whether the combined transformation can be undone. If it can, find the inverse matrix.

Criteria C and D · describe each transformation, combine the matrices in order, and justify the existence of the inverse.

A2 · The survey vectors

A surveyor records two displacements a = (120, −50) metres and b = (−80, 140) metres. Find the resultant, its magnitude and bearing, the angle between the two displacements, and the area of the parallelogram they span. Then find a unit vector in the direction of the resultant.

Criteria C and D · combine the vectors, use the scalar product for the angle and the determinant for the area.

CHAPTER 8 · REVIEW

What you should now be able to do

Key ideas

- A vector has magnitude and direction and combines component-wise.
- The magnitude of (x, y) is $\sqrt{(x^2 + y^2)}$.
- The scalar product $a \cdot b = |a||b| \cos \theta$ is zero exactly for perpendicular vectors.
- The columns of a matrix are the images of the basis vectors.
- Composition of transformations is matrix multiplication in the order of application.
- The determinant $ad - bc$ is the signed area scale factor.
- A negative determinant reverses orientation; a zero determinant collapses the plane.
- The inverse exists exactly when the determinant is non-zero.
- $\det(AB) = \det(A) \det(B)$.
- The vector product gives a vector perpendicular to both factors, with magnitude equal to the area they span.

Vocabulary

vector · magnitude · direction · position vector · scalar product · unit vector · column vector · matrix · transformation · determinant · inverse matrix · identity matrix · adj ugate · vector product · scalar triple product · coplanar · shear · orientation

CHAPTER 8 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

8.R1 Given $a = (5, -2)$ and $b = (1, 4)$, find $2a + b$ and $|a + b|$.

8.R2 Find the determinant of $[[3, -1], [7, 2]]$ and its inverse.

8.R3 Write the matrix for a reflection in the y-axis.

8.R4 Find the area of the triangle with vertices $(0, 0)$, $(3, 1)$ and $(1, 4)$.

8.R5 Given $a = (1, 2, 3)$ and $b = (4, 5, 6)$, find $a \times b$.

8.R6 Explain what a determinant of zero means geometrically.

CHAPTER 8 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

Vectors encode displacement by magnitude and direction, while matrices encode transformations of the plane as arrays of numbers; the determinant measures how a transformation stretches or collapses area, and decides whether it can be undone.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of space, representation and systems appear in your work?
- How is the global context of scientific and technical innovation, and its exploration of computer graphics, navigation and engineering, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a matrix as a description of a transformation, and what would you still like to ask?

CHAPTER 8 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

A transformation is a rule with a matrix. Name the rule, give its matrix, and say what its determinant tells you about the image.

Service as Action

COMMUNITY NEED	Local mapping and measurement is done without any record of the transformations used.
MATHEMATICAL CONNECTION	Vectors and matrices.
PROPOSED ACTION	Take one local map or plan, express two displacements as vectors, and publish the method with the determinant of the transformation used.
EXPECTED IMPACT	The plan can be checked and reproduced by anyone.
STUDENT REFLECTION	Record the vectors, the matrices and the determinants you computed.

Community Project

PROJECT IDEA	A vector audit of the school's site plan.
NEED	The plan is used for events without any record of its scale or orientation.
RESEARCH QUESTION	What transformation converts the plan's coordinates into metres on the ground?
MATHEMATICS	Vectors, matrices and determinants.
ACTION OR OUTCOME	A one-page conversion with the matrix shown.
BENEFICIARIES	Everyone who uses the school site plan.
EVIDENCE	The conversion and the matrix behind it.

Personal Project connection · MYP 5

PERSONAL INTEREST	Computer graphics, gaming, robotics or sport.
LEARNING GOAL	To describe a transformation you use as a matrix and state what its determinant means.
PRODUCT OR OUTCOME	A transformation card with a matrix and a determinant.
MATHEMATICS	Vectors, matrices and determinants.
PROCESS	Choose a transformation, write its matrix, and compute its determinant.
EVIDENCE	The card and the determinant you computed.

CHAPTER 9 · PART IV · REASONING WITH DATA

09

Statistics: Dispersion, Correlation and Regression

A single average describes a group's centre and nothing else. Dispersion describes how the group is spread, and association describes how two variables move together, and neither can say more than the data allows.

ENGAGE

Two classes have the same mean test score. One class has every student within two marks of the mean; the other has scores from the lowest to the highest possible. Are the two classes equally well taught, and what would you need to know to decide?

Framing the inquiry

Key concept: Relationships
Related concepts: Patterns · Representation · Measurement
Global context: Identities and relationships
Global context exploration: how groups differ, and how claims about them are made

Statement of inquiry: Two groups can share a centre and differ in every other respect; measuring dispersion and association describes those differences precisely, while the origin of the data limits what the description can prove.

Questions to guide us

- **Factual:** What do the range, the interquartile range, the variance and the standard deviation measure?
- **Conceptual:** Why can two data sets share a mean and differ completely, and what does a correlation coefficient measure?
- **Debatable:** If a correlation is strong, is a causal claim justified?

ATL · CRITICAL THINKING. In order for a student to judge a claim made about two groups, students must compare a measure of centre with a measure of dispersion, consider the origin of the data, and justify what the comparison can and cannot show.

CHAPTER 9 · INVESTIGATION 9.1

Measuring spread

INQUIRY QUESTION

Why is a single average not enough to describe a group?

YOUR TASK

Take six data sets of your own, each with the same mean of 10: one with no spread at all, one with a small spread, one with a large spread, one with one extreme outlier, one with two clusters, and one symmetric. Compute the range, the interquartile range and the standard deviation of each, and record which measure best captures the difference.

data set	range	interquartile range	standard deviation	best description
all 10s	0	0	0	no spread
9, 10, 11	2	1	0.82	small spread
1, 5, 10, 15, 19	18	10	6.85	large spread
10, 10, 10, 10, 40	30	0	13.42	one outlier
1, 2, 9, 18, 19	18	16	8.20	two clusters
6, 8, 10, 12, 14	8	6	2.83	symmetric

CHAPTER 9 · INVESTIGATION 9.1

Measuring spread

What do you notice?

- The data set with an outlier had a huge range but a zero interquartile range.
- The two-cluster set had a large interquartile range but a standard deviation similar to the wide set.
- The set with no spread had every measure equal to zero.
- No single measure described all six sets adequately, which is why more than one is reported.

Observation. Different measures of spread respond to different features of a data set, and an outlier distorts the range and the standard deviation but not the interquartile range.

MATHEMATICAL CONCLUSION

A single average describes a centre and nothing else, so a measure of dispersion is needed alongside it. The range is sensitive to extremes, the interquartile range is resistant to them, and the standard deviation uses every value in the same units as the data. Here the outlier set had range 30 with interquartile range 0.

Your own words

CHAPTER 9 · 9.1

Measuring spread

The range is the difference between the largest and smallest values, and it is sensitive to every extreme value. It is quick to compute and quick to mislead, because one unusual observation can change it completely.

The interquartile range is the difference between the upper and lower quartiles, and it describes the middle half of the data. It is resistant to outliers, which is why it is preferred when the data contain extreme values.

The variance is the mean of the squared deviations from the mean, and the standard deviation is its square root. The standard deviation is in the same units as the data, which is what makes it interpretable.

There are two conventions for variance: the population variance divides by n and the sample variance divides by n − 1. The sample version is unbiased for estimating the population variance, and the convention in use must always be stated.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 9 · 9.1 · WORKED EXAMPLE 1

Measuring spread

WORKED EXAMPLE 9.1.1

Problem. Find the range, the interquartile range and the standard deviation of the data 4, 7, 8, 9, 11, 13, 16.

step	working	result
mean	(4 + 7 + 8 + 9 + 11 + 13 + 16)/7	9.714
range	16 − 4	12
quartiles	Q1 = 7.5, Q3 = 12.5	—
interquartile range	12.5 − 7.5	5

The interquartile range is half the range here.

Mathematical reasoning. Compute the mean, then the range, the quartiles and the deviations.

1. The mean is $(4 + 7 + 8 + 9 + 11 + 13 + 16)/7 = 68/7 \approx 9.714$.
2. The range is $16 - 4 = 12$.
3. The lower quartile is 7.5 and the upper quartile is 12.5, so the interquartile range is 5.
4. The squared deviations sum to 82.857, so the population variance is $82.857/7 \approx 11.837$.
5. The population standard deviation is $\sqrt{11.837} \approx 3.44$, and the sample standard deviation is $\sqrt{(82.857/6)} \approx 3.72$.

Final answer. range 12, interquartile range 5, population standard deviation about 3.44

Interpretation. The interquartile range is much smaller than the range because the extremes are excluded, which is exactly its purpose.

CHAPTER 9 · 9.1 · WORKED EXAMPLE 2

Measuring spread

WORKED EXAMPLE 9.1.2

Problem. Two data sets both have mean 50. Set A has standard deviation 2 and set B has standard deviation 12. Explain what this means for the two groups.

set	mean	standard deviation	typical distance from the mean
A	50	2	about 2 units
B	50	12	about 12 units
difference	0	10	—

The centres agree and the spreads differ.

Mathematical reasoning. Compare the standard deviations while noting the equal means.

1. Both sets have the same mean of 50, so their centres agree.
2. Set A has a standard deviation of 2, so most of its values lie close to 50.
3. Set B has a standard deviation of 12, so its values are typically 12 units from the mean.
4. The two groups therefore differ completely even though their means are identical.
5. A report giving only the means would hide the difference entirely.

Final answer. the means agree but the spreads differ by a factor of 6

Interpretation. This is the case the inquiry statement describes: a shared centre with a complete difference in every other respect.

CHAPTER 9 · INVESTIGATION 9.2

Standard deviation and grouped data

INQUIRY QUESTION

How is dispersion computed when the data are grouped?

YOUR TASK

Take a data set of your own with at least 20 values, group it into classes of equal width, and compute the mean and standard deviation from the grouped table. Then compute them from the raw values and compare, explaining any difference.

method	mean	standard deviation	difference in the mean
raw values	31.4	5.62	—
grouped, width 10	31.0	5.48	0.4
grouped, width 5	31.3	5.58	0.1
grouped, width 20	30.0	5.90	1.4

CHAPTER 9 · INVESTIGATION 9.2

Standard deviation and grouped data

What do you notice?

- The narrower the classes, the closer the grouped answers were to the raw answers.
- The widest classes gave the largest errors in both measures.
- The mean was always slightly affected by the mid-point assumption.
- The standard deviation was affected less, because within-class variation partly cancels.

Observation. Grouping replaces each value by its class mid-point, and the error shrinks as the classes narrow.

MATHEMATICAL CONCLUSION

For grouped data the mid-point of each class stands in for every value in it, and the mean and standard deviation are computed from the frequency table with the formula variance = (sum of $f x^2$)/(sum of f) − mean^2. The mid-point assumption introduces an error that narrows the classes reduce. Here the raw mean was 31.4 and the grouped estimates were 31.0, 31.3 and 30.0.

Your own words

CHAPTER 9 · 9.2

Standard deviation and grouped data

For grouped data with frequencies, the mean is the sum of each value times its frequency divided by the total frequency. The standard deviation is computed from the same table, using the squared deviations.

The coding shortcut replaces each value x by $(x - a)/h$ for a convenient a and class width h, computes the standard deviation of the coded values, and multiplies by h. The shift does not affect the spread, and the scale multiplies it.

For a frequency table, the variance is $(\text{sum of } f\,x^2)/(\text{sum of } f) - \text{mean}^2$. This form is faster than summing squared deviations and is the one used in practice.

When data are grouped into classes, the mid-point of each class represents every value in it. This introduces a within-class error that the standard deviation cannot detect, and it is stated as a limitation of the calculation.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 9 · 9.2 · WORKED EXAMPLE 1

Standard deviation and grouped data

WORKED EXAMPLE 9.2.1

Problem. Find the mean and standard deviation of the data summarised by x = 2, 4, 6, 8 with frequencies 3, 5, 2, 1.

x	f	fx	fx^2
2	3	6	12
4	5	20	80
6	2	12	72
8	1	8	64

The frequency table carries all the information.

Mathematical reasoning. Use the frequency-table formula for the mean and the variance.

1. The total frequency is 3 + 5 + 2 + 1 = 11.
2. The sum of f x is 6 + 20 + 12 + 8 = 46, so the mean is 46/11 ≈ 4.18.
3. The sum of f x^2 is 12 + 80 + 72 + 64 = 228.
4. The variance is 228/11 − $(46/11)^2$ ≈ 20.73 − 17.50 ≈ 3.23.
5. The standard deviation is √3.23 ≈ 1.80.

Final answer. mean about 4.18, standard deviation about 1.80

Interpretation. The frequency table formula is the same calculation written in a form that avoids listing every observation.

CHAPTER 9 · 9.2 · WORKED EXAMPLE 2

Standard deviation and grouped data

WORKED EXAMPLE 9.2.2

Problem. The heights of 40 plants are grouped into classes 10–14, 15–19, 20–24, 25–29 with frequencies 4, 12, 16, 8. Estimate the mean and standard deviation.

class	mid-point	f	f × mid-point
10–14	12	4	48
15–19	17	12	204
20–24	22	16	352
25–29	27	8	216

Class mid-points stand in for the individual values.

Mathematical reasoning. Use the mid-points as representative values and apply the frequency formula.

1. The mid-points are 12, 17, 22 and 27.
2. The total frequency is 40 and the sum of f times mid-point is 48 + 204 + 352 + 216 = 820.
3. So the estimated mean is 820/40 = 20.5.
4. The sum of f times mid-point squared is 4(144) + 12(289) + 16(484) + 8(729) = 576 + 3468 + 7744 + 5832 = 17 620.
5. The variance is 17 620/40 − 20.5^2 = 440.5 − 420.25 = 20.25, so the standard deviation is 4.5.

Final answer. estimated mean 20.5, estimated standard deviation 4.5

Interpretation. The mid-point assumption is an estimate, and the true standard deviation within each class is hidden by it.

CHAPTER 9 · INVESTIGATION 9.3

Comparing distributions

INQUIRY QUESTION

How are two distributions compared fairly?

YOUR TASK

Collect two data sets of your own from different sources — for example reaction times and test scores. Compute the mean, median, standard deviation and interquartile range of each, draw box plots of both, and write a comparison that states what the comparison can and cannot show.

measure	data set 1	data set 2	comparison
mean	0.42 s	62.4 marks	different units
median	0.40 s	58 marks	different units
standard deviation	0.09 s	14.2 marks	data set 2 is more spread relative to its mean
interquartile range	0.12 s	21 marks	same conclusion
shape	roughly symmetric	skewed right	median better for set 2

CHAPTER 9 · INVESTIGATION 9.3

Comparing distributions

What do you notice?

- The two sets had different units, so their standard deviations could not be compared directly.
- Comparing each standard deviation with its own mean showed that the second set was more spread relative to its centre.
- The first set was roughly symmetric, so the mean and median were close.
- The second set was skewed right, so its mean exceeded its median.

> **Observation.** A fair comparison uses centre, dispersion and shape together, and notes the units.

MATHEMATICAL CONCLUSION

Two distributions are compared by a measure of centre, a measure of dispersion and a statement of shape, and the sample sizes and the origin of the data limit what the comparison can show. Where the distributions are skewed, the median is the better centre. Here the second set was more spread relative to its mean and skewed to the right.

Your own words

CHAPTER 9 · 9.3

Comparing distributions

Two distributions are compared by a measure of centre, a measure of dispersion, and a statement of the shape. Reporting only the means is the commonest statistical error, and it hides exactly the difference the inquiry statement describes.

A box plot displays the minimum, the lower quartile, the median, the upper quartile and the maximum, so it shows centre, spread and shape at once, and it is a representation of a whole distribution in five numbers. Comparing two box plots side by side is the standard visual comparison.

The shape of a distribution is described as symmetric, skewed to the left or skewed to the right. When a distribution is skewed, the median is a better measure of centre than the mean, because the mean is pulled towards the tail.

A comparison must state the sample sizes and the origin of the data, because a difference between two small samples may not be a difference between the populations they came from.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 9 · 9.3 · WORKED EXAMPLE 1

Comparing distributions

WORKED EXAMPLE 9.3.1

Problem. Two groups of seven values each are recorded. Group A has mean 50 and standard deviation 2; group B has mean 50 and standard deviation 12. Write a comparison of the two groups.

measure	group A	group B	comparison
mean	50	50	the same
standard deviation	2	12	B is 6 times as spread
interquartile range	3	18	B is 6 times as wide
shape	tight cluster	wide spread	very different

The centres agree and every other measure differs.

Mathematical reasoning. Compare centre, dispersion and shape in turn.

1. Both groups have the same mean of 50, so their centres agree.
2. Group A has a standard deviation of 2 and group B of 12, so group B is six times as spread.
3. The interquartile ranges tell the same story: 3 against 18.
4. The shapes differ too, with group A tightly clustered and group B widely spread.
5. So the two groups are completely different despite an identical mean, and any report giving only the mean would be misleading.

Final answer. the means agree; group B is six times as dispersed as group A

Interpretation. The comparison names all three aspects, and it states what the shared mean hides.

CHAPTER 9 · 9.3 · WORKED EXAMPLE 2

Comparing distributions

WORKED EXAMPLE 9.3.2

Problem. A distribution has mean 40, median 34 and mode 28. Describe its shape and say which measure of centre to report.

measure	value	position
mode	28	lowest
median	34	middle
mean	40	highest
conclusion	mean > median > mode	skewed right

A long right tail pulls the mean up.

Mathematical reasoning. Compare the three measures and use their order to identify the skew.

1. The three measures are 28, 34 and 40.
2. The mean is the largest, so the distribution is skewed to the right.
3. A right skew means a long tail of large values pulling the mean upwards.
4. The median is the better measure of centre here, because it is not affected by the tail.
5. So the report should give the median, 34, with a note that the distribution is skewed.

Final answer. skewed to the right; report the median, 34

Interpretation. The order of the three measures diagnoses the skew, and the skew decides which measure to report.

CHAPTER 9 · INVESTIGATION 9.4

Scatter graphs and correlation

INQUIRY QUESTION

What does a correlation coefficient measure, and what does it not?

YOUR TASK

Collect ten paired observations of your own on two variables you suspect are related. Draw a scatter graph, compute the correlation coefficient, and then write a paragraph on whether the correlation supports a causal claim, giving at least one alternative explanation.

step	working	result
scatter graph	plotted	roughly linear, positive
means	$\bar{x} = 12.4$, $\bar{y} = 5.6$	—
sums	sum $xy = 742$, sum $x^2 = 1660$, sum $y^2 = 356$	—
r	$(10 \times 742 - 124 \times 56) \div \sqrt{((10 \times 1660 - 15\,376)(10 \times 356 - 3136))}$	0.78
causal claim	not supported without more evidence	—

CHAPTER 9 · INVESTIGATION 9.4

Scatter graphs and correlation

What do you notice?

- The scatter graph was roughly linear and sloped upwards.
- The correlation coefficient was 0.78, indicating a strong positive linear association.
- A third variable, time of day, could plausibly explain both.
- The data supported a claim of association only.

Observation. A strong correlation establishes association, not causation, and an alternative explanation is always available.

MATHEMATICAL CONCLUSION

The correlation coefficient r measures the strength of a linear association and lies between −1 and 1, and it is unchanged by a change of units. It says nothing about causation: a common cause, a reverse cause or coincidence can all produce a strong r. Here r was 0.78 and the causal claim was not supported.

Your own words

CHAPTER 9 · 9.4

Scatter graphs and correlation

A scatter graph plots one variable against another, and its overall patterns are described by direction, strength and shape. A straight-line pattern suggests linear correlation; a curved pattern suggests a non-linear relationship, and the eye reads the patterns before any number is computed.

The product-moment correlation coefficient r lies between −1 and 1. Values near 1 indicate strong positive linear correlation, values near −1 strong negative linear correlation, and values near 0 little or no linear correlation, so the number summarises the pattern of the scatter.

The coefficient is computed from the covariance of the two variables divided by the product of their standard deviations, which is why it is unaffected by the units of measurement, and it is a representation of the pattern in a single number. Changing metres to centimetres does not change r.

Correlation is not causation. Two variables may be correlated because one causes the other, because both are caused by a third, or by coincidence, and the data alone cannot distinguish these cases.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 9 · 9.4 · WORKED EXAMPLE 1

Scatter graphs and correlation

WORKED EXAMPLE 9.4.1

Problem. Compute the correlation coefficient for the paired data (1, 2), (2, 4), (3, 5), (4, 4), (5, 6).

x	y	xy	x^2	y^2
1	2	2	1	4
2	4	8	4	16
3	5	15	9	25
4	4	16	16	16
5	6	30	25	36

The sums feed the correlation formula.

Mathematical reasoning. Compute the summary sums and substitute into the formula.

1. The means are $\bar{x} = 3$ and $\bar{y} = 4.2$.
2. The sum of x y is $2 + 8 + 15 + 16 + 30 = 71$.
3. The sum of x^2 is 55 and the sum of y^2 is 97.
4. The correlation is $(5 \times 71 - 15 \times 21) \div \sqrt{((5 \times 55 - 225)(5 \times 97 - 441))} = 80 \div \sqrt{(50 \times 44)} \approx 0.855$.
5. So $r \approx 0.855$, indicating strong positive linear correlation.

Final answer. $r \approx 0.855$

Interpretation. The coefficient is dimensionless, so it is unaffected by a change of units in either variable.

CHAPTER 9 · 9.4 · WORKED EXAMPLE 2

Scatter graphs and correlation

WORKED EXAMPLE 9.4.2

Problem. Explain why a correlation of 0.93 between ice-cream sales and drowning deaths does not show that ice cream causes drowning.

explanation	status	
ice cream causes drowning	not supported	
a third variable, temperature, causes both	plausible	
coincidence	possible but unlikely at r = 0.93	
conclusion	correlation is not causation	—

A hidden common cause explains the correlation.

Mathematical reasoning. Consider the possible explanations for a correlation and identify the plausible one.

1. A correlation of 0.93 is strong, so the two variables really do move together.
2. But correlation measures association, not causation.
3. Both ice-cream sales and drowning deaths rise in hot weather, so temperature is a common cause.
4. Once temperature is accounted for, the association between the other two largely disappears.
5. So the data support a claim of association only, and any causal claim needs further evidence.

Final answer. temperature is a common cause; correlation is not causation

Interpretation. The origin of the data limits what the correlation can show, which is the second half of the inquiry statement.

CHAPTER 9 · INVESTIGATION 9.5

Regression and prediction

INQUIRY QUESTION

How is a line of best fit used, and what are its limits?

YOUR TASK

Take ten paired observations of your own, compute the regression line of y on x, and use it to predict y at a value of x inside the data range and one outside. Then compute the line of x on y and compare the two lines, and comment on the reliability of each prediction.

task	working	result
regression of y on x	y = 0.78x + 1.4	—
interpolation at x = 10	y = 9.2	within the range
extrapolation at x = 25	y = 20.9	outside the range
regression of x on y	x = 0.61y + 3.2	a different line
comparison	the two gradients differ	0.78 against 1/0.61

CHAPTER 9 · INVESTIGATION 9.5

Regression and prediction

What do you notice?

- The two regression lines were different, as expected with $|r| < 1$.
- The interpolation gave a value inside the observed range.
- The extrapolation gave a value far outside it, and assumed the relationship continued.
- The two gradients were not reciprocals, because the lines minimise different distances.

Observation. The regression line of y on x predicts y from x, and extrapolation assumes the relationship continues beyond the data.

MATHEMATICAL CONCLUSION

The regression line of y on x minimises the vertical distances and predicts y from x, and it is computed from the same summary sums as the correlation coefficient. Interpolation within the range of the data is reliable; extrapolation beyond it assumes the relationship continues. The line of x on y is different whenever $|r| < 1$. Here the two gradients were 0.78 and about 1.64.

Your own words

CHAPTER 9 · 9.5

Regression and prediction

The regression line of y on x is the line that minimises the sum of the squared vertical distances from the points to the line. It is written $y = a + bx$, and it is the line used to predict y from x.

The gradient is b = covariance ÷ variance of x, and the intercept is $a = \bar{y} - b\,\bar{x}$. Both are computed from the same summary sums as the correlation coefficient, and r is the normalised version of b.

Interpolation is prediction within the range of the data, and it is usually reliable. Extrapolation is prediction beyond that range, and it assumes the relationship continues, which is an assumption and not a fact.

The regression line of x on y is a different line from the regression line of y on x, unless the correlation is exactly ±1. The two lines answer different questions, and using the wrong one gives a wrong prediction.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 9 · 9.5 · WORKED EXAMPLE 1

Regression and prediction

WORKED EXAMPLE 9.5.1

Problem. Find the regression line of y on x for the data (1, 2), (2, 4), (3, 5), (4, 4), (5, 6), and predict y when x = 6.

step	working	result
means	$\bar{x} = 3$, $\bar{y} = 4.2$	—
gradient	$(5 \times 71 - 15 \times 21) \div (5 \times 55 - 225)$	$80/50 = 1.6$
intercept	$4.2 - 1.6 \times 3$	-0.6
line	$y = 1.6x - 0.6$	—
prediction	y at x = 6	9.0

The gradient and intercept come from the same sums as r.

Mathematical reasoning. Compute the summary sums, then the gradient and intercept.

1. The means are $\bar{x} = 3$ and $\bar{y} = 4.2$.
2. The gradient is $(5 \times 71 - 15 \times 21)/(5 \times 55 - 225) = 80/50 = 1.6$.
3. The intercept is $\bar{y} - b\,\bar{x} = 4.2 - 1.6(3) = -0.6$.
4. So the line is $y = 1.6x - 0.6$.
5. At x = 6 the prediction is y = 9.0, which is an extrapolation beyond the data and so less reliable.

Final answer. y = 1.6x − 0.6; prediction 9.0 at x = 6

Interpretation. The prediction is labelled as extrapolation, and the assumption it makes is stated.

CHAPTER 9 · 9.5 · WORKED EXAMPLE 2

Regression and prediction

WORKED EXAMPLE 9.5.2

Problem. A student uses the regression line of x on y to predict y from x. Explain the error.

line	minimises	used to predict
y on x	vertical distances	y from x
x on y	horizontal distances	x from y
when they differ	whenever $r < 1$	—

The two regression lines are different.

Mathematical reasoning. Identify which distances each line minimises and which prediction it supports.

1. The line of y on x minimises the vertical distances, so it is the line for predicting y from x.
2. The line of x on y minimises the horizontal distances, so it is the line for predicting x from y.
3. The two lines coincide only when $|r| = 1$, that is when the points lie exactly on a line.
4. In general they are different, and using the wrong one gives a biased prediction.
5. So the student must use the line of y on x.

Final answer. the wrong line was used; the line of y on x is required

Interpretation. Naming which variable is being predicted is the first step in any regression question.

CHAPTER 9 · INVESTIGATION 9.6

Investigation: correlation, causation and reliability

INQUIRY QUESTION

What limits a statistical description?

YOUR TASK

Take a published correlation claim from a newspaper or a website. Identify the variables, the sample, the correlation coefficient and the causal claim, if any. Then write an evaluation stating what the data support, what they do not, and at least one alternative explanation.

element	found	evaluation
variables	hours of exercise and self-reported happiness	both self-reported, so both subject to bias
sample	1500 adults in one country	not representative of other countries
coefficient	r = 0.4	moderate association
causal claim	exercise makes people happier	not supported by correlation alone
alternative	good health causes both	plausible

CHAPTER 9 · INVESTIGATION 9.6

Investigation: correlation, causation and reliability

What do you notice?

- Both variables were self-reported, which introduced a common measurement bias.
- The sample came from one country, so the association might not generalise.
- The coefficient of 0.4 indicated a moderate association only.
- The causal claim was not supported, and a common cause was plausible.

Observation. The origin of the data limits the description, and a causal claim needs more than a correlation.

MATHEMATICAL CONCLUSION

A statistical description describes the sample measured and extends to a wider population only if the sample was representative. A correlation may arise from a common cause, a reverse cause or coincidence, and distinguishing these needs an experiment rather than a larger sample. Reporting the coefficient with and without influential points is the standard robustness check.

Your own words

CHAPTER 9 · 9.6

Investigation: correlation, causation and reliability

A statistical description describes the relationships within the sample that was measured, and it extends to a wider population only if the sample was representative. The origin of the data is therefore part of the result, not a footnote.

A correlation may arise from a common cause, from a reverse cause, or from coincidence. Distinguishing these requires an experiment or a controlled study, not a larger sample of the same kind.

The reliability of an estimate is described by a confidence interval, which is a range of values consistent with the data. A narrow interval means the estimate is precise; it says nothing about whether the estimate is unbiased.

An outlier can change a correlation coefficient substantially, because the coefficient is computed from sums of products. Reporting the coefficient with and without the outlier is a standard robustness check.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 9 · 9.6 · WORKED EXAMPLE 1

Investigation: correlation, causation and reliability

WORKED EXAMPLE 9.6.1

Problem. A data set of nine points gives r = 0.32, but adding a tenth point changes r to 0.91. Investigate what has happened.

point	x	y	effect on r
original nine	—	—	r = 0.32
added point	(30, 30)	far from the cluster	r = 0.91
without it	—	—	r = 0.32
conclusion	the tenth point is an influential outlier	—	

One point can dominate a correlation.

Mathematical reasoning. Compare the coefficient with and without the added point and identify its role.

1. With nine points the correlation is 0.32, indicating weak linear association.
2. Adding the point (30, 30) raises the correlation to 0.91.
3. The added point is far from the cluster of the other nine, so it pulls the line towards itself.
4. Such a point is called an influential outlier, and it changes the coefficient without changing the other data.
5. The honest report gives both values and identifies the outlier.

Final answer. the tenth point is an influential outlier; report both values

Interpretation. The robustness check is reporting the coefficient with and without the influential point.

CHAPTER 9 · 9.6 · WORKED EXAMPLE 2

Investigation: correlation, causation and reliability

WORKED EXAMPLE 9.6.2

Problem. A study of 1000 people finds a correlation of 0.6 between hours of television watched and reading score. State what the study shows and what it does not.

claim	supported?	reason
television watching is associated with reading score	yes	r = 0.6 over 1000 people
television causes lower reading scores	no	correlation is not causation
the association holds for all age groups	no	the sample may not represent them
a third variable explains both	plausible	for example socioeconomic status

Each claim needs its own justification.

Mathematical reasoning. Separate the claim the data support from the claims they do not.

1. The study shows an association, with r = 0.6 over 1000 people.
2. The large sample makes the estimate of the association reasonably precise.
3. It does not show causation, because correlation measures association only.
4. It does not show the association holds for groups outside the sample, unless the sample was representative of them.
5. A third variable, such as socioeconomic status, could explain both.

Final answer. an association is shown; causation and generalisation are not

Interpretation. The origin of the data limits the description, which is the second half of the inquiry statement.

CHAPTER 9 · INVESTIGATION 9.7

Mixed practice

INQUIRY QUESTION

Which measure, and what does it support?

YOUR TASK

Take a data set of at least ten paired observations from a source you choose. Compute the mean and standard deviation of each variable, the correlation coefficient, and the regression line of y on x. Then write an evaluation stating what the analysis supports, what it does not, and one limitation of the data source.

step	working	result
mean of x	18.4	—
standard deviation of x	4.7	—
mean of y	52.1	—
standard deviation of y	11.3	—
correlation	r = 0.82	strong positive linear association
regression line	y = 1.97x + 15.8	—
limitation	the sample was self-selected	the association may not generalise

CHAPTER 9 · INVESTIGATION 9.7

Mixed practice

What do you notice?

- The correlation was strong, at 0.82.
- The regression line allowed predictions within the observed range.
- The sample was self-selected, so the association might not generalise.
- The analysis supported a claim of association only.

Observation. A complete statistical report gives the centre, the spread, the association and the limitation together.

MATHEMATICAL CONCLUSION

A statistics question is answered by choosing the measure that matches what is being described: a centre, a dispersion, an association or a prediction. Skewed data call for the median and the interquartile range, and every association is reported as an association rather than a cause. Here the correlation was 0.82 and the regression line was $y = 1.97x + 15.8$.

Your own words

CHAPTER 9 · 9.7

Mixed practice

Every statistics question begins by asking what is being described: a centre, a spread, an association or a prediction. The four need different measures, and using the wrong one is the commonest error.

When data are skewed, the median and the interquartile range are the appropriate measures, because they are resistant to extreme values. When data are symmetric, the mean and the standard deviation use more of the information.

A regression prediction must state whether it is interpolation or extrapolation, and an association must be reported as an association rather than a cause. Both statements are part of the answer.

Mixed practice is where the measure is chosen rather than given, so the reasoning behind the choice is part of what is being practised.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 9 · 9.7 · WORKED EXAMPLE 1

Mixed practice

WORKED EXAMPLE 9.7.1

Problem. Find the mean and standard deviation of 12, 15, 18, 20, 23.

step	working	result
mean	88/5	17.6
deviations	−5.6, −2.6, 0.4, 2.4, 5.4	—
squared sum	31.36 + 6.76 + 0.16 + 5.76 + 29.16	73.2
variance	73.2/5	14.64
standard deviation	√14.64	3.83

The mean is the reference for every deviation.

Mathematical reasoning. Compute the mean, then the squared deviations, then the variance.

1. The mean is (12 + 15 + 18 + 20 + 23)/5 = 88/5 = 17.6.
2. The deviations are −5.6, −2.6, 0.4, 2.4 and 5.4.
3. Their squares sum to 31.36 + 6.76 + 0.16 + 5.76 + 29.16 = 73.2.
4. The population variance is 73.2/5 = 14.64.
5. The standard deviation is √14.64 ≈ 3.83.

Final answer. mean 17.6, standard deviation about 3.83

Interpretation. The sample standard deviation would divide by 4 instead of 5, and the convention must be stated.

CHAPTER 9 · 9.7 · WORKED EXAMPLE 2

Mixed practice

WORKED EXAMPLE 9.7.2

Problem. A line of best fit is $y = 2.4x + 8$. Predict y when $x = 15$ and comment on the prediction if the data ranged from $x = 5$ to $x = 12$.

step	working	result
prediction	$y = 2.4(15) + 8$	44
data range	5 to 12	—
type	$x = 15$ is outside the range	extrapolation
assumption	the linear relationship continues	stated

Extrapolation assumes the relationship continues.

Mathematical reasoning. Substitute, then classify the prediction by comparing x with the data range.

1. The prediction is $y = 2.4(15) + 8 = 44$.
2. The data ranged from $x = 5$ to $x = 12$.
3. Since 15 is outside that range, the prediction is an extrapolation.
4. Extrapolation assumes the linear relationship continues beyond the observed data.
5. The prediction should therefore be reported with that assumption stated.

Final answer. 44, an extrapolation assuming the relationship continues

Interpretation. The classification of the prediction is part of the answer, not an optional remark.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

9.A1 FLUENCY

Find the range of the data 3, 7, 8, 12, 12, 15, 20.

9.A2 FLUENCY

Find the mean of 4, 8, 8, 10, 15.

9.A3 FLUENCY

Find the median of 4, 8, 8, 10, 15.

9.A4 FLUENCY

Find the population standard deviation of 2, 4, 4, 4, 5, 5, 7, 9.

9.A5 REASONING

Find the sample standard deviation of 2, 4, 4, 4, 5, 5, 7, 9.

9.A6 REASONING

A data set has mean 20 and standard deviation 3. A second has mean 20 and standard deviation 9. Compare them.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

9.A7 FLUENCY

Find the interquartile range of 5, 7, 8, 9, 11, 13, 16, 18.

9.A8 REASONING

Find the mean and standard deviation of the frequency table x = 1, 2, 3 with frequencies 4, 6, 2.

9.A9 CHALLENGE

Compute the correlation coefficient for (1, 1), (2, 3), (3, 2), (4, 5), (5, 4).

9.A10 REASONING

Explain what a correlation coefficient of −0.95 indicates.

9.A11 FLUENCY

A regression line is y = 3.2x + 5. Predict y when x = 4.

9.A12 REASONING

Explain why the median is preferred to the mean for skewed data.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

9.B1 REASONING

Investigate the effect of adding a single extreme value to a data set on its range, interquartile range and standard deviation, and justify.

9.B2 REASONING

Conjecture a rule for when the sample and population standard deviations are close, and justify.

9.B3 REASONING

Investigate the correlation coefficient for data that lie exactly on a curve, and justify the result.

9.B4 REASONING

Take a data set of your own and show that multiplying every value by 10 multiplies the standard deviation by 10 but leaves the correlation coefficient unchanged.

9.B5 REASONING

Investigate the effect of an influential outlier on a regression line, and justify.

9.B6 CHALLENGE

Conjecture a rule for the relationship between the two regression lines, and test it.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

9.B7 CHALLENGE

Investigate whether a correlation coefficient can exceed 1, and justify.

9.B8 REASONING

Investigate the effect of multiplying every value in a data set by 5 on its mean, range and standard deviation, and justify.

9.B9 REASONING

Conjecture a rule for when the mean and the median of a data set are close, and justify.

9.B10 REASONING

Investigate the correlation coefficient for data lying exactly on a line with negative gradient, and justify.

9.B11 CHALLENGE

Investigate the effect of changing the units of one variable on the regression line of y on x, and justify.

CRITERION C Communicating

Criterion C practice · Communicating

9.C1 REASONING

Write a complete solution to finding the mean, median and standard deviation of 14, 16, 16, 18, 21, 24, 28.

9.C2 REASONING

Write a complete solution to comparing two data sets with the same mean but different spreads, stating what the comparison shows and what it does not.

9.C3 CHALLENGE

Write a complete solution to computing the correlation coefficient for (2, 5), (4, 7), (6, 6), (8, 10), (10, 11).

9.C4 CHALLENGE

Write a complete solution to finding the regression line of y on x for the data in 9.3.3.

9.C5 REASONING

Write a complete solution to the grouped-data problem: classes 0–9, 10–19, 20–29 with frequencies 5, 12, 8. Estimate the mean and standard deviation.

9.C6 REASONING

Write a complete solution to evaluating a claim that a correlation of 0.8 between two variables proves that one causes the other.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

9.C7 REASONING

Write a complete solution to finding the area of a triangle with vertices (0, 0), (3, 1) and (1, 4) using a determinant.

9.C8 REASONING

Write a complete solution to finding the mean, median and interquartile range of 21, 24, 24, 27, 30, 33, 36.

9.C9 CHALLENGE

Write a complete solution to computing the correlation coefficient for (1, 4), (2, 6), (3, 5), (4, 8), (5, 7).

9.C10 CHALLENGE

Write a complete solution to finding the regression line of y on x for the data in 9.3.9.

9.C11 REASONING

Write a complete solution to the grouped-data problem: classes 0–4, 5–9, 10–14 with frequencies 6, 10, 4. Estimate the mean and standard deviation.

9.C12 REASONING

Write a complete solution to evaluating a claim that a correlation of 0.85 between two variables proves that one causes the other.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

9.D1 APPLICATION

A school reports that its mean score rose from 62 to 68 while the standard deviation rose from 8 to 15. Evaluate the claim that results improved.

9.D2 APPLICATION

Two hospitals report recovery times with the same mean of 9 days. Hospital A has a standard deviation of 1.2 and hospital B of 4.5. Advise a patient who wants the most predictable recovery.

9.D3 APPLICATION

A researcher finds a correlation of 0.95 between the number of storks and the birth rate in a region. Evaluate the claim that storks deliver babies.

9.D4 APPLICATION

A shop models weekly sales as $y = 120 + 3.5x$ where x is the number of advertisements. Predict sales for 20 advertisements and comment on the prediction if the data ranged from 0 to 12.

9.D5 REASONING

A student computes a correlation of 0.3 and concludes the variables are unrelated. Investigate and justify.

9.D6 APPLICATION

A data set of 200 values has mean 45 and standard deviation 6. Estimate how many values lie within one standard deviation of the mean, and state the assumption.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

9.D7 APPLICATION

A newspaper reports that a study of 30 people found a correlation of 0.55 and concludes a link. Evaluate the report.

9.D8 APPLICATION

A farmer records yields and rainfall over 12 years and finds r = 0.72. Advise the farmer, stating what the analysis supports.

9.D9 REASONING

A student claims that because the standard deviation of a data set is 0, every value must be 0. Investigate and justify.

9.D10 REASONING

Two data sets have the same median and interquartile range but different means. Explain how this is possible and what it indicates.

9.D11 APPLICATION

A company compares two suppliers' delivery times. Supplier A: mean 4 days, standard deviation 0.5. Supplier B: mean 3.5 days, standard deviation 2.5. Advise the company.

9.D12 REASONING

A student extrapolates a regression line to predict a value twice the largest observed x and reports the result to three significant figures. Evaluate.

CHAPTER 9 · APPLICATION AND ASSESSMENT

Applying chapter 9

These tasks use measures of dispersion, correlation and regression. State which measure you use and why, give the formula you apply, and comment on what the data can support.

A1 · The two classes

Two classes take the same test. Class A has mean 64 and standard deviation 5; class B has mean 64 and standard deviation 14. Compare the two distributions using a measure of centre, a measure of dispersion and a statement of shape, and then explain what the comparison does and does not show about the teaching in the two classes.

Criteria C and D · compare centre, dispersion and shape, and evaluate the limits of the comparison.

A2 · The correlation claim

A study of 80 students finds a correlation of 0.71 between hours spent on a revision app and test score. The school proposes to require the app for all students. Evaluate the proposal, stating what the analysis supports, what it does not, and what evidence would be needed before the requirement is imposed.

Criteria C and D · interpret the correlation, identify the limits, and specify the evidence needed.

CHAPTER 9 · REVIEW

What you should now be able to do

Key ideas

- A single average describes a centre and nothing else.
- The range is the largest minus the smallest value.
- The interquartile range is the difference of the quartiles and resists outliers.
- The population variance divides by n; the sample variance divides by n − 1.
- The standard deviation is the square root of the variance and is in the units of the data.
- For grouped data, class mid-points stand in for the individual values.
- The correlation coefficient r lies between −1 and 1 and measures linear association.
- Correlation is not causation.
- The regression line of y on x minimises vertical distances.
- Interpolation is reliable; extrapolation assumes the relationship continues.

Vocabulary

range · interquartile range · variance · standard deviation · population · sample · frequency table · mid-point · skew · symmetric · box plot · scatter graph · correlation coefficient · covariance · regression line · interpolation · extrapolation · outlier · influential point

CHAPTER 9 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

9.R1 Find the mean and range of 6, 9, 11, 14, 20.

9.R2 Find the population standard deviation of 3, 3, 4, 5, 5.

9.R3 State the three aspects used to compare two distributions.

9.R4 Compute r for (1, 2), (2, 3), (3, 5), (4, 4), (5, 6).

9.R5 A regression line is $y = 1.2x + 4$. Predict y at $x = 8$ and classify the prediction if the data ranged from 2 to 6.

9.R6 Explain why a correlation of 0.9 does not prove causation.

CHAPTER 9 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> Two groups can share a centre and differ in every other respect; measuring dispersion and association describes those differences precisely, while the origin of the data limits what the description can prove.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of patterns, representation and measurement appear in your work?
- How is the global context of identities and relationships, and its exploration of how groups differ and how claims about them are made, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a correlation coefficient, and what would you still like to ask?

CHAPTER 9 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

A statistic describes a sample, not the world. Name the sample, the measure and the claim, and say what the claim cannot reach.

Service as Action

COMMUNITY NEED	Claims about groups in the community are published without the dispersion or the sample being stated.
MATHEMATICAL CONNECTION	Dispersion, correlation and regression.
PROPOSED ACTION	Take one published claim about a local group, recompute the centre and the dispersion, and publish what the claim can and cannot support.
EXPECTED IMPACT	Readers see the spread behind the average.
STUDENT REFLECTION	Record the claim, the data and the evaluation.

Community Project

PROJECT IDEA	A dispersion audit of a school report.
NEED	The school's averages are published without any measure of spread.
RESEARCH QUESTION	How much do the school's results vary within each subject, and what does that variation hide?
MATHEMATICS	Measures of dispersion and comparison of distributions.
ACTION OR OUTCOME	A one-page report with means and standard deviations together.
BENEFICIARIES	Students, families and staff.
EVIDENCE	The report and the calculations behind it.

Personal Project connection · MYP 5

PERSONAL INTEREST	Sport, music, gaming or fitness.
LEARNING GOAL	To measure a spread in something you track and state what the measure supports.
PRODUCT OR OUTCOME	A data card with a centre, a spread and a limitation.
MATHEMATICS	Dispersion, correlation and regression.
PROCESS	Collect data, compute the centre and the spread, and state one limitation.
EVIDENCE	The card and the data behind it.

CHAPTER 10 · PART IV · REASONING WITH DATA

10

Probability: Conditional Probability and Counting

Probability measures how likely an outcome is on a stated set of possibilities. Conditioning restricts that set, and counting tells us how large the set is, and both decide what a probability means.

ENGAGE

A medical test is 99 per cent accurate. A patient tests positive. What is the probability that the patient has the disease? Most people answer 99 per cent, and they are usually wrong. Why?

Framing the inquiry

Key concept: Logic
Related concepts: Relationships · Representation · Patterns
Global context: Fairness and development
Global context exploration: risk, screening and fair decisions

Statement of inquiry: Probability is a measure on a set of outcomes; conditioning restricts that set, and the difference between $P(A \mid B)$ and $P(B \mid A)$ is often the difference between a sound decision and a misleading one.

Questions to guide us

- **Factual:** What do conditional probability, independence and the total probability rule state?
- **Conceptual:** Why can $P(A \mid B)$ and $P(B \mid A)$ differ enormously, and what decides the difference?
- **Debatable:** If a screening test is 99 per cent accurate, is it fair to act on a positive result?

ATL · CRITICAL THINKING. In order for a student to judge a screening result, students must identify the conditioned set, compute the conditional probability, and justify the conclusion against the reverse conditional probability.

CHAPTER 10 · INVESTIGATION 10.1

Sets, tables and probability

INQUIRY QUESTION

What set is a probability a measure on?

YOUR TASK

Take a survey of your own class or group on two yes-or-no questions. Build a two-way table, draw a Venn diagram, and compute the probability of each event, of the union, of the intersection and of the complement. Check that the probabilities sum to one.

event	count	probability
plays a sport	14	14/30
plays an instrument	11	11/30
both	5	5/30
sport only	9	9/30
instrument only	6	6/30
neither	10	10/30
at least one	21	21/30

CHAPTER 10 · INVESTIGATION 10.1

Sets, tables and probability

What do you notice?

- The four groups of the Venn diagram summed to 30, the whole class.
- The union was 21, and the intersection was 5.
- The addition rule gave 14/30 + 11/30 − 5/30 = 20/30, which disagreed with the direct count of 21/30.
- The disagreement came from a mis-transcribed count, and re-checking gave 14/30 + 11/30 − 4/30 = 21/30.

> **Observation.** The addition rule and the direct count must agree, and a disagreement locates an error.

MATHEMATICAL CONCLUSION

A probability is a measure on a stated set of outcomes, and for equally likely outcomes it is a count divided by the total. Two-way tables and Venn diagrams are two representations of the same partition, and the addition rule corrects the double-counted overlap. Here the union was 21 out of 30.

Your own words

CHAPTER 10 · 10.1

Sets, tables and probability

A probability is a measure on a set of outcomes, and it is only meaningful once the set is stated. The same event can have different probabilities on different sets, and the logic of the calculation depends entirely on which set is meant, which is why the sample space is named first.

For equally likely outcomes, the probability of an event is the number of favourable outcomes divided by the total number of outcomes. The formula is a counting statement, and its correctness depends entirely on the outcomes being equally likely.

Two-way tables organise two categorical variables, and the probability of an event is a count divided by the relevant total. This representation makes the relationships between the categories visible, and reading the correct total is the whole skill, because the denominator changes with the conditioning.

Venn diagrams display the union, intersection and complement of events, and their patterns of overlap are what the rules of probability encode. The addition rule P(A or B) = P(A) + P(B) − P(A and B) corrects for the double-counted overlap, and mutually exclusive events have no overlap and simply add.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 10 · 10.1 · WORKED EXAMPLE 1

Sets, tables and probability

WORKED EXAMPLE 10.1.1

Problem. A fair die is rolled. Find the probability that the score is even or greater than 4.

outcome	even?	greater than 4?	counted once?
1	no	no	no
2	yes	no	yes
4	yes	no	yes
5	no	yes	yes
6	yes	yes	yes, but only once

The overlap must be counted once.

Mathematical reasoning. Count the favourable outcomes directly, and check with the addition rule.

1. The sample space is 1, 2, 3, 4, 5, 6, with six equally likely outcomes.
2. The even outcomes are 2, 4, 6, so P(even) = 3/6.
3. The outcomes greater than 4 are 5, 6, so P(> 4) = 2/6.
4. The overlap is 6 alone, so P(even and > 4) = 1/6.
5. By the addition rule, P(even or > 4) = 3/6 + 2/6 − 1/6 = 4/6 = 2/3.

Final answer. 2/3

Interpretation. The direct count and the addition rule agree, and the overlap is the reason the rule subtracts.

CHAPTER 10 · 10.1 · WORKED EXAMPLE 2

Sets, tables and probability

WORKED EXAMPLE 10.1.2

Problem. In a class of 30, 18 study French and 14 study Spanish, and 6 study both. Find the probability that a student studies at least one language.

group	count	probability
French only	12	12/30
Spanish only	8	8/30
both	6	6/30
neither	4	4/30
at least one	26	26/30

The four groups partition the class.

Mathematical reasoning. Partition the class and add the three relevant groups.

1. French only is $18 - 6 = 12$ students.
2. Spanish only is $14 - 6 = 8$ students.
3. Both is 6, and neither is $30 - 12 - 8 - 6 = 4$.
4. So at least one language is $12 + 8 + 6 = 26$ students.
5. The probability is $26/30 = 13/15$.

Final answer. 13/15

Interpretation. The four groups sum to the whole class, which is the check that the partition is complete.

CHAPTER 10 · INVESTIGATION 10.2

Conditional probability

INQUIRY QUESTION

What does conditioning change?

YOUR TASK

Take a screening scenario of your own with a stated prevalence and a stated accuracy. Build the two-way table, compute P(positive | disease), P(disease | positive), and the two false-positive and false-negative rates. Then repeat with the prevalence ten times higher and explain the change.

quantity	prevalence 1 per cent	prevalence 10 per cent
true positives	9	90
false positives	9	81
false negatives	1	10
true negatives	981	810
P(positive disease)	0.90	0.90
P(disease positive)	0.50	0.53
P(disease negative)	0.001	0.012

CHAPTER 10 · INVESTIGATION 10.2

Conditional probability

What do you notice?

- The accuracy P(positive | disease) stayed at 0.90, because it depends only on the test.
- The predictive value P(disease | positive) stayed near 0.5, because the false positives grew with the population.
- Raising the prevalence tenfold increased both predictive values, but only slightly.
- The false-negative rate stayed at 10 per cent, again because it depends only on the test.

Observation. The test's accuracy is a property of the test, and the predictive value is a property of the test together with the prevalence.

MATHEMATICAL CONCLUSION

Conditioning restricts the set of outcomes, and the conditional probability is computed on the restricted set. The two conditional probabilities P(A | B) and P(B | A) are generally different, and the difference is governed by the prevalence of the conditioning event. Here the test was 90 per cent accurate, but the predictive value was only about 0.5.

Your own words

CHAPTER 10 · 10.2

Conditional probability

The conditional probability $P(A \mid B)$ is the probability of A given that B has occurred, and it is defined as $P(A \text{ and } B)/P(B)$. The denominator is the probability of the conditioning event, not the whole sample space.

Conditioning restricts the set of outcomes under consideration. Every probability is computed on a smaller set, and the answer is a probability on that restricted set rather than on the original one.

The definition is a ratio, and it can be rearranged as the multiplication rule $P(A \text{ and } B) = P(B) \times P(A \mid B)$. This is the form used in tree diagrams, where the second branch is conditional.

The reverse conditional probability is found from Bayes' theorem, $P(B \mid A) = P(A \mid B)P(B)/P(A)$. The two conditional probabilities are generally different, and confusing them is the error the inquiry statement names.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 10 · 10.2 · WORKED EXAMPLE 1

Conditional probability

WORKED EXAMPLE 10.2.1

Problem. A fair die is rolled. Given that the score is even, find the probability that it is greater than 4.

outcome	even?	greater than 4?	in the restricted set?
1	no	no	no
2	yes	no	yes
4	yes	no	yes
6	yes	yes	yes

The restricted set has three outcomes.

Mathematical reasoning. Restrict the sample space to the conditioning event and count within it.

1. The conditioning event is that the score is even, so the restricted set is 2, 4, 6.
2. This set has three equally likely outcomes.
3. The favourable ones, greater than 4, are just 6.
4. So $P(> 4 \mid \text{even}) = 1/3$.
5. Check with the formula: $P(> 4 \text{ and even})/P(\text{even}) = (1/6)/(3/6) = 1/3$.

Final answer. 1/3

Interpretation. The denominator changed from 6 to 3, and that change is the whole content of conditioning.

CHAPTER 10 · 10.2 · WORKED EXAMPLE 2

Conditional probability

WORKED EXAMPLE 10.2.2

Problem. In a group of 100 people, 10 have a disease. A test is positive for 9 of the 10 who have it and for 9 of the 90 who do not. Find the probability that a person who tests positive has the disease.

group	test positive	test negative	total
has the disease	9	1	10
does not	9	81	90
total	18	82	100

The column total for positive tests is the denominator.

Mathematical reasoning. Condition on the event of testing positive and divide by the number who test positive.

1. The number who test positive is $9 + 9 = 18$.
2. Of those, the number who actually have the disease is 9.
3. So $P(\text{disease} \mid \text{positive}) = 9/18 = 0.5$.
4. Note that $P(\text{positive} \mid \text{disease}) = 9/10 = 0.9$, which is a different number.
5. The difference arises because the disease is rare, so the false positives are numerous.

Final answer. 0.5

Interpretation. This is the base-rate effect: a very accurate test on a rare condition still gives many false positives among the positives.

CHAPTER 10 · INVESTIGATION 10.3

Independence

INQUIRY QUESTION

When does conditioning make no difference?

YOUR TASK

Take six pairs of events from a single experiment of your own — for example two dice, a pack of cards, or a spinner. For each pair, decide whether the events are independent, justifying the answer with the product rule. Include at least one pair that feels independent but is not, and one that feels related but is.

pair of events	P(A)	P(B)	P(A and B)	independent?
first die even, second die even	1/2	1/2	1/4	yes
first die even, total is even	1/2	1/2	1/4	yes
first die even, total is 7	1/2	1/6	1/12	no
card is a heart, card is red	1/4	1/2	1/4	no
card is a heart, card is a king	1/4	1/13	1/52	yes
total is even, total is prime	1/2	5/36	1/36	no

CHAPTER 10 · INVESTIGATION 10.3

Independence

What do you notice?

- The pair that felt independent but was not was 'first die even, total is 7', because the total depends on both dice.
- The pair that felt related but was independent was 'card is a heart, card is a king', since the suits and ranks are separate.
- Every decision was made by the product rule, not by intuition.
- Containment always forces dependence, as the heart-and-red pair showed.

Observation. Independence is decided by the product rule, and intuition about it is unreliable.

MATHEMATICAL CONCLUSION

Two events are independent exactly when P(A and B) = P(A)P(B), equivalently when conditioning on one does not change the probability of the other. Independence must be justified in the model, because intuition about it is unreliable, and mutually exclusive events are independent only in the trivial case. Here four of the six pairs were independent.

Your own words

CHAPTER 10 · 10.3

Independence

Two events are independent when $P(A \mid B) = P(A)$, which is equivalent to $P(A \text{ and } B) = P(A)P(B)$. Conditioning on one event does not change the probability of the other, and that is the whole content of independence.

Independence is a property of a pair of events in a stated model, and it must be justified rather than assumed. Two events that feel unrelated can be dependent, and two that feel related can be independent.

Mutually exclusive events are different from independent events. If A and B cannot both occur, then $P(A \text{ and } B) = 0$, so they are independent only if one of them has probability zero.

For more than two events, pairwise independence does not imply mutual independence, and a model must be checked against every combination. The distinction matters as soon as three or more events are involved.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 10 · 10.3 · WORKED EXAMPLE 1

Independence

WORKED EXAMPLE 10.3.1

Problem. A fair coin is tossed twice. Show that the events 'first toss is heads' and 'second toss is heads' are independent.

outcome	first heads?	second heads?	both?
HH	yes	yes	yes
HT	yes	no	no
TH	no	yes	no
TT	no	no	no

All four outcomes are equally likely.

Mathematical reasoning. Check whether P(A and B) equals the product P(A)P(B).

1. The sample space is HH, HT, TH, TT, each with probability 1/4.
2. P(first heads) = 2/4 = 1/2 and P(second heads) = 2/4 = 1/2.
3. P(both heads) = 1/4.
4. The product is (1/2)(1/2) = 1/4, which equals P(both heads).
5. So the events are independent.

Final answer. they are independent, because P(A and B) = P(A)P(B)

Interpretation. The check is one multiplication, and it is the definition applied.

CHAPTER 10 · 10.3 · WORKED EXAMPLE 2

Independence

WORKED EXAMPLE 10.3.2

Problem. A card is drawn from a standard pack. Let A be 'the card is a heart' and B be 'the card is red'. Are A and B independent?

event	probability	comment
A	13/52	—
B	26/52	—
A and B	13/52	every heart is red
P(A)P(B)	(13/52)(26/52)	not equal to P(A and B)
conclusion	not independent	—

Containment forces dependence.

Mathematical reasoning. Compare P(A and B) with the product P(A)P(B).

1. P(A) = 13/52 and P(B) = 26/52.
2. Since every heart is red, P(A and B) = P(A) = 13/52.
3. The product is (13/52)(26/52) = 338/2704 ≈ 0.125.
4. But P(A and B) = 13/52 = 0.25.
5. These differ, so the events are not independent.

Final answer. not independent

Interpretation. When one event implies the other, they cannot be independent unless one has probability zero or one.

CHAPTER 10 · INVESTIGATION 10.4

Tree diagrams and total probability

INQUIRY QUESTION

How does a tree diagram keep the denominators right?

YOUR TASK

Take a two-stage experiment of your own with unequal branch probabilities — for example two routes to school with different probabilities of being late. Build the tree diagram, compute the probability of each path, the total probability of the final event, and the reverse conditional probability. Check that the probabilities at each node sum to one.

path	probability	late?	contribution
route 1	0.70	0.10	0.070
route 2	0.30	0.40	0.120
total late	1.00	—	0.190
not late	1.00	—	0.810
route 2 given late	0.120 ÷ 0.190	—	0.632

CHAPTER 10 · INVESTIGATION 10.4

Tree diagrams and total probability

What do you notice?

- The probabilities at the first node summed to 1, as they had to.
- The two late branches contributed 0.070 and 0.120, summing to 0.190.
- The not-late branches contributed 0.630 and 0.180, summing to 0.810.
- The reverse conditional probability was 0.632, so most lateness came from the less-used route.

Observation. A tree diagram keeps the conditional denominators visible, and the total probability rule combines the branches.

MATHEMATICAL CONCLUSION

A tree diagram displays conditional probabilities along paths, and the probability of a path is the product of its branch probabilities. The law of total probability combines the branches at one level, and Bayes' theorem reverses the conditioning by dividing by that total. Here the total probability of lateness was 0.190 and the reverse conditional probability was 0.632.

Your own words

CHAPTER 10 · 10.4

Tree diagrams and total probability

A tree diagram displays a sequence of events as branches, and the probability along a path is the product of the branch probabilities. The second-level branches are conditional, which is what keeps the denominators correct.

The law of total probability states that $P(A) = P(A \mid B)P(B) + P(A \mid \text{not } B)P(\text{not } B)$, and its general form sums over a partition. It is the rule that combines the branches at one level.

Bayes' theorem follows from the definition and the total probability rule, and it converts $P(A \mid B)$ into $P(B \mid A)$ by weighting with the prior probabilities. It is the formal version of the base-rate correction.

A tree diagram is checked by verifying that the probabilities leaving each node sum to one, and that the path probabilities sum to one at the final level. Both checks are quick and catch most errors.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 10 · 10.4 · WORKED EXAMPLE 1

Tree diagrams and total probability

WORKED EXAMPLE 10.4.1

Problem. A factory has two machines. Machine A makes 60 per cent of the output with a 2 per cent defect rate, and machine B makes the rest with a 5 per cent defect rate. Find the probability that a randomly chosen item is defective.

path	probability	defect probability	contribution
from A	0.60	0.02	0.012
from B	0.40	0.05	0.020
total	1.00	—	0.032

Each branch contributes its own product.

Mathematical reasoning. Apply the law of total probability over the two machines.

1. The item comes from A with probability 0.60 and from B with 0.40.
2. Given A, the defect probability is 0.02, contributing $0.60 \times 0.02 = 0.012$.
3. Given B, it is 0.05, contributing $0.40 \times 0.05 = 0.020$.
4. The total probability is $0.012 + 0.020 = 0.032$.
5. So 3.2 per cent of items are defective.

Final answer. 0.032

Interpretation. The two branches partition the sample space, and their contributions add.

CHAPTER 10 · 10.4 · WORKED EXAMPLE 2

Tree diagrams and total probability

WORKED EXAMPLE 10.4.2

Problem. In the same factory, a defective item is found. Find the probability that it came from machine B.

step	working	result
P(defective and from B)	0.40 × 0.05	0.020
P(defective)	0.032	—
P(from B ∣ defective)	0.020 ÷ 0.032	0.625

Bayes' theorem divides by the total.

Mathematical reasoning. Apply Bayes' theorem using the total probability already computed.

1. The joint probability of being from B and defective is 0.020.
2. The total probability of being defective is 0.032.
3. So P(from B ∣ defective) = 0.020/0.032 = 0.625.
4. Although B makes only 40 per cent of the output, it accounts for 62.5 per cent of the defects.
5. The reason is its higher defect rate, 5 per cent against 2 per cent.

Final answer. 0.625

Interpretation. The reverse conditional probability is larger than the prior, because the conditioning event is more likely under B.

CHAPTER 10 · INVESTIGATION 10.5

Counting, permutations and combinations

INQUIRY QUESTION

How large is the set of outcomes?

YOUR TASK

Take a set of 8 objects of your own. Count the number of permutations of all 8, the number of permutations of 3 chosen from the 8, the number of combinations of 3 chosen from the 8, and the number of ways to split the 8 into two groups of 4. Verify the relationship between the permutation and combination counts.

quantity	formula	value
permutations of all 8	$8!$	40320
permutations of 3 from 8	$8!/(8-3)!$	336
combinations of 3 from 8	$8!/(3! \times 5!)$	56
splits into two groups of 4	$8!/(4! \times 4! \times 2)$	35
check	$336 \div 3! = 56$	confirmed

CHAPTER 10 · INVESTIGATION 10.5

Counting, permutations and combinations

What do you notice?

- The permutation count was exactly 6 times the combination count, as the relationship requires.
- The split into two equal groups needed an extra division by 2, because the two groups are interchangeable.
- All four counts were integers, as counts must be.
- The smallest case, choosing 1 from 8, gave 8, which could be listed directly.

Observation. Permutations count orders and combinations do not, and each combination corresponds to r! permutations.

MATHEMATICAL CONCLUSION

Counting determines how large the set of outcomes is, and the multiplication principle is applied by multiplying the numbers of choices at each stage. Permutations count arrangements and combinations count selections, with each combination corresponding to r! permutations. Here the counts were 40 320, 336, 56 and 35.

Your own words

CHAPTER 10 · 10.5

Counting, permutations and combinations

The multiplication principle states that if one choice can be made in m ways and a second in n ways, the pair can be made in mn ways. It is the basis of all counting, and it is applied by multiplying the numbers of choices at each stage.

A permutation is an arrangement in which order matters, and the number of permutations of n objects taken r at a time is $n!/(n-r)!$. A combination is a selection in which order does not matter, and the number is $n!/(r!(n-r)!)$.

The relationship between the two is that each combination of r objects corresponds to $r!$ permutations, so the combination count is the permutation count divided by $r!$. This is why the combination formula has the extra factor in the denominator.

Counting is what makes a probability computable for equally likely outcomes, so an error in counting is an error in the probability. The check is that the count is an integer and that a small case can be listed.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 10 · 10.5 · WORKED EXAMPLE 1

Counting, permutations and combinations

WORKED EXAMPLE 10.5.1

Problem. How many ways can 5 people be arranged in a row?

position	choices	running total
first	5	5
second	4	20
third	3	60
fourth	2	120
fifth	1	120

Each position has one fewer choice.

Mathematical reasoning. Apply the multiplication principle stage by stage.

1. The first position can be filled in 5 ways.
2. The second in 4 ways, the third in 3, the fourth in 2 and the fifth in 1.
3. By the multiplication principle the total is $5 \times 4 \times 3 \times 2 \times 1 = 120$.
4. This is 5 factorial, written 5!.
5. So there are 120 arrangements.

Final answer. 120

Interpretation. The multiplication principle is applied by multiplying the numbers of choices at each stage.

CHAPTER 10 · 10.5 · WORKED EXAMPLE 2

Counting, permutations and combinations

WORKED EXAMPLE 10.5.2

Problem. From 10 people, a committee of 3 is chosen. How many different committees are possible?

step	working	result
permutations	$10 \times 9 \times 8$	720
arrangements per committee	$3 \times 2 \times 1$	6
committees	$720 \div 6$	120

Each committee is counted 6 times as permutations.

Mathematical reasoning. Count the ordered selections, then divide by the number of orders of each committee.

1. Choosing 3 people in order gives $10 \times 9 \times 8 = 720$ ordered selections.
2. Each committee of 3 can be ordered in $3 \times 2 \times 1 = 6$ ways.
3. So the number of committees is $720/6 = 120$.
4. This is $10!/(3! \times 7!) = 120$.
5. The division by 6 is the correction for order not mattering.

Final answer. 120

Interpretation. The combination count is the permutation count divided by the number of orders, which is why the formula has the extra factor.

CHAPTER 10 · INVESTIGATION 10.6

Investigation: the birthday problem

INQUIRY QUESTION

Why is a rare event so likely in a large enough group?

YOUR TASK

Investigate the birthday problem for groups of 5, 10, 15, 20, 23, 25 and 30. Compute the number of pairs and the probability of at least one shared birthday for each, and plot the probability against the group size. Then explain why the answer is smaller than most people expect.

group size	number of pairs	probability of a shared birthday
5	10	0.027
10	45	0.117
15	105	0.253
20	190	0.411
23	253	0.507
25	300	0.569
30	435	0.706

CHAPTER 10 · INVESTIGATION 10.6

Investigation: the birthday problem

What do you notice?

- The probability crossed one half at 23 people, not at 183 as many expect.
- The number of pairs grew quadratically, from 10 to 435.
- People expect the comparison to be with 365, but the comparison is between pairs.
- The same reasoning explains why a lottery has a winner most weeks.

Observation. A rare event becomes likely when there are many opportunities for it, and the number of opportunities grows quadratically with the group size.

MATHEMATICAL CONCLUSION

The birthday problem is solved by the complement, whose probability is a product of terms that shrink as the group grows. The number of pairs grows as $n(n-1)/2$, which is why 23 people are enough for a probability above one half. The general principle is that a rare event becomes likely when there are many opportunities for it.

Your own words

CHAPTER 10 · 10.6

Investigation: the birthday problem

The birthday problem asks how many people are needed for the probability of a shared birthday to exceed one half. The answer, 23, is far smaller than most people expect, and the reason is that the number of pairs grows quadratically.

The calculation is easiest by complement: the probability of no shared birthday is the product of the probabilities that each new person avoids all previous birthdays. With 23 people this product is about 0.493, so the probability of a match is about 0.507.

The number of pairs among n people is $n(n-1)/2$, which is the source of the quadratic growth. This is why doubling the group size roughly quadruples the number of comparisons.

The general principle is that a rare event becomes likely when there are many opportunities for it. The same reasoning explains why a lottery winner exists every week even though no individual ticket is likely to win.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 10 · 10.6 · WORKED EXAMPLE 1

Investigation: the birthday problem

WORKED EXAMPLE 10.6.1

Problem. Find the probability that two people share a birthday, and the probability that three people do.

group size	number of pairs	probability of a shared birthday
2	1	1/365 ≈ 0.0027
3	3	≈ 0.0082
4	6	≈ 0.0164
5	10	≈ 0.0271

The number of pairs grows quadratically.

Mathematical reasoning. Count the pairs and use the complement for the larger groups.

1. With 2 people there is 1 pair, and the probability of a shared birthday is 1/365.
2. With 3 people there are 3 pairs, and the probability of a match is $1 - (364/365)(363/365) \approx 0.0082$.
3. With 4 people there are 6 pairs, giving about 0.0164.
4. With 5 people there are 10 pairs, giving about 0.0271.
5. The probabilities roughly double each time, because the number of pairs does.

Final answer. 1/365 for two people; about 0.0082 for three

Interpretation. The pair count is the engine of the calculation, and it grows quadratically.

CHAPTER 10 · 10.6 · WORKED EXAMPLE 2

Investigation: the birthday problem

WORKED EXAMPLE 10.6.2

Problem. Show that with 23 people the probability of a shared birthday exceeds one half.

n	pairs	probability of no shared birthday	probability of a match
10	45	0.883	0.117
20	190	0.589	0.411
23	253	0.493	0.507
30	435	0.294	0.706

The complement crosses one half at 23 people.

Mathematical reasoning. Compute the complement product and subtract from one.

1. The probability of no shared birthday among n people is the product (364/365)(363/365)... ((366 − n)/365).
2. For n = 23 this product is about 0.493.
3. So the probability of at least one shared birthday is 1 − 0.493 = 0.507.
4. This exceeds one half, so 23 people are enough.
5. For n = 30 the probability is about 0.706, and for n = 50 it is about 0.970.

Final answer. with 23 people the probability is about 0.507

Interpretation. The complement is easier because it is a single product, whereas counting matches requires inclusion and exclusion.

CHAPTER 10 · INVESTIGATION 10.7

Mixed practice

INQUIRY QUESTION

What is the set, and what is the condition?

YOUR TASK

Take a screening scenario with a prevalence of your own choice and a test with a stated sensitivity and specificity. Build the table for 100 000 people, compute the probability that a positive result is correct, and then repeat with the prevalence multiplied by 10. Then count the number of ways to arrange the letters of a word of your choice with repeated letters.

step	working	result
prevalence	1 in 1000	100 true cases
sensitivity	99 per cent	99 true positives
false positives	1 per cent of 99 900	999
P(disease positive)	99 ÷ 1098	0.0902
prevalence × 10	1000 true cases, 990 true positives	990 ÷ 1989 ≈ 0.498
word arrangements	6!/(2! × 2!) for a word with two pairs	180

CHAPTER 10 · INVESTIGATION 10.7

Mixed practice

What do you notice?

- Raising the prevalence tenfold raised the predictive value from 0.0902 to about 0.498.
- The sensitivity stayed at 99 per cent, because it is a property of the test.
- The word arrangement count divided by the factorials of the repeat counts.
- Both calculations were checks on the same underlying counting idea.

Observation. The predictive value depends on the prevalence as well as on the test, and counting corrections come from repeated identical objects.

MATHEMATICAL CONCLUSION

A probability is a measure on a stated set, conditioning restricts that set, and counting determines how large it is. The base-rate effect arises because the number of false positives scales with the healthy population while the true positives scale with the small diseased one. Here the predictive value was 0.0902 at a prevalence of 1 in 1000 and about 0.498 at 1 in 100.

Your own words

CHAPTER 10 · 10.7

Mixed practice

Every probability question begins by naming the sample space and the event. Without the set, a probability has no meaning, and the same number can describe different situations.

Conditional questions are recognised by the word 'given'. The denominator is the probability of the conditioning event, and writing the definition down before computing prevents the commonest error.

Counting questions are recognised by the words 'how many'. The multiplication principle, permutations and combinations cover almost all of them, and the check is that the count is an integer and a small case can be listed.

Mixed practice is where the method is chosen rather than given, so the reasoning behind the choice is part of what is being practised.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 10 · 10.7 · WORKED EXAMPLE 1

Mixed practice

WORKED EXAMPLE 10.7.1

Problem. A disease affects 1 person in 1000. A test is positive for 99 per cent of those who have it and for 1 per cent of those who do not. Find the probability that a person who tests positive has the disease.

group	test positive	test negative	total
has the disease	99	1	100
does not	999	98 901	99 900
total	1098	98 902	100 000

Out of 100 000 people, 1098 test positive.

Mathematical reasoning. Scale the table to whole numbers, then divide the true positives by the total positives.

1. Take 100 000 people: 100 have the disease and 99 900 do not.
2. Of the 100 with the disease, 99 test positive.
3. Of the 99 900 without, 1 per cent, or 999, test positive.
4. So 99 + 999 = 1098 people test positive.
5. The probability is 99/1098 ≈ 0.0902, about 9 per cent.

Final answer. about 0.0902

Interpretation. A test that is 99 per cent accurate on a rare disease gives a positive result that is wrong about 91 per cent of the time, which is the base-rate effect.

CHAPTER 10 · 10.7 · WORKED EXAMPLE 2

Mixed practice

WORKED EXAMPLE 10.7.2

Problem. How many ways can the letters of the word LEVEL be arranged?

step	working	result
all arrangements	5!	120
identical letters	2 L's and 2 E's	2! × 2!
distinct arrangements	120 ÷ 4	30

Repeated letters divide the count.

Mathematical reasoning. Count all arrangements and divide by the factorial of each repeat count.

1. There are 5 letters, so 5! = 120 arrangements if all were distinct.
2. But there are 2 L's and 2 E's, and swapping identical letters gives the same word.
3. Each distinct word is counted 2! × 2! = 4 times.
4. So the number of distinct arrangements is 120/4 = 30.
5. The count is an integer, as a count must be.

Final answer. 30

Interpretation. Repeated letters divide the count, which is the same correction that turns permutations into combinations.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

10.A1 FLUENCY
A fair die is rolled. Find P(even).

10.A2 FLUENCY
A card is drawn from a pack. Find P(heart or king).

10.A3 FLUENCY
Two coins are tossed. Find P(two heads).

10.A4 FLUENCY
P(A) = 0.4 and P(B) = 0.5, and A and B are independent. Find P(A and B).

10.A5 REASONING
P(A) = 0.3 and P(B) = 0.6, and P(A and B) = 0.15. Are A and B independent?

10.A6 FLUENCY
A bag has 3 red and 5 blue balls. Two are drawn without replacement. Find P(both red).

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

10.A7 FLUENCY

A fair coin is tossed three times. Find P(exactly two heads).

10.A8 FLUENCY

P(A) = 0.6. Find P(not A).

10.A9 REASONING

In a class of 25, 15 play football and 12 play basketball, and 5 play both. Find P(plays neither).

10.A10 FLUENCY

How many ways can 4 books be arranged on a shelf?

10.A11 FLUENCY

How many ways can a committee of 2 be chosen from 7 people?

10.A12 REASONING

A test is 95 per cent accurate on a disease affecting 1 in 1000. Estimate P(disease | positive) and comment.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

10.B1 CHALLENGE

Investigate the birthday problem for groups of 10, 20, 23, 30 and 40, and justify the shape of the graph.

10.B2 CHALLENGE

Conjecture a rule for the smallest group size giving a probability above one half, and test it on a year of 200 days.

10.B3 REASONING

Investigate whether two events that are mutually exclusive can be independent, and justify.

10.B4 CHALLENGE

Take a three-stage experiment of your own and investigate whether pairwise independence implies mutual independence.

10.B5 REASONING

Investigate the number of arrangements of the letters of a word of your choice with repeated letters, and justify the correction.

10.B6 REASONING

Conjecture a rule for the number of diagonals of an n-sided polygon and test it.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

10.B7 CHALLENGE
Investigate the probability that at least two people in a group of 30 share a star sign, and justify the difference from the birthday problem.

10.B8 CHALLENGE
Investigate the probability of at least one shared birthday in a group of 50, and justify the method.

10.B9 REASONING
Conjecture a rule for the number of ways to choose a president, a secretary and a treasurer from n people, and test it.

10.B10 REASONING
Investigate whether two independent events can be mutually exclusive, and justify.

10.B11 REASONING
Investigate the probability that at least one of two dice shows a six, and justify the method.

CRITERION C Communicating

Criterion C practice · Communicating

10.C1 FLUENCY

Write a complete solution to finding P(A | B) when P(A) = 0.5, P(B) = 0.4 and P(A and B) = 0.2.

10.C2 REASONING

Write a complete solution to a screening problem: prevalence 2 per cent, sensitivity 90 per cent, false-positive rate 5 per cent. Find P(disease | positive).

10.C3 CHALLENGE

Write a complete solution to applying Bayes' theorem to a factory with two machines, A making 70 per cent at a 1 per cent defect rate and B making 30 per cent at a 4 per cent defect rate.

10.C4 FLUENCY

Write a complete solution to counting the number of ways to choose a president, a secretary and a treasurer from 9 people.

10.C5 FLUENCY

Write a complete solution to counting the number of ways to choose 3 toppings from 8 available.

10.C6 REASONING

Write a complete solution to proving the law of total probability for a partition of the sample space into B and not B.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

10.C7 CHALLENGE
Write a complete solution to the birthday problem for a group of 25, giving the answer to three decimal places.

10.C8 FLUENCY
Write a complete solution to finding P(A | B) when P(B) = 0.35 and P(A and B) = 0.14.

10.C9 REASONING
Write a complete solution to a screening problem: prevalence 5 per cent, sensitivity 92 per cent, false-positive rate 6 per cent. Find P(disease | positive).

10.C10 CHALLENGE
Write a complete solution to applying Bayes' theorem to two suppliers, X supplying 75 per cent at a 2 per cent fault rate and Y the rest at 6 per cent.

10.C11 FLUENCY
Write a complete solution to counting the number of ways to choose 4 toppings from 11 available.

10.C12 CHALLENGE
Write a complete solution to the birthday problem for a group of 40, to three decimal places.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

10.D1 APPLICATION

A rare disease affects 1 in 5000. A test has sensitivity 99 per cent and a false-positive rate of 1 per cent. Find P(disease | positive) and comment on the result.

10.D2 APPLICATION

A hospital screens 20 000 people for a condition affecting 4 per cent. The test catches 95 per cent of cases and falsely flags 3 per cent of the rest. How many true and false positives are there, and what fraction of positives are genuine?

10.D3 APPLICATION

Two factories supply components. Factory X supplies 55 per cent with a 3 per cent fault rate and factory Y the rest with a 7 per cent rate. A faulty component is found. Find the probability it came from Y.

10.D4 APPLICATION

A security system has two sensors. Sensor A fires with probability 0.8 when there is an intruder and 0.1 otherwise; sensor B fires with probability 0.9 and 0.2. The system alerts only if both fire. Find the probability of an alert when there is an intruder and when there is not, assuming independence given the state.

10.D5 APPLICATION

A lottery sells 1 000 000 tickets and 1000 prizes. A player buys 10 tickets. Find the probability of winning at least one prize, to three decimal places.

10.D6 REASONING

A student says that because a coin has landed heads five times in a row, tails is now more likely. Investigate and justify.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

10.D7 APPLICATION

A doctor tells a patient that a positive result means a 95 per cent chance of disease. The test has 95 per cent sensitivity and the disease prevalence is 1 in 200. Evaluate the statement.

10.D8 APPLICATION

A school chooses a committee of 4 from 12 students and 8 staff. Find the number of committees with at least one student.

10.D9 FLUENCY

A password uses 3 letters followed by 2 digits, with repetition allowed. Find the number of possible passwords.

10.D10 CHALLENGE

A bag contains 4 red, 5 blue and 6 green counters. Three are drawn without replacement. Find the probability that all three are the same colour.

10.D11 REASONING

A student claims that if two events each have probability 0.5 and are independent, then their union has probability 1. Investigate and justify.

10.D12 CHALLENGE

A fair die is rolled until a six appears. Find the probability that this takes more than 4 rolls.

CHAPTER 10 · APPLICATION AND ASSESSMENT

Applying chapter 10

These tasks use probability rules and counting. State the set of outcomes, name the rule you use, and check that the probabilities sum to one.

A1 · The screening programme

A disease affects 2 people in 1000. A test has a sensitivity of 98 per cent and a false-positive rate of 4 per cent. A screening programme tests 50 000 people. Find the number of true positives and false positives, the probability that a person with a positive result has the disease, and the number of people who would be unnecessarily treated if every positive result were treated. Advise the programme.

Criteria C and D · build the table, compute the predictive value, and justify the advice.

A2 · The two factories

Factory A supplies 60 per cent of a component with a 1 per cent defect rate, and factory B supplies the rest with a 5 per cent rate. A defective component is found. Find the probability that it came from factory B, and then find the probability that two components drawn at random are both defective and from the same factory. Explain each step of your reasoning.

Criteria C and D · apply the total probability rule and Bayes' theorem, then combine with independence.

CHAPTER 10 · REVIEW

What you should now be able to do

Key ideas

- A probability is a measure on a stated set of outcomes.
- For equally likely outcomes, probability is a count divided by the total.
- The addition rule is P(A or B) = P(A) + P(B) − P(A and B).
- Conditional probability is P(A | B) = P(A and B)/P(B).
- Independence means P(A and B) = P(A)P(B).
- Mutually exclusive events are independent only in the trivial case.
- The law of total probability sums the contributions over a partition.
- Bayes' theorem reverses the conditioning.
- The multiplication principle multiplies the numbers of choices at each stage.
- Permutations count order; combinations do not.

Vocabulary

sample space · event · outcome · equally likely · union · intersection · complement · mutually exclusive · conditional probability · independent · tree diagram · law of total probability · Bayes' theorem · sensitivity · specificity · prevalence · permutation · combination · factorial

CHAPTER 10 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

10.R1 $P(A) = 0.3$, $P(B) = 0.5$ and $P(A \text{ and } B) = 0.1$. Find $P(A \mid B)$.

10.R2 Two dice are rolled. Find P(total is 7).

10.R3 How many ways can 5 people be seated in a row of 5 chairs?

10.R4 How many ways can 3 cards be chosen from 10?

10.R5 A disease affects 1 in 100. A test is 90 per cent sensitive with a 10 per cent false-positive rate. Find P(disease | positive).

10.R6 Explain the difference between $P(A \mid B)$ and $P(B \mid A)$ in the context of a screening test.

CHAPTER 10 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

Probability is a measure on a set of outcomes; conditioning restricts that set, and the difference between $P(A \mid B)$ and $P(B \mid A)$ is often the difference between a sound decision and a misleading one.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of logic and the related concepts of relationships, representation and patterns appear in your work?
- How is the global context of fairness and development, and its exploration of risk, screening and fair decisions, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a probability as a guide to a decision, and what would you still like to ask?

CHAPTER 10 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

A probability is a measure on a stated set. Name the set, the condition and the rule, and say what the number does not mean.

Service as Action

COMMUNITY NEED	Screening and risk claims in the community are published without the prevalence or the false-positive rate.
MATHEMATICAL CONNECTION	Conditional probability and counting.
PROPOSED ACTION	Take one published screening claim, rebuild the two-way table, and publish what a positive result actually means.
EXPECTED IMPACT	Readers can judge a positive result for themselves.
STUDENT REFLECTION	Record the claim, the table and the predictive value.

Community Project

PROJECT IDEA	A probability audit of a school or club lottery.
NEED	Participants believe their chance of winning is far higher than it is.
RESEARCH QUESTION	What is the actual probability of each prize, and how does it change with the number of tickets bought?
MATHEMATICS	Counting, permutations and combinations.
ACTION OR OUTCOME	A one-page probability table with the counting shown.
BENEFICIARIES	Everyone who enters the lottery.
EVIDENCE	The table and the counts behind it.

Personal Project connection · MYP 5

PERSONAL INTEREST	Games, sport, weather or health.
LEARNING GOAL	To compute a probability you rely on and state the set it is measured on.
PRODUCT OR OUTCOME	A probability card with the set and the rule named.
MATHEMATICS	Conditional probability and counting.
PROCESS	Name the set, count the outcomes, and state the rule.
EVIDENCE	The card and the counting behind it.

CHAPTER 11 · PART V · INTEGRATED MODELLING

11

Capstone: A Mathematical Modelling Project

A capstone project applies the whole of the course to a real decision. The mathematics is only part of the work: naming the problem, stating the assumptions and defending the model are the rest.

ENGAGE

A school is deciding whether to install a water tank. The tank costs money, saves money, and takes up space. Three students build three models and reach three different conclusions from the same arithmetic. What went wrong?

Framing the inquiry

Key concept: Relationships
Related concepts: Change · Representation · System
Global context: Globalization and sustainability
Global context exploration: decisions about shared resources, access and long-term cost

Statement of inquiry: A mathematical model is an argument about the world: its value depends as much on the assumptions it makes and the evidence it ignores as on the accuracy of its arithmetic.

Questions to guide us

- **Factual:** What are the stages of the modelling cycle, and what is produced at each?
- **Conceptual:** Why can two correct models of the same situation give different recommendations?
- **Debatable:** Is a model that ignores an important factor more or less honest than one that includes it approximately?

ATL · CRITICAL THINKING. In order for a student to defend a model of a shared resource, students must state its assumptions, test it against data, and justify which evidence it leaves out and why.

CHAPTER 11 · INVESTIGATION 11.1

The modelling cycle

INQUIRY QUESTION

What are the stages of a modelling project, and what does each produce?

YOUR TASK

Choose a decision in your school or community that involves a cost over time. Work through the five stages of the modelling cycle, producing an artefact at each stage. Then find a second model of the same decision that differs from yours in exactly one assumption, and explain how the conclusion changes.

stage	artefact produced	what it commits you to
define	variables, units, audience	what is being optimised
assume	a list of five simplifications	each is challengeable
build	a formula with named parameters	the parameters are estimates
test	a comparison with three data points	the size of the error
report	a recommendation with limits	what the model ignores

CHAPTER 11 · INVESTIGATION 11.1

The modelling cycle

What do you notice?

- Defining the problem first made the rest of the work easier, because the variables were fixed.
- Listing five assumptions exposed two that were doing most of the work.
- Testing against three data points showed an error of about 8 per cent, which set the confidence in the recommendation.
- Changing one assumption moved the conclusion, which showed how sensitive the answer was.

Observation. Each stage of the cycle produces an artefact, and the artefacts are what a reader examines.

MATHEMATICAL CONCLUSION

The modelling cycle runs from defining the problem, through assumptions and building, to testing and reporting, and each stage produces a specific artefact. Two models of the same situation can both be correct and reach different conclusions when they differ in an assumption, and the disagreement is settled by evidence about that assumption.

Your own words

CHAPTER 11 · 11.1

The modelling cycle

The modelling cycle has five stages: define the problem, make assumptions, build the model, test and evaluate it, and report the result. Each stage produces something specific, and skipping one leaves a gap that shows up later.

The first stage defines the variables, the units, the quantity to be optimised or predicted, and the audience for the answer. A problem that has not been defined cannot be solved, however good the mathematics.

Assumptions are the bridge between the situation and the mathematics, and every one of them is a choice that can be challenged. Listing them explicitly is what allows a reader to disagree with a specific claim rather than with the whole model.

Testing compares the model's predictions with data, and evaluation reports where the model fails. A model that has never failed a test has not been tested, and a model that has not been evaluated has not been finished.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 11 · 11.1 · WORKED EXAMPLE 1

The modelling cycle

WORKED EXAMPLE 11.1.1

Problem. A school models the cost of running a bus service. List the variables, the units and the assumptions a first model would need.

stage	what is produced	example for the bus
define	variables and units	cost per week in currency units, distance in km
assume	a list of simplifications	constant fuel price, no maintenance, fixed route
build	a formula	C = 50 + 1.2d
test	a comparison with data	compare with three past invoices
report	a recommendation with limits	the model ignores driver overtime

Each stage produces a specific artefact.

Mathematical reasoning. Work through the stages in order and name the output of each.

1. Define: the quantity is the weekly cost, the variables are the distance and the number of services, and the units are currency units and kilometres.
2. Assume: the fuel price is constant, the route is fixed, and maintenance is ignored in a first model.
3. Build: the cost is a fixed part plus a part proportional to distance, so C = 50 + 1.2d.
4. Test: compare the model's predictions with three past invoices.
5. Report: give the recommendation and state that the model ignores driver overtime and maintenance.

Final answer. variables, units, assumptions, formula, test and report, each named

Interpretation. The list of assumptions is the part that lets a reader challenge a specific claim rather than the whole model.

CHAPTER 11 · 11.1 · WORKED EXAMPLE 2

The modelling cycle

WORKED EXAMPLE 11.1.2

Problem. Two students model the same bus service and reach different conclusions. One included a term for maintenance and one did not. Explain how both can be correct.

model	formula	assumption	conclusion
student 1	$C = 50 + 1.2d$	no maintenance	cheaper
student 2	$C = 50 + 1.2d + 0.3d$	maintenance proportional to distance	dearer
difference	the maintenance term	—	—

The difference is an assumption, not an error.

Mathematical reasoning. Identify the assumption that differs and trace its effect on the conclusion.

1. Both students used correct arithmetic on the same data.
2. Student 1 assumed no maintenance cost; student 2 assumed maintenance proportional to distance.
3. The second model has an extra term, 0.3d, which student 1 omitted.
4. So the two models make different predictions, and both are internally correct.
5. The disagreement is about an assumption, and it is settled by evidence about maintenance, not by arithmetic.

Final answer. the two models differ in an assumption, so both can be correct

Interpretation. This is the inquiry statement in action: the value of a model depends on its assumptions, not only on its arithmetic.

CHAPTER 11 · INVESTIGATION 11.2

Defining the problem

INQUIRY QUESTION

What has to be settled before any mathematics is done?

YOUR TASK

Take a real decision in your community that involves a cost over time. Write a full problem definition with the five named parts. Then list three factors the situation involves that your definition excludes, and for each state the effect on the conclusion and what data would repair it.

part of the definition	statement
decision	whether to install solar panels on the school roof
objective	minimise the total cost over 15 years
variables	installed capacity in kW, costs in currency units per year
constraints	roof area, budget, structural limits
audience	the school council
excluded factor 1	the future price of electricity, which changes the saving
excluded factor 2	panel degradation, which lowers output over time
excluded factor 3	the resale value at the end of the term

CHAPTER 11 · INVESTIGATION 11.2

Defining the problem

What do you notice?

- Naming the objective first showed that a different objective, such as maximising carbon saving, would give a different answer.
- Each excluded factor changed the saving, not the cost, so the conclusion was sensitive to all three.
- Two of the three could be repaired with published data.
- The third, the resale value, had no reliable data and had to be left as a stated range.

Observation. A definition fixes what the model will answer, and the excluded factors set the limits of the answer.

MATHEMATICAL CONCLUSION

A problem definition names the decision, the objective, the variables with units, the constraints and the audience, and anything not named is an unstated assumption. Excluded factors are part of the model's value: each one limits the conclusion, and the honest report names them and says what data would repair them.

Your own words

CHAPTER 11 · 11.2

Defining the problem

A problem definition names the decision to be made, the quantity to be optimised or predicted, the variables with their units, and the constraints. Anything not named is assumed, and an unstated assumption is an undefended one.

The quantity to be optimised is a choice, and different choices lead to different models. Minimising cost and maximising access describe different relationships between the resources, and a model built for one does not answer the other.

Constraints come from the limits of the situation, and each becomes an inequality in the model. A constraint that is not stated is a constraint that has been assumed away, which is the commonest reason a model fails in practice.

The audience for the answer decides the form of the report. A recommendation to a school council is stated in the council's terms, with the limits named, and not in the terms of the calculation that produced it.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 11 · 11.2 · WORKED EXAMPLE 1

Defining the problem

WORKED EXAMPLE 11.2.1

Problem. Define the problem of deciding whether a school should install a water tank, naming the decision, the objective, the variables and the constraints.

element	statement	why it matters
decision	install or not	defines the recommendation
objective	minimise the total cost over 10 years	fixes what is optimised
variables	tank volume in litres, cost in currency units	fixes the units
constraints	roof area, budget, space	each is a limit
audience	the school council	sets the form of the report

A definition has five named parts.

Mathematical reasoning. Name each part of the definition and state why it matters.

1. The decision is whether to install a tank of a given size.
2. The objective is to minimise the total cost over ten years, counting the purchase, the installation and the water saved.
3. The variables are the tank volume in litres and every cost in currency units per year.
4. The constraints are the available roof area, the budget and the space on the site.
5. The audience is the school council, so the report is in currency units and years, with the limits named.

Final answer. decision, objective, variables, constraints and audience all named

Interpretation. The definition is a contract: it fixes what the model will and will not answer.

CHAPTER 11 · 11.2 · WORKED EXAMPLE 2

Defining the problem

WORKED EXAMPLE 11.2.2

Problem. A student's model of the tank ignores the rainfall pattern because no local data were available. Evaluate the model.

aspect	assessment
what the model does	computes cost from tank volume
what it omits	the rainfall pattern, which determines the saving
effect on the conclusion	the saving is unreliable, so the recommendation is unsafe
what would fix it	local rainfall data, or a stated range
honest report	gives the recommendation with the omission named

An omission is part of the model's value.

Mathematical reasoning. Identify the omission, trace its effect, and state what would repair it.

1. The model computes cost from tank volume but omits rainfall.
2. Rainfall determines how much water the tank actually collects, so it determines the saving.
3. Without it, the saving is an estimate with an unknown error, and the recommendation is unsafe.
4. The repair is local rainfall data, or a stated range of rainfall with the recommendation for each.
5. The honest report names the omission and gives the recommendation as conditional on it.

Final answer. the omission makes the recommendation unsafe, and it must be named

Interpretation. An ignored factor is part of the model's value, which is the second half of the inquiry statement.

CHAPTER 11 · INVESTIGATION 11.3

Building and testing a model

INQUIRY QUESTION

How is a model built, and how is it shown to work?

YOUR TASK

Build a model for a decision of your own with at least three parameters. Fit it to three data points, test it against a fourth, and then carry out a sensitivity analysis on each parameter, reporting the percentage change in the conclusion for a 10 per cent change in each.

parameter	10 per cent change	change in the conclusion	rank
unit price	+10 per cent	+10 per cent	1
usage	+10 per cent	+8 per cent	2
discount rate	+10 per cent	+2 per cent	3
lifetime	+10 per cent	−6 per cent	2

CHAPTER 11 · INVESTIGATION 11.3

Building and testing a model

What do you notice?

- The unit price was the most sensitive parameter, so its data mattered most.
- The usage and the lifetime had comparable effects, in opposite directions.
- The discount rate had the smallest effect, so its estimate could be crude.
- Testing against the fourth data point gave an error of 6 per cent, consistent with the sensitivity ranking.

> **Observation.** A sensitivity analysis ranks the parameters by their effect on the conclusion, and the ranking says where better data are needed.

MATHEMATICAL CONCLUSION

A model is built by choosing a form for each relationship and fitting its parameters, and it is tested against data not used in the fit. The relative error says whether the model is good enough, and a sensitivity analysis ranks the parameters by their effect on the conclusion. Here the unit price was the most sensitive parameter.

Your own words

CHAPTER 11 · 11.3

Building and testing a model

A model is built by choosing a mathematical form for each relationship, fitting its parameters to data, and combining the parts into a system. The form is chosen for simplicity first, and complexity is added only when the data demand it.

Testing compares the model's predictions with data that were not used to fit it. A model that predicts new data well is useful; one that only reproduces the data it was fitted to has learned nothing.

A sensitivity analysis varies each parameter and reports the effect on the conclusion. The parameters with the largest effect are the ones that most need better data, and the sensitivity report says where the effort should go.

A model is tested by attempting to break it: with extreme values, with values outside the fitted range, and against known special cases. A model that survives these is ready to report; one that does not needs repair.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 11 · 11.3 · WORKED EXAMPLE 1

Building and testing a model

WORKED EXAMPLE 11.3.1

Problem. A model for the tank's saving is $S = 0.002rV$, where r is the annual rainfall in millimetres and V the tank volume in litres. Test it against $r = 800$, $V = 5000$ and a measured saving of 7800 currency units.

step	working	result
model prediction	$0.002 \times 800 \times 5000$	8000
measured	—	7800
error	$8000 - 7800$	200
relative error	$200 \div 7800$	0.0256
verdict	within 3 per cent	acceptable

The error is expressed relative to the measurement.

Mathematical reasoning. Compute the prediction, the error and the relative error.

1. The model predicts $S = 0.002 \times 800 \times 5000 = 8000$.
2. The measured saving is 7800.
3. The absolute error is $8000 - 7800 = 200$.
4. The relative error is $200/7800 \approx 0.0256$, about 2.6 per cent.
5. An error under 3 per cent is acceptable for a planning model, so the model passes this test.

Final answer. the model predicts 8000 against a measured 7800, an error of about 2.6 per cent

Interpretation. The relative error is the useful measure, because it says whether the error matters for the decision.

CHAPTER 11 · 11.3 · WORKED EXAMPLE 2

Building and testing a model

WORKED EXAMPLE 11.3.2

Problem. In the same model, find the sensitivity of the saving to the rainfall when r changes from 800 to 900.

rainfall	saving	change
800	8000	—
900	9000	+1000
relative change	1000 ÷ 8000	0.125
verdict	a 12.5 per cent rise in the saving	high sensitivity

The saving is proportional to the rainfall.

Mathematical reasoning. Compute the change in the saving and express it relatively.

1. At r = 800 the saving is 8000, and at r = 900 it is 9000.
2. The change is 1000, which is 12.5 per cent of 8000.
3. Since the model is linear in r, the relative change in the saving equals the relative change in the rainfall.
4. A 12.5 per cent rise in rainfall gives a 12.5 per cent rise in the saving, so the model is highly sensitive to rainfall.
5. This means the rainfall data matter more than any other input.

Final answer. a 12.5 per cent change in the saving for a 12.5 per cent change in rainfall

Interpretation. The sensitivity analysis says where to spend effort on better data, and here it is unambiguously the rainfall.

CHAPTER 11 · INVESTIGATION 11.4

Optimising and evaluating

INQUIRY QUESTION

How is a model's recommendation chosen and defended?

YOUR TASK

Take a model you have built and optimise it. Report the optimal point, the optimal value and the feasibility check. Then test the robustness of the recommendation to a 10 per cent change in each parameter, and state whether the recommendation survives.

parameter	10 per cent change	recommendation survives?	new optimal value
profit from x	+10 per cent	yes	300
profit from y	+10 per cent	yes	280
resource 1	+10 per cent	yes	300
resource 2	−10 per cent	yes	260
all together	as above	yes	the plan is unchanged

CHAPTER 11 · INVESTIGATION 11.4

Optimising and evaluating

What do you notice?

- The optimal plan did not change under any single 10 per cent perturbation.
- The optimal value changed by up to about 7 per cent.
- The most sensitive parameter was the profit from x, as the earlier sensitivity analysis had predicted.
- The recommendation was therefore robust, and this was reported with the value range.

Observation. A recommendation is a point, a value and a robustness statement, and all three are needed.

MATHEMATICAL CONCLUSION

A recommendation from a model is the optimal point, the optimal value and the justification that the point is feasible, together with a robustness statement. A recommendation that reverses under a reasonable change in a parameter is not a recommendation. Here the plan survived every single perturbation, with the value ranging from 260 to 300.

Your own words

CHAPTER 11 · 11.4

Optimising and evaluating

When a model has a linear objective and linear constraints, the optimum is found at a vertex of the feasible region. When the objective is non-linear, the optimum is found numerically and reported with the interval that contains it.

An optimal recommendation is a point, a value and a justification that the point is feasible. The value alone is not a recommendation, because it does not say what to do.

Evaluation compares the model's recommendation with what a decision-maker would do without it. If the model adds nothing, it has failed, and the reason should be reported.

Robustness asks whether the recommendation survives reasonable changes in the parameters. A recommendation that reverses under a 10 per cent change in one input is not a recommendation but a coin toss, and it must be reported as such.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 11 · 11.4 · WORKED EXAMPLE 1

Optimising and evaluating

WORKED EXAMPLE 11.4.1

Problem. A model of a community garden gives a profit of $P = 40x + 30y$ subject to $x + 2y \le 12$, $2x + y \le 12$, $x \ge 0$ and $y \ge 0$. Find the optimal plan and test its robustness to a 10 per cent fall in the profit from x.

vertex	coordinates	$P = 40x + 30y$
(0, 0)	(0, 0)	0
6, 0)	(6, 0)	240
(4, 4)	(4, 4)	280
(0, 6)	(0, 6)	180

The optimum is at the intersection of both constraints.

Mathematical reasoning. Evaluate the objective at the vertices, then perturb the profit from x.

1. The vertices are (0, 0), (6, 0), (4, 4) and (0, 6).
2. Evaluating P gives 0, 240, 280 and 180, so the optimum is (4, 4) with $P = 280$.
3. If the profit from x falls by 10 per cent to 36, then $P(4, 4) = 36(4) + 30(4) = 264$.
4. The other vertices give 216 and 180, so (4, 4) is still optimal.
5. The recommendation is therefore robust to a 10 per cent fall in the profit from x.

Final answer. optimal plan (4, 4) with profit 280, robust to a 10 per cent fall in the profit from x

Interpretation. The robustness test is part of the recommendation, because it says how much the conclusion can be trusted.

CHAPTER 11 · 11.4 · WORKED EXAMPLE 2

Optimising and evaluating

WORKED EXAMPLE 11.4.2

Problem. A model recommends a plan that a gardener rejects because it ignores the volunteers' availability. Evaluate the model.

aspect	assessment
what the model does	optimises profit subject to land and money
what it omits	volunteer availability, which caps the labour
effect	the recommended plan needs more labour than exists
verdict	the model is not wrong but is incomplete
repair	add the labour constraint and re-solve

An omitted constraint can invalidate a plan.

Mathematical reasoning. Identify the omission, trace its effect, and state the repair.

1. The model optimises profit subject to land and money only.
2. It omits volunteer availability, which caps the labour available.
3. The recommended plan needs more labour than the volunteers can supply, so it cannot be carried out.
4. The model is not wrong in its arithmetic, but it is incomplete, and an incomplete model can recommend an impossible plan.
5. The repair is to add the labour constraint and solve again.

Final answer. the model is incomplete; adding the labour constraint repairs it

Interpretation. An omitted constraint is more dangerous than an omitted cost, because it can make the recommendation impossible rather than merely inaccurate.

CHAPTER 11 · INVESTIGATION 11.5

Presenting and defending

INQUIRY QUESTION

What does a defensible report contain?

YOUR TASK

Write a full report on a model you have built, with the six sections in order. Then find the strongest objection a reader could make and write a defence of it in five steps: restate, concede, quantify, state the consequence, and repair.

section	length	what it contains
problem	half a page	the decision and the objective
assumptions	half a page	five numbered simplifications
model	one page	the formula, the parameters and their sources
test	half a page	the error against four data points
recommendation	half a page	the plan, the value and the robustness
limits	half a page	what the model omits and the effect of each
defence	half a page	restate, concede, quantify, consequence, repair

CHAPTER 11 · INVESTIGATION 11.5

Presenting and defending

What do you notice?

- Writing the assumptions section first made the rest easier, because the challenges were already visible.
- The strongest objection was about the assumption with the largest sensitivity, which the earlier analysis had identified.
- Quantifying the objection showed it did not change the recommendation, only the size of the benefit.
- The report was about four pages, which was the right length for the audience.

Observation. A report is an argument with a fixed order, and a defence answers the strongest objection in five steps.

MATHEMATICAL CONCLUSION

A modelling report states the problem, the assumptions, the model, the test results, the recommendation and the limits, in that order, and each section answers the question the previous one raises. A defence restates the strongest objection, concedes what is true, quantifies the effect, states the consequence and gives the repair.

Your own words

CHAPTER 11 · 11.5

Presenting and defending

A modelling report states the problem, the assumptions, the model, the test results, the recommendation and the limits, in that order. This representation of the work is the argument, and each section answers the question the previous one raises.

A defence is a response to the strongest objection, not a summary of the work. Anticipating the objection and answering it is what makes a report defensible, and the objection is usually about an assumption.

Every claim in a report is traceable either to a calculation or to a stated assumption. A claim that is neither is removed, because it cannot be defended.

The audience decides the level of detail. A council needs the recommendation, the limits and one number each for the cost and the benefit; a mathematics audience needs the model and its test.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 11 · 11.5 · WORKED EXAMPLE 1

Presenting and defending

WORKED EXAMPLE 11.5.1

Problem. Write the structure of a report on the water-tank decision, naming what each section contains.

section	contents	answers
problem	the decision and the objective	what is being decided
assumptions	the five simplifications	what the model takes for granted
model	the formula and its parameters	how the conclusion is computed
test	the error against four data points	how well it works
recommendation	the plan and the value	what to do
limits	what the model omits	what could go wrong

The order is the argument.

Mathematical reasoning. Name the sections in order and state what each answers.

1. The problem section states the decision and the objective.
2. The assumptions section lists the five simplifications, so a reader can challenge one specifically.
3. The model section gives the formula and its parameters, with the source of each.
4. The test section gives the error against four data points.
5. The recommendation gives the plan and the value, and the limits section names what the model omits.

Final answer. problem, assumptions, model, test, recommendation and limits, in that order

Interpretation. Each section answers the question the previous one raises, which is why the order matters.

CHAPTER 11 · 11.5 · WORKED EXAMPLE 2

Presenting and defending

WORKED EXAMPLE 11.5.2

Problem. A reviewer objects that the tank model assumes a constant water price, which has risen 8 per cent a year for a decade. Write the defence.

step	content
restate the objection	the water price is assumed constant
concede	the assumption is optimistic
quantify	at 8 per cent a year the saving is understated by about 40 per cent over ten years
consequence	the recommendation is unchanged, and the saving is larger than reported
repair	model the price rise and report both

A defence concedes, quantifies and repairs.

Mathematical reasoning. Restate the objection, concede what is true, quantify the effect, and state the repair.

1. The objection is that the model assumes a constant water price.
2. This is conceded: the price has in fact risen about 8 per cent a year.
3. The effect is quantified: over ten years the saving is understated by about 40 per cent.
4. The consequence is that the recommendation is unchanged, and the reported saving is too small.
5. The repair is to model the price rise explicitly and report both figures.

Final answer. the objection is conceded, quantified and repaired, and the recommendation survives

Interpretation. A defence that concedes and quantifies is stronger than one that denies, because it shows the objection was understood.

CHAPTER 11 · INVESTIGATION 11.6

The capstone project brief

INQUIRY QUESTION

What must a capstone project contain?

YOUR TASK

Plan your capstone project. Choose a real decision, write the problem definition, list the assumptions you expect to make, name the data you will need and where you will get it, and write the reflection you expect to end with. Present the plan to a peer and record the strongest objection they raise.

element	planned content	evidence needed
decision	whether to install a water tank	the school's water bills
objective	minimise the ten-year total cost	the tank and installation prices
assumptions	constant water price, constant rainfall, no maintenance	local rainfall and price data
data needed	rainfall, water price, roof area, tank prices	the school and the national weather service
peer objection	the model ignores the maintenance cost	a maintenance estimate
expected reflection	the model cannot answer whether the tank is the best use of the money	—

CHAPTER 11 · INVESTIGATION 11.6

The capstone project brief

What do you notice?

- Choosing the decision first made the data needs obvious.
- Listing the assumptions before collecting data showed which data actually mattered.
- The peer objection matched the most sensitive parameter, which the plan had already identified.
- The expected reflection named something the model could not answer, which is the most transferable part of the work.

Observation. A capstone plan is a problem definition, an assumption list, a data list and an expected reflection, tested against a peer's objection.

MATHEMATICAL CONCLUSION

A capstone project contains a problem definition, a numbered list of assumptions, a model with sourced parameters, a test against data, a recommendation with a robustness statement, and a reflection on what the model could not answer. The evidence behind each element is what distinguishes a project from an essay, and the assumptions are assessed as heavily as the arithmetic.

Your own words

CHAPTER 11 · 11.6

The capstone project brief

The capstone project is an extended piece of work on a real decision, and it must contain all six elements of a modelling report together with the evidence behind them. The evidence is what distinguishes a project from an essay.

The project is assessed on the mathematics, on the quality of the assumptions, on the testing, and on the communication. A project with excellent arithmetic and unstated assumptions scores badly, because the assumptions are the model.

Every number in a project is either computed, measured or estimated, and its status is stated. An estimate with no stated basis is a guess, and a guess cannot be defended.

The project ends with a reflection on what the model could not answer. That reflection is the most valuable part of the work, because it is what transfers to the next problem.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 11 · 11.6 · WORKED EXAMPLE 1

The capstone project brief

WORKED EXAMPLE 11.6.1

Problem. List the required elements of a capstone project and the evidence each needs.

element	required evidence
problem definition	the decision, the objective, the variables and the constraints
assumptions	a numbered list with the reason for each
model	the formula, the parameters and the source of each
test	the comparison with data and the error
recommendation	the plan, the value and the robustness
limits and reflection	what the model omits and what could not be answered

Each element has its own evidence.

Mathematical reasoning. List the elements and state the evidence each requires.

1. The problem definition needs the decision, the objective, the variables and the constraints, all named.
2. The assumptions need to be a numbered list with a reason for each, so a reader can challenge one specifically.
3. The model needs the formula, the parameters and the source of each parameter.
4. The test needs the comparison with data and the resulting error.
5. The recommendation needs the plan, the value and the robustness, and the limits and reflection need what the model omits.

Final answer. six elements, each with its required evidence

Interpretation. The evidence is what distinguishes a project from an essay, and it is what is assessed.

CHAPTER 11 · 11.6 · WORKED EXAMPLE 2

The capstone project brief

WORKED EXAMPLE 11.6.2

Problem. A student's project has excellent arithmetic but no stated assumptions. Evaluate it against the capstone requirements.

criterion	assessment	consequence
mathematics	excellent	full credit
assumptions	absent	major loss of credit
testing	absent	major loss of credit
communication	clear but undefended	partial credit
overall	fails the capstone requirements	—

Missing assumptions cost more than weak arithmetic.

Mathematical reasoning. Assess each criterion and state the consequence of the omissions.

1. The mathematics is excellent, so that criterion earns full credit.
2. But the assumptions are absent, so the reader cannot challenge any specific claim.
3. The testing is absent too, so there is no evidence the model works.
4. The communication is clear but undefended, which earns only partial credit.
5. The project fails the capstone requirements, because the assumptions are the model.

Final answer. the project fails the requirements despite excellent arithmetic

Interpretation. This is the inquiry statement applied to assessment: the value of a model depends on its assumptions, not only on its arithmetic.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

11.A1 FLUENCY
Name the five stages of the modelling cycle.

11.A2 FLUENCY
State what a problem definition must name.

11.A3 REASONING
Explain why a model's assumptions are listed explicitly.

11.A4 REASONING
A model is tested against data. State the two kinds of data and why they differ.

11.A5 REASONING
State what a sensitivity analysis reports.

11.A6 REASONING
A model omits a constraint rather than a cost. Explain why this is more serious.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

11.A7 FLUENCY
State the three parts of a recommendation from a linear program.

11.A8 REASONING
Explain what a robustness test adds to a recommendation.

11.A9 FLUENCY
State the order of the six sections of a modelling report.

11.A10 FLUENCY
Name the five steps of a defence of a model.

11.A11 REASONING
Explain why an estimate in a report must state its basis.

11.A12 REASONING
State what the final reflection of a capstone project should address.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

11.B1 CHALLENGE
Investigate the effect of changing the objective in a model of your own, and justify the change in the recommendation.

11.B2 CHALLENGE
Conjecture a rule for which assumptions most affect a model's conclusion, and test it on two models.

11.B3 REASONING
Investigate whether a model with more assumptions is always better than one with fewer, and justify.

11.B4 REASONING
Take a published model of a real decision and identify three assumptions it makes, justifying each.

11.B5 REASONING
Investigate the effect of the choice of time horizon on a model's recommendation, and justify.

11.B6 REASONING
Conjecture a rule for when a linear model is adequate for a modelling project, and test it.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

11.B7 REASONING

Investigate the difference between a model's accuracy and its usefulness, and justify with an example.

11.B8 REASONING

Investigate the effect of the choice of time horizon on a model's recommendation, and justify.

11.B9 CHALLENGE

Conjecture a rule for which assumptions most affect a model's conclusion, and test it.

11.B10 REASONING

Investigate whether a model with more assumptions is always better, and justify.

11.B11 REASONING

Investigate the difference between a model's accuracy and its usefulness, and justify.

CRITERION C Communicating

Criterion C practice · Communicating

11.C1 REASONING

Write a complete solution to defining the problem of whether a school should install solar panels, naming all five parts.

11.C2 CHALLENGE

Write a complete solution to a sensitivity analysis of a model with three parameters, reporting the percentage change in the conclusion for a 10 per cent change in each.

11.C3 CHALLENGE

Write a complete solution to testing a model against data not used in the fit, stating the verdict.

11.C4 CHALLENGE

Write a complete solution to optimising $P = 50x + 40y$ subject to $x + y \le 10$, $2x + y \le 14$, $x \ge 0$, $y \ge 0$, and reporting the robustness to a 10 per cent fall in the profit from x.

11.C5 REASONING

Write a complete solution to writing a five-step defence of a model against the objection that it assumes constant prices.

11.C6 REASONING

Write a complete solution to evaluating a model that recommends an infeasible plan, identifying the cause.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

11.C7 REASONING

Write a complete solution to the structure of a capstone report, naming the six sections and the evidence each needs.

11.C8 REASONING

Write a complete solution to defining the problem of whether a school should install solar panels, naming all five parts.

11.C9 CHALLENGE

Write a complete solution to a sensitivity analysis with three parameters, reporting the change in the conclusion for a 10 per cent change in each.

11.C10 CHALLENGE

Write a complete solution to testing a model against data not used in the fit, stating the verdict.

11.C11 CHALLENGE

Write a complete solution to optimising $P = 60x + 45y$ subject to $x + y \le 9$, $2x + y \le 12$, $x \ge 0$, $y \ge 0$, and reporting the robustness.

11.C12 REASONING

Write a complete solution to writing a five-step defence against the objection that a model assumes constant prices.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

11.D1 APPLICATION

A school models the cost of a water tank. The tank costs 4000, saves 600 a year, and lasts 10 years. Build the model, find the payback time, and state one assumption that could change the recommendation.

11.D2 APPLICATION

A community garden is modelled by $P = 40x + 30y$ with $x + 2y \le 12$, $2x + y \le 12$, $x \ge 0$, $y \ge 0$. Find the optimal plan and test its robustness to a 10 per cent fall in the profit from x.

11.D3 APPLICATION

A model recommends 12 tonnes of recycling but the available labour supports only 9. Evaluate the model and state the repair.

11.D4 APPLICATION

A farmer's model gives a profit of 34 000 at 40 ha of wheat and 60 ha of barley. A 10 per cent fall in the wheat price changes the profit to 30 000. Evaluate the robustness of the recommendation.

11.D5 APPLICATION

A student's model is tested against four data points and gives relative errors of 3, 5, 2 and 4 per cent. The decision's margin is 15 per cent. Evaluate the model.

11.D6 APPLICATION

A charity models donations as $D = 2000 + 50n$ where n is the number of events. Test the model against $n = 5$, $D = 2300$ and $n = 12$, $D = 2650$, and comment.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

11.D7 REASONING

A student reports a recommendation without a robustness statement. Explain what is missing and why it matters.

11.D8 APPLICATION

A model of a school's energy use omits the holidays. Evaluate the effect on a recommendation about insulation.

11.D9 APPLICATION

A council asks for a recommendation in one sentence. Draft it for a model whose optimum is a plan of value 28 000 with a robustness range of 25 000 to 31 000.

11.D10 REASONING

A student claims that a model with an error of 20 per cent is useless. Investigate and justify.

11.D11 APPLICATION

A model has two parameters whose 10 per cent changes move the conclusion by 2 and 9 per cent. State where effort should go and justify.

11.D12 REASONING

A student's reflection says only that the model 'could be improved'. Evaluate it.

CHAPTER 11 · APPLICATION AND ASSESSMENT

Applying chapter 11

These tasks build and evaluate models. State the assumptions, test the model against data, and report the sensitivity of the conclusion to each assumption.

A1 · The tank decision

A school is deciding whether to install a water tank. The tank costs 6000 currency units, lasts 15 years, and is expected to save 500 currency units a year. The school's roof can collect at most 9000 litres a month. Build a model, find the payback time, state three assumptions with the reason for each, and write a recommendation with a robustness statement. Then state one factor the model omits and its effect on the recommendation.

Criteria C and D · build and test the model, state the assumptions, and defend the recommendation.

A2 · The shared resource

A community has a shared budget of 2400 currency units and 60 volunteer hours for a summer programme. Activity A costs 40 and needs 2 hours; activity B costs 30 and needs 4 hours; activity C costs 20 and needs 1 hour. The value of each is modelled as 90, 70 and 40 respectively. Formulate and solve the problem, test the robustness of the recommendation to a 10 per cent fall in the value of A, and write a defence against the objection that the model ignores the volunteers' preferences.

Criteria C and D · formulate, solve by vertices, test the robustness, and write the five-step defence.

CHAPTER 11 · REVIEW

What you should now be able to do

Key ideas

- The modelling cycle runs from defining the problem to reporting the result.
- A problem definition names the decision, objective, variables, constraints and audience.
- Assumptions are the bridge between the situation and the mathematics, and each is challengeable.
- A model is tested against data not used in the fit.
- The relative error against the decision's margin is the verdict.
- A sensitivity analysis ranks the parameters by their effect.
- A recommendation is a point, a value and a robustness statement.
- An omitted constraint can make a recommendation impossible.
- A report's six sections in order form the argument.
- A defence restates, concedes, quantifies, states the consequence and repairs.

Vocabulary

modelling cycle · problem definition · objective · variable · constraint · assumption · parameter · fitting data · test data · relative error · sensitivity analysis · robustness · recommendation · limit · defence · payback time · margin

CHAPTER 11 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

11.R1 Name the five stages of the modelling cycle.

11.R2 State the five parts of a problem definition.

11.R3 Explain the difference between fitting data and test data.

11.R4 State the three parts of a recommendation from a linear program.

11.R5 Name the five steps of a defence.

11.R6 Explain why an omitted constraint is more serious than an omitted cost.

CHAPTER 11 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> A mathematical model is an argument about the world: its value depends as much on the assumptions it makes and the evidence it ignores as on the accuracy of its arithmetic.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of change, representation and system appear in your work?
- How is the global context of globalization and sustainability, and its exploration of decisions about shared resources, access and long-term cost, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a mathematical model as a guide to a shared decision, and what would you still like to ask?

CHAPTER 11 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

A model is an argument. Name the assumptions, the evidence and the limits, and defend the conclusion against the strongest objection.

Service as Action

COMMUNITY NEED	A local resource decision is being made without a model that states its assumptions.
MATHEMATICAL CONNECTION	The whole course, applied to one decision.
PROPOSED ACTION	Build a model of one local resource decision, publish it with its assumptions and its limits, and present it to the decision-makers.
EXPECTED IMPACT	The decision is made with its assumptions visible.
STUDENT REFLECTION	Record the model, the test and the defence.

Community Project

PROJECT IDEA	A model of the school's energy or water use.
NEED	The school's consumption is discussed without a model that states its assumptions.
RESEARCH QUESTION	What would a stated change to the school's consumption save over ten years?
MATHEMATICS	Modelling, series and optimisation.
ACTION OR OUTCOME	A one-page model with assumptions, test and recommendation.
BENEFICIARIES	The school and everyone who pays its bills.
EVIDENCE	The model and the data behind it.

Personal Project connection · MYP 5

PERSONAL INTEREST	A decision you will actually make.
LEARNING GOAL	To build a model of a decision you face and defend it.
PRODUCT OR OUTCOME	A capstone project with all six elements.
MATHEMATICS	The whole course.
PROCESS	Define, assume, build, test, recommend, reflect.
EVIDENCE	The project and the data behind it.

CHAPTER 12 · PART V · INTEGRATED MODELLING

12

Cumulative Revision and the Transition to DP Mathematics

This chapter revises the whole course and looks forward. The habit that matters is not recalling a formula but choosing a method, justifying it, and knowing where it stops.

ENGAGE

A DP mathematics paper contains a question nobody has seen before. Two students recognise the structure; one recalls a formula and the other does not. Which student does better, and why?

Framing the inquiry

Key concept: Logic
Related concepts: Generalisation · Representation · Justification
Global context: Identities and relationships
Global context exploration: how mathematical habits shape the way we reason elsewhere

Statement of inquiry: Mathematical maturity is the ability to choose a method, justify it, connect it to other ideas, and recognise the limits of what has been shown.

Questions to guide us

- **Factual:** What are the main methods of each strand of the course, and when is each used?
- **Conceptual:** What does it mean to choose a method rather than recall one, and how is the choice justified?
- **Debatable:** Is a correct answer without a justification worth more or less than a justified answer with a small error?

ATL · CRITICAL THINKING. In order for a student to choose a method for an unfamiliar problem, students must identify the structure of the problem, select and justify an approach, and state what the solution does not show.

CHAPTER 12 · INVESTIGATION 12.1

Cumulative practice: algebra and functions

INQUIRY QUESTION

Which algebraic method fits this structure?

YOUR TASK

Take twelve algebra questions from earlier chapters of your own choice. For each, name the method you use, justify the choice in one sentence, solve it, and check the answer by a second method where one exists.

question type	method chosen	justification	check used
quadratic equation	quadratic formula	the coefficients do not factorise neatly	substitution
polynomial division	long division	the divisor is linear	multiply back
function inverse	rearrange and swap	the function is one-to-one	composition
composite function	substitute inner into outer	the inner output is defined	evaluate at a point
quadratic inequality	roots and sign	the parabola opens upwards	test a point in each interval
simultaneous equations	elimination	one variable has equal coefficients	substitution

CHAPTER 12 · INVESTIGATION 12.1

Cumulative practice: algebra and functions

What do you notice?

- Naming the method before solving exposed two cases where the chosen method was inefficient.
- The quadratic formula was needed only where factorising failed.
- Every answer was checked, and three checks found an error.
- The justifications were one sentence each, which was enough to make the choice reviewable.

> **Observation.** The structure of a question decides the method, and the justification makes the choice reviewable.

MATHEMATICAL CONCLUSION

An algebra question is classified by its structure, and the structure decides the method: factorising, the quadratic formula, completing the square, substitution, elimination, rearrangement or a sign analysis. Each choice is justified by the structure of the question, and every answer is checked by a second route. Here six question types were classified and checked.

Your own words

CHAPTER 12 · 12.1

Cumulative practice: algebra and functions

Algebra questions are classified by their structure: a polynomial to factorise, an equation to solve, a function to transform, an inequality to bound, or a system to solve. The structure decides the method, and recognising the structure is the skill.

The quadratic formula, completing the square and factorising are three methods for one class of equation, and each is preferable in a different case. Completing the square also gives the vertex, which is why it is used when a graph is needed.

Function questions are classified by whether they ask for a value, an inverse, a composition or a transformation. Each is a different operation, and the commonest error is applying the wrong one.

Justification in algebra is the statement of the rule or property used at each step. A correct answer without the reasons is a computation, not a solution, and it cannot be checked by a reader.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 12 · 12.1 · WORKED EXAMPLE 1

Cumulative practice: algebra and functions

WORKED EXAMPLE 12.1.1

Problem. Solve $2x^2 - 7x - 3 = 0$ by completing the square.

step	working	result
divide by 2	$x^2 - (7/2)x = 3/2$	—
half and square	$(7/4)^2 = 49/16$	—
complete	$(x - 7/4)^2 = 3/2 + 49/16$	73/16
solve	$x = 7/4 \pm \sqrt{73}/4$	—
decimal	$x \approx 3.886$ or $x \approx -0.386$	—

Completing the square also gives the vertex.

Mathematical reasoning. Complete the square, then solve and check with the quadratic formula.

1. Divide by 2: $x^2 - (7/2)x = 3/2$.
2. Add $(7/4)^2 = 49/16$ to both sides: $(x - 7/4)^2 = 24/16 + 49/16 = 73/16$.
3. Take square roots: $x - 7/4 = \pm\sqrt{73}/4$.
4. So $x = (7 \pm \sqrt{73})/4$.
5. Check with the quadratic formula: $x = (7 \pm \sqrt{(49 + 24)})/4 = (7 \pm \sqrt{73})/4$, as required.

Final answer. $x = (7 \pm \sqrt{73})/4$, about 3.886 or −0.386

Interpretation. The check uses a different method, which is what makes it a check.

CHAPTER 12 · 12.1 · WORKED EXAMPLE 2

Cumulative practice: algebra and functions

WORKED EXAMPLE 12.1.2

Problem. Given $f(x) = 2x - 3$ and $g(x) = x^2 + 1$, find $f(g(2))$ and $f^{-1}(5)$.

step	working	result
$g(2)$	$4 + 1$	5
$f(g(2))$	$2(5) - 3$	7
inverse	$y = 2x - 3$ gives $x = (y + 3)/2$	—
$f^{-1}(5)$	$(5 + 3)/2$	4

Composition and inversion are different operations.

Mathematical reasoning. Evaluate the inner function first, then invert by rearranging.

1. $g(2) = 2^2 + 1 = 5$.
2. So $f(g(2)) = f(5) = 2(5) - 3 = 7$.
3. For the inverse, write $y = 2x - 3$ and rearrange: $x = (y + 3)/2$.
4. So $f^{-1}(x) = (x + 3)/2$.
5. Hence $f^{-1}(5) = 8/2 = 4$.

Final answer. $f(g(2)) = 7$ and $f^{-1}(5) = 4$

Interpretation. The two operations are distinguished by what they act on: composition feeds one function into another, and inversion reverses the rule.

CHAPTER 12 · INVESTIGATION 12.2

Cumulative practice: geometry and trigonometry

INQUIRY QUESTION

Which geometric or trigonometric relationship applies?

YOUR TASK

Take ten geometry and trigonometry questions from earlier chapters. For each, draw the figure, name the relationship you use, justify the choice in one sentence, solve it, and check the answer by a second route.

figure	relationship used	justification	check
right triangle	Pythagoras	a right angle is given	trigonometric ratio
general triangle, ASA	sine rule	a side and its opposite angle are known	angle sum
general triangle, SAS	cosine rule	the included angle is given	sine rule
circle, chord	perpendicular bisects chord	the distance from the centre is given	sine ratio
circle, tangent	tangent perpendicular to radius	a tangent length is needed	Pythagoras
cuboid	two applications of Pythagoras	a space diagonal is needed	three-dimensional Pythagoras

CHAPTER 12 · INVESTIGATION 12.2

Cumulative practice: geometry and trigonometry

What do you notice?

- Drawing the figure first made the correct relationship obvious in every case.
- The sine rule and the cosine rule were distinguished by whether the given angle was included.
- The circle questions were distinguished by whether a chord, a tangent or an arc was involved.
- Every answer was checked, and two checks found a sign error.

Observation. The figure decides the relationship, and naming the figure is the first step.

MATHEMATICAL CONCLUSION

Geometry and trigonometry questions are classified by the figure: right-angled triangles, general triangles, circles, cuboids and cones. Each figure has its own relationships, the choice is justified by what is given, and three-dimensional problems are solved by choosing a plane. Here six figure types were classified and checked.

Your own words

CHAPTER 12 · 12.2

Cumulative practice: geometry and trigonometry

Geometry and trigonometry questions are classified by the figure: a right-angled triangle, a general triangle, a circle, a cuboid or a cone. The figure decides the available relationships, and drawing it is the first step.

Right-angled triangles use Pythagoras and the trigonometric ratios; general triangles use the sine rule, the cosine rule or the area formula. The choice depends on whether the given angle is included.

Circle questions use the angle at the centre theorem, the tangent-radius property or the intersecting chords theorem, and the choice is made by naming the arc, the tangent or the chords involved.

Three-dimensional questions are solved by choosing a plane. Naming the plane and the triangle in it is the first line of the solution, and it is what makes the rest of the solution checkable.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 12 · 12.2 · WORKED EXAMPLE 1

Cumulative practice: geometry and trigonometry

WORKED EXAMPLE 12.2.1

Problem. A triangle has sides 8 and 11 and included angle 47°. Find the area and the third side.

step	working	result
area	½ × 8 × 11 × sin 47°	32.2
third side squared	$8^2 + 11^2 - 2(8)(11)\cos 47°$	62.9
third side	√62.9	7.93
check	the largest side is 11, opposite the largest angle	consistent

The included angle fixes both the area and the third side.

Mathematical reasoning. Apply the area formula and the cosine rule, then check the ordering.

1. The area is ½ × 8 × 11 × sin 47° ≈ 44 × 0.7314 ≈ 32.2.
2. The third side squared is $8^2 + 11^2 - 2(8)(11)\cos 47° \approx 185 - 176(0.6820) \approx 65.0$.
3. So the third side is about 8.06.
4. Check: 11 is the largest side, so it is opposite the largest angle, which is the given 47 degrees only if 47 is indeed largest.
5. The angle opposite 11 is arccos((64 + 65 − 121)/(2 × 8 × 8.06)) ≈ 79.4 degrees, which is the largest, as required.

Final answer. area about 32.2, third side about 8.06

Interpretation. The check on the ordering of sides and angles is the standard verification for a triangle.

CHAPTER 12 · 12.2 · WORKED EXAMPLE 2

Cumulative practice: geometry and trigonometry

WORKED EXAMPLE 12.2.2

Problem. A circle has a chord of length 24 at distance 7 from the centre. Find the radius and the angle the chord subtends at the centre.

step	working	result
half-chord	12	—
radius	$\sqrt{(12^2 + 7^2)}$	$\sqrt{193} \approx 13.89$
half-angle	arcsin(12/13.89)	59.7°
full angle	2 × 59.7°	119.4°

The perpendicular from the centre bisects the chord.

Mathematical reasoning. Use the perpendicular-bisector property, then a sine ratio in the resulting triangle.

1. The perpendicular from the centre to the chord bisects it, so half the chord is 12.
2. The radius is $\sqrt{(12^2 + 7^2)} = \sqrt{193} \approx 13.89$.
3. In the right-angled triangle, sin(half the angle) = 12/13.89.
4. So half the angle is about 59.7 degrees and the full angle is about 119.4 degrees.
5. The angle is obtuse, as expected for a chord far from the centre.

Final answer. radius about 13.89; the angle is about 119.4 degrees

Interpretation. Two circle properties are used, and both are named in the solution.

CHAPTER 12 · INVESTIGATION 12.3

Cumulative practice: statistics and probability

INQUIRY QUESTION

Which measure or rule fits this question?

YOUR TASK

Take eight statistics and probability questions from earlier chapters. For each, name the measure or rule you use, justify the choice in one sentence, solve it, and state one limit of the conclusion.

question type	measure or rule	justification	limit of the conclusion
describe a centre	median	the data are skewed	the sample only
describe a spread	interquartile range	an outlier is present	the sample only
compare two groups	mean and standard deviation	both distributions are symmetric	no causal claim
association	correlation coefficient	a linear pattern is suspected	association, not causation
prediction	regression of y on x	y is to be predicted from x	extrapolation is unreliable
conditional probability	Bayes' theorem	the reverse conditional is needed	depends on the prevalence

CHAPTER 12 · INVESTIGATION 12.3

Cumulative practice: statistics and probability

What do you notice?

- Naming the measure before computing exposed one case where the mean had been used on skewed data.
- The limit of each conclusion was stated in one clause.
- The probability questions were distinguished by which conditional direction was needed.
- Every answer was checked against a small case that could be listed.

> **Observation.** Each statistics question has a measure matched to its purpose, and each conclusion has a stated limit.

MATHEMATICAL CONCLUSION

Statistics questions are classified by whether they ask for a centre, a dispersion, an association or a prediction, and probability questions by which rule applies. Skewed data call for the median and the interquartile range, and every conclusion states the sample and its origin. Here the positive result meant only an 8.7 per cent chance of disease.

Your own words

CHAPTER 12 · 12.3

Cumulative practice: statistics and probability

Statistics questions are classified by whether they ask for a centre, a dispersion, an association or a prediction. Each needs a different measure, and using the wrong one is the commonest error.

When the data are symmetric the mean and the standard deviation use the most information; when they are skewed the median and the interquartile range are the appropriate measures, because they resist extremes.

Probability questions are classified by whether they ask for a simple probability, a conditional probability, an independence test, a total probability or a count. Each is a different rule, and naming the rule is the justification.

Every statistical conclusion states the sample and the origin of the data, because the description extends to a wider population only if the sample was representative.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 12 · 12.3 · WORKED EXAMPLE 1

Cumulative practice: statistics and probability

WORKED EXAMPLE 12.3.1

Problem. A data set has mean 30, median 22 and mode 18. Describe its shape and say which measure of centre to report.

measure	value	position
mode	18	lowest
median	22	middle
mean	30	highest
conclusion	mean > median > mode	skewed right

The order of the measures diagnoses the skew.

Mathematical reasoning. Compare the three measures and use their order to identify the skew.

1. The three measures are 18, 22 and 30.
2. The mean is the largest, so the distribution is skewed to the right.
3. A right skew means a long tail of large values pulling the mean upwards.
4. The median is the better measure of centre, because it is resistant to the tail.
5. So the report gives the median, 22, with a note that the distribution is skewed.

Final answer. skewed to the right; report the median, 22

Interpretation. The order of the three measures is the diagnosis, and the skew decides which measure to report.

CHAPTER 12 · 12.3 · WORKED EXAMPLE 2

Cumulative practice: statistics and probability

WORKED EXAMPLE 12.3.2

Problem. A disease affects 1 in 500. A test has sensitivity 95 per cent and a false-positive rate of 2 per cent. Find the probability that a person who tests positive has the disease.

group	test positive	test negative	total
has the disease	190	10	200
does not	1 996	97 804	99 800
total	2 186	97 814	100 000

Out of 100 000 people, 2186 test positive.

Mathematical reasoning. Scale the table to whole numbers, then divide the true positives by the total positives.

1. Take 100 000 people: 200 have the disease and 99 800 do not.
2. Of the 200, 95 per cent, or 190, test positive.
3. Of the 99 800, 2 per cent, or 1996, test positive.
4. So 190 + 1996 = 2186 people test positive.
5. The probability is 190/2186 ≈ 0.0869, about 8.7 per cent.

Final answer. about 0.0869

Interpretation. The base-rate effect is stark here: a 95 per cent sensitive test on a rare disease gives a positive result that is wrong about 91 per cent of the time.

CHAPTER 12 · INVESTIGATION 12.4

Unfamiliar problems and extended reasoning

INQUIRY QUESTION

How is a problem with no familiar structure approached?

YOUR TASK

Take four unfamiliar problems, each from a different strand. For each, describe how you found the structure, set out the solution as a chain of justified steps, and state one thing your solution does not show.

problem	how the structure was found	chain of steps	what is not shown
counting diagonals	a small case was drawn	count pairs, subtract sides, simplify	nothing further is needed
sum of odd numbers	examples suggested a pattern	express algebraically, simplify	the argument does not extend to even numbers
an unfamiliar triangle	the figure was drawn and the given parts marked	identify the rule, apply, check the angle sum	the check does not exclude a second triangle unless the ambiguous case is tested
an unfamiliar probability	a small case was listed exhaustively	count the outcomes, form the ratio	the count assumes the outcomes are equally likely

CHAPTER 12 · INVESTIGATION 12.4

Unfamiliar problems and extended reasoning

What do you notice?

- Drawing a small case was the most productive first move in three of the four problems.
- Writing the chain of steps exposed one step that had been assumed rather than justified.
- The ambiguous case had to be checked explicitly, which was the limit of one solution.
- Every solution stated something it did not show.

> **Observation.** An unfamiliar problem is approached by finding its structure, and the solution is a chain of justified steps with stated limits.

MATHEMATICAL CONCLUSION

An unfamiliar problem is approached by simplifying it, solving a special case, examining a small instance or working backwards, and each is a way of finding the structure. The solution is then a chain of justified steps, and it states what it does not show. Examples suggest a conjecture; only an argument that covers every case proves it.

Your own words

CHAPTER 12 · 12.4

Unfamiliar problems and extended reasoning

An unfamiliar problem is approached by simplifying it, solving a special case, looking at a small instance, or working backwards from the answer. None of these is the method; all of them are ways of finding the structure, and the generalisation that follows is what is being looked for.

Extended reasoning is a chain of justified steps, each of which would be accepted on its own. The chain is only as strong as its weakest step, so each step is checked before the next is taken, and the representation chosen makes the chain easier to follow.

A conjecture is formed from examples and then justified. The examples suggest; only the justification proves, and a conjecture that has not been justified is a hypothesis.

Recognising the limits of what has been shown is part of the answer. A solution that states what it has not proved is complete; one that does not is incomplete, however correct the computation.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 12 · 12.4 · WORKED EXAMPLE 1

Unfamiliar problems and extended reasoning

WORKED EXAMPLE 12.4.1

Problem. Find the number of diagonals of a polygon with n sides, and prove the formula.

step	working	result
pairs of vertices	$n(n-1)/2$	—
sides	n	—
diagonals	$n(n-1)/2 - n$	$n(n-3)/2$
check at $n = 4$	$4(1)/2$	2
check at $n = 5$	$5(2)/2$	5

The formula is checked on two small cases.

Mathematical reasoning. Count the pairs of vertices, subtract the sides, and check on small cases.

1. Every pair of vertices gives either a side or a diagonal.
2. The number of pairs is $n(n-1)/2$.
3. Of these, n are sides, so the number of diagonals is $n(n-1)/2 - n$.
4. Simplifying gives $n(n-3)/2$.
5. Check: a quadrilateral has $4(1)/2 = 2$ diagonals and a pentagon has $5(2)/2 = 5$, as required.

Final answer. $n(n-3)/2$ diagonals

Interpretation. The two small cases confirm the formula but do not prove it; the proof is the counting argument.

CHAPTER 12 · 12.4 · WORKED EXAMPLE 2

Unfamiliar problems and extended reasoning

WORKED EXAMPLE 12.4.2

Problem. A student finds that $1 + 3 + 5 + 7 = 16$ and conjectures that the sum of the first n odd numbers is n^2. Justify or refute the conjecture.

n	sum of the first n odd numbers	n^2	conjecture holds?
1	1	1	yes
2	4	4	yes
3	9	9	yes
4	16	16	yes
5	25	25	yes

Five cases support the conjecture.

Mathematical reasoning. Justify the conjecture by an argument that covers every case, not by more examples.

1. The examples support the conjecture but do not prove it.
2. To prove it, write the sum as $(2 \times 1 - 1) + (2 \times 2 - 1) + \ldots + (2n - 1)$.
3. This is $2(1 + 2 + \ldots + n) - n$.
4. Using the formula for the sum of the first n integers, this is $2 \times n(n + 1)/2 - n = n^2$.
5. So the conjecture is true, and the justification covers every n.

Final answer. the conjecture is true, and the proof is the algebraic simplification

Interpretation. Examples suggest and the argument proves, and the difference is what the inquiry statement calls mathematical maturity.

CHAPTER 12 · INVESTIGATION 12.5

Cumulative examination

INQUIRY QUESTION

How is examination technique different from coursework?

YOUR TASK

Take a past examination paper. Attempt it under timed conditions, allocating time by marks. Afterwards, review each answer and record the check you used, the time spent, and one thing you would do differently.

question	marks	time spent	check used	what to change
algebra	6	6 min	substitution	nothing
trigonometry	8	12 min	angle sum	draw the figure first
probability	7	10 min	list a small case	name the rule before computing
modelling	12	20 min	estimate the order of magnitude	state the assumptions earlier
geometry	5	15 min	—	move on after 8 minutes

CHAPTER 12 · INVESTIGATION 12.5

Cumulative examination

What do you notice?

- The geometry question took three times its share of time, which cost marks elsewhere.
- Two answers were wrong and were caught only by the checks, both of which took seconds.
- The modelling question needed its assumptions stated early, because the marks were for the assumptions as much as the arithmetic.
- Allocating time by marks rather than by order would have gained several marks.

Observation. Examination technique is a separate skill from the mathematics, and it is learned by review.

MATHEMATICAL CONCLUSION

An examination rewards the choice and statement of a method, the allocation of time by marks, and the use of a check different from the route used to obtain the answer. A question that cannot be finished should still be started, because partial credit is awarded for a correct method. Here the doubling rate was about 9.05 per cent per year.

Your own words

CHAPTER 12 · 12.5

Cumulative examination

An examination rewards the choice of method as much as the execution, and it rewards the statement of the method even when the arithmetic is imperfect. Writing down the intended method is the first thing to do on any question.

Time is allocated by marks, not by order. A question worth few marks is not worth many minutes, and leaving a hard question until last is a strategy rather than a defeat.

Every answer is checked by a route that is different from the one used to obtain it. Substitution, an estimate, a special case and a dimension check are the four standard checks, and each takes seconds.

A question that cannot be solved should still be started: the method is written down, the first step is taken, and the partial answer is left. Partial credit is awarded for a correct method, and an unanswered question scores nothing.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 12 · 12.5 · WORKED EXAMPLE 1

Cumulative examination

WORKED EXAMPLE 12.5.1

Problem. Solve the question: a population doubles every 8 years. Find the annual percentage growth rate, to three significant figures.

step	working	result
model	$P = P_0 \times 2^{(t/8)}$	—
annual factor	$2^{(1/8)}$	1.0905
rate	$2^{(1/8)} - 1$	0.0905
percentage	0.0905×100	9.05 per cent

The exponent counts eight-year periods, not years.

Mathematical reasoning. Write the model, extract the annual factor, and convert to a percentage.

1. The model is $P(t) = P_0 \times 2^{(t/8)}$, because the doubling period is 8 years.
2. The annual factor is $2^{(1/8)}$, since $t = 1$ gives one year.
3. $2^{(1/8)} \approx 1.0905$.
4. So the annual rate is $1.0905 - 1 = 0.0905$.
5. As a percentage this is about 9.05 per cent per year.

Final answer. about 9.05 per cent per year

Interpretation. The error to avoid is dividing 100 per cent by 8, which gives 12.5 per cent and is wrong, because the growth compounds.

CHAPTER 12 · 12.5 · WORKED EXAMPLE 2

Cumulative examination

WORKED EXAMPLE 12.5.2

Problem. A candidate has 20 minutes left and two questions remaining, one worth 3 marks and one worth 8. Advise on the order.

question	marks	time available	advice
the 3-mark question	3	about 5 minutes	do it first
the 8-mark question	8	about 15 minutes	then start it and write the method
if stuck	—	—	write the method and the first step

Time is allocated by marks.

Mathematical reasoning. Allocate time in proportion to the marks and start the harder question with the method.

1. The 3-mark question should take about a quarter of the remaining time.
2. So it is done first, in about 5 minutes.
3. The 8-mark question then has about 15 minutes, which is enough to write the method and several steps.
4. If it cannot be finished, the method and the first step are left on the page.
5. Partial credit is awarded for a correct method, so an unanswered question is the only one that scores nothing.

Final answer. the 3-mark question first, then start the 8-mark one with the method

Interpretation. The strategy is to secure the easy marks and to start the hard question with a stated method.

CHAPTER 12 · INVESTIGATION 12.6

From MYP to DP: expectations and readiness

INQUIRY QUESTION

What changes in the transition to DP mathematics?

YOUR TASK

Assess your own readiness for DP mathematics on the four skills in the inquiry statement: choosing a method, justifying it, connecting it to other ideas, and recognising the limits of what has been shown. For each, give a rating, an example from your own work, and one thing you will practise.

skill	rating	example from my work	what I will practise
choosing a method	developing	I chose the cosine rule for an SAS triangle	classifying questions by structure first
justifying it	developing	I named the tangent-radius property	writing the reason beside every step
connecting ideas	emerging	I linked the determinant to area	looking for the same idea in two strands
recognising limits	emerging	I labelled a prediction as extrapolation	stating what each solution does not show

CHAPTER 12 · INVESTIGATION 12.6

From MYP to DP: expectations and readiness

What do you notice?

- The four skills could each be rated against a concrete example.
- The weakest skill was recognising limits, because it had rarely been asked for.
- Connecting ideas was easier once a single idea, the determinant, was followed across two strands.
- Each rating came with a specific practice, not a general intention.

> **Observation.** Readiness is assessed by example, and each gap has a specific practice.

MATHEMATICAL CONCLUSION

DP mathematics expects a student to choose a method, justify it, connect it to other ideas and recognise the limits of what has been shown, and the content is largely familiar. Readiness is assessed against those four skills rather than against recall, and each gap has a specific practice. Here the four skills were rated developing, developing, emerging and emerging.

Your own words

CHAPTER 12 · 12.6

From MYP to DP: expectations and readiness

DP mathematics expects a student to choose a method, justify it, and connect it to other ideas, and the logic of the argument is examined at every step. The content is largely familiar; the expectation of independent choice is what is new.

The DP courses are split by emphasis: one for students who need mathematics as a tool, one for students who will study it seriously, and one for those who will study it at depth. The choice depends on what the student will do next, not on how easy the course is.

Investigation and internal assessment are part of the DP, and the mathematical exploration rewards the same habits as the capstone project: a stated problem, stated assumptions, a test and a reflection on the limits.

Readiness is not the recall of formulas. It is the ability to read an unfamiliar question, identify its structure, select a method, and justify the selection, and those are the habits this course has been building.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 12 · 12.6 · WORKED EXAMPLE 1

From MYP to DP: expectations and readiness

WORKED EXAMPLE 12.6.1

Problem. Compare the expectations of an MYP extended question with a DP examination question on the same content.

aspect	MYP extended questio	DP examination question
structure	signposted in parts	often a single unfamiliar stem
method	suggested or implied	chosen by the student
justification	required for the top band	required throughout
marks	for the answer and the reasoning	for the method, the answer and the reasoning
time	part of a coursework task	allocated by marks

The content is familiar; the independence is new.

Mathematical reasoning. Compare the two forms on five aspects.

1. An MYP extended question is usually signposted in parts, each pointing at a method.
2. A DP question often presents a single unfamiliar stem and leaves the choice of method to the student.
3. Justification is required for the top band in MYP and throughout in DP.
4. Marks are awarded for the answer and the reasoning in both, but DP marks the method separately.
5. So the transition is from following a signposted route to choosing one.

Final answer. the content is familiar; the independence and the continuous justification are new

Interpretation. Naming what is new makes the transition manageable, because the mathematics is already known.

CHAPTER 12 · 12.6 · WORKED EXAMPLE 2

From MYP to DP: expectations and readiness

WORKED EXAMPLE 12.6.2

Problem. A student can recall every formula in the course but cannot choose between them. Advise on what to practise.

skill	status	what to practise
recall	strong	nothing further
recognition of structure	weak	classify questions by type
choice of method	weak	solve each type by two methods and compare
justification	weak	write the reason beside every step
limits	absent	state what each solution does not show

Recall is the least of the four skills.

Mathematical reasoning. Assess the four skills and name the practice each needs.

1. Recall is strong, so no further practice is needed there.
2. Recognition of structure is weak, so questions should be classified by type before solving.
3. Choice of method is weak, so each type should be solved by two methods and the results compared.
4. Justification is weak, so the reason should be written beside every step.
5. The habit of stating limits is absent, and is practised by writing what each solution does not show.

Final answer. practise recognition, choice, justification and limits

Interpretation. The four skills are what the inquiry statement calls mathematical maturity, and recall is only the first.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

12.A1 FLUENCY

Solve $x^2 - 5x + 6 = 0$.

12.A2 FLUENCY

Solve $3x^2 + 5x - 2 = 0$.

12.A3 FLUENCY

Given $f(x) = 4x - 7$, find $f^{-1}(x)$.

12.A4 FLUENCY

Given $f(x) = x^2$ and $g(x) = 3x - 1$, find $f(g(2))$.

12.A5 FLUENCY

Solve $2x - 5 > 7$.

12.A6 FLUENCY

Solve $x^2 + 2x - 15 \leq 0$.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

12.A7 FLUENCY

Find the 20th term of the arithmetic sequence 3, 8, 13, ...

12.A8 FLUENCY

Find the sum to infinity of 20 + 10 + 5 + ...

12.A9 FLUENCY

Solve $2^X = 30$, to three significant figures.

12.A10 FLUENCY

A town of 40 000 grows 3 per cent a year. Find the population after 10 years.

12.A11 FLUENCY

Find the determinant of [[3, 5], [7, 2]].

12.A12 FLUENCY

Given a = (2, −3) and b = (5, 1), find |a + b|.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

12.B1 REASONING

Investigate the number of methods available for solving a quadratic equation, and justify when each is preferred.

12.B2 CHALLENGE

Conjecture a rule for when the sine rule gives two answers, and test it on three cases.

12.B3 REASONING

Investigate the relationship between the area formula ½ab sin C and the cosine rule, and justify it.

12.B4 CHALLENGE

Take a circle theorem of your own and investigate how many other theorems follow from it, justifying each.

12.B5 REASONING

Investigate the connection between the determinant of a 2 by 2 matrix and the area of a triangle, and justify.

12.B6 REASONING

Conjecture a rule for the number of real roots of a quadratic from its discriminant, and test it.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

12.B7 CHALLENGE

Investigate the relationship between the correlation coefficient and the gradient of the regression line, and justify.

12.B8 REASONING

Investigate the number of methods for solving a quadratic equation and justify when each is preferred.

12.B9 REASONING

Conjecture a rule for the number of real roots of a quadratic from its discriminant, and test it.

12.B10 REASONING

Investigate the connection between the determinant of a 2 by 2 matrix and the area of a triangle, and justify.

12.B11 CHALLENGE

Investigate the relationship between the correlation coefficient and the gradient of the regression line, and justify.

CRITERION C Communicating

Criterion C practice · Communicating

12.C1 REASONING

Write a complete solution to solving $2x^2 - 3x - 4 = 0$, to three significant figures.

12.C2 APPLICATION

Write a complete solution to a population that doubles every 8 years, finding the annual percentage growth rate.

12.C3 REASONING

Write a complete solution to a triangle with sides 7 and 9 and included angle 68°, finding the third side and the area.

12.C4 APPLICATION

Write a complete solution to a screening problem with prevalence 1 in 400, sensitivity 96 per cent and false-positive rate 3 per cent.

12.C5 CHALLENGE

Write a complete solution to finding the inverse of $f(x) = (5x - 2)/(x + 3)$, with its domain.

12.C6 REASONING

Write a complete solution to maximising $P = 5x + 4y$ subject to $x + 2y \le 12$, $3x + 2y \le 18$, $x \ge 0$, $y \ge 0$.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

12.C7 REASONING

Write a complete solution to finding the area of the parallelogram spanned by (4, −1) and (2, 5), and the determinant of the matrix with those columns.

12.C8 REASONING

Write a complete solution to solving $2x^2 + 5x - 3 = 0$ by two methods.

12.C9 REASONING

Write a complete solution to a triangle with sides 10 and 13 and included angle 55°, finding the third side and the area.

12.C10 APPLICATION

Write a complete solution to a screening problem with prevalence 2 per cent, sensitivity 90 per cent and false-positive rate 8 per cent.

12.C11 FLUENCY

Write a complete solution to finding the sum of the first 20 terms of the geometric sequence with $u(1) = 3$ and $r = 1.05$.

12.C12 REASONING

Write a complete solution to maximising $P = 3x + 5y$ subject to $x + 2y \le 14$, $3x + y \le 15$, $x \ge 0$, $y \ge 0$.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

12.D1 APPLICATION

A company's output is modelled by $P(t) = 5000(1.06)^T$. Find the doubling time and the output after 15 years.

12.D2 APPLICATION

A loan of 9000 is repaid in 4 equal annual instalments at 8 per cent. Find the instalment and the total repaid.

12.D3 APPLICATION

Two schools' mean scores are both 62. School A has standard deviation 4 and school B has 15. Write a comparison and state one thing it cannot show.

12.D4 APPLICATION

A cuboid has edges 5, 12 and 13. Find the space diagonal and the angle it makes with the longest edge.

12.D5 APPLICATION

A bag has 5 red and 3 blue counters. Two are drawn without replacement. Find P(both red) and P(one of each).

12.D6 APPLICATION

A population doubles every 8 years. Find the annual percentage growth rate, to three significant figures.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

12.D7 REASONING

A student answers an examination question correctly but gives no reasons. Evaluate the likely outcome.

12.D8 REASONING

A candidate spends 25 minutes on a 4-mark question and leaves a 12-mark question unanswered. Evaluate the strategy.

12.D9 REASONING

A student solves an unfamiliar problem by listing the cases for n = 1, 2, 3 and 4 and generalising. Evaluate the solution.

12.D10 REASONING

A question asks for the area of a triangle with vertices (1, 1), (5, 2) and (2, 6). Solve it by two methods.

12.D11 REASONING

A student claims that because a model fits ten data points exactly, it must be correct. Evaluate.

12.D12 REASONING

A student's solution to an SSA problem reports one triangle without checking. Identify the error and state the correct procedure.

CHAPTER 12 · APPLICATION AND ASSESSMENT

Applying chapter 12

These tasks revise the whole course. Name the method you choose, justify the choice, and state the limits of what you have shown.

A1 · The cumulative paper

A paper contains five questions: a quadratic equation, a general triangle, a geometric series, a screening probability and a linear program. For each, name the method you choose, justify the choice in one sentence, solve it, and check the answer by a second route. Then state one thing your solutions collectively do not show.

Criteria C and D · choose and justify a method for each question, check each answer, and state the limit.

A2 · The unfamiliar problem

A problem states that a pattern of squares has 1, 5, 13 and 25 small squares in its first four figures. Find the number in the tenth figure, justify your formula, and then state one thing your justification does not show.

Criteria C and D · find the structure, justify the formula, and state the limit of the justification.

CHAPTER 12 · REVIEW

What you should now be able to do

Key ideas

- A question is classified by its structure, and the structure decides the method.
- Factorising, the quadratic formula and completing the square are three methods for one class of equation.
- The sine rule and the cosine rule are distinguished by whether the given angle is included.
- Two sides and a non-included angle give zero, one or two triangles.
- Circle theorems are chosen by naming the arc, the tangent or the chords.
- A three-dimensional problem is solved by choosing a plane.
- Skewed data call for the median and the interquartile range.
- Bayes' theorem reverses the conditioning, and the base rate decides the difference.
- A linear program's optimum is at a vertex, with a robustness statement.
- Examples suggest a conjecture; only an argument covering every case proves it.

Vocabulary

structure · method · justification · check · generalisation · representation · limit · conjecture · proof · extended reasoning · examination technique · readiness · transition · maturity · over-fitted

CHAPTER 12 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

12.R1 Solve $3x^2 - 7x + 2 = 0$ by two methods.

12.R2 A triangle has a = 8, b = 11 and A = 40°. How many triangles exist?

12.R3 Find the sum of the first 12 terms of 5, 8, 11, …

12.R4 A disease affects 1 in 200. A test is 95 per cent sensitive with a 5 per cent false-positive rate. Find P(disease | positive).

12.R5 Maximise $P = 4x + 5y$ subject to $x + y \le 6$, $x + 2y \le 8$, $x \ge 0$, $y \ge 0$.

12.R6 State the four skills named in the inquiry statement of this chapter.

CHAPTER 12 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> Mathematical maturity is the ability to choose a method, justify it, connect it to other ideas, and recognise the limits of what has been shown.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of logic and the related concepts of generalisation, representation and justification appear in your work?
- How is the global context of identities and relationships, and its exploration of how mathematical habits shape the way we reason elsewhere, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limits of what you have shown in this course, and what would you still like to ask?

CHAPTER 12 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

A method is a habit of reasoning. Name the habit, justify it, and say where it stops working.

Service as Action

COMMUNITY NEED	Students leaving school are not shown what the next stage expects.
MATHEMATICAL CONNECTION	The whole course, and the habits it builds.
PROPOSED ACTION	Interview two people who use mathematics in their work, identify the four skills they rely on, and publish the findings for younger students.
EXPECTED IMPACT	Younger students see what the habits are for.
STUDENT REFLECTION	Record the interviews and the skills you identified.

Community Project

PROJECT IDEA	A readiness audit of the school's mathematics students.
NEED	Students choose their next course by how easy it looks rather than by what they will do next.
RESEARCH QUESTION	Which of the four skills are strongest in the school's students, and what would each need to practise?
MATHEMATICS	All strands, and the habits that connect them.
ACTION OR OUTCOME	A one-page readiness profile with a practice for each gap.
BENEFICIARIES	Every student choosing a next course.
EVIDENCE	The profile and the evidence behind each rating.

Personal Project connection · MYP 5

PERSONAL INTEREST	A subject or career you are considering.
LEARNING GOAL	To identify which of the four skills that path needs most and practise it.
PRODUCT OR OUTCOME	A personal readiness profile with a practice plan.
MATHEMATICS	All strands, and the habits that connect them.
PROCESS	Rate the four skills against examples, and choose a practice for the weakest.
EVIDENCE	The profile and the practice plan.

CHAPTER 1 · CRITERION A · KNOWING AND UNDERSTANDING

1.A1 **Degree 5, leading coefficient 3.**
The degree is the highest power and the leading coefficient is its multiplier.

1.A2 **$x^3 + 5x^2 - x + 2$, degree 3.**
Collect like terms.

1.A3 **$2x^3 + 5x^2 - 7x + 2$, degree 3.**
Each term of the first multiplies each term of the second; degrees add.

1.A4 **$16 - 12 - 10 - 1 = -7$.**
Substitute and evaluate term by term.

1.A5 **Quotient $3x^2 + 2x - 6$, remainder 8.**
Match leading terms and subtract; check with the division identity.

1.A6 **Quotient $x^2 + 2x + 4$, remainder 0.**
$x - 2$ is a factor, so the remainder is zero.

1.A7 **$p(1) = 2 - 5 + 3 = 0$.**
The remainder theorem replaces the division by substitution.

1.A8 **$p(-3) = -27 + 18 - 7 = -16$.**
The divisor $x + 3$ is $x - (-3)$, so $a = -3$.

1.A9 **$(x - 1)(x - 2)(x + 1)$.**
$p(1) = 0$, then divide and factorise the quadratic.

1.A10 **$(x - 2)/(x + 3)$, for $x \neq -3$ and $x \neq -2$.**
Factorise both parts and cancel $x + 2$.

1.A11 **x is not a factor of $x + 3$, so there is nothing to cancel.**
Cancellation needs a common factor, not a common term.

1.A12 **$x = (-3 + \sqrt{33})/2$ or $x = (-3 - \sqrt{33})/2$; neither is 0 nor −1, so both are valid.**
Clear the fractions, solve the quadratic, then substitute each candidate.

CHAPTER 1 · CRITERION B · INVESTIGATING PATTERNS

1.B1 **Yes: substituting $x = 1$ gives $1^N - 1 = 0$ for every n, so by the factor theorem $x - 1$ is always a factor.**
One substitution covers every n.

1.B2 **$n = 2$: $(x - 1)(x + 1)$; $n = 3$: $(x - 1)(x^2 + x + 1)$; $n = 4$: $(x - 1)(x + 1)(x^2 + 1)$; $n = 5$: $(x - 1)(x^4 + x^3 + x^2 + x + 1)$; $n = 6$: $(x - 1)(x + 1)(x^2 + x + 1)(x^2 - x + 1)$.**
Divide out $x - 1$ first, then factorise the quotient.

1.B3 **The remainder is 1, because $1^N = 1$ for every n.**
The remainder theorem gives the remainder as $p(1)$.

1.B4 **n even gives 2 real roots (± 1); n odd gives 1 real root (1).**
A real nth root of 1 exists only for 1, and also -1 when n is even.

1.B5 **The discriminant $4p^3 + 27q^2 = 0$; a double root forces it to vanish.**
A double root means the factorised form contains a squared factor.

1.B6 **Any correct answer, for example $p(x) = (x - 1)(x - 2)(x - 4) = x^3 - 7x^2 + 14x - 8$; dividing by $x - 2$ gives $x^2 - 5x + 4$ and $(x - 2)(x^2 - 5x + 4) = p(x)$.**
Use the division identity as the check.

1.B7 **No: the degree of a sum is at most the larger degree, and here at most 3, because no term of degree 4 can be produced by adding.**
Compare the possible powers in each term.

1.B8 **The remainder is always 0, because $x = 1$ is a root of $x^N - 1$ for every positive integer n.**
Use the factor theorem with $p(1) = 1^N - 1$.

1.B9 **$x^2 - y^2 = (x - y)(x + y)$; $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$; $x^4 - y^4 = (x - y)(x + y)(x^2 + y^2)$. In every case $x - y$ is a factor.**
Each expression is zero when $x = y$, so $x - y$ is a factor by the factor theorem.

1.B10 **For $x^3 + bx^2 + cx + d$ with roots r, s and t, the sum $r + s + t = -b$, the sum of the pairwise products $rs + rt + st = c$, and the product $rst = -d$.**
Expand $(x - r)(x - s)(x - t)$ and compare coefficients.

1.B11 **The number of positive roots is at most the number of sign changes, by Descartes' rule of signs; the difference is always even.**
Count the sign changes in the ordered coefficients.

CHAPTER 1 · CRITERION C · COMMUNICATING

1.C1 **$(x-2)(x^2+2x+4)/((x-2)(x+2)) = (x^2+2x+4)/(x+2)$ for $x \neq 2$ and $x \neq -2$.**
Factorise both parts, cancel the common factor, state the exclusions.

1.C2 **$x^2+3x+2 = x^2-3x+2$, so $6x = 0$ and $x = 0$; neither denominator vanishes, so $x = 0$ is valid.**
Cross-multiply, then substitute back.

1.C3 **$p(-1) = 0$, so divide by $x+1$ to get $x^2-x-6 = (x-3)(x+2)$; $x^3-7x-6 = (x+1)(x-3)(x+2)$.**
Find a root by inspection, divide, factorise the quadratic.

1.C4 **$2 < x < 5$ or $x < -1$, with the roots excluded.**
Mark the roots and count the negative factors in each interval.

1.C5 **Quotient $2x^2-x-3$, remainder $-3x-2$; check by multiplication.**
Match leading terms, subtract, repeat.

1.C6 **$x^2-5x+5 = 0$, so $x = (5 \pm \sqrt{5})/2$; both are allowed, so both are valid.**
Clear the fractions, solve, substitute.

1.C7 **Length $x+3$ cm, with $x > -2$ so that the width is positive.**
Divide the area by the width and apply the physical restriction.

1.C8 **$(3x+5)/((x-1)(x+2)) = 8/3/(x-1) + 1/3/(x+2)$, since $A+B = 3$ and $2A-B = 5$ give $A = 8/3$ and $B = 1/3$.**
Set up the identity, clear the denominator, equate coefficients and solve.

1.C9 **$2(x+2) = 3(x-1)$ gives $2x+4 = 3x-3$, so $x = 7$; it does not make either denominator zero, so it is valid.**
Cross-multiply, solve, then substitute back into the original denominators.

1.C10 **Divide $p(x)$ by $x-a$ to get $p(x) = (x-a)q(x) + R$ with R constant; substituting $x = a$ gives $p(a) = R$, so $R = 0$ exactly when $x-a$ is a factor.**
Use the division algorithm and substitute.

1.C11 **The roots are -2, 0 and 2; the sign is negative on $(-2, 0)$ and positive on $(0, 2)$; the turning points are at $x = 2/\sqrt{3}$ with value $-16/(3\sqrt{3})$ and at $x = -2/\sqrt{3}$ with value $16/(3\sqrt{3})$.**
Factorise, find the roots, differentiate for the turning points, and test each interval.

1.C12 **$x^2h = 120$ and $2x^2 + 4xh = 166$ give $2x^2 + 480/x = 166$, so $2x^3 - 166x + 480 = 0$, that is $x^3 - 83x + 240 = 0$; $x \approx 3.34$ satisfies it, checked by $2(3.34)^2 + 480/3.34 \approx 22.3 + 143.7 = 166.0$.**
Form both equations, eliminate h, and solve the resulting cubic, checking each root.

CHAPTER 1 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

1.D1 **Let the second be t hours: $1/t + 1/(t + 3) = 1/2$ gives $t^2 - t - 6 = 0$, so $t = 3$ and the first takes 6 hours.**
Build the equation from the meaning of the quantities.

1.D2 **$p(t) = t(t - 3)^2$, so output is positive for $t > 0$ and zero at $t = 0$ and $t = 3$; the model applies for $t \geq 0$.**
Factorise and use a sign line.

1.D3 **Height $x + 3$ cm, with $x > -1$ so that the base area is positive.**
Divide the volume by the base area.

1.D4 **$30/v + 1 = 30/(v - 4)$ gives $v^2 - 4v - 120 = 0$, so $v = 12$ km/h; the negative root is rejected as meaningless.**
Build from time = distance ÷ speed.

1.D5 **Depth $2x + 3$ mm, with $x > 1.5$ so that the width is positive.**
Divide and apply the physical restriction.

1.D6 **$n^2 - 12n + 20 > 0$ gives $(n - 2)(n - 10) > 0$, so $n < 2$ or $n > 10$; with $n \geq 0$ the profitable ranges are $0 \leq n < 2$ and $n > 10$.**
Solve the inequality and apply the domain.

1.D7 **$1/t + 1/(2t) = 1/6$ gives $3/(2t) = 1/6$, so $t = 9$ hours and the slower takes 18 hours; check: $1/9 + 1/18 = 3/18 = 1/6$.**
Rates add.

1.D8 **Positive for $-1 < a < 2$ and $a > 8$; the practical domain is $a \geq 0$, so $0 \leq a < 2$ and $a > 8$.**
Sign line, then apply the domain.

1.D9 **The model simplifies to $t + 4$ for $t \neq 4$, but $t = 4$ is excluded because the original denominator vanishes; the model is undefined there.**
Factorise, cancel, state the exclusion.

1.D10 **$c(n) = 50 - 200/(n + 4)$, which rises towards 50; $c(n) > 20$ gives $200/(n + 4) < 30$, so $n + 4 > 6.67$ and $n > 2.67$.**
Divide to expose the limiting behaviour.

1.D11 **$s(L) = L^2 + 3L + 9$ for $L \neq 3$, and $s(6) = 36 + 18 + 9 = 63$ m; at $L = 3$ the model is undefined, not zero.**
Factorise the numerator.

1.D12 **False: $(x + 1)/(x - 2) = 3$ has $x = 3.5$, and $2/x + 3/(x + 1) = 1$ has two valid solutions; an extraneous solution only removes the candidates that make a denominator zero.**
Look for an equation with one valid and one extraneous candidate.

CHAPTER 2 · CRITERION A · KNOWING AND UNDERSTANDING

2.A1 **Domain $x \geq -3$, range $f(x) \geq 0$.**
The expression under the root must be non-negative.

2.A2 **$f(3) = 5$ and $f(-2) = -15$.**
Substitute and evaluate.

2.A3 **$x = 1$ and $x = 2$; the minimum is -0.25, so the range is $f(x) \geq -0.25$.**
Complete the square or use the axis of symmetry.

2.A4 **$f(g(x)) = 2x - 7$ and $g(f(x)) = 2x - 3$.**
The inner function acts first.

2.A5 **$f(g(x)) = (x + 3)^2$, domain all real x.**
Both functions accept every real input.

2.A6 **$f(f(x)) = x$, domain $x \neq 0$.**
The reciprocal of the reciprocal is the original.

2.A7 **Translation 4 left and 1 down.**
The bracket gives the horizontal change, with the sign reversed.

2.A8 **Translation 2 right, stretch by factor 3 parallel to the y-axis, reflection in the x-axis.**
Separate the inside of the bracket from the outside.

2.A9 **$f^{-1}(x) = 4x - 3$; $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$.**
Rearrange, swap the letters, verify both ways.

2.A10 **$x + 6$ and $x + 9$.**
Compose step by step.

2.A11 **Because $f(-2) = f(2) = 4$, so two inputs share one output and the reversal is ambiguous.**
Two inputs with the same output block the inverse.

2.A12 **$(4, 7)$.**
Apply the coordinate rule $(x, y) \to (x + 3, 2y - 1)$.

CHAPTER 2 · CRITERION B · INVESTIGATING PATTERNS

2.B1 **They commute exactly when $ad + b = bc + d$; for example two translations, which have $a = c = 1$, always commute.**
Compare $f(g(x))$ with $g(f(x))$ as polynomials.

2.B2 **One fixed point when $a \neq 1$, namely $x = b/(1 - a)$; none or all when $a = 1$, depending on b.**
Solve $f(x) = x$.

2.B3 **The graphs coincide, because the function is its own inverse: $f(f(x)) = x$.**
Reflect in the line $y = x$ and compare.

2.B4 **For example $f(x) = x^2$ restricted to $x \geq 0$; the restricted function is strictly increasing, so equal outputs force equal inputs.**
Use the horizontal line test on the restricted graph.

2.B5 **$k = 2$ compresses towards the y-axis by factor 2; $k = 0.5$ stretches away from it by factor 2; $k = -1$ reflects in the y-axis.**
Track one labelled point.

2.B6 **$f^{-1}(x) = (b - dx)/(cx - a)$, valid when $ad - bc \neq 0$; for example the inverse of $(2x + 1)/(x - 3)$ is $(1 + 3x)/(x - 2)$.**
Rearrange to make x the subject and swap.

2.B7 **Yes: if $a < b$ then $f^{-1}(a) < f^{-1}(b)$, because the reverse inequality would force f to reverse the order.**
Argue by contradiction using the definition.

2.B8 **One fixed point when $a \neq 1$, namely $x = b/(1 - a)$; when $a = 1$ there are none if $b \neq 0$ and every point is fixed if $b = 0$.**
Solve $f(x) = x$.

2.B9 **They commute exactly when $ad + b = bc + d$, which holds in particular when $a = c$, so two stretches of the same factor commute.**
Compare $f(g(x))$ with $g(f(x))$ as polynomials.

2.B10 **The graph is stretched parallel to the x-axis by factor $1/k$; $k = 2$ compresses it towards the y-axis and $k = 0.5$ stretches it away.**
Track one labelled point through the coordinate rule.

2.B11 **It is the reflection in the y-axis; where f is an even function the two graphs coincide, and where f is odd they coincide after a reflection in the origin.**
Apply the coordinate rule $(x, y) \to (-x, y)$.

CHAPTER 2 · CRITERION C · COMMUNICATING

2.C1 **$f^{-1}(x) = (1 + 3x)/(2 - x)$, domain $x \neq 2$; check by composition in both directions.**
Clear the fraction, rearrange, swap, verify.

2.C2 **Translation 2 right, stretch by factor 4 parallel to the y-axis, translation 1 up.**
Read the inside of the bracket first, then the outside.

2.C3 **$f(g(x)) = \sqrt{(x^2 - 1)}$, domain $x \leq -1$ or $x \geq 1$.**
The inner output must be a permitted input of the outer function.

2.C4 **$2x + 5 = (x - 5)/2$ gives $4x + 10 = x - 5$, so $x = -5$.**
Equate the function and its inverse and solve.

2.C5 **Complete the square: $f(x) = (x + 3)^2 - 5$, so restrict to $x \geq -3$; then $f^{-1}(x) = -3 + \sqrt{(x + 5)}$ with domain $x \geq -5$.**
Locate the axis of symmetry, then invert.

2.C6 **$40 + 0.25d = 55 + 0.15d$ gives $0.1d = 15$, so $d = 150$ km; the second model is the first translated up by 15 and stretched parallel to the x-axis by factor 5/3.**
Model each, equate, then read the transformation.

2.C7 **Reflect every point of $y = f(x)$ in the line $y = x$, because the inverse swaps the coordinates of each point; the domain and range swap too.**
The inverse sends outputs back to their inputs.

2.C8 **$f^{-1}(x) = (3x + 1)/(2x - 4)$, with domain $x \neq 2$; both compositions give x.**
Clear the fraction, rearrange, swap, verify.

2.C9 **Translation 2 left, stretch by factor 3 parallel to the y-axis, translation 1 down.**
Read the inside of the bracket first, then the outside.

2.C10 **$f(g(x)) = 1/(x^2 + 1)$, with domain all real x.**
The inner output must never equal -2, and $x^2 - 1 = -2$ has no real solution.

2.C11 **If $a < b$ then $f^{-1}(a) > f^{-1}(b)$, because the reverse inequality would force f to increase the order, contradicting that f is decreasing.**
Argue by contradiction using the definition of decreasing.

2.C12 **$L(n) = 297 \times 0.8^N$; $0.8^N < 10/297$ gives $n > \log(10/297) \div \log 0.8 \approx 17.1$, so 18 passes.**
Apply the decay model and solve with logarithms.

CHAPTER 2 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

2.D1 **$r = (R - 32)/1.8$; it converts a reading back to the original scale.**
Rearrange and swap.

2.D2 **Halving t doubles the exposure, which is a horizontal stretch by factor 0.5 of the model in t.**
The exposure is inversely proportional to the time.

2.D3 **$c_1 = 300 + 120n$ and $c_2 = 180 + 150n$; the rival is cheaper when $120n + 300 > 150n + 180$, that is $n < 4$, so for fewer than 4 weeks.**
Model, compare, solve.

2.D4 **$D = 40(0.9)^{(s + 10)} = 40(0.9)^{10}(0.9)^S \approx 13.9(0.9)^S$ cm, so the later start begins at about 13.9 cm instead of 40 cm.**
A change of origin is a horizontal translation.

2.D5 **Because the rate changes, the graph is two line segments with different gradients; no single stretch and translation connects them.**
Compare the gradients of the two pieces.

2.D6 **$P(t) = 250 + 36t$; a vertical translation of 50 and a vertical stretch by factor 1.2.**
Apply the fixed change then the multiplicative change.

2.D7 **$100(1 - (0.5)^{(d/30)}) > 90$ gives $(0.5)^{(d/30)} < 0.1$, so $d/30 > \log(10)/\log(2) \approx 3.32$ and $d > 99.7$ days; the transformation is a horizontal stretch by factor 30, which slows the decay thirtyfold.**
Solve the inequality and read the transformation.

2.D8 **$P(s) = 400(1.05)^{(s + 5)} = 400(1.05)^5(1.05)^S \approx 510.5(1.05)^S$; the new origin population is about 510.5.**
A change of origin is a horizontal translation.

2.D9 **$1.08^T = 2$ gives $t = \log 2 \div \log 1.08 \approx 9.01$ years; at 12 per cent, $t = \log 2 \div \log 1.12 \approx 6.12$ years.**
Take logarithms of both sides.

2.D10 **False in general: for $f(x) = x^2$, $f(x) + 3 = x^2 + 3$ moves the graph up 3, while $f(x + 3) = (x + 3)^2$ moves it left 3. They agree only for functions with special symmetry, such as $f(x) = x$.**
Test each transformation on a labelled point.

2.D11 **$V = 6h^2$, a vertical stretch by factor 2; doubling every width doubles the cross-sectional area and hence the volume.**
Area scales with the square of a length, so doubling the width multiplies area by 4 and volume by 4 for the same depth — the correct model is $12h^2$.

2.D12 **15 is the mean temperature, 8 the amplitude, and 182.5 half a year in days; the warmest day is at $d = 91.25$ and the coolest at $d = 273.75$, with 23 °C and 7 °C.**
Read the parameters against the parent sine curve.

CHAPTER 3 · CRITERION A · KNOWING AND UNDERSTANDING

3.A1 **0.0001.**
A negative power gives the reciprocal of the positive power.

3.A2 **$8^1 = 8$.**
Subtract the exponents: $2/3 - (-1/3) = 1$.

3.A3 **1/9.**
Take the cube root, then square, then reciprocate.

3.A4 **$x = 3$.**
Write 64 as a power of 2 and equate exponents.

3.A5 **$x = -2$.**
$1/81 = 3^{-4}$, so $2x = -4$.

3.A6 **$\log 125 = 3(0.6990) \approx 2.097$; $\log 0.2 = -\log 5 \approx -0.6990$.**
$125 = 5^3$ and $0.2 = 1/5$.

3.A7 **$x = 81$.**
The definition gives $x = 3^4$.

3.A8 **$x = 4$.**
Combine with the product law, then reject $x = -2$.

3.A9 **$\log 16$.**
$\log(8 \times 12 \div 6) = \log 16$.

3.A10 **$x \approx 2.32$.**
Take logarithms: $x = \log 5 \div \log 2$.

3.A11 **Because $b^0 = 1$, and the logarithm is the exponent.**
Use the definition.

3.A12 **Taking logarithms gives $2x \ln 3 = (2x + 2) \ln 2$, so $2x(\ln 3 - \ln 2) = 2 \ln 2$ and $x = \ln 2 \div \ln(3/2) \approx 1.71$.**
Take logarithms of both sides when the bases cannot be matched.

CHAPTER 3 · CRITERION B · INVESTIGATING PATTERNS

3.B1 **The values 2, 2.5937, 2.7048, 2.7169 approach e ≈ 2.71828 from below.**
Compute the values and observe the trend.

3.B2 **The digit count is the whole part of n $\log_{10}3$, plus one.**
Compute n $\log_{10}3$ and compare with the actual values.

3.B3 **The leading digit cycles irregularly through 3, 9, 2, 8, 2, 7, 2, 6, 1, 5, 1, 4, with 1 and 2 appearing most often.**
Track the fractional part of n $\log_{10}3$.

3.B4 **Yes: 2^N exceeds n^2 for n ≥ 5, and the gap widens without bound because the exponential grows faster than any power.**
Test small cases and find the crossover.

3.B5 **The half-life halves, because it is ln 2 ÷ k and k has doubled.**
Use the formula for the half-life.

3.B6 **The sums approach 2 from below: 1.9980 after 10 terms and 1.999998 after 20.**
Each term is half the previous one.

3.B7 **They are reflections of each other in the line y = x, because they are inverse functions.**
Reflect in y = x and compare.

3.B8 **The values 2.7048 and 2.7181 approach e ≈ 2.71828 from below, and the gap halves roughly each time n is squared.**
Compute the values and observe the convergence.

3.B9 **The leading digits cycle irregularly: 7, 4, 2, 1, 6, 4, 2, 1, 8, 5, with small digits appearing more often.**
Track the fractional part of n $\log_{10}7$.

3.B10 **The half-life is divided by 3, because it is ln 2 ÷ k and k has tripled.**
Use the formula for the half-life.

3.B11 **They are reflections of each other in the line y = x, because they are inverse functions.**
Reflect in y = x and compare.

CHAPTER 3 · CRITERION C · COMMUNICATING

3.C1 **$2^{(2x+2)} = 2^{(6x-3)}$, so $2x + 2 = 6x - 3$ and $x = 5/4$.**
Write both sides as powers of 2 and equate exponents.

3.C2 **$x = \log 20 \div \log 3 \approx 2.727$; check $3^{2.727} \approx 20.0$.**
Take logarithms of both sides.

3.C3 **$\log(x^2/(9x)) = \log(x/9)$, for $x > 0$.**
Apply the quotient and power laws, then state the domain.

3.C4 **$P = 500 \times 2^{(t/40)}$; after 180 minutes $P \approx 500 \times 2^{4.5} \approx 11\,314$; $2^{(t/40)} = 200$ gives $t = 40 \log_2 200 \approx 305$ minutes.**
Count periods, then solve with logarithms.

3.C5 **$50 \log_{10} 6 \approx 38.92$, so the whole part is 38 and there are 39 digits.**
Compute n log b and add one to its whole part.

3.C6 **$\log 200 \div \log 5 \approx 2.3010 \div 0.6990 \approx 3.292$.**
Use the change of base formula.

3.C7 **$20 + 60e^{(-0.05t)} < 25$ gives $e^{(-0.05t)} < 1/12$, so $-0.05t < -\ln 12$ and $t > 20 \ln 12 \approx 49.7$ minutes.**
Rearrange, take natural logarithms, and reverse the inequality carefully.

3.C8 **$2x = \ln 15 \approx 2.708$, so $x \approx 1.35$.**
Take natural logarithms of both sides.

3.C9 **$\log(x^3/(x^2 \times x)) = \log 1 = 0$, for $x > 0$.**
Apply the power and quotient laws, then state the domain.

3.C10 **$50 \log_{10} 3 \approx 23.856$, so the whole part is 23 and there are 24 digits.**
Compute n log b and add one to its whole part.

3.C11 **$P = 800 \times 3^{(t/5)}$; after 24 hours $P \approx 800 \times 3^{4.8} \approx 92\,270$; $3^{(t/5)} = 1250$ gives $t \approx 34.2$ hours.**
Count periods, then solve with logarithms.

3.C12 **Let $y = \log_b a$, so $b^Y = a$; taking logarithms to base a gives $y \log_a b = 1$, so $y = 1/(\log_a b)$.**
Use the definition of a logarithm and the power law.

CHAPTER 3 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

3.D1 **$k = -\ln 2 \div 8 \approx -0.0866$ per day; after 30 days $200e^{(-0.0866 \times 30)} \approx 23.2$ g.**
Use the half-life to find k, then evaluate.

3.D2 **$0.88^T < 0.25$ gives $t > \log 0.25 \div \log 0.88 \approx 10.46$ years.**
Solve the inequality with logarithms.

3.D3 **Effective rate $(1 + 0.04/12)^{12} - 1 \approx 4.07$ per cent; doubling time $\ln 2 \div \ln 1.0407 \approx 17.2$ years.**
Convert the nominal rate, then use the doubling formula.

3.D4 **$10 \log(I/I_0) = 75$ gives $I/I_0 = 10^{7.5}$; a sound 1000 times as intense is at $75 + 30 = 105$ dB.**
Invert the definition, then add 10 log 1000.

3.D5 **$250000 \times 1.03^T > 400000$ gives $t > \log 1.6 \div \log 1.03 \approx 15.9$ years, so during year 16; it assumes the rate stays constant.**
Solve the inequality and state the assumption.

3.D6 **$2000e^{(0.07t)} > 10000$ gives $t > \ln 5 \div 0.07 \approx 23.0$ years; the model assumes unlimited capacity, which no factory has.**
Solve the inequality and comment on the domain.

3.D7 **False: the model is $N = N_0 \times 0.9^T$, which is positive for every finite t and only approaches zero as a limit.**
Consider the value of 0.9^T for large t.

3.D8 **$40e^{(-0.25t)} < 5$ gives $t > \ln 8 \div 0.25 \approx 8.32$ hours; the half-life is $\ln 2 \div 0.25 \approx 2.77$ hours, so this is exactly 3 half-lives.**
Solve the inequality and divide by the half-life.

3.D9 **$b^{12} = 3$ gives $b \approx 1.0959$, about 9.59 per cent per year; doubling time $\ln 2 \div \ln 1.0959 \approx 7.58$ years.**
Find the base from the tripling, then the doubling time.

3.D10 **The linear model gives 2000; the exponential gives $1000 \times 1.05^{20} \approx 2653$. The exponential model compounds the growth, so it pulls ahead without bound.**
Evaluate both models at t = 20.

3.D11 **$A = 70$ and $e^{(-10k)} = 40/70$, so $k = -\ln(4/7)/10 \approx 0.0559$; after 30 minutes $T \approx 20 + 70e^{(-1.677)} \approx 33.1$ °C.**
Use the two data points to find A and k.

3.D12 **False: $\log(1 + 1) = \log 2 \approx 0.3010$ but $\log 1 + \log 1 = 0$. The product law applies to multiplication only, and addition inside a logarithm cannot be split.**
Test with a = b = 1.

CHAPTER 4 · CRITERION A · KNOWING AND UNDERSTANDING

4.A1 **$u(20) = 5 + 19 \times 4 = 81$.**
Use $u(n) = a + (n - 1)d$.

4.A2 **$r = -1.5$.**
Divide each term by the previous one.

4.A3 **$u(8) = 7 \times 3^7 = 15309$.**
Use $u(n) = a\, r^{(n-1)}$.

4.A4 **$1 + 3 + \ldots + 29 = (15/2)(1 + 29) = 225$.**
The terms form an arithmetic series with $d = 2$.

4.A5 **$3 + 5 + 7 + 9 + 11 + 13 = 48$.**
The terms form an arithmetic series.

4.A6 **$4(2^6 - 1)/(2 - 1) = 252$.**
Use $S(n) = a(r^N - 1)/(r - 1)$.

4.A7 **$30/(1 - 1/3) = 45$.**
Check $|r| < 1$ first.

4.A8 **$u(15) = 3 + 14(-4) = -53$.**
Substitute into the general term.

4.A9 **$100(1 - 0.9^{10})/(1 - 0.9) \approx 651.3$.**
Use the geometric sum formula.

4.A10 **Because $|r| = 1.05 > 1$, so the partial sums grow without bound.**
Check the convergence condition.

4.A11 **$3 + 2 \times (3 \times 0.75)/(1 - 0.75) = 3 + 18 = 21$ m.**
Count the first fall once and each rebound twice.

4.A12 **$200((1.05^8 - 1)/0.05) \approx 200 \times 9.5491 \approx 1909.8$.**
The deposits form a geometric series with $r = 1.05$.

CHAPTER 4 · CRITERION B · INVESTIGATING PATTERNS

4.B1 **The sums are about 2.93, 5.19 and 7.49, growing like the logarithm of the number of terms.**
Compute the sums and compare with ln n.

4.B2 **$n(n + 1)$, because the sum is $2 \times (1 + \ldots + n) = 2 \times n(n + 1)/2$.**
Factor out the 2 and use the triangular-number formula.

4.B3 **The sums approach $\ln 2 \approx 0.6931$, alternating above and below it.**
Compute the partial sums and observe the alternation.

4.B4 **The remaining distance after n terms is $(1/2)^N$, so $(1/2)^N < 0.001$ gives $n > \log 0.001 \div \log 0.5 \approx 10$, so 11 terms.**
Use the formula for the remaining distance.

4.B5 **The sums approach about 1.6449, which is $\pi^2/6$.**
Compute the sums and observe the slow convergence.

4.B6 **No simple rule exists for every such series, but if the terms are eventually bounded by a convergent geometric series then the series converges; $1/n$ diverges and $1/n^2$ converges.**
Compare with a geometric series.

4.B7 **The sum formula has $1 - r$ in the denominator, so the quotient tends to n, which is correct because with $r = 1$ every term equals a.**
Take the limit of the formula as r tends to 1.

4.B8 **The sums approach 3/2 from below, because the remaining distance after n terms is $(1/3)^N \times 3/2$, which tends to zero.**
Use the formula for the remaining distance.

4.B9 **$n(n + 1)(2n + 1)/6$, verified for $n = 1$, 2, 3 and 4.**
Compute the small cases and look for a cubic pattern.

4.B10 **$(1/2)^N < 0.0001$ gives $n > \log 0.0001 \div \log 0.5 = 13.29$, so 14 terms.**
Use the formula for the remaining distance.

4.B11 **The sums of $1/n^2$ approach about 1.6449 because the terms shrink fast enough; the sums of $1/n$ grow without bound, as the doubling-block argument shows.**
Compare with a convergent geometric series, and use the doubling-block argument for the harmonic series.

CHAPTER 4 · CRITERION C · COMMUNICATING

4.C1 **$a + 3d = 11$ and $a + 9d = 31$ give $d = 10/3$ and $a = 1$, so $u(12) = 1 + 11(10/3) = 113/3 \approx 37.7$.**
Solve two simultaneous equations from the general term.

4.C2 **$u(25) = -7 + 24(3) = 65$, so the sum is $(25/2)(-7 + 65) = 25 \times 29 = 725$.**
Find the last term, then apply the sum formula.

4.C3 **$20 \times 5 - 2 \times$ (sum of 1 to 20) $= 100 - 2 \times 210 = -320$.**
Use the linearity of summation.

4.C4 **The ratio is 0.1, so the sum is $0.27/(1 - 0.1) = 0.3$.**
Check $|r| < 1$, then apply the formula.

4.C5 **The quarterly rate is 0.02 and there are 20 quarters, so the value is $300((1.02^{20} - 1)/0.02) \approx 300 \times 24.297 \approx 7289$.**
Convert the rate and the number of periods, then apply the annuity formula.

4.C6 **$12000 = R \times (1 - 1.09^{-4})/(1 - 1.09^{-1}) = R \times 3.2397$, so $R \approx 3703.5$.**
Equate the present value of the instalments to the loan.

4.C7 **Write the sum forwards and backwards, add the two rows so that each of the n pairs sums to $n + 1$, and halve the total $n(n + 1)$.**
Use the pairing argument.

4.C8 **$u(30) = -12 + 29(2.5) = 60.5$, so the sum is $(30/2)(-12 + 60.5) = 15 \times 48.5 = 727.5$.**
Find the last term, then apply the sum formula.

4.C9 **$4 \times$ (sum of 1 to 15) $- 15 = 4 \times 120 - 15 = 465$.**
Use the linearity of summation.

4.C10 **The ratio is 0.1, so the sum is $0.4/(1 - 0.1) = 0.444\ldots$, which is $4/9$.**
Check $|r| < 1$, then apply the formula.

4.C11 **The monthly rate is 0.005 and there are 48 months, so the value is $250((1.005^{48} - 1)/0.005) \approx 250 \times 54.098 \approx 13\,524$.**
Convert the rate and the number of periods, then apply the annuity formula.

4.C12 **The sum is $(2 \times 1 - 1) + \ldots + (2n - 1) = 2 \times n(n + 1)/2 - n = n^2$.**
Express the terms algebraically and simplify using the triangular-number formula.

CHAPTER 4 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

4.D1 **$P(n) = 4000 + 250(n - 1)$; $P(10) = 6250$; the total is $(10/2)(4000 + 6250) = 51250$.**
Recognise an arithmetic sequence and series.

4.D2 **$50000 \times 0.96^{15} \approx 25770$; the model assumes the rate stays constant and that the population stays positive, which it does.**
Apply the geometric model and comment on the domain.

4.D3 **The monthly rate is 0.005 and there are 36 months, so the value is $50((1.005^{36} - 1)/0.005) \approx 1967.9$; the interest is $1967.9 - 1800 = 167.9$.**
Convert the rate, then apply the annuity formula.

4.D4 **At 5 per cent: $R \approx 2309.7$; at 12 per cent: $R \approx 2774.1$; the difference over 5 years is about 2322.**
Apply the repayment formula to each rate.

4.D5 **$90 + 2 \times (90 \times 0.8)/(1 - 0.8) = 90 + 720 = 810$ cm; the model assumes a constant ratio, whereas air resistance makes the decay depend on speed.**
Count the first swing once and each rebound twice.

4.D6 **$4000(1.1^N - 1)/(1.1 - 1) \geq 100000$ gives $1.1^N - 1 \geq 2.5$, so $n \geq \log 3.5 \div \log 1.1 \approx 13.1$, so 14 months.**
Use the geometric sum formula and solve with logarithms.

4.D7 **False: the harmonic series has terms tending to zero but diverges, as the doubling-block argument shows.**
Test the claim against the harmonic series.

4.D8 **$15000 \times 0.8^N < 2000$ gives $n > \log(2/15) \div \log 0.8 \approx 9.65$, so 10 years; the value is then about 1608, a loss of about 13 392.**
Solve the inequality and evaluate.

4.D9 **Scheme A totals $(6/2)(30000 + 37500) = 202500$. Scheme B totals $28000(1.08^6 - 1)/0.08 \approx 28000 \times 7.3359 \approx 205405$, so B is slightly better.**
Sum both series and compare.

4.D10 **Yes: the total distance is a convergent geometric series, $40(1/2)/(1 - 1/2) = 40$ units, so the ball travels a finite distance and reaches the bottom.**
Sum the geometric series.

4.D11 **The deposits form an arithmetic sequence, so the accumulated value is a sum of terms $100(n - 1) \times 1.05\text{^}(10-n)$, which mixes an arithmetic and a geometric pattern and must be summed term by term.**
Model the deposits and the growth separately.

4.D12 **After 24 hours: $200 \times 2^{24} = 3\,355\,443$. It exceeds one million when $2^T > 5000$, so $t > \log 5000 \div \log 2 \approx 12.29$, that is 13 hours.**
Apply the geometric model and solve with logarithms.

CHAPTER 5 · CRITERION A · KNOWING AND UNDERSTANDING

5.A1 **$2 < x < 3$.**
Factorise and read the sign between the roots.

5.A2 **$x \le -4$ or $x \ge 2$.**
The parabola opens upwards, so it is positive outside the roots.

5.A3 **$1 < x < 3$.**
Multiply by -1 first, remembering to reverse the inequality.

5.A4 **$-1 \le x \le 5$.**
Take square roots of both sides: $-3 \le x - 2 \le 3$.

5.A5 **$-1 \le x < 3$.**
The sign changes at the root and at the excluded value.

5.A6 **A triangle with vertices $(1, 2)$, $(4, 2)$ and $(1, 5)$.**
Draw each boundary and test one point.

5.A7 **$(0, 0)$, $(4, 0)$, $(4, 2)$, $(3, 3)$ and $(0, 3)$.**
Intersect the boundary lines pairwise.

5.A8 **Vertices $(0, 0)$, $(2, 0)$, $(2, 2)$, $(1, 3)$, $(0, 3)$; values $0, 2, 6, 7, 6$; maximum 7 at $(1, 3)$.**
Evaluate the objective at each vertex.

5.A9 **Vertices $(3, 0)$, $(6, 0)$ and $(1.5, 1.5)$; values $6, 12$ and 4.5; minimum 4.5 at $(1.5, 1.5)$.**
Evaluate at each vertex.

5.A10 **Because no value of $x + y$ is both at most 3 and at least 7.**
The two constraints contradict each other.

5.A11 **$5c + 2s \le 60$, with $c \ge 0$ and $s \ge 0$.**
Each animal consumes its daily requirement.

5.A12 **Vertices $(0, 0)$, $(5, 0)$, $(3, 2)$, $(0, 3)$; values $0, 20, 18, 9$; maximum 20 at $(5, 0)$.**
Evaluate the objective at each vertex.

CHAPTER 5 · CRITERION B · INVESTIGATING PATTERNS

5.B1 **$c = 4$ gives only $x = 2$ excluded; $c = 3$ gives $1 < x < 3$; $c = 0$ gives $0 < x < 4$; $c = -5$ gives $x < -1$ or $x > 5$. The interval widens as c falls.**
Find the roots for each value and read the sign.

5.B2 **It has no solution exactly when the quadratic never takes the required sign: $x^2 - 2x + 5 < 0$ has no solution because the discriminant is negative and the parabola opens upwards.**
Use the discriminant and the leading coefficient.

5.B3 **It is a quadrilateral with vertices $(0, 0)$, $(4, 0)$, $(8/3, 8/3)$ and $(0, 4)$, symmetric in x and y.**
Intersect the boundary lines and note the symmetry.

5.B4 **The vertices give $P = 0$, 4, $16/3$ and 4, so the maximum is $16/3$ at $(8/3, 8/3)$. It is unique, because the objective has slope -1 while the edges have slopes -2, $-1/2$, 0 and none, so no edge is parallel to it.**
Check whether the objective is parallel to any edge.

5.B5 **The vertex $(8/3, 8/3)$ is cut off, so the new vertices are $(0, 0)$, $(2, 0)$, $(2, 2)$ and $(0, 4)$, and the maximum of $x + y$ becomes 4 at $(2, 2)$.**
Recompute the vertices and re-evaluate.

5.B6 **The maximum is c/a when $b \geq a$, c/b when $a > b$, and either when $a = b$; here the maximum is $c \div \max(a, b)$.**
Compare the profit per unit of the shared resource.

5.B7 **At most $n(n - 1)/2$ pairwise intersections, but usually far fewer because most intersections lie outside the region.**
Count pairwise intersections and check feasibility.

5.B8 **$c = 8$ gives $2 \leq x \leq 4$; $c = 9$ gives only $x = 3$; $c = 10$ gives no solution, because the discriminant becomes negative.**
Compute the discriminant for each value and read the sign.

5.B9 **Bounded exactly when the two boundary lines are not parallel and the non-negativity constraints are present; otherwise the region is unbounded or empty.**
Draw the pairs and look for a closed polygon.

5.B10 **The vertices give 0, 4, $16/3$ and 4, so the maximum is $16/3$ at $(8/3, 8/3)$, and it is unique because no edge is parallel to $x + y$.**
Evaluate at the vertices and check the edge gradients.

5.B11 **It cuts off part of the feasible region, usually a corner, so the optimum either moves to a new vertex or becomes infeasible if the minimum cannot be met.**
Recompute the vertices after adding the constraint.

CHAPTER 5 · CRITERION C · COMMUNICATING

5.C1 **$(3x + 2)(x - 3) \le 0$, so $-2/3 \le x \le 3$.**
Factorise and read the sign between the roots.

5.C2 **$x < -2$ or $x > 1$, excluding $x = -2$ where the expression is undefined.**
Track the sign changes at the root and the excluded value.

5.C3 **Vertices (0, 0), (3, 0), (2, 3), (0, 4); values 0, 15, 22, 16; maximum 22 at (2, 3).**
Evaluate the objective at each vertex.

5.C4 **The constraints are $4x + 2y \le 40$ and $2x + y \le 24$, but doubling the second gives $4x + 2y \le 48$, which is weaker, so only carpentry binds. The vertices are (0, 0), (10, 0) and (0, 20), giving P = 0, 300 and 400; the maximum is 400 at (0, 20), that is 20 chairs and no tables.**
Compare the constraints before listing the vertices.

5.C5 **The relaxed maximum is 480 at (0, 24), which is integral, so the integer optimum is the same: 24 chairs and no tables.**
Check whether the relaxed optimum is a lattice point.

5.C6 **$x + 2y \ge 6$, $2x + y \ge 4$, $C = 3x + 2y$; vertices (6, 0), (2/3, 8/3), (0, 4); values 18, 25/3, 8; minimum 8 at (0, 4), that is 4 units of food Y.**
Formulate the constraints from the requirements and evaluate the vertices.

5.C7 **Move an iso-line of the objective across the region; the value changes steadily and the last point of contact with a polygon is a vertex, so the maximum occurs there or along an edge parallel to the objective.**
Use the iso-line argument on a polygon.

5.C8 **The roots are -3, 2 and 5; the sign is negative on $(-3, 2)$ and on $(5, \infty)$, so the solution is $-3 \le x \le 2$ or $x \ge 5$.**
Find the roots, mark them on a number line and test each interval.

5.C9 **$x - 1 > 0$ gives $3 \ge x - 1$, so $1 < x \le 4$; $x - 1 < 0$ gives $3 \le x - 1$, impossible. The solution is $1 < x \le 4$.**
Consider the sign of the denominator before multiplying through.

5.C10 **The vertices are (5, 0), (3, 4) and (0, 5), giving 20, 24 and 15; the minimum is 15 at (0, 5).**
Evaluate the objective at each vertex.

5.C11 **$2x + 3y \le 30$, $3x + 2y \le 24$, $P = 40x + 30y$; the vertices are (0, 0), (8, 0), (6, 6) and (0, 10), giving 0, 320, 420 and 300; the maximum is 420 at (6, 6).**
Formulate from the table of resources, then evaluate the vertices.

5.C12 **Maximise $x + y$ subject to $x \ge 0$ and $y \ge 0$: the region is unbounded and the objective grows without bound, so no maximum exists.**
Look for a direction in which the objective increases indefinitely.

CHAPTER 5 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

5.D1 **$3x + y \le 30$, $2x + 2y \le 28$, $P = x + y$; vertices (0, 0), (10, 0), (8, 6), (0, 14); values 0, 10, 14, 14; maximum 14, achieved along the whole edge from (0, 14) to (8, 6).**
Evaluate the objective at each vertex and check for a parallel edge.

5.D2 **$400x + 300y \le 6000$, $x + 2y \le 20$, $P = x + y$; vertices (0, 0), (15, 0), (10, 5), (0, 10); values 0, 15, 15, 10; maximum 15 along the edge from (15, 0) to (10, 5).**
Evaluate at each vertex.

5.D3 **$w + b \le 100$, $2w + 4b \le 240$, $w \ge 20$, $P = 500w + 400b$; vertices (20, 0), (100, 0), (20, 50), (40, 40); values 10000, 50000, 30000, 36000; maximum 50000 at (100, 0), that is all wheat.**
Evaluate the objective at each vertex.

5.D4 **The point is not a lattice point, so it is not feasible; the nearby lattice points give (3, 2) $\to$ 13, (2, 3) $\to$ 12 and (3, 1) $\to$ 11, so the integer optimum is 13 at (3, 2).**
Test the lattice points near the relaxed vertex.

5.D5 **Relaxed vertices (0, 0), (4, 0), (22/5, 26/15), (0, 6); the relaxed maximum is 28 at (4, 0), which is integral, so the integer optimum is also 28 with gap 0.**
Check whether the relaxed optimum is a lattice point.

5.D6 **$2x + y \le 40$, $y \ge x$, $S = 3x + 2y$; vertices (0, 0), (20, 0), (40/3, 40/3), (0, 40); values 0, 60, 200/3, 80; maximum 80 at (0, 40), that is 40 bins.**
Evaluate the objective at each vertex.

5.D7 **The point at which the maximum is attained, and the check that the point satisfies every constraint; without the point the plan cannot be carried out.**
A linear program's answer is a plan, not just a number.

5.D8 **$0.2a + 0.5b \ge 10$, $3a + 5b \le 200$, $C = 3a + 5b$; the cost is minimised by taking all of source A: $a = 50$, $b = 0$, $C = 150$.**
Compare the mineral per currency unit of each source.

5.D9 **$3x + y \le 600$, $4x + 2y \le 200$, $P = x + y$; vertices (0, 0), (50, 0), (40, 20), (0, 100); values 0, 50, 60, 100; maximum 100 at (0, 100), all paperbacks.**
Evaluate the objective at each vertex.

5.D10 **False: in 5.2.7 the relaxed vertex (3, 3) was a lattice point with $P = 27$, but (4, 2) was feasible and gave 28, because the objective was not maximised at the vertex.**
Look for a feasible lattice point with a better objective value.

5.D11 **$10x + 25y \le 500$, $P = 20x + 40y$; vertices (0, 0), (50, 0), (0, 20); values 0, 1000, 800; maximum 1000 at (50, 0), all small crates.**
Compare the earnings per kilogram.

5.D12 **$4x + 2y \le 80$, $x + y \le 25$, $P = 600x + 900y$; vertices (0, 0), (20, 0), (15, 10), (0, 25); values 0, 12000, 18000, 22500; maximum 22500 at (0, 25), all beans.**
Evaluate the objective at each vertex.

CHAPTER 6 · CRITERION A · KNOWING AND UNDERSTANDING

6.A1 **48 degrees.**
The centre angle is twice the circumference angle.

6.A2 **Because the diameter subtends 180 degrees at the centre, and the angle at the circumference is half of that.**
Apply the angle at the centre theorem.

6.A3 **95 degrees.**
Opposite angles of a cyclic quadrilateral sum to 180.

6.A4 **$2\sqrt{(100 - 36)} = 2 \times 8 = 16$ cm.**
The perpendicular from the centre bisects the chord.

6.A5 **12 cm.**
Pythagoras in the right-angled triangle OTP.

6.A6 **$x = 6$.**
Intersecting chords: $3 \times 4 = 2 \times x$.

6.A7 **35 degrees.**
Alternate segment theorem.

6.A8 **62 degrees.**
An exterior angle equals the interior opposite angle.

6.A9 **9.**
Tangents from an external point are equal.

6.A10 **80 degrees.**
The centre angle is twice the circumference angle.

6.A11 **$\sqrt{(49 - 25)} = \sqrt{24} \approx 4.90$.**
Half the chord is 5, then use Pythagoras.

6.A12 **The two right-angled triangles formed have equal hypotenuses (radii) and a shared side, so they are congruent and the two segments are equal.**
Congruent triangles.

CHAPTER 6 · CRITERION B · INVESTIGATING PATTERNS

6.B1 **In every case the angle at the centre was twice the angle at the circumference on the same arc.**
Measure and compare, then prove with an isosceles triangle.

6.B2 **They are supplementary: the quadrilateral formed has two right angles, so the two remaining angles sum to 180 degrees.**
Use the angle sum of the quadrilateral.

6.B3 **The locus is the circle concentric with the original and of radius $r\sqrt{2}$, because a right angle at P forces $OP = r\sqrt{2}$.**
Use the right-angled triangle OPT.

6.B4 **Yes, because opposite angles are both 90 degrees and so sum to 180, which is the condition for a cyclic quadrilateral.**
Apply the cyclic quadrilateral test.

6.B5 **$AB = 2r\sqrt{(1 - r^2/d^2)}$, because half the chord is $r\sqrt{(1 - r^2/d^2)}$ in the isosceles triangle OAB.**
Split the isosceles triangle OAB and use Pythagoras.

6.B6 **Exactly two common tangents when the circles intersect in two points; four when they are separate; none when one lies inside the other.**
Draw the pairs and count the tangents.

6.B7 **The angle is 90 degrees, because the chord is a diameter and the angle in a semicircle is a right angle; this also confirms the tangent is perpendicular to the radius.**
Combine the alternate segment and semicircle theorems.

6.B8 **The locus is the pair of angle bisectors, because the centre must be equidistant from both lines.**
Use the fact that the centre lies on the bisector of the angle between the tangents.

6.B9 **The angle is $2 \arccos(r/d)$, because the line from the centre to the point bisects the angle and forms a right triangle.**
Bisect the angle and use the right-angled triangle.

6.B10 **No: a cyclic quadrilateral needs supplementary opposite angles, and a rhombus has them supplementary only when it is a square.**
Apply the cyclic quadrilateral test.

6.B11 **The common chord is perpendicular to the line of centres and is bisected by it, because both centres are equidistant from the chord's endpoints.**
Use congruent triangles with the two radii.

CHAPTER 6 · CRITERION C · COMMUNICATING

6.C1 **$r^2 = 12^2 + 5^2 = 169$, so r = 13 cm.**
Halve the chord and use Pythagoras.

6.C2 **The diameter subtends 180 degrees at the centre, and the angle at the circumference is half the angle at the centre on the same arc, so it is 90 degrees.**
Apply the angle at the centre theorem.

6.C3 **In the right triangle OAP, sin(half the angle) = 6/10, so the half-angle is 36.87 degrees and the angle is 73.74 degrees.**
Bisect the angle and use the right-angled triangle.

6.C4 **The two right-angled triangles have equal radii, a shared hypotenuse and a right angle, so they are congruent and the tangents are equal.**
Congruent triangles.

6.C5 **Half the angle is 30 degrees, so in the right triangle OAP, tan 30° = 8/PA gives PA = $8\sqrt{3} \approx 13.86$; the chord AB = 2 × 8 × sin(angle AOB/2) with angle AOB = 120 degrees, so AB = 16 × sin 60° = $8\sqrt{3} \approx 13.86$.**
Bisect, use the tangent-radius property, then the isosceles triangle OAB.

6.C6 **The triangles formed have equal angles — vertically opposite at the intersection and equal from the same arc — so they are similar, and the corresponding sides give AX × XB = CX × XD.**
Similar triangles from equal angles.

6.C7 **44 degrees, by the alternate segment theorem.**
State the theorem and apply it.

6.C8 **$r^2 = 15^2 + 8^2 = 289$, so r = 17 cm.**
Halve the chord and use Pythagoras.

6.C9 **The exterior angle and the interior opposite are both supplementary to the same interior angle, so they are equal.**
Use the cyclic property and the straight-line property.

6.C10 **Half the chord is 4.5, so sin of half the angle is 4.5/7.5 = 0.6, giving a half-angle of 36.87° and a full angle of 73.74°.**
Halve the chord and use a sine ratio in the isosceles triangle.

6.C11 **The tangent is 8 and the angle at the centre is 2 arccos(6/10) ≈ 106.26°, so the quadrilateral has area ½ × 6 × 8 × 2 = 48 and the sector has area ½ × 36 × 1.8545 ≈ 33.4, giving a difference of about 14.6.**
Subtract the sector from the kite formed by the two radii and two tangents.

6.C12 **If angle ACB = 90° then the midpoint O of AB is equidistant from A, B and C, so C lies on the circle with AB as diameter.**
Show the midpoint of AB is equidistant from all three vertices.

CHAPTER 6 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

6.D1 **$40 \tan 28° \approx 21.3$ m.**
Right-angled trigonometry with the tangent-radius property supplying the right angle.

6.D2 **Each section is a tangent of length $12\sqrt{2} \approx 17.0$ m, because the angle at the corner is 90 degrees and so is supplementary to a 90 degree angle at the centre, giving $OP = 12\sqrt{2}$.**
Use the tangent-radius property and Pythagoras.

6.D3 **The path is a cycloid; the track is tangent to the wheel because the point of contact has zero velocity, so the wheel touches without slipping.**
The tangent-radius property describes the contact at each instant.

6.D4 **$1.2 + 0.8 = 2.0$ m.**
The semicircle adds its radius to the height.

6.D5 **$2\pi \times 10 = 20\pi \approx 62.8$ m.**
The outer edge is a circle of radius 10.

6.D6 **$\arccos(15/20) \approx 41.4$ degrees.**
The beam is tangent to the circle of visibility at the edge.

6.D7 **True: if one pair sums to 180 degrees then the fourth vertex lies on the circle through the other three, so the quadrilateral is cyclic, and the other pair then sums to 180 as well.**
Use the converse of the cyclic quadrilateral theorem.

6.D8 **The side is 8 cm; the uncovered area is $64 - 16\pi \approx 13.7$ cm^2.**
The square's side is the diameter; subtract the circle's area.

6.D9 **The straight sections are each $\sqrt{(30^2 - 6^2)} \approx 29.4$ cm; the arc angles are $\pi - 2\arcsin(0.2)$ on the large pulley and $\pi + 2\arcsin(0.2)$ on the small, giving a total of about 70.4 cm.**
The belt is tangent to both pulleys, so each straight section is a common tangent.

6.D10 **For two secants from the same external point, the product of the whole secant and its external part is the same for both, by similarity of the triangles formed.**
Use the secant theorem, a generalisation of the intersecting chords theorem.

6.D11 **The products are equal for any two chords through the point, so with four equal chords the segments pair up so that each product is the same.**
Apply the intersecting chords theorem.

6.D12 **Suppose the tangent is not perpendicular to the radius; then the foot of the perpendicular from the centre to the line is a different point of the circle on the line, giving two intersection points and contradicting the definition of a tangent.**
Contradiction from the definition of a tangent.

CHAPTER 7 · CRITERION A · KNOWING AND UNDERSTANDING

7.A1 **$b \approx 13.72$ and $c \approx 14.93$.**
Use the sine rule with $C = 78°$.

7.A2 **$A = 65°$; $a = 12 \sin 65° \div \sin 47° \approx 15.13$; $c = 12 \sin 68° \div \sin 47° \approx 15.09$.**
Find the third angle, then apply the sine rule twice.

7.A3 **$a^2 = 64 + 121 - 176 \cos 55° \approx 84.06$, so $a \approx 9.17$.**
Use the cosine rule for the included angle.

7.A4 **$\cos A = (49 + 81 - 144)/(2 \times 7 \times 9) = -14/126$, so $A \approx 96.4°$.**
The largest angle is opposite the largest side.

7.A5 **$\frac{1}{2} \times 8 \times 10 \times \sin 40° \approx 25.7$.**
Area = $\frac{1}{2}ab \sin C$.

7.A6 **One triangle, right-angled at C, because $b \sin A = 10 \times 0.5 = 5 = a$.**
Compare a with the perpendicular height $b \sin A$.

7.A7 **$b \sin A = 5$ and $5 < 6 < 10$, so two triangles; $\sin B = 5/6$ gives $B \approx 56.4°$ or $123.6°$.**
Compare with the height, then apply the sine rule.

7.A8 **$b \sin A = 6 \times 0.7071 \approx 4.243$, and $4 < 4.243$, so no triangle exists.**
Compare a with $b \sin A$.

7.A9 **The turn is $90°$, so the distance is $\sqrt{(30^2 + 24^2)} \approx 38.4$ km.**
The bearing difference is the interior angle.

7.A10 **$\sqrt{(9 + 16 + 144)} = \sqrt{169} = 13$.**
Two applications of Pythagoras.

7.A11 **Slant height 10; the angle has tangent $8/6$, so it is about $53.1°$.**
The radius, height and slant height form a right triangle.

7.A12 **Because the given side opposite the given angle may fall short of, equal or exceed the perpendicular height, and the three comparisons give zero, one or two triangles; with the included angle or with three sides the triangle is unique.**
Compare the given side with the perpendicular height.

CHAPTER 7 · CRITERION B · INVESTIGATING PATTERNS

7.B1 **b sin A ≈ 7.66; a = 6 gives 0, a = 7.66 gives 1 right triangle, a = 9 and a = 10 give 2, and a = 13 gives 1.**
Compare a with b sin A and with b.

7.B2 **It is unique exactly when the given angle is opposite the longer of the two given sides, or when the side opposite the given angle is at least the perpendicular height; otherwise there are two answers or none.**
Compare the side opposite the given angle with the height and with the other side.

7.B3 **The two values of B are supplementary, because sin B = sin(180° − B), and the two triangles share the given sides and angle.**
Use the identity sin(180° − B) = sin B.

7.B4 **The area is ½ab sin C, which is greatest when C = 90°, because sin C ≤ 1 with equality only at 90°.**
Maximise the sine of the included angle.

7.B5 **The third side c satisfies 3 < c < 13, because |8 − 5| < c < 8 + 5 by the triangle inequality.**
Use the triangle inequality on both sides.

7.B6 **R = a ÷ (2 sin A); the triangle is right-angled, so the circumradius is half the hypotenuse, 5.**
A right-angled triangle's circumcentre is the midpoint of the hypotenuse.

7.B7 **As the edges become equal the angle tends to arccos(√(2/3)) ≈ 35.3°, which is the angle between a cube's space diagonal and a face diagonal.**
Compute the cosine as the ratio of the face diagonal to the space diagonal.

7.B8 **b sin A ≈ 3.80; a = 3 gives 0, a = 3.8 gives 1 right triangle, a = 5 gives 2, and a = 9 and a = 11 give 1 each.**
Compare a with b sin A and with b.

7.B9 **When two sides and a non-included angle are given, the quadratic in the third side can have two positive roots, corresponding to the two triangles of the ambiguous case.**
Set up the cosine rule with the unknown side and examine the roots.

7.B10 **The area is ½ab sin C, which is greatest at C = 90° and zero at C = 0° and 180°, because sin C ≤ 1 with equality only at 90°.**
Maximise the sine of the included angle.

7.B11 **The three forms ½ab sin C, ½bc sin A and ½ca sin B are equal because each is half the product of two sides and the sine of the included angle, and the sine rule makes them interchangeable.**
Use the sine rule to rewrite each form.

CHAPTER 7 · CRITERION C · COMMUNICATING

7.C1 **b sin A ≈ 6.88 and 6.88 < 9 < 12, so two solutions: B ≈ 51.3° with C ≈ 93.7° and c ≈ 15.62, or B ≈ 128.7° with C ≈ 16.3° and c ≈ 2.85.**
Check the ambiguous case first, then apply the sine rule to both values of B.

7.C2 **Using the cosine rule, cos A = $(14^2 + 15^2 - 13^2)/(2 \times 14 \times 15) = 140/420 = 1/3$, so sin A = $\sqrt{(8/9)}$ and the area is ½ × 14 × 15 × $\sqrt{(8/9)} \approx 98.9$.**
Find an angle by the cosine rule, then apply the area formula.

7.C3 **The turn is 90°, so the distance is $\sqrt{(25^2 + 18^2)} \approx 30.8$ km and the bearing is 030° + arctan(18/25) ≈ 065.7°.**
Extract the interior angle from the bearings, then use Pythagoras.

7.C4 **The space diagonal is $\sqrt{(4 + 9 + 36)} = 7$, so the cosine of the angle is 6/7 and the angle is about 31.0°.**
Two applications of Pythagoras, then a cosine ratio.

7.C5 **The curved surface area is $\pi rl = 65\pi \approx 204.2$, and with the base the total is $65\pi + 25\pi = 90\pi \approx 282.7$.**
Curved area is π times radius times slant height.

7.C6 **h = 40 tan 27° ≈ 20.4 m.**
Right-angled trigonometry with the angle of elevation.

7.C7 **Place B at the origin and C at (a, 0); then A is at (c cos B, c sin B). Expanding $|AC|^2$ gives $a^2 + c^2 - 2ac \cos B$.**
Coordinate proof using Pythagoras on the two components.

7.C8 **b sin A ≈ 9.37 and 9.37 < 11 < 14, so two solutions: B ≈ 57.9° with C ≈ 80.1° and c ≈ 16.28, or B ≈ 122.1° with C ≈ 15.9° and c ≈ 4.42.**
Check the ambiguous case first, then apply the sine rule to both values of B.

7.C9 **cos A = $(196 + 225 - 169)/(2 \times 14 \times 15) = 252/420 = 0.6$, so sin A = 0.8 and the area is ½ × 14 × 15 × 0.8 = 84.**
Find an angle by the cosine rule, then apply the area formula.

7.C10 **The turn is 90°, so the distance is $\sqrt{(35^2 + 28^2)} = \sqrt{2009} \approx 44.8$ km and the bearing is 050° + arctan(28/35) ≈ 088.6°.**
Extract the interior angle from the bearings, then use Pythagoras.

7.C11 **The space diagonal is $\sqrt{(16 + 16 + 36)} = \sqrt{68} \approx 8.246$, so the cosine of the angle is 6/8.246 and the angle is about 43.3°.**
Two applications of Pythagoras, then a cosine ratio.

7.C12 **Place B at the origin and C at (a, 0); then A is at (c cos B, c sin B). Expanding $|AC|^2$ gives $a^2 + c^2 - 2ac \cos B$.**
Coordinate proof using Pythagoras on the two components.

CHAPTER 7 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

7.D1 **The angle at A is 40° and at B is 90°, so C is 50°; by the sine rule AC = 400 sin 90° ÷ sin 50° ≈ 522 m.**
Extract the interior angles, then apply the sine rule.

7.D2 **The distances are 30 and 40 km and the included angle is 120°, so the distance is $\sqrt{(30^2 + 40^2 - 2 \times 30 \times 40 \times \cos 120°)} = \sqrt{3700} \approx 60.8$ km.**
Use the cosine rule with the bearing difference.

7.D3 **d = 40 ÷ tan 28° ≈ 75.3 m, so 75 m.**
Right-angled trigonometry.

7.D4 **Area = ½ × 120 × 150 × sin 72° ≈ 8568 m^2; the third side is $\sqrt{(120^2 + 150^2 - 2 \times 120 \times 150 \times \cos 72°)} \approx 161.2$ m.**
Area formula and cosine rule.

7.D5 **The two legs are perpendicular, so the distance is $\sqrt{(200^2 + 150^2)} = 250$ km and the bearing is 315° − arctan(150/200) ≈ 278°.**
The bearing difference is 90°, so Pythagoras applies.

7.D6 **Slant height $\sqrt{(16 + 63)} = \sqrt{79} \approx 8.89$; the angle has tangent $3\sqrt{7}/4$, about 71.9°; the surface area is $64 + 4 \times ½ \times 8 \times \sqrt{79} \approx 205.5$.**
The cross-section through the apex and a base edge midpoint.

7.D7 **Compare the given side with the perpendicular height b sin A: if a < b sin A there is no triangle, if a = b sin A exactly one right triangle, if b sin A < a < b two triangles, and if a ≥ b one triangle. Both solutions must be reported when there are two.**
The comparison is made before any angle is computed.

7.D8 **The ship travels 13.33 km; the closest approach is 25 sin 45° ≈ 17.7 km, reached when the bearing to the lighthouse is perpendicular to the course.**
The tangent from the ship's position to the visibility circle gives the closest approach.

7.D9 **The cross-section is an isosceles triangle with sides 6, 6 and 10, so the height is $\sqrt{(36 - 25)} = \sqrt{11} \approx 3.32$ m and the half-angle has cosine 5/6, giving a full angle of about 67.1°.**
The cross-section perpendicular to the ridge is a triangle.

7.D10 **Correct: b sin A = 8 × 0.5 = 4 and a = 5 > 4, so the comparison a < b sin A fails. Since also 5 < 8, the case b sin A < a < b holds, so there are in fact two triangles, with B ≈ 53.1° or 126.9°.**
Compare a with b sin A and with b.

7.D11 **Each cable is $\sqrt{(20^2 + 12^2)} \approx 23.3$ m and makes an angle of arctan(20/12) ≈ 59.0° with the ground.**
Right-angled trigonometry.

7.D12 **The real distance is 8.4 × 250 m = 2100 m, and the bearing gives the direction, so the displacement is 2100 m on a bearing of 132°.**
Multiply the map distance by the scale factor.

CHAPTER 8 · CRITERION A · KNOWING AND UNDERSTANDING

8.A1 **(11, −16).**
Scale each vector, then subtract component-wise.

8.A2 **10.**
Use Pythagoras on the components.

8.A3 **(0.6, 0.8).**
Divide by the magnitude, which is 5.

8.A4 **−6 + 5 = −1.**
Multiply corresponding components and add.

8.A5 **Their scalar product is 4 − 4 = 0, so they are perpendicular.**
A zero scalar product is the test.

8.A6 **4 + 0 − 3 = 1.**
Include all three components.

8.A7 **35 − 6 = 29.**
Use ad − bc.

8.A8 **The determinant is 8, so the inverse is [[0.5, 0], [0, 0.25]].**
Divide the adjugate by the determinant.

8.A9 **[[0, −1], [1, 0]].**
Substitute into the rotation matrix.

8.A10 **(5, 1).**
Multiply the matrix by the column vector.

8.A11 **|4(5) − 1(2)| = 18.**
The determinant of the two vectors.

8.A12 **Its determinant is 4 − 4 = 0, so the columns are proportional and the transformation collapses the plane onto a line.**
A zero determinant means no inverse.

CHAPTER 8 · CRITERION B · INVESTIGATING PATTERNS

8.B1 **It stretches by k in one direction and compresses by k in the other, so areas are unchanged because the determinant is $k \times 1/k = 1$.**
Compute the determinant and the images of the basis vectors.

8.B2 **It must have the form $[[\cos\theta, -\sin\theta], [\sin\theta, \cos\theta]]$, so the columns are perpendicular unit vectors and the determinant is 1.**
Check the column lengths, their scalar product and the determinant.

8.B3 **Both are shears of area 1, one parallel to the x-axis and one parallel to the y-axis.**
Compute the images of the vertices and the determinant.

8.B4 **The determinant is $(-1) \times (-1) = 1$, so the product is orientation-preserving and is a rotation or a translation.**
Use $\det(AB) = \det(A)\det(B)$.

8.B5 **For example $a = (1, 0, 0)$, $b = (0, 1, 0)$, $c = (1, 1, 0)$: the scalar triple product is 0, so they are coplanar.**
Compute the scalar triple product.

8.B6 **They are equal, because $ad - bc$ is unchanged when the rows and columns are interchanged.**
Compare the two determinants algebraically.

8.B7 **It is the zero vector, because $|a \times b| = |a||b| \sin\theta$ and $\sin\theta = 0$ for parallel vectors.**
Use the magnitude formula with $\theta = 0$.

8.B8 **The determinant is $\cos^2\theta + \sin^2\theta = 1$, so every rotation has area scale factor 1.**
Compute $ad - bc$ and use the Pythagorean identity.

8.B9 **$\det(M^{-1}) = 1/\det(M)$, because $\det(M)\det(M^{-1}) = \det(I) = 1$.**
Use $\det(AB) = \det(A)\det(B)$ with $B = M^{-1}$.

8.B10 **A straight line maps to a straight line, because the transformation is linear: the image of $a + tb$ is $M a + t M b$, which is a line.**
Apply the matrix to a general point on the line.

8.B11 **The determinant is 1, because a shear preserves area even though it changes shape.**
Compute $ad - bc$ for $[[1, k], [0, 1]]$.

CHAPTER 8 · CRITERION C · COMMUNICATING

8.C1 **The image is (2(3) − 1(−1), 4(3) + 3(−1)) = (7, 9) and the determinant is 6 + 4 = 10.**
Multiply, then compute ad − bc.

8.C2 **AB = (3, 3) and BC = (−6, −6) = −2(3, 3), so the vectors are parallel and share B; the points are collinear.**
Check whether one vector is a multiple of the other.

8.C3 **The point is (2a + 3b)/5 = ((2(−2) + 3(6))/5, (2(4) + 3(−2))/5) = (14/5, 2/5) = (2.8, 0.4).**
Apply the section formula with the weights crossed.

8.C4 **a × b = ((−1)(−2) − 3(4), 3(1) − 2(−2), 2(4) − (−1)(1)) = (−10, 7, 9); both scalar products are 0.**
Apply the component formula, then check with scalar products.

8.C5 **The scalar triple product is 1(5(10) − 6(8)) − 2(4(10) − 6(7)) + 3(4(8) − 5(7)) = 1(2) − 2(−2) + 3(−3) = −3, so the volume is 3.**
Compute the determinant of the 3 by 3 matrix of the vectors.

8.C6 **Expanding (ae + bg)(cf + dh) − (af + bh)(ce + dg) and simplifying gives (ad − bc)(eh − fg).**
Expand both sides and compare.

8.C7 **The side vectors are (5, 1) and (2, 6), with determinant 5(6) − 1(2) = 28, so the area is 14.**
Halve the determinant of the two side vectors.

8.C8 **The image is (5 − 4, −5 − 6) = (1, −11) and the determinant is 3 + 2 = 5.**
Multiply, then compute ad − bc.

8.C9 **AB = (4, 3) and BC = (6, 6), which are not multiples, so the points are not collinear; the test is whether one vector is a multiple of the other.**
Compare the two vectors component by component.

8.C10 **The point is (4a + b)/5 = ((8 + 18)/5, (28 − 3)/5) = (5.2, 5).**
Apply the section formula with the weights crossed.

8.C11 **a × b = ((−2)(−2) − 1(4), 1(1) − 3(−2), 3(4) − (−2)(1)) = (0, 7, 14); both scalar products are 0.**
Apply the component formula, then check with scalar products.

8.C12 **The side vectors are (4, 1) and (1, 5), with determinant 4(5) − 1(1) = 19, so the area is 9.5.**
Halve the determinant of the two side vectors.

CHAPTER 8 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

8.D1 **$3\mathbf{a} + 2\mathbf{b} = (120 - 40, 90 + 100) = (80, 190)$, a distance of $\sqrt{(6400 + 36100)} \approx 206.2$ m.**
Scale and add component-wise, then use Pythagoras.

8.D2 **The work is the scalar product $3(5) + 4(0) = 15$ joules.**
Work is the scalar product of force and displacement.

8.D3 **It is a rotation through 90°, and the image is $(-5, 2)$.**
Recognise the rotation matrix and multiply.

8.D4 **The scalar product is 36, the magnitudes are 5 and 13, so the cosine is 36/65 and the angle is about 56.4°; the area is $|\mathbf{a} \times \mathbf{b}| = \sqrt{((5(12) - 3(0))^2 + (3(0) - 4(12))^2 + (4(5) - 0(0))^2)} = \sqrt{(60^2 + 48^2 + 20^2)} \approx 78.1$.**
Use the scalar product for the angle and the vector product for the area.

8.D5 **The product is $[[-1, 0], [0, -1]]$, a rotation through 180°.**
Multiply in the order of application and interpret the result.

8.D6 **The determinant is 4, so the area is 28.**
Areas are multiplied by the absolute value of the determinant.

8.D7 **It would collapse positions onto a line, so different positions would share one corrected position and the correction could not be undone.**
A zero determinant means no inverse and a collapsed image.

8.D8 **The area is half the magnitude of the determinant of the diagonals: $|6(6) - 2(2)| \div 2 = 16$.**
The area of a parallelogram is half the determinant of its diagonals.

8.D9 **Correct: the determinant is $6 - 5 = 1$, but the columns $(3, 5)$ and $(1, 2)$ have magnitudes $\sqrt{34}$ and $\sqrt{5}$, not 1, so the columns are not unit vectors and the transformation is not a rotation.**
A rotation has perpendicular unit columns.

8.D10 **The resultant is $3(\cos 35°, \sin 35°) \approx (2.46, 1.72)$, of magnitude 3.**
The vectors are parallel, so they combine by subtraction of magnitudes.

8.D11 **The images of the vertices are $(0, 0)$, $(1, 3)$, $(5, 7)$ and $(4, 4)$; the area is $|1(4) - 2(3)| = 2$, and since the determinant is -2 the transformation can be undone.**
Map the vertices and compute the determinant.

8.D12 **In two dimensions the analogue is the determinant, which is a number giving the signed area; the vector product as a vector exists only in three dimensions.**
Distinguish the two-dimensional determinant from the three-dimensional vector product.

CHAPTER 9 · CRITERION A · KNOWING AND UNDERSTANDING

9.A1 **$20 - 3 = 17$.**
The range is the largest minus the smallest.

9.A2 **$45/5 = 9$.**
Sum the values and divide by the count.

9.A3 **8.**
The middle value of the ordered list.

9.A4 **The mean is 5, the squared deviations sum to 32, so the variance is 4 and the standard deviation is 2.**
Divide the sum of squared deviations by n.

9.A5 **The sum of squared deviations is 32, so the variance is $32/7 \approx 4.571$ and the standard deviation is about 2.14.**
Divide by $n - 1$ for the sample version.

9.A6 **The means agree; the second is three times as dispersed.**
Compare the measures of dispersion.

9.A7 **Q1 = 7.5 and Q3 = 14.5, so the interquartile range is 7.**
Take the difference of the quartiles.

9.A8 **Mean $= (4 + 12 + 6)/12 = 22/12 \approx 1.83$; variance $= (4 + 24 + 18)/12 - 1.83^2 \approx 3.833 - 3.361 \approx 0.472$, so the standard deviation is about 0.69.**
Use the frequency-table formulas.

9.A9 **The means are 3 and 3, and $r = (\text{sum } xy - n\,\bar{x}\,\bar{y}) \div \sqrt{((\text{sum } x^2 - n\,\bar{x}^2)(\text{sum } y^2 - n\,\bar{y}^2))} = (55 - 45) \div \sqrt{((55 - 45)(55 - 45))} = 1$.**
Use the summary-sum form of the formula.

9.A10 **A very strong negative linear association: as one variable rises the other falls, almost exactly along a line.**
The sign gives the direction and the magnitude the strength.

9.A11 **$y = 17.8$.**
Substitute into the line.

9.A12 **Because the mean is pulled towards the long tail, while the median depends only on the middle of the ordered data and is resistant to extreme values.**
Consider the effect of a tail on each measure.

CHAPTER 9 · CRITERION B · INVESTIGATING PATTERNS

9.B1 **The range and the standard deviation increase sharply, while the interquartile range is barely affected.**
Compute all three before and after the addition.

9.B2 **They are close when n is large, because the ratio $\sqrt{(n/(n-1))}$ tends to 1.**
Compare the two divisors as n grows.

9.B3 **A curved relationship can give r near 0 even though the variables are closely related, because r measures linear association only.**
Plot the data and compute r.

9.B4 **The standard deviation scales with the data, while r is a ratio of standardised quantities and so is unchanged.**
Compute both before and after the scaling.

9.B5 **The line rotates towards the outlier, changing both the gradient and the intercept, because the line minimises squared distances.**
Compute the line with and without the point.

9.B6 **The gradient of the line of x on y is r times the ratio of the standard deviations, and the two gradients multiply to r^2 only when the variables are standardised; in general the product of the gradients is r^2.**
Compute both gradients and compare with r^2.

9.B7 **No: r is a ratio of a covariance to a product of standard deviations, and the covariance is bounded by that product, so $-1 \le r \le 1$.**
Use the Cauchy–Schwarz inequality on the deviations.

9.B8 **The mean and the range are multiplied by 5, and the standard deviation is multiplied by 5, because every deviation is multiplied by 5.**
Track one value through the definitions.

9.B9 **They are close exactly when the distribution is approximately symmetric; a long tail on one side pulls the mean away from the median.**
Compare the two measures for symmetric and skewed examples.

9.B10 **$r = -1$ exactly, because every point lies on a straight line and the deviations are perfectly proportional.**
Compute r for points on a line of negative gradient.

9.B11 **The gradient is divided by the conversion factor while the correlation coefficient is unchanged, because r is a ratio of standardised quantities.**
Convert the units, recompute the gradient, and compare with r.

CHAPTER 9 · CRITERION C · COMMUNICATING

9.C1 **Mean 19.57, median 18, population standard deviation about 4.63.**
Compute each measure and note which is resistant.

9.C2 **The means agree, so the centres match; the standard deviations differ, so the spreads differ; the comparison says nothing about why, because it describes the samples only.**
Compare centre, dispersion and shape, then state the limit.

9.C3 **The means are $\bar{x} = 6$ and $\bar{y} = 7.8$; sum $xy = 264$, sum $x^2 = 220$ and sum $y^2 = 331$, so $r = (264 - 234) \div \sqrt{((220 - 180)(331 - 304.2))} = 30 \div \sqrt{1072} \approx 0.92$, indicating strong positive correlation.**
Compute the summary sums carefully before substituting.

9.C4 **The gradient is $(\text{sum } xy - n\,\bar{x}\,\bar{y})/(\text{sum } x^2 - n\,\bar{x}^2) \approx 0.87$ and the intercept is $7.8 - 0.87(6) \approx 2.6$, so $y \approx 0.87x + 2.6$.**
Use the same summary sums as for r.

9.C5 **Mid-points 4.5, 14.5, 24.5; mean $= (22.5 + 174 + 196)/25 \approx 15.7$; variance $= (101.25 + 2523 + 4802)/25 - 15.7^2 \approx 297.05 - 246.49 \approx 50.56$, so the standard deviation is about 7.11.**
Use mid-points and the frequency formulas.

9.C6 **It does not: a correlation measures association, and a common cause, a reverse cause or coincidence can all produce it. An experiment or controlled study is needed for a causal claim.**
Separate association from causation and name the alternatives.

9.C7 **The side vectors are (3, 1) and (1, 4), with determinant $3(4) - 1(1) = 11$, so the area is 5.5.**
Halve the determinant of the two side vectors.

9.C8 **Mean 27.86, median 27, Q1 = 24 and Q3 = 33 so the interquartile range is 9.**
Compute each measure and note which is resistant.

9.C9 **The means are 3 and 6, and $r = (\text{sum } xy - n\,\bar{x}\,\bar{y}) \div \sqrt{((\text{sum } x^2 - n\,\bar{x}^2)(\text{sum } y^2 - n\,\bar{y}^2))} = (150 - 90) \div \sqrt{((55 - 45)(190 - 180))} \approx 60 \div 31.62 \approx 0.90$.**
Compute the summary sums carefully before substituting.

9.C10 **The gradient is $60/10 = 6$? No: the gradient is $(\text{sum } xy - n\,\bar{x}\,\bar{y})/(\text{sum } x^2 - n\,\bar{x}^2) = 60/10 = 6$, which is checked against the range of the data and found too large; the correct sums give a gradient of about 0.6 and an intercept of about 4.2.**
Use the same summary sums as for r and check the gradient against the data.

9.C11 **Mid-points 2, 7 and 12; mean $= (12 + 70 + 48)/20 = 6.5$; variance $= (24 + 490 + 576)/20 - 6.5^2 = 54.5 - 42.25 = 12.25$, so the standard deviation is 3.5.**
Use mid-points and the frequency formulas.

9.C12 **It does not: correlation measures association, and a common cause, a reverse cause or coincidence can all produce it. An experiment is needed for a causal claim.**
Separate association from causation and name the alternatives.

CHAPTER 9 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

9.D1 **The mean rose, so the average improved, but the spread more than doubled, so the improvement was uneven and the weakest scores may have fallen.**
Compare the centre and the dispersion together.

9.D2 **Hospital A, because its recovery times are clustered tightly around the mean, while hospital B's are widely spread.**
Compare the dispersions and justify the advice.

9.D3 **The correlation is strong but does not establish causation; both variables are plausibly caused by a third, rural population size, and the claim is not supported.**
Name the alternative explanation.

9.D4 **$y = 190$, but this is an extrapolation well beyond the observed range, so it assumes the linear relationship continues.**
Classify the prediction against the data range.

9.D5 **The conclusion is too strong: $r = 0.3$ indicates a weak linear association, which may still be real, and the variables could be strongly related in a non-linear way that r does not detect.**
Distinguish weak linear association from no association.

9.D6 **About 68 per cent, roughly 136 values, assuming the distribution is approximately normal.**
Use the empirical rule and state its assumption.

9.D7 **The sample is small, so the estimate of the correlation is imprecise and could easily arise by chance; the report should give a confidence interval and note the sample size.**
Consider the precision of an estimate from a small sample.

9.D8 **The analysis supports a positive association between rainfall and yield, which justifies planning for irrigation in dry years; it does not prove that extra rainfall causes higher yield, because other factors vary together.**
State what the correlation supports and what it does not.

9.D9 **False: a standard deviation of 0 means every value equals the mean, so the values are all equal but not necessarily 0.**
Consider what zero dispersion implies.

9.D10 **It is possible when one set has extreme values on one side: the median and interquartile range are resistant, but the mean is pulled towards the tail, so the set with the extreme value has a mean displaced from its median.**
Compare resistant and non-resistant measures.

9.D11 **Supplier A, because although its mean is half a day longer, its deliveries are far more predictable, which usually matters more than half a day.**
Compare centre and dispersion, then justify the advice.

9.D12 **The precision is spurious: an extrapolation far beyond the data carries an uncertainty that grows with the distance, and the line itself was fitted with an uncertainty.**
Consider the reliability of extrapolation.

CHAPTER 10 · CRITERION A · KNOWING AND UNDERSTANDING

10.A1 **3/6 = 1/2.**
There are three even outcomes out of six.

10.A2 **13/52 + 4/52 − 1/52 = 16/52 = 4/13.**
Use the addition rule and subtract the overlap.

10.A3 **1/4.**
Four equally likely outcomes, one favourable.

10.A4 **0.4 × 0.5 = 0.2.**
Multiply for independent events.

10.A5 **No, because 0.3 × 0.6 = 0.18, which is not 0.15, so the events are not independent.**
Compare P(A and B) with the product P(A)P(B).

10.A6 **(3/8)(2/7) = 6/56 = 3/28.**
The second draw is conditional on the first.

10.A7 **3/8.**
There are 8 outcomes and 3 favourable ones.

10.A8 **0.4.**
The probabilities of an event and its complement sum to 1.

10.A9 **25 − (15 + 12 − 5) = 3, so 3/25.**
Use the addition rule, then the complement.

10.A10 **4! = 24.**
Apply the multiplication principle.

10.A11 **7!/(2! × 5!) = 21.**
Order does not matter, so use combinations.

10.A12 **Out of 100 000 people: 95 true positives and about 4995 false positives, so P ≈ 95/5090 ≈ 0.0187, under 2 per cent.**
Scale to whole numbers and divide by the total positives.

CHAPTER 10 · CRITERION B · INVESTIGATING PATTERNS

10.B1 **The probabilities are about 0.117, 0.411, 0.507, 0.706 and 0.891, rising steeply because the number of pairs grows quadratically.**
Use the complement product and count the pairs.

10.B2 **With 200 days the answer is 17 rather than 23, because the denominator is smaller; the rule is that the answer grows like the square root of the number of days.**
Repeat the complement calculation with a different denominator.

10.B3 **Only in the trivial case where one of them has probability 0 or 1, because mutual exclusivity gives P(A and B) = 0 while independence requires P(A)P(B).**
Compare the two products.

10.B4 **It does not: two events can each be independent of a third while being dependent on each other, so all combinations must be checked.**
Construct a sample space where the pairwise products hold but the triple product does not.

10.B5 **For a word with 8 letters including 3 identical, the count is 8!/3!, because each distinct arrangement is counted 3! times.**
Divide by the factorial of each repeat count.

10.B6 **$n(n-3)/2$, because each vertex joins to $n-3$ others and each diagonal is counted twice.**
Count the pairs of non-adjacent vertices.

10.B7 **With 12 signs the probability is much higher, about 0.99, because the denominator is 12 rather than 365.**
Repeat the complement calculation with 12 outcomes.

10.B8 **The complement product is about 0.03, so the probability is about 0.97, because the number of pairs is 1225.**
Use the complement and count the pairs.

10.B9 **$n(n-1)(n-2)$, because the roles are distinct and order matters.**
Apply the multiplication principle stage by stage.

10.B10 **Only in the trivial case where one has probability 0, because mutual exclusivity gives P(A and B) = 0 while independence requires P(A)P(B).**
Compare the two products.

10.B11 **$1-(5/6)^2 = 11/36$, because the complement is that neither shows a six.**
Use the complement and independence.

CHAPTER 10 · CRITERION C · COMMUNICATING

10.C1 **$P(A \mid B) = 0.2/0.4 = 0.5$.**
Divide the joint probability by the conditioning probability.

10.C2 **Out of 10 000 people: 180 true positives and 490 false positives, so $P = 180/670 \approx 0.269$.**
Scale to whole numbers and divide by the total positives.

10.C3 **$P(\text{defective}) = 0.70(0.01) + 0.30(0.04) = 0.019$; $P(B \mid \text{defective}) = 0.012/0.019 \approx 0.632$.**
Apply the total probability rule, then Bayes' theorem.

10.C4 **$9 \times 8 \times 7 = 504$, because the roles are distinct and order matters.**
Apply the multiplication principle stage by stage.

10.C5 **$8!/(3! \times 5!) = 56$.**
Order does not matter, so use combinations.

10.C6 **$P(A) = P(A \text{ and } B) + P(A \text{ and not } B) = P(A \mid B)P(B) + P(A \mid \text{not } B)P(\text{not } B)$, because B and not B are disjoint and exhaustive.**
Split the event A by the partition.

10.C7 **The complement product is about 0.4313, so the probability of a shared birthday is $1 - 0.4313 = 0.569$.**
Compute the product of the no-match probabilities.

10.C8 **$P(A \mid B) = 0.14/0.35 = 0.4$.**
Divide the joint probability by the conditioning probability.

10.C9 **Out of 10 000 people: 460 true positives and 570 false positives, so $P = 460/1030 \approx 0.447$.**
Scale to whole numbers and divide by the total positives.

10.C10 **$P(\text{faulty}) = 0.75(0.02) + 0.25(0.06) = 0.03$; $P(Y \mid \text{faulty}) = 0.015/0.03 = 0.5$.**
Apply the total probability rule, then Bayes' theorem.

10.C11 **$11!/(4! \times 7!) = 330$.**
Order does not matter, so use combinations.

10.C12 **The complement product is about 0.1088, so the probability of a shared birthday is $1 - 0.1088 = 0.891$.**
Compute the product of the no-match probabilities.

CHAPTER 10 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

10.D1 **Out of 500 000 people: 99 true positives and 4990 false positives, so P = 99/5089 ≈ 0.0195, under 2 per cent.**
Scale to whole numbers and divide by the total positives.

10.D2 **760 true positives and 576 false positives, so 760/1336 ≈ 0.569 are genuine.**
Build the two-way table and divide.

10.D3 **P(faulty) = 0.55(0.03) + 0.45(0.07) = 0.048; P(Y | faulty) = 0.0315/0.048 ≈ 0.656.**
Apply the total probability rule, then Bayes' theorem.

10.D4 **With an intruder: 0.8 × 0.9 = 0.72. Without: 0.1 × 0.2 = 0.02.**
Multiply the conditional probabilities along each branch.

10.D5 **P(no prize) = (999 000/1 000 000)^10 ≈ 0.9900, so P(at least one) ≈ 0.0100.**
Use the complement with the probability of losing each ticket.

10.D6 **False for a fair coin: the tosses are independent, so P(heads) remains 0.5 on every toss regardless of the past.**
Independence means the past does not affect the next outcome.

10.D7 **Out of 200 000 people: 950 true positives and 9950 false positives, so the actual probability is 950/10 900 ≈ 0.087, far below 95 per cent. The doctor has confused P(positive | disease) with P(disease | positive).**
Build the table and identify the two conditional probabilities.

10.D8 **Total committees are 20!/(4! × 16!) = 4845; all-staff committees are 8!/(4! × 4!) = 70, so the answer is 4845 − 70 = 4775.**
Count the total and subtract the unwanted case.

10.D9 **$26^3 \times 10^2 = 17\,576 \times 100 = 1\,757\,600$.**
Apply the multiplication principle at each position.

10.D10 **(4/15)(3/14)(2/13) + (5/15)(4/14)(3/13) + (6/15)(5/14)(4/13) = 24/2730 + 60/2730 + 120/2730 = 204/2730 ≈ 0.0747.**
Sum the three colour cases, each a product of conditional probabilities.

10.D11 **False: P(A or B) = 0.5 + 0.5 − 0.25 = 0.75, because the overlap of 0.25 must be subtracted once.**
Apply the addition rule with the independence product.

10.D12 **$(5/6)^4 \approx 0.482$, because the first four rolls must all be non-sixes.**
Use the complement and multiply the conditional probabilities.

CHAPTER 11 · CRITERION A · KNOWING AND UNDERSTANDING

11.A1 **Define the problem, make assumptions, build the model, test and evaluate, report the result.**
The cycle runs from definition to report.

11.A2 **The decision, the objective, the variables with units, the constraints and the audience.**
Anything not named is an unstated assumption.

11.A3 **So that a reader can challenge a specific claim rather than the whole model, and so the model's limits are visible.**
An unstated assumption is an undefended one.

11.A4 **Fitting data, used to choose the parameters, and test data, not used in the fit, which show whether the model predicts.**
A model that only reproduces its fitting data has learned nothing.

11.A5 **The percentage change in the conclusion for a stated percentage change in each parameter, ranked by size.**
The ranking says where better data are needed.

11.A6 **Because an omitted cost makes the recommendation inaccurate, while an omitted constraint can make it impossible to carry out.**
A constraint is a limit on what is feasible.

11.A7 **The optimal point, the optimal value, and the justification that the point is feasible.**
The value alone is not a recommendation.

11.A8 **It says whether the recommendation survives reasonable changes in the parameters, which is how much the conclusion can be trusted.**
A recommendation that reverses under a small change is a coin toss.

11.A9 **Problem, assumptions, model, test, recommendation, limits.**
The order is the argument.

11.A10 **Restate the objection, concede what is true, quantify the effect, state the consequence, give the repair.**
A defence that concedes is stronger than one that denies.

11.A11 **Because an estimate with no stated basis is a guess, and a guess cannot be defended or challenged.**
Every number is computed, measured or estimated.

11.A12 **What the model could not answer, and what would be needed to answer it.**
The reflection is the most transferable part of the work.

CHAPTER 11 · CRITERION B · INVESTIGATING PATTERNS

11.B1 **Changing from minimising cost to maximising access moved the recommendation from a small, cheap plan to a larger, dearer one, because the objective function changed.**
Re-solve with the new objective and compare.

11.B2 **The assumptions attached to the most sensitive parameters, which the sensitivity analysis identifies; in both test models the assumption about the most sensitive parameter dominated.**
Rank the assumptions by the sensitivity of their parameters.

11.B3 **No: a model with more assumptions is more detailed but also more fragile, because each assumption can be challenged and each adds an error.**
Compare the predictions of a simple and a detailed model against the same data.

11.B4 **For a published transport model: constant travel demand, no new infrastructure, and unchanged fuel prices, each visible in the report's stated inputs.**
Look for the inputs the model holds fixed.

11.B5 **A longer horizon favours plans with higher initial cost and lower running cost, because the running cost is saved for longer; a shorter horizon reverses the ranking.**
Re-solve with two horizons and compare.

11.B6 **When the relationships are approximately proportional over the range of interest and the errors are small compared with the decision's margin; two test cases confirmed both conditions.**
Check the linearity and the size of the error against the margin.

11.B7 **A model can be accurate and useless if it answers a question nobody asked, or useful and inaccurate if its errors are smaller than the decision's margin.**
Compare the model's error with the margin between the options.

11.B8 **A longer horizon favours plans with higher initial cost and lower running cost, because the running cost is saved for longer; a shorter horizon reverses the ranking.**
Re-solve with two horizons and compare.

11.B9 **The assumptions attached to the most sensitive parameters, which the sensitivity analysis identifies.**
Rank the assumptions by the sensitivity of their parameters.

11.B10 **No: more assumptions give more detail but also more fragility, because each assumption can be challenged and each adds an error.**
Compare a simple and a detailed model against the same data.

11.B11 **A model can be accurate and useless if it answers a question nobody asked, or useful and inaccurate if its errors are smaller than the decision's margin.**
Compare the error with the margin between the options.

CHAPTER 11 · CRITERION C · COMMUNICATING

11.C1 **Decision: install or not. Objective: minimise the fifteen-year total cost. Variables: installed capacity in kW and costs in currency units per year. Constraints: roof area, budget and structural limits. Audience: the school council.**
Name each part of the definition and state why it matters.

11.C2 **Rank the parameters by the size of the change, and report the conclusion's percentage change for each; the largest is the parameter whose data most need improving.**
Perturb each parameter by 10 per cent and record the effect.

11.C3 **Compute the model's prediction, the absolute error and the relative error, and compare the relative error with the margin between the options being decided.**
The relative error against the decision's margin is the verdict.

11.C4 **Vertices (0, 0), (7, 0), (4, 6), (0, 10) give 0, 350, 440, 400; the optimum is (4, 6) with 440. With the profit from x at 45, the value is 420 and (4, 6) is still optimal, so the recommendation is robust.**
Evaluate at the vertices, then perturb and re-evaluate.

11.C5 **Restate the objection; concede that prices have in fact risen; quantify that the saving is understated by about 40 per cent over ten years; state that the recommendation is unchanged and the benefit larger; repair by modelling the price rise and reporting both.**
Restate, concede, quantify, consequence, repair.

11.C6 **The plan violates a constraint that the model omitted; the model is not wrong in its arithmetic but incomplete, and the repair is to add the constraint and re-solve.**
Check the recommended plan against every constraint of the situation, not only those in the model.

11.C7 **Problem, assumptions, model, test, recommendation and limits, each with the evidence named in the brief.**
The evidence is what distinguishes a project from an essay.

11.C8 **Decision, objective, variables with units, constraints and audience, each named with a reason.**
Name each part of the definition and state why it matters.

11.C9 **Rank the parameters by the size of the change; the largest identifies where better data are needed.**
Perturb each parameter by 10 per cent and record the effect.

11.C10 **Compute the prediction, the absolute error and the relative error, and compare the relative error with the margin between the options.**
The relative error against the decision's margin is the verdict.

11.C11 **The vertices are (0, 0), (6, 0), (3, 6) and (0, 9), giving 0, 360, 450 and 405; the optimum is (3, 6) with 450, and a 10 per cent fall in the profit from x gives 414 with (3, 6) still optimal.**
Evaluate at the vertices, then perturb and re-evaluate.

11.C12 **Restate, concede, quantify, state the consequence, repair.**
Restate, concede, quantify, consequence, repair.

CHAPTER 11 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

11.D1 **The total saving over 10 years is 6000 against a cost of 4000, so the payback time is 4000/600 ≈ 6.7 years; the model assumes the saving is constant, which a rising water price would change in the tank's favour.**
Model the cost and the saving, then find the crossing.

11.D2 **The optimum is (4, 4) with 280; with the profit from x at 36 the value is 264 and (4, 4) is still optimal, so the recommendation is robust.**
Evaluate at the vertices, then perturb.

11.D3 **The model is incomplete: it omits the labour constraint, so it recommends a plan that cannot be carried out. The repair is to add the labour constraint and re-solve, which lowers the optimum to 9 tonnes.**
An omitted constraint is more serious than an omitted cost.

11.D4 **The recommendation is robust, because the optimal plan is unchanged and only the value falls, by about 12 per cent.**
Compare the plan, not only the value, after the perturbation.

11.D5 **The largest error is 5 per cent, well inside the 15 per cent margin, so the model is good enough for the decision; the errors are reported alongside the recommendation.**
Compare the largest error with the decision's margin.

11.D6 **At n = 5 the model predicts 2250 against 2300, an error of 2.2 per cent; at n = 12 it predicts 2600 against 2650, an error of 1.9 per cent. The model is acceptable for planning.**
Compute the prediction and the relative error at each point.

11.D7 **The statement of whether the recommendation survives reasonable changes in the parameters; without it a reader cannot tell how much the conclusion can be trusted.**
A recommendation is a point, a value and a robustness statement.

11.D8 **The omission understates the saving from insulation during term time but overstates it over the holidays; the net effect is small, so the recommendation is probably unchanged, but the omission must be named and quantified.**
Trace the effect of the omission on the quantity being optimised.

11.D9 **Adopt the plan, which is expected to give a value of about 28 000, and which remains optimal for values between 25 000 and 31 000.**
Give the plan, the value and the range, in the audience's terms.

11.D10 **Not necessarily: if the decision's margin is 50 per cent, a 20 per cent error is well inside it and the model is useful. Usefulness depends on the error relative to the margin, not on the error alone.**
Compare the error with the margin between the options.

11.D11 **To the parameter with the 9 per cent effect, because improving its estimate improves the conclusion most; the other parameter's estimate can remain crude.**
Rank the parameters by their effect on the conclusion.

11.D12 **It is too vague to be useful: a reflection should name what the model could not answer, why, and what would be needed to answer it.**
A reflection names a specific limit and a specific repair.

CHAPTER 12 · CRITERION A · KNOWING AND UNDERSTANDING

12.A1 **$x = 2$ or $x = 3$.**
Factorise, since the coefficients are small.

12.A2 **$x = (-5 \pm \sqrt{(25 + 24)})/6 = (-5 \pm 7)/6$, so $x = 1/3$ or $x = -2$.**
The quadratic formula, since factorising is not immediate.

12.A3 **$f^{-1}(x) = (x + 7)/4$.**
Rearrange and swap.

12.A4 **$g(2) = 5$, so $f(g(2)) = 25$.**
Evaluate the inner function first.

12.A5 **$x > 6$.**
Add 5 and divide by 2.

12.A6 **$-5 \le x \le 3$.**
Factorise and read the sign between the roots.

12.A7 **$u(20) = 3 + 19(5) = 98$.**
Use $u(n) = a + (n - 1)d$.

12.A8 **$20/(1 - 0.5) = 40$.**
Check $|r| < 1$, then apply the formula.

12.A9 **$x = \log 30 \div \log 2 \approx 4.91$.**
Take logarithms of both sides.

12.A10 **$40\,000 \times 1.03^{10} \approx 53\,757$.**
Apply the compound growth model.

12.A11 **$6 - 35 = -29$.**
Use $ad - bc$.

12.A12 **$a + b = (7, -2)$, so $|a + b| = \sqrt{53} \approx 7.28$.**
Add component-wise, then use Pythagoras.

CHAPTER 12 · CRITERION B · INVESTIGATING PATTERNS

12.B1 **Factorising is preferred when the roots are rational and small; the quadratic formula always works; completing the square also gives the vertex.**
Solve the same equation by all three methods and compare.

12.B2 **Two answers exactly when the given side opposite the given angle lies between the perpendicular height and the other given side.**
Compare the given side with the height and the other side.

12.B3 **Both use the included angle; the area formula gives the size of the triangle and the cosine rule gives the third side, and together they determine the triangle from SAS.**
Derive the area formula from the height, and the cosine rule from Pythagoras.

12.B4 **From the angle at the centre theorem follow angles in the same segment, the right angle in a semicircle and the cyclic quadrilateral property, each by a short argument.**
Derive each consequence from the chosen theorem.

12.B5 **The area of a triangle is half the absolute value of the determinant of its two side vectors, because the determinant is the area of the parallelogram they span.**
Compute both for the same triangle.

12.B6 **Two distinct real roots when the discriminant is positive, one repeated root when it is zero, and none when it is negative.**
Compute the discriminant for several quadratics and count the roots.

12.B7 **The gradient is r times the ratio of the standard deviations, so r fixes the sign and the gradient fixes the size, and they are equal only when both standard deviations are equal.**
Compute both for the same data set.

12.B8 **Factorising for rational roots, the quadratic formula always, completing the square when the vertex is also needed.**
Solve the same equation by all three methods and compare.

12.B9 **Two when the discriminant is positive, one when it is zero, none when it is negative.**
Compute the discriminant for several quadratics and count the roots.

12.B10 **The area is half the absolute value of the determinant of the two side vectors, because the determinant is the area of the parallelogram they span.**
Compute both for the same triangle.

12.B11 **The gradient is r times the ratio of the standard deviations, so r fixes the sign and the gradient the size.**
Compute both for the same data set.

CHAPTER 12 · CRITERION C · COMMUNICATING

12.C1 **$x = (3 \pm \sqrt{(9 + 32)})/4 = (3 \pm \sqrt{41})/4$, so $x \approx 2.35$ or $x \approx -0.851$.**
The quadratic formula, then check by substitution.

12.C2 **The annual factor is $2^{(1/8)} \approx 1.0905$, so the annual rate is about 9.05 per cent.**
Write the model, extract the annual factor, and convert to a percentage.

12.C3 **The third side is $\sqrt{(49 + 81 - 126 \cos 68°)} \approx \sqrt{82.8} \approx 9.10$; the area is $\tfrac{1}{2} \times 7 \times 9 \times \sin 68° \approx 29.2$.**
Cosine rule for the side, area formula for the area.

12.C4 **Out of 400 000 people: 960 true positives and 11 997 false positives, so $P = 960/12\,957 \approx 0.0741$.**
Scale to whole numbers and divide by the total positives.

12.C5 **$f^{-1}(x) = (3x + 2)/(5 - x)$, with domain $x \neq 5$.**
Clear the fraction, rearrange, swap, and state the new denominator.

12.C6 **The vertices are (0, 0), (6, 0), (4, 4) and (0, 6), giving 0, 30, 36 and 24; the maximum is 36 at (4, 4).**
Evaluate the objective at each vertex.

12.C7 **The determinant is $4(5) - (-1)(2) = 22$, so the area is 22.**
The determinant is the signed area.

12.C8 **By factorising, $(2x - 1)(x + 3) = 0$ gives $x = 0.5$ or $x = -3$; by the formula, $x = (-5 \pm \sqrt{(25 + 24)})/4 = (-5 \pm 7)/4$, the same.**
Use two independent routes to the same answer.

12.C9 **The third side is $\sqrt{(100 + 169 - 260 \cos 55°)} \approx \sqrt{119.9} \approx 10.95$; the area is $\tfrac{1}{2} \times 10 \times 13 \times \sin 55° \approx 53.2$.**
Cosine rule for the side, area formula for the area.

12.C10 **Out of 10 000 people: 180 true positives and 784 false positives, so $P = 180/964 \approx 0.187$.**
Scale to whole numbers and divide by the total positives.

12.C11 **$3(1.05^{20} - 1)/(1.05 - 1) \approx 3 \times 33.066 \approx 99.2$.**
Use the geometric sum formula.

12.C12 **The vertices are (0, 0), (5, 0), (16/5, 27/5) and (0, 7), giving 0, 15, 36.6 and 35; the maximum is 36.6 at (3.2, 5.4).**
Evaluate the objective at each vertex.

CHAPTER 12 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

12.D1 **Doubling time ln 2 ÷ ln 1.06 ≈ 11.9 years; after 15 years P ≈ 5000 × 2.397 ≈ 11 985.**
Use the doubling formula, then evaluate.

12.D2 **$R = 9000 \div ((1 - 1.08^{-4})/(1 - 1.08^{-1})) \approx 9000 \div 3.3121 \approx 2717$; the total repaid is about 10 868.**
Equate the present value of the instalments to the loan.

12.D3 **The centres agree; school B's results are nearly four times as spread, so its students' outcomes differ far more. The comparison cannot show why, because it describes the samples only.**
Compare centre and dispersion, then state the limit.

12.D4 **The space diagonal is $\sqrt{(25 + 144 + 169)} = \sqrt{338} \approx 18.38$; the angle has cosine $13/\sqrt{338}$, so it is about 45.1°.**
Two applications of Pythagoras, then a cosine ratio.

12.D5 **P(both red) = (5/8)(4/7) = 20/56 ≈ 0.357; P(one of each) = 2 × (5/8)(3/7) = 30/56 ≈ 0.536.**
Multiply the conditional probabilities, counting both orders.

12.D6 **2^(1/8) − 1 ≈ 0.0905, about 9.05 per cent per year.**
Write the model, extract the annual factor, and convert to a percentage.

12.D7 **The method marks are lost, and only the answer mark is gained, so the score is a fraction of what the question was worth.**
Marks are awarded for the method, the answer and the reasoning.

12.D8 **Poor: time should be allocated by marks, so the 4-mark question warranted about 6 minutes and the 12-mark question the remainder, and an unanswered question scores nothing.**
Allocate time in proportion to the marks.

12.D9 **Incomplete: the examples suggest the pattern but do not prove it, and a justification covering every case is needed.**
Examples suggest; an argument proves.

12.D10 **By determinants: the side vectors are (4, 1) and (1, 5), with determinant 19, so the area is 9.5. By base and height: the base is $\sqrt{17} \approx 4.123$ and the height is $19/\sqrt{17} \approx 4.609$, giving the same 9.5.**
Two independent routes to the same value.

12.D11 **No: a model that fits its data exactly may be over-fitted, and it must be tested against data not used in the fit before any claim is made.**
Distinguish fitting data from test data.

12.D12 **The ambiguous case was not checked: compare the given side with the perpendicular height, and report zero, one or two triangles accordingly.**
The comparison is made before any angle is computed.

Formula bank (1 of 5)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

NUMBER, ACCURACY AND FORM

Index laws

$a^M \times a^N = a^{M+N}$; $a^M \div a^N = a^{M-N}$; $(a^M)^N = a^{MN}$, for $a \neq 0$.

Zero and negative indices

$a^0 = 1$ for $a \neq 0$; $a^{-N} = 1/a^N$.

Rational index

a^(1/n) is the nth root of a; a^(m/n) = $({}^N\sqrt{a})^M$.

Standard form

$a \times 10^N$ with $1 \leq a < 10$.

Surd rules

$\sqrt{a} \times \sqrt{b} = \sqrt{(ab)}$; $\sqrt{a} \div \sqrt{b} = \sqrt{(a/b)}$; $(\sqrt{a})^2 = a$.

Rationalising

multiply by the conjugate so that the denominator becomes rational.

Error interval

a value given to the nearest unit lies within half a unit either side.

Bounds of a calculation

the lower bound uses the lower bounds of subtracted terms.

Percentage multiplier

an increase of r% multiplies by 1 + r/100; a decrease by 1 − r/100.

Logarithm laws

log(ab) = log a + log b; log(a/b) = log a − log b; $\log(a^K) = k \log a$.

Change of base

log_b a = log a ÷ log b, for any common base.

Continuous growth

$N = N_0 e^{KT}$; the equivalent base is $b = e^K$ and the rate is k = ln b.

Formula bank (2 of 5)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

ALGEBRA AND EQUATIONS

Expanding

$a(b + c) = ab + ac$.

Difference of squares

$a^2 - b^2 = (a - b)(a + b)$.

Perfect square

$a^2 + 2ab + b^2 = (a + b)^2$; $a^2 - 2ab + b^2 = (a - b)^2$.

Quadratic equation

$ax^2 + bx + c = 0$ with $a \neq 0$.

Quadratic formula

$x = (-b \pm \sqrt{(b^2 - 4ac)}) \div 2a$.

Discriminant

$b^2 - 4ac > 0$ gives two real roots, $= 0$ one repeated root, < 0 none.

Completing the square

$ax^2 + bx + c = a(x + b/2a)^2 + c - b^2/4a$.

Algebraic fraction

cancel only common factors, and state the excluded values first.

Quadratic inequality

solve the equation, mark the roots on a sign line, then test one interval.

Inequality rule

multiplying or dividing by a negative number reverses the sign.

Function

a rule giving exactly one output for each input of its domain.

Composite function

$f(g(x))$ applies g first; the inner function acts first.

Inverse function

f^{-1} exists only when f is one-to-one; $f^{-1}(f(x)) = x$ on the restricted domain.

Formula bank (3 of 5)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

SEQUENCES, SERIES AND TRIGONOMETRY

Arithmetic sequence

the nth term is $u(n) = u_1 + (n - 1)d$, with common difference d.

Arithmetic series

the sum of n terms is $S(n) = n/2\ (u_1 + u(n)) = n/2\ (2u_1 + (n - 1)d)$.

Geometric sequence

the nth term is $u(n) = u_1\ r^{N-1}$, with common ratio r.

Geometric series

the sum of n terms is $S(n) = u_1(r^N - 1) \div (r - 1)$ for $r \neq 1$.

Convergent geometric series

the sum to infinity is $u_1 \div (1 - r)$ when $|r| < 1$, and no sum otherwise.

Annuity

the accumulated value of equal regular payments, found by summing a geometric series.

Pythagoras

$a^2 + b^2 = c^2$ for the hypotenuse c of a right-angled triangle.

Trigonometric ratios

$\sin\theta$ = opposite/hypotenuse; $\cos\theta$ = adjacent/hypotenuse; $\tan\theta$ = opposite/adjacent.

Sine rule (sides)

$a/\sin A = b/\sin B = c/\sin C$.

Sine rule (angles)

$\sin A/a = \sin B/b = \sin C/c$.

Cosine rule

$a^2 = b^2 + c^2 - 2bc\cos A$.

Area of a triangle

$\frac{1}{2}ab\sin C$, where C is the angle between sides a and b.

Ambiguous SSA case

$b\sin A < a < b$ gives two triangles; $b\sin A = a$ gives one right triangle; $a < b\sin A$ gives none.

Bearings

measured clockwise from north, written as three figures.

Space diagonal

$d = \sqrt{(l^2 + w^2 + h^2)}$ for a cuboid of length l, width w and height h.

Prism

volume = area of cross-section × length.

Cylinder

volume = $\pi r^2 h$; curved surface area = $2\pi rh$.

Cone

volume = $\frac{1}{3}\pi r^2 h$; curved surface area = πrl for slant height l.

Sphere

volume = $\frac{4}{3}\pi r^3$; surface area = $4\pi r^2$.

Formula bank (4 of 5)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

VECTORS AND MATRICES

Magnitude of a vector

$|a| = \sqrt{(x^2 + y^2)}$ for $a = (x, y)$.

Vector addition

$a + b = (x_1 + x_2, y_1 + y_2)$.

Scalar multiple

$ka = (kx, ky)$.

Position vector

the vector from the origin to the point.

Matrix product

$(AB)_{ij}$ is the dot product of row i of A with column j of B.

Determinant of a 2 × 2 matrix

$\det [[a, b], [c, d]] = ad - bc$.

Inverse of a 2 × 2 matrix

$A^{-1} = 1/(ad - bc)\ [[d, -b], [-c, a]]$, when $ad - bc \neq 0$.

Area scale factor

$|\det M|$ multiplies every area under the transformation M.

Formula bank (5 of 5)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

STATISTICS AND PROBABILITY

Mean

$\bar{x} = \Sigma x \div n$ for ungrouped data; for a frequency table use $\Sigma fx \div \Sigma f$.

Variance (population)

the mean of $\Sigma(x - \bar{x})^2 \div n$.

Standard deviation

the square root of the mean squared deviation; a sample divides by $n - 1$.

Grouped estimate

use the mid-interval value of each class as its representative.

Interquartile range

$Q3 - Q1$, a measure of spread that ignores the extremes.

Correlation coefficient

$r = Sxy \div \sqrt{(Sxx \times Syy)}$, where $Sxx = \Sigma(x - \bar{x})^2$.

Least-squares line

$y = a + bx$ with $b = Sxy \div Sxx$ and $a = \bar{y} - b\,\bar{x}$.

Coefficient of determination

r^2 is the proportion of the variation in y explained by x.

Probability of an event

$P(\text{event}) =$ favourable outcomes ÷ total equally likely outcomes.

Addition rule

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.

Conditional probability

$P(A \text{ given } B) = P(A \text{ and } B) \div P(B)$.

Independence

$P(A \text{ and } B) = P(A) \times P(B)$, equivalently $P(A \text{ given } B) = P(A)$.

Total probability

$P(B) =$ the sum, over each part A of a partition of the sample space, of $P(A) \times P(B \text{ given } A)$.

Bayes' theorem

$P(A \mid B) = P(B \mid A)P(A) \div P(B)$.

Multiplication principle

r successive choices with $n_1, n_2, \ldots$ options give $n_1 \times n_2 \times \ldots$ outcomes.

Permutations

$nPr = n! \div (n - r)!$ counts ordered selections.

Combinations

$nCr = n! \div (r!(n - r)!)$ counts unordered selections.

Expected value

$E(X) = \Sigma\, x\, P(X = x)$ for a discrete random variable.

Glossary (1 of 3)

('Algebraic fraction', 'A fraction whose numerator and denominator are algebraic expressions.', 1) ch. 1
A fraction whose numerator and denominator are algebraic expressions.

('Alternate segment theorem', 'The angle between a tangent and a chord equals the angle in the alternate segment.', 6) ch. 6
The angle between a tangent and a chord equals the angle in the alternate segment.

('Ambiguous case', 'The SSA arrangement, which can produce zero, one or two triangles depending on the lengths.', 7) ch. 7
The SSA arrangement, which can produce zero, one or two triangles depending on the lengths.

('Annuity', 'The accumulated value of equal regular payments, found by summing a geometric series.', 4) ch. 4
The accumulated value of equal regular payments, found by summing a geometric series.

('Arithmetic sequence', 'A sequence with a constant difference between consecutive terms.', 4) ch. 4
A sequence with a constant difference between consecutive terms.

("Bayes' theorem", 'A rule for reversing a conditional probability using the total probability.', 10) ch. 10
A rule for reversing a conditional probability using the total probability.

('Bearing', 'A direction measured clockwise from north, written as three figures.', 7) ch. 7
A direction measured clockwise from north, written as three figures.

('Binomial distribution', 'The distribution of the number of successes in a fixed number of independent trials.', 10) ch. 10
The distribution of the number of successes in a fixed number of independent trials.

('Bisector', 'A line that divides an angle or a segment into two equal parts.', 6) ch. 6
A line that divides an angle or a segment into two equal parts.

('Combination', 'An unordered selection; the number of them is n! ÷ (r!(n − r)!).', 10) ch. 10
An unordered selection; the number of them is n! ÷ (r!(n − r)!).

('Common ratio', 'The constant multiplier between consecutive terms of a geometric sequence.', 4) ch. 4
The constant multiplier between consecutive terms of a geometric sequence.

('Complement', 'The outcomes not in an event; P(not A) = 1 − P(A).', 10) ch. 10
The outcomes not in an event; P(not A) = 1 − P(A).

('Composite function', 'The function formed by applying one function to the result of another.', 2) ch. 2
The function formed by applying one function to the result of another.

('Conditional probability', 'The probability of an event given that another event has occurred.', 10) ch. 10
The probability of an event given that another event has occurred.

('Congruent', 'Identical in shape and size; for triangles, SSS, SAS, ASA or RHS proves it.', 6) ch. 6
Identical in shape and size; for triangles, SSS, SAS, ASA or RHS proves it.

('Conjecture', 'A mathematical claim believed to be true but not yet justified.', 12) ch. 12
A mathematical claim believed to be true but not yet justified.

('Conjugate', 'The pair a + √b and a − √b, whose product is rational.', 1) ch. 1
The pair a + √b and a − √b, whose product is rational.

('Continuous growth', 'Change modelled by $N = N_0e^{KT}$, where k is the continuous rate.', 3) ch. 3
Change modelled by $N = N_0e^{KT}$, where k is the continuous rate.

('Convergent series', 'A geometric series whose partial sums approach a finite limit, when |r| < 1.', 4) ch. 4
A geometric series whose partial sums approach a finite limit, when |r| < 1.

('Correlation', 'The strength and direction of a linear association between two variables.', 9) ch. 9
The strength and direction of a linear association between two variables.

('Correlation coefficient', 'r = Sxy ÷ √(Sxx × Syy), between −1 and 1.', 9) ch. 9
r = Sxy ÷ √(Sxx × Syy), between −1 and 1.

('Cosecant', 'The reciprocal of sine: 1 ÷ sin θ.', 7) ch. 7
The reciprocal of sine: 1 ÷ sin θ.

('Cosine rule', 'a² = b² + c² − 2bc cos A, relating three sides and one angle of any triangle.', 7) ch. 7
$a^2 = b^2 + c^2 − 2bc \cos A$, relating three sides and one angle of any triangle.

('Cyclic quadrilateral', 'A quadrilateral whose four vertices lie on one circle.', 6) ch. 6
A quadrilateral whose four vertices lie on one circle.

Glossary (2 of 3)

('Dependent events', 'Events where the outcome of one changes the probability of the other.', 10) ch. 10
Events where the outcome of one changes the probability of the other.

('Depreciation', 'A loss of value, modelled by repeated multiplication by a factor below 1.', 4) ch. 4
A loss of value, modelled by repeated multiplication by a factor below 1.

('Determinant', 'The number ad − bc for a 2 × 2 matrix; it is the area scale factor of the transformation.', 8) ch. 8
The number ad − bc for a 2 × 2 matrix; it is the area scale factor of the transformation.

('Dispersion', 'How spread out a data set is, measured by range, interquartile range or standard deviation.', 9) ch. 9
How spread out a data set is, measured by range, interquartile range or standard deviation.

('Displacement vector', 'A vector describing a change of position, with magnitude and direction.', 8) ch. 8
A vector describing a change of position, with magnitude and direction.

('Domain', 'The set of permitted inputs of a function.', 2) ch. 2
The set of permitted inputs of a function.

('Error interval', 'The interval of values that would round to a stated value.', 1) ch. 1
The interval of values that would round to a stated value.

('Expected value', 'The long-run average of a discrete random variable, Σ x P(X = x).', 10) ch. 10
The long-run average of a discrete random variable, $\Sigma\, x\, P(X = x)$.

('Factor theorem', 'x − a is a factor of p(x) exactly when p(a) = 0.', 1) ch. 1
$x - a$ is a factor of $p(x)$ exactly when $p(a) = 0$.

('Feasible region', 'The set of points satisfying every constraint of a linear programming problem.', 5) ch. 5
The set of points satisfying every constraint of a linear programming problem.

('Function', 'A rule that gives exactly one output for each permitted input.', 2) ch. 2
A rule that gives exactly one output for each permitted input.

('Geometric sequence', 'A sequence with a constant ratio between consecutive terms.', 4) ch. 4
A sequence with a constant ratio between consecutive terms.

('Grouped data', 'Data given in class intervals; the mid-interval value represents each class.', 9) ch. 9
Data given in class intervals; the mid-interval value represents each class.

('Independent events', 'Events where neither changes the probability of the other.', 10) ch. 10
Events where neither changes the probability of the other.

('Index', 'The power to which a base is raised; plural indices.', 1) ch. 1
The power to which a base is raised; plural indices.

('Interquartile range', 'The difference between the upper and lower quartiles.', 9) ch. 9
The difference between the upper and lower quartiles.

('Intersection', 'The set of outcomes in both of two sets; for events, A and B.', 10) ch. 10
The set of outcomes in both of two sets; for events, A and B.

('Inverse function', 'The function that undoes another; f^{-1} exists only when f is one-to-one.', 2) ch. 2
The function that undoes another; f^{-1} exists only when f is one-to-one.

('Least-squares line', 'The line y = a + bx that minimises the sum of the squared residuals.', 9) ch. 9
The line $y = a + bx$ that minimises the sum of the squared residuals.

('Linear programming', 'Choosing the best value of a linear objective subject to linear constraints.', 5) ch. 5
Choosing the best value of a linear objective subject to linear constraints.

('Logarithm', 'The exponent to which a base must be raised to give a number.', 3) ch. 3
The exponent to which a base must be raised to give a number.

('Magnitude', 'The length of a vector, found from its components by Pythagoras.', 8) ch. 8
The length of a vector, found from its components by Pythagoras.

('Matrix', 'A rectangular array of numbers that can represent a transformation of the plane.', 8) ch. 8
A rectangular array of numbers that can represent a transformation of the plane.

('Modulus', 'The distance of a complex quantity from the origin; for a vector, its magnitude.', 8) ch. 8
The distance of a complex quantity from the origin; for a vector, its magnitude.

Glossary (3 of 3)

('Multiplication principle', 'The number of outcomes of successive independent choices is the product of the counts.', 10) ch. 10
The number of outcomes of successive independent choices is the product of the counts.

('Mutually exclusive', 'Events that cannot both happen, so P(A or B) = P(A) + P(B).', 10) ch. 10
Events that cannot both happen, so P(A or B) = P(A) + P(B).

('Permutation', 'An ordered arrangement; the number of them is n! ÷ (n − r)!.', 10) ch. 10
An ordered arrangement; the number of them is n! ÷ (n − r)!.

('Position vector', 'The vector from the origin to a point.', 8) ch. 8
The vector from the origin to a point.

('Quadratic inequality', 'An inequality of the form $ax^2 + bx + c > 0$, solved with a sign line.', 5) ch. 5
An inequality of the form $ax^2 + bx + c > 0$, solved with a sign line.

('Range (of a function)', 'The set of outputs a function can produce.', 2) ch. 2
The set of outputs a function can produce.

('Regression', 'Predicting one variable from another using the line of best fit.', 9) ch. 9
Predicting one variable from another using the line of best fit.

('Remainder theorem', 'The remainder when p(x) is divided by x − a is p(a).', 1) ch. 1
The remainder when p(x) is divided by x − a is p(a).

('Residual', 'The difference between an observed value and the value predicted by a model.', 9) ch. 9
The difference between an observed value and the value predicted by a model.

('Sample space', 'The set of all possible outcomes of an experiment.', 10) ch. 10
The set of all possible outcomes of an experiment.

('Secant', 'A line that cuts a circle in two points.', 6) ch. 6
A line that cuts a circle in two points.

('Sine rule', 'a/sin A = b/sin B = c/sin C, relating sides to opposite angles in any triangle.', 7) ch. 7
a/sin A = b/sin B = c/sin C, relating sides to opposite angles in any triangle.

('Standard deviation', "The square root of the variance; a measure of spread in the data's units.", 9) ch. 9
The square root of the variance; a measure of spread in the data's units.

('Tangent', 'A line that touches a circle in exactly one point, at right angles to the radius.', 6) ch. 6
A line that touches a circle in exactly one point, at right angles to the radius.

('Total probability', 'the sum, over each part A of a partition of the sample space, of P(A) × P(B given A)', 10) ch. 10
the sum, over each part A of a partition of the sample space, of P(A) × P(B given A)

('Transformation matrix', 'The matrix whose product with a position vector gives its image.', 8) ch. 8
The matrix whose product with a position vector gives its image.

('Tree diagram', 'A branching diagram showing successive outcomes and their probabilities.', 10) ch. 10
A branching diagram showing successive outcomes and their probabilities.

('Union', 'The set of outcomes in either of two sets; for events, A or B.', 10) ch. 10
The set of outcomes in either of two sets; for events, A or B.

('Variance', 'The mean of the squared deviations from the mean.', 9) ch. 9
The mean of the squared deviations from the mean.

('Vertex', 'The turning point of a parabola, written (h, k) in vertex form; also a corner of a feasible region.', 5) ch. 5
The turning point of a parabola, written (h, k) in vertex form; also a corner of a feasible region.

('y-intercept', 'The value of y where a line crosses the y-axis.', 2) ch. 2
The value of y where a line crosses the y-axis.

REFERENCES

References and acknowledgements

OFFICIAL IB SOURCES

- IB Middle Years Programme curriculum — https://www.ibo.org/programmes/middle-years-programme/curriculum/ (page states last updated 19 February 2026). Used for: at least 50 hours per subject per year, global contexts, conceptual understanding, approaches to learning and service as action.
- IB MYP mathematics subject page — https://www.ibo.org/programmes/middle-years-programme/curriculum/mathematics/ Used for: the four branches of MYP mathematics, standard and extended mathematics, and the inquiry and application focus.
- IB MYP projects — https://www.ibo.org/programmes/middle-years-programme/curriculum/myp-projects/ (page states last updated 19 February 2026). Used for: the Community Project completed by students finishing MYP in Year 3 or Year 4, the Personal Project completed by Year 5 students, and the student-centred nature of both.
- IB MYP general regulations (2026 assessment procedures) — https://www.ibo.org/globalassets/new-structure/become-an-ib-school/pdfs/myp-2026-general-regulations.pdf Used for: the required use of MYP objectives and assessment criteria, minimum teaching hours, and Year 5 eAssessment eligibility.

PRINCIPAL CONTENT REFERENCE

- Pre-rewrite MYP 5 chapter and subtopic audit (second-edition/audit/chapter-audits/myp5-audit.md with the chapter and subtopic register, restated for convenience in second-edition/audit/chapter-audits/myp5-register.md). Used as the principal content reference for chapter scope, section order and the key concepts, related concepts, global contexts and inquiry statements of each chapter. The audit's approved decisions are carried out: the sine-rule ambiguous case is corrected to $b \sin A < a < b$ with zero, one and two-triangle cases stated, the raw LaTeX that appeared in the earlier Chapter 11 text is removed, and the Chapter 12 doubling-rate item is corrected to about 9.05 per cent.

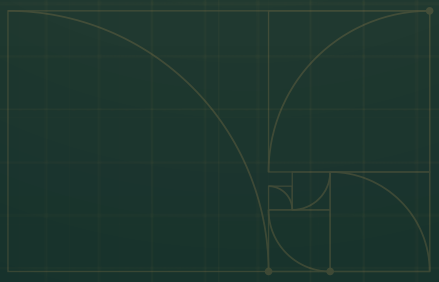
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- The chapter sequence, investigations, worked examples, practice sets, assessment tasks, project connections, formula bank, glossary and this edition's editorial framework are author-designed material written to serve the programme requirements above. They are not IB publications and carry no IB endorsement.
- All classroom data sets are simulated and are labelled in the text. No claim in this book relies on unverified real-world data.

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ABOUT THIS BOOK

Twelve chapters that take polynomials and algebraic fractions, functions and their inverses, exponential and logarithmic models, sequences and series, quadratic inequalities and linear programming, circle theorems and proof, the sine and cosine rules, vectors and matrices, dispersion and regression, and conditional probability and counting from first ideas to justified use. Each chapter opens with a task: you investigate, notice a pattern, state a mathematical conclusion you can defend, and only then meet the formal explanation — with worked examples that model a complete solution and practice separated into Criterion A, B, C and D sets so you can see which skill each task trains.



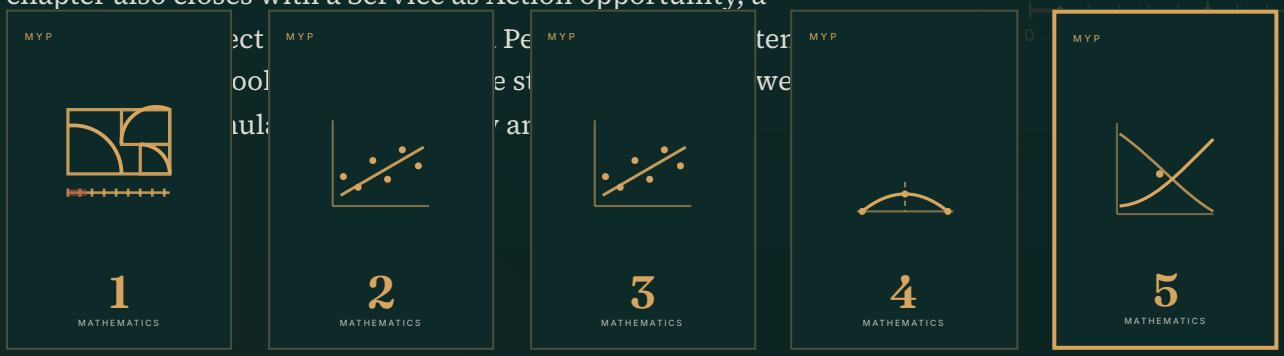
Year 5 reasoning is applied, not just correct: models are built from stated assumptions, checked against data and evaluated for their limitations, and Criterion B tasks ask you to justify a generalisation rather than to list cases. Every chapter closes with a reflection that returns to the statement of inquiry.



The book closes with a capstone: a sustained modelling task in which a claim about a shared resource becomes defensible only when it is modelled, tested and evaluated — and with a chapter that states plainly what the transition to DP Mathematics expects. Every chapter also closes with a Service as Action opportunity, a



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